

Nature and Spirit

A Contribution to a Philosophy of Nature in Accord with Physics

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Introduction.

The influence of the science of nature upon the cultural life of the present day is perceptible in every direction. To give a description of the great practical results to which the discoveries of the latest age have led is superfluous: steamships, railways, telegraphs are signs so eloquent that men from the youth of Napoleon the First, were they to wake in their graves and look up, would believe themselves set down in a world of wonders. A science that reaches so mightily into all actual relations cannot, of course, confine itself to serving merely finite interests and promoting what is useful in the civil sense; its origin and its goal are of a higher kind, it wills a knowledge of truth: its significance is essentially theoretical. It is natural science that has opened men's eyes to what the wisest of antiquity still saw only as through mists — the laws of nature, the connection of things, the secret kinship of the forces. The visible universe has become another, for the illumination has become another. The arrangement of the solar system, the development of the terrestrial globe, the natural descent of the living genera and species, all this presents itself in broad strokes, with distinct outlines, in a new light, reality's own daylight. There are, to be sure, still those who harbour a secret mistrust of the reliability of research, but there is no one able to withdraw himself from its influence. Those who hold to the old strive against it in vain; the new picture of the world is actually there; it is glimpsed in everything, they meet it everywhere, they see it even when they close their eyes.

But precisely the circumstance that the natural sciences together make up a spiritual great power, to whose authority, where real insight into real things is at issue, the civilized

world so willingly bows, invites us to consider whether the common consciousness does not at the same time attach to it an unwarranted afterthought, a misunderstanding that may have grave consequences both for science itself and for life.

It is then, in the first place, a question whether the science of nature, so long as it keeps within the bounds of experience, can own as its task: to solve the riddle of existence by fathoming what nature is in itself, whether it has any origin and temporal beginning or exists from eternity, whether it is matter that brings forth Spirit or the reverse. If the science of experience, by its character and method, neither can nor will concern itself with the solution of such tasks, then it is surely put in a false position when one ascribes to it an endeavour it does not have, and sets it up as judge over what it cannot judge. Next, it is doubtful which of the many mutually combating world-views — materialism, pantheism, atheism, nihilism, etc., all of which appeal to “Science” — can properly be said to have the full support of the scientific study of nature. It is well known that the investigators of nature themselves, where “the theories” are concerned, express themselves in different spirits and proceed from highly different hypotheses. Even though an investigator with a famous name holds it probable that man is a modified ape, such a way of conceiving may not on that account be confused, without more ado, with an infallible fact and exalted into a dogma. In assumptions of this kind a distinction must surely be made between the construing of the facts and the facts themselves, between the investigator’s individual opinion and his scientific proofs. The science of nature leans upon hypotheses; but it treats its hypotheses critically, for it knows that their nature is ambiguous. Every hypothesis has, so to speak, two faces: one severe and earnest, turned toward the facts and aiming unswervingly at the discovery of their hidden connection; another smiling, more ingratiating, which addresses the general public and would win itself a place among its favourite doctrines.

From whichever side one regards the matter, it is as impossible by the way of hypotheses as by continued observation of facts — with whose infinite multiplicity research never comes to an end — to get clear about those fundamental relations to which all knowledge of nature must essentially be led back; the investigation of these fundamental relations, so far as it can be carried through on the terms of human knowing, falls not under the science of experience but under the philosophy of nature.

The philosophy of nature is indeed dependent on the science of experience, inasmuch as it must use the yield of the observations and keep as exactly as possible in the track of research; but it has for all that its own tasks, its own method, its own goal, and, in the knowing consciousness, its own standpoint.

It is difficult in few words to give a generally intelligible account of the course and the essence of the philosophy of nature; a closer understanding depends on its being seen in the doing how its work is carried out. So much can be said in advance, that it is to build neither upon hypotheses nor upon abstract speculations.

As concerns the present contribution, the following propositions may serve as an

introduction. The collected impression of all contemplation of nature is an impression of power: nature is a realm of power. The eternal power that reveals itself in the order of nature is visible and invisible, visible in corporeal things, invisible in the concepts of things. From the sensuous side power shows itself as object-power, a visible power in the series of the phenomena, in a concatenation of members of which the one incessantly presupposes the other and points to the other, thus as otherness-power, outward power, *power in another*. From the non-sensuous side, by contrast, power shows itself as power in the inward, as the invisible power of thoughts, of concepts, of wisdom, as the power of ideas, as *self-power*. Out of this doubling in the essence of power the philosophy of nature develops two fundamental concepts, rich in content and decisive for all existence: out of power in another the concept *nature*, out of the essence of self-power the concept *Spirit*.

It goes without saying that the philosophy of nature does not set out to exhaust the doctrine of Spirit; it does not consider nature for the sake of Spirit, but conversely, Spirit for the sake of nature. The science of experience presents nature in nature's own light; the philosophy of nature presents it in the light of Spirit. Will the altered illumination be able to open to us new insights into the mechanical, elemental and organic spheres of nature? That must depend upon an attempt.

Part One.

Mechanical Nature.

Chapter One.

Foundational Concepts.

A.

Space and Time.

Since all outward things exist in space and could not possibly exist if space were not; since things arise and pass away in time and could not possibly either arise or pass away were there no time — it surely follows from this that space and time must

belong to the essence of natural things, that they must, in other words, be regarded as the first universal, all-embracing forms of nature. But although space and time are universally valid forms of nature, it will nevertheless appear in what follows that the natural subsistence of these forms has a spiritual ground; that they are object-forms only by being selfhood-forms, objective only by being objectified by the eternal Spirit.

a) *Space*. The most important contributions to a deeper understanding of the nature of space one will, in more recent philosophy, chiefly seek in Kant and Hegel; but the two ways of seeing must be treated critically, for, viewed superficially, the one contradicts the other.

Kant places himself, as is well known, upon an unconditionally subjective standpoint and regards space as an a priori form of sense, a merely human form of consciousness. Space is no concept borrowed from experience; in order to be able to regard things in space, we must ourselves bring the intuition of space with us. One cannot possibly represent to oneself that there is no space; but one can very well represent to oneself that there are no objects in space. When there is talk of many spaces, this is not to be understood as though there were talk of parts out of which a whole was composed: the many parts flow together into a single space given with our necessary intuition of sense. From this, then, the conclusion is drawn that it is not the thing in itself but only the phenomenon of the thing, that is, the thing as a thing of sight, that is apprehended in space; which may also be expressed thus, that *space is only a selfhood-form, but no otherness-form valid in itself*.

In sharp and decisive opposition to Kant, Hegel stands upon an absolutely objective standpoint. For him space is, in its “abstract universality,” the first form of nature, the immediately necessary form of *outward being*. In order to make clear what lies in the concept of space’s “ideal being-beside-one-another,” “being-outside-one-another,” and so forth, he chooses to show how space looks in our representation. In order to prove that space corresponds to our thought, we must, he says, compare our representations of space with the determinations that lie in our concept. The many *Heres* and *Theres* in space are beside one another; the one *Here* is completely the same as the other, and this abstract plurality — without true interruption and limit — is precisely outwardness. Space with its innumerable points is sheer discreteness; but this discreteness is again a flowing unity, unity in “outside-one-another,” complete continuity. The point has significance only insofar as it occupies a space, and so is an outward over against itself and over against another; in the point itself there is then again an above, a below, a right, a left, for a last *Here* is not given. Beyond its limit space is constantly “at home with itself”: the continuous is discrete, the discrete continuous. The unity of both moments, discreteness and continuity, is “the objectively determined concept of space.” On this way of seeing it is then so far from being the case that objective space should rest upon any subjective intuition whatever, that on the contrary, as “a non-sensible sensibility” resting upon its own concept, it furnishes the foundation of natural things: *space is*

otherness-form without being selfhood-form.

By holding the two conceptions sharply against one another one will easily be convinced that both are one-sided.

That Kant rightly makes the form of space a human form of consciousness can be seen expressly from Hegel's demonstration; for this demonstration supports itself point by point upon human intuiting, human thinking and knowing. Our insight into the validity of objective space depends absolutely upon its validity for us. Space is determined as the "*abstract*" form of outward being; but it is we — not nature — who abstract the form. The many different "*Heres*" and "*Theres*" must be posited and cancelled; but it is we — not nature — who posit them and cancel them. The unity of "discreteness" and "continuity" is to think in difference, and the difference again in unity; but it is we — not nature — who must think this unity of difference: for space to show itself as objective concept, it is we who must conceive.

But that Hegel is nonetheless right in making space an objective concept independent of us becomes recognizable, conversely, from the Kantian demonstration. Kant holds fast that the phenomena, the sensible things, cannot possibly be apprehended without space; herewith he has then at the same time conceded that space is a necessary, objective form for the things of sight. Things in nature do not, after all, become sensible by our sensing them; rather we sense them because they are in themselves sensible. As little as it is our sight that makes things visible, or our thinking that makes them thinkable, just as little is it our intuition of space that gives them existence in space. If the partition wall Kant has raised between the essence of things in themselves and their phenomena were more than a critical fancy, then it would be altogether unthinkable that this unknown essence, incompatible with space, should be able to furnish any content of experience and to fill the empty intuition from without with actual objects.

Thus: every time the form of space is made exclusively an otherness-form, it transforms itself and becomes a selfhood-form; and when it is held fast exclusively as a selfhood-form, it transforms itself again and becomes a form for another. This shows plainly enough that space could not possibly be a truly real form of nature if this form of nature did not from eternity receive objective being in divine intuiting. The Kantian error does not consist in his making space a form of intuition, but in his arbitrarily restricting the form of space to being a merely human selfhood-form, and precisely thereby coming to deny its objective validity. Hegel's fault is not that he asserts the objectivity of space; the fundamental fault is rather that in his philosophy of nature, as in his logic, he constantly overlooks that nothing objective exists except what is objectified, and that no objectification is possible without an objectifying self. That human consciousness ascribes outward being to space in nature rests at bottom upon the tacit presupposition that the world-space which we in our contemplation of nature take to be objective must from the beginning be — objectified by the creating Spirit. The starry heaven makes a symbolic impression; it is human intuiting that, in its self-knowing, is met in the infinity

of space by the divine, as eye meets eye.

How matters stand more closely, as regards the objectification of space, with the reciprocal determination of selfhood and otherness can be seen by a consideration of the dimensions of space. Length, breadth and height are in space different and yet not different. A distinction must necessarily be made between the principal directions just named; for without such a distinction there would be no determinations in the intuition, and then no objectively determined space either. But upon what does the difference rest? Length is height, and height is again breadth: the one dimension is what the other is, and can be exchanged for the other. They are posited simultaneously and cancelled simultaneously; they presuppose one another to infinity. From this it is seen that the three dimensions, in their outward reciprocity, are double-sided determinations, determinations of selfhood and determinations for another. As determinations of selfhood they are arbitrary, as determinations of otherness accidental, as objectified — and so objective — determinations of reciprocity mutually necessary. If we choose, e.g., a rectangular coordinate system with the axes X , Y and Z , the objectification is easy to illustrate. The three axes must have a point of intersection. Now where is this point of intersection in space itself? Everywhere and nowhere. The point is objective only by being objectified. To be objectified, it must be posited; to be posited, it must be posited arbitrarily, and so by a self-determination. The arbitrariness is necessary; for whatever place we choose, we could just as well have chosen another place. By being posited the point has become objective; it has received a represented existence and now furnishes a determination for another, namely for the axes. In this objective determinateness the point is to be taken as given; now it is precisely here, not there. Its existence is accidental, for the axes could just as well have cut one another at any other point whatever. But the accidentality is necessary, for whatever point were given, another could with equal right have been given. What holds of the point of intersection holds in part of the axes. It is arbitrary to make Z the axis of height, X that of length and Y that of breadth; but once arbitrariness has given them this signification, then height, length and breadth, as determinations for one another mutually, and so as determinations of otherness, have thereby also obtained an objective, accidental existence. The accidentality in this existence is just as unavoidable, that is, just as necessary, as the arbitrariness. Neither the arbitrary nor the accidental does any damage to necessity; on the contrary, these foundational determinations may rightly be said to furnish presuppositions, indispensable conditions for an exact objectification of conditioned necessity. If a point M be assumed whose coordinates in the given system are x , y and z , and whose distance from the point of intersection may be designated by r , then an equation such as:

$$r^2 = x^2 + y^2 + z^2$$

will involuntarily recall that manifold of spatial determinations which, in their exact

necessity, are all conditioned — subjectively by arbitrary, objectively by accidental presuppositions.

The three coexisting axes are, in their outward, immediate accidentality, without relations to one another; only with knowing do the relations come, and with the relations necessity. Here, if we take all knowing away, there are not even three axes, for threeness demands that a knowing of the two unite with a knowing of the third. The representation of an abstract objective threeness derives from a co-knowing which the objectifying consciousness, without thinking of it, itself slips underneath. It is incontestably certain that the equation cited, answering to the axes, has objective validity — provided, be it noted, that it is objectified in knowing; but when there is no knowing, the equation is meaningless. *All objective relation is subjectively grounded in reflexion*; that is a law of objectification, as strict as any law of nature.

In the Schellingian philosophy of nature there is mystically profound talk of an intuiting dwelling within nature, an objective phantasy of nature, an unconscious knowing of nature, and the like. How groundless such turns of phrase are shows itself every time the gaze is fixed upon something determinate. Were the three dimensions of space posited by an unconscious knowing, then in this knowing there would have to be a co-knowing of all three, a knowing in which each single one of them was determined with respect to the two others. The unconscious knowing would then have to be *a knowing of respects*; here is the contradiction. A knowing of respects is a knowing that hovers over the single determinations, i.e., a self-knowing. In order that *X* can be determined with respect to *Y*, and *Y* with respect to *X*, the determining knowing must necessarily hover over both *X* and *Y*. Such a hovering is possible only when the same self that knows *X* also knows *Y*. But since knowing of another is, by the nature of the case, inseparable from self-knowing, profundities such as “an unconscious knowing of nature,” “a self-intuiting space” — on a par with “a self-thinking air,” “a self-conceiving matter” — are to be referred, not to “strict science,” but to the utopian speculations.

To determine space is to posit limits in space. Here, then, a qualitative difference shows itself between the limit, the negative, and the limited space, the positive. The negative is for thought, the positive for intuition. The first absolutely negative is the point. In the determination of the point a peculiar conflict arises between intuiting and thinking. Thought demands that the point be without extension; intuition demands that it have an extension, however small. If thought is not satisfied, the point becomes a surface, no point; if intuition is not satisfied, the point can occupy no place: a pure point of thought is no point in space. The resolution of the contradiction is a reflexion in which intuition and thought alternately cancel and confirm one another. Insofar as thought cancels intuition, the element of space, the positive, is consumed by the negative and posited as point. Insofar as intuition cancels thought, the negative is converted into the positive, and the point posited as element. For thought the point is the limit of the line, but for intuition it is the element of the line; for thought the line is the limit of the

surface, but for intuition the element of the surface; and so the surface too is again both the limit of the body and its element. The element of the line can be integrated into a line, that of the surface into a surface, that of the body into a body; but the limit itself, the absolutely negative, cannot be integrated.

As the negative, the non-being, the limit can therefore set no barrier to the being of space; there is space on both sides of the limit; the limit is itself the negative unity in which the limited are united and separated. Everywhere that thought posits a limit in space, intuition goes out beyond the limit and discovers a new space. Space is full of limits, for the limit is everywhere, and yet limitless, for the limit is everywhere overstepped. To exchange a limit that is overstepped for a new limit that is again overstepped — that is the secret: infinite space is objective only for an infinite objectification.

b) Time. The infinite extension of space is grounded in the infinite becoming of the limit. This becoming, which posits a difference between the vanished, the present and the coming limit, points to another form of nature, distinct from space: the succession hidden in extension belongs to time. Indifferent toward its limit, space is indifferent toward its becoming; the space that comes when the limit is overstepped was already there beforehand, and that which has been is there still; the difference is only a shadow of time. In time itself, by contrast, the difference between what has been, what is, and what will come is an essential, though as yet outward, difference.

That the form of time is originally a selfhood-form can be seen from the reflexion into which intuition is resolved. Time has only one dimension, *length*; but this is so broken in the tripartition of past, present and future that only a thinking that runs through them is able to hold the unity fast. The past exists in recollection, but recollection is an act of selfhood; the future exists in expectation, but expectation is an act of selfhood. The part of time that might be said to have an immediate being, independent of a recollecting and expecting subjectivity, would then be the present. But the present is, when we look more closely, just as reflected as past and future. If the present is apprehended as the absolute instant, then it is apprehended not as time but as limit of time, the limit in which past and future touch one another. This limit, the pure now, is a non-being, a mathematical zero point. If the present is apprehended as relative instant, as element of time, then it is a duration which intuition slips underneath, in ceaseless flight from pastness to futurity, i.e., from non-being to non-being. A time-element can indeed be apprehended as constant, $dt = dt$, when it is applied as a measure for an element of space; but the proposition “now is now” becomes a self-contradicting judgment, for the now of the subject is past when we come to that of the predicate. If the present be held fast as finite, existing time, then it is solely the recollection and foresight of subjectivity which, in reflected unity, are able to let intuition stretch itself along an arbitrary present length of time.

Hegel’s dialectic of what he calls “the dimensions of time” is, from the objective side, fine and sharp. It is becoming, whose abstract moments of unity are, each for

itself, posited as the whole, but under opposite determinations. The two determinations, past and future, are thus each of them a unity of being and nothing; but they are also different. This difference can only be that between arising and passing away. “The one time, in pastness (Hades), being is the foundation from which the start is made; the past has actually been, as world-history, as events of nature, but posited under the determination of non-being, which supervenes. The second time it is the reverse; in time to come non-being is the first determination, being the later, though not in point of time. The middle is the indifferent unity of both,” and so forth. That is excellent. But in glaring contradiction with these objectified conceptual determinations — determinations manifestly owing to a self-conscious, objectifying subjectivity — stands a declaration such as this, that the difference between objectivity and a consciousness subjective over against it does not concern the concept of time at all. “Time,” it is said, “is the same principle as the $I = I$ of pure self-consciousness: but the same principle, or the simple concept, still in its complete outwardness and abstraction, as intuited mere *becoming* — pure being-in-itself as a sheer coming-outside-itself.” — Under such turns of phrase the clear objectification of time, an objectification possible only for the knowing Spirit, is pushed aside, and the fixed idea of a self-intuiting, self-thinking form of nature, a self-conceiving time, befogs consciousness.

As the form of becoming, time has something becoming as its presupposition. The reciprocal determination of time’s becoming and of what becomes in time throws light upon the relation between the objective and the subjective. From the objective side time goes together into one with what goes on in time: the becoming content lays out its essence in the form of becoming and imparts to the form its predicates. Time becomes, time hastens, runs; the times are changed (*tempora mutantur*), there are good times, bad times, etc. From the subjective side, by contrast, the form is held fast in abstract universality for itself, in distinction from the content. Everything finite begins in time and ends in time, but time itself has neither beginning nor end. If a first, absolute point of beginning be posited, thought easily discovers the contradiction that there must then have been a time when there was as yet no time. If a last, absolute end point be posited, the consequence must be that at last there will come a time when there is no time. If Hegel does wrong in holding fast to objective time without regard to an objectifying self-knowing, then Kant also does wrong in letting the subjective and the objective fall apart into two pieces. How such a parcelling out must become a snare in which the finite understanding catches itself is shown most clearly in the “antinomy” of the origin of the world.

“*The world*,” says the thesis, “*has a beginning in time*”; for if one assumed the opposite, then at every given point of time a whole eternity must have elapsed, and an infinite series of successive states have been completed. *The world*, says the antithesis, *has no beginning in time*; for in the contrary case there would have to be an empty time prior to the beginning of the world, but in an empty time, destitute of all conditions, nothing can

begin.

According to the thesis, the infinity of a series consists in its never letting itself be “completed by successive synthesis.” What does this Never mean? That the completion is postponed into an infinite time. The proposition can therefore be expressed positively thus: to conclude an infinite series by successive synthesis, an infinite time is required; or: by continuing the successive synthesis through an infinite time, representation forms a series of infinitely many members, an infinite series. The formation of the series is the same whether we begin in finitude and approach infinity, or from infinity approach finitude.^{*)} Between a point of time given in temporality and a beginning in eternity there lies, then, an infinite time. There is accordingly no contradiction whatever in an infinite series of states having elapsed “at every given point of time.” If the proof of the thesis were to signify anything, then it would have to be proved either that an infinite series cannot possibly be formed in an infinite time, or that between a beginning in eternity and a temporal point of time there cannot possibly lie an infinite time. But to prove such impossibilities is impossible.

As concerns the antithesis, the emphasis must fall upon the relativity of the beginning. Every finite beginning presupposes a preceding one and furnishes the presupposition for one that follows. This, on Kant’s train of thought, is supposed to prove the impossibility of a *first* beginning. But no. Let the finite beginnings correspond to the members in an infinite series; there is nothing whatever to prevent the series from beginning with a first finite member, for this member has, precisely, the same general presupposition as all the following ones: the infinite series is a successive unfolding of the function that encloses the ground of infinity.^{**)} Were the proof of the antithesis to have any true validity, then it would have to be proved that a beginning in time cannot possibly consist with an infinite succession of finite beginnings, be it past or future. But this, for good reasons, has not been proved.

That the conflict between the finite and the infinite on which the antinomy is built rests essentially upon a critical rupture between consciousness itself and objective being is easy to see. To what, then, does finitude answer? To that objectivity which sensation can grasp and representation hold fast. In this respect it is quite right that an infinite series of time can neither be sensed nor represented. How do we then come to infinity? By the way of thought. We go, as it is called in the language of mathematics, to the limit; that is to say: thought dissolves finitude into its uttermost limit, posits the negation of the finite. It happens by a leap. If to the infinity thus posited there is now also to

^{*)}In the series $\frac{1}{1-x} = 1 + x + x^2 + \dots + x^{n-1} + x^n + \frac{x^n + 1}{1-x}$, x^n is the general member; setting $n = 0$, we get $x^0 = 1$; setting $n = \infty$, we get x^∞ ; but the way from 1 to x^∞ and from x^∞ to 1 is equally long: it is infinite.

^{**)} The infinite series $1 + x^2 + x^3 + \dots$ has its ground and presupposition in the function $F(x) = \frac{1}{1-x}$; the infinite series $1 + \frac{x}{1} + \frac{x^2}{1.2} + \frac{x^3}{1.2.3} + \dots$ has its ground and presupposition in the function $f(x) = e^x$; there are examples enough in mathematics.

answer an objectivity of its own, then it is surely clear that objectivity in the infinite must come into conflict with objectivity in the finite. But instead of laying the blame on the intuition of time, we should on the contrary recognize that it is precisely time which, in the natural sense, strives to reconcile the finite with the infinite. Time is succession, and succession an outward transition which satisfies representation and thought in equal degree. By succession the finite follows upon the finite; each finite member comes to be successively for representation. Representation can then demand no more than that the succession shall further continue. But the continuance of the succession is precisely the coming-to-be of infinity; the infinity that comes to be must then likewise satisfy thought. Objectively regarded, succession is a becoming; but when becoming in time is the outward manner in which the finite passes incessantly over into the infinite, and conversely, then both, the finite and the infinite in time as time, have equal objectivity.

To sharpen the sting of the antinomy Kant has thrown time and eternity together into one and yet set them up in the sharpest opposition. Since the world, for the sake of the infinite series of time, cannot be from eternity, it must have a beginning in time; that is the thesis. Since otherwise an infinite, empty time (an empty eternity) would have to lie prior to a beginning in time, the world must be from eternity; that is the antithesis. Here there is an alternation between time and eternity, and so time is after all eternity, and eternity time — to infinity. The fact is that at the ground of all becoming there must lie a being that does not become. This being reflected into itself, the essence, is namely reflected in its other, in the phenomenon. In virtue of this double-sided reflexion the essence has its inner infinity in the form of selfhood, which is opposed to the outer infinity of the otherness-forms. The outer infinity answers to time, the inner to eternity. There is a qualitative difference between time and eternity.

Upon this difference rests in turn the difference between *beginning* and *origin*. The beginning happens in time, but the origin issues from eternity. The absolute essence is eternal, unoriginated, exalted above time; the relative essence has, by contrast, its origin from the absolute and is by its relativity subject to time. Origin points to the self-essence's objectification of its essential other, to Spirit's objectification of nature, to God's objectification of the world; origin is transition from eternity to time. But beginning is a category of time; the beginning presupposes the origin, for it belongs to time, the infinite as well as the finite: infinite time is a succession of nothing but finite beginnings. Spirit is eternal in time, above time; but the temporal world is not eternal, even though its becoming is from eternal time.

c) *The Business between Space and Time.* If the concepts of nature, space and time, were able in an unconscious knowing to conceive themselves, each for itself, and one another mutually; if by a common conceiving effort they really availed to bring forth something so tangible as what we call matter, then modern nature-panteism could undeniably boast of a sure footing in "strict science"; but the fixed idea becomes more recognizable with every step the development of the concept takes.

The Hegelian deduction of place, motion and matter is instructive in this respect. The words are obscure, but the thought is clear. Space, with its indifferent discreteness and its equally indifferent continuity, is a contradiction that resolves itself in time. But time for itself is also a contradiction; for the present is an unarrestable conversion of the past into the future: the separated moments dissolve into immediate indifference. The concrete point in which the contradictions, converging from both sides, fall together is “*place*.” Place is unity of “*here*” and “*now*,” and thereby expressly posited as unity of space and time. The place determined by here and now points, however, out beyond itself to another place determined by here and now; this becoming of place is “*motion*.” But here and now are everywhere; the determinate place is, in all its indeterminacy, only the universal place: the contradictory moments of motion fall together in immediate unity; this immediately existing unity is “*matter*.” Matter is *impenetrable* and offers *resistance*, is something tangible, visible, etc. These predicates are nothing else than that matter is partly for determinate observation, in general *for another*, but partly also *for itself*. Both are determinations which it has just as the identity of space and time, of the immediate *outside-one-another* and *negativity*, or of singularity that is for itself.”

This deduction and its dialectic are characteristic of the philosophy of immanence. “Space,” it is said, “is not adequate to its concept; it is therefore the concept of space itself that in matter procures existence for itself.” The doctrine of the objective concept’s miracle-working power cannot be uttered more plainly. But the sorcery vanishes, and the miracle is called back again, by an explanation such as the following: “One has,” it is added, “often begun with matter, and then regarded space and time as its forms. What is right in this is that matter is the real element in space and time. But these must, on account of their abstraction, appear here as the first, and then it must show itself that matter is their truth.” The last of these statements we subscribe to unconditionally; but it stands in screaming contradiction with the first. If it is really the concept of space which, in union with that of time, has availed to give matter existence, then matter, the real, cannot possibly be the first; it must, in the subjective as in the objective sense, be the second, the derived. If, on the other hand, matter is in itself to be the first, because it is the concrete which takes space and time up into itself as abstract forms, then it is surely an insane assertion that the abstract should have brought forth that from which it is itself abstracted, that space, spurred on by its own restless concept abstracted from matter, should have created matter.

Abstraction, contradiction, negativity, infinite negativity, etc., are categories of selfhood which assuredly have validity in infinite objectification; but they are misunderstood, misused and taken in vain when they are laid without more ado into the objective, so that outward being is ascribed to them. Space and time are forms whose abstract being must be integrated by a more complete existence: they must have a *business between them*, which is taken up into them and in which they are taken up. But abstraction and integration are acts of thought, functions of objectifying selfhood. What reality the

required business between them is to have will then depend upon the objectification. For abstract objectification, insofar as it is still only a question of a general determining of place and mechanical motion, a represented business between them is sufficient.

No motion is possible without a something that can move. Motion goes on in space, but the moving thing is distinct from space, for space does not move. Change of place is a change in space, but without space being changed. The change happens with the moved thing, which passes over from place to place; and yet it is not the moved thing itself, but only the place, that is changed. The place does not belong to the moved thing, for it belongs to space; and yet it does not belong to space for itself, for it first becomes place, in the motion, by being occupied by the moved thing. The place is neither straightforwardly existent nor non-existent; it is an objective determination of relation, objectively reflected. Motion goes on in time, and what is moved in time is distinct from time; but time enters into motion in another manner than space. The places of space are simultaneously existent, namely according to possibility; every place traversed can for that reason be occupied again. But the instants of time vanish, and every vanished instant is irrevocably lost. With the movable as the business between them, space and time each receive their reflected share in the motion. Determinate motion is velocity, and velocity a quotient in which the element of space as numerator and the element of time as denominator reflect one another objectively, $\frac{ds}{dt} = v$; but on the terms of objectification: the objectifying reflexion is subjective.

What confusion of concepts can arise when, in the phenomenon of motion, one will absolutely think the objective without considering its relation to the objectifying subjectivity is well known. The arrow of the Eleatics, flying in infinite space, is now here, now here, always now here! and does not get off the spot. It is an objective extreme that cancels itself. For motion to come to be, the observer must recollect, gather together and intuit — namely the places the moved thing has traversed and goes on traversing. No wonder that the objective extreme in the Eleatics struck over into a subjective one, so that those inflexible thinkers declared all motion outright to be semblance and deception. But that the opposite too can happen, that the subjective, namely, can by a logical self-deception strike over into the objective, is evident from the Hegelian dialectic. While Zeno, in order to get the arrow to stand still, points to the infinite space in which all relativity has vanished, Hegel holds the determination of place fast in thought. “Here are three places: the one that is, the next following, and the one abandoned. But it is at the same time only one place, the universal place of those places, an unchanged amid all changes.” One learns from this how abstraction, contradiction and negation drive one another forward. That the three places are one place rests upon an abstraction, in that abstraction is made from that difference in outward being which is precisely characteristic of space. It is at the same time a contradiction; but this contradiction, conditioned by the abstraction, is resolved by the unity negated in the threeness being negated again in the universal unity, the abiding, so that “duration (*die Dauer*) is according to its concept

motion.” That nature does not institute such reflexions on its own account, but leaves it to subjectivity, must surely be conceded. None the less the subjectively reflected concept of motion is straightforwardly declared to be a not subjective but merely objective concept of nature, and is moreover made one with motion itself.

As abstract forms, hovering between being and non-being, space and time are without particular independence; their existence depends upon the business between them in which they are united, and so upon the content with which they are filled: in this dependence their ideality is seen. If space and time are filled with dream-images, they have only a dreamt existence; if they are filled with a world of poetic figures, they attain an aesthetic existence; only with a content of actual things do they become actual. If it holds of the forms, as we have seen, that their objective being rests upon objectification, then the same must necessarily hold of the content with which they are filled.

B.

Matter and Force.

The actuality of space coincides with its actual filling. What fills space, matter, and what is at work in space, force, belong to nature but are posited by Spirit; the task is to show how the two foundational determinations — origin from Spirit and natural reality — presuppose one another.

a) Matter. Nature’s first immediate real is not a concept, but a sensible existent in which a concept exists. The concept of existence is a concept of reality; it is then first and chiefly a matter of seeing into the relation between concept and reality.

To presuppose reality and exclude the concept is characteristic of materialism. The one-sidedness of this way of seeing does not consist in its ascribing reality to matter — for if matter were only a semblance, the ground of nature would be hollowed out; the one-sidedness is rather that matter is made into something absolute in itself, into an eternal, unconditioned existent out of whose fundamental parts everything is composed and formed. These fundamental parts, the atoms, are apprehended at once both as parts of mass and as absolute real points. That is a contradiction; for if the atoms are parts of mass, then they are relative; if, on the other hand, they are absolute, then all actual relation between them is impossible. To ascribe properties to the atoms is to make them relative, for the property of the one atom acquires significance, in attraction and repulsion, only through the property of the other. To presuppose the absolute unity of the atoms with their properties is to concede their absolute relativity and to reduce them to moments for one another mutually. With this consequence materialism has dissolved itself.

To posit the concept as reality and reality as concept is characteristic of the one-sidedness of idealism. Idealism is powerless; it mistakes the opposition of the ideal and

the real, of Spirit and matter, and does not see that to a mighty inner there must answer a mighty outer. Idealism would transform the concept of matter into actual matter; that is a barren speculation. The opposition of concept and matter is an opposition in power, an opposition so hard and strong that the concept of their unity coincides with the concept of omnipotence, of absolutely overreaching self-power. Self-power's inner original opposition is object-power, power in another; the first outer for power in another is matter. Hence the swarm of atoms: nothing but points of otherness, points of reality. The qualitative opposition of Spirit and matter is marked by the issuing of power. The issuing is not blind; it is a fundamental act in virtue of an omnipotent conceiving: the concept is the ground of reality, reality the existence of the concept.

An immediate object-power without a corresponding concept is just as impossible as an existing concept of power without an object. One cannot divide the existent into two absolutely different kinds: an existent *with* a ground and an existent *without* a ground; for the essence of existence is to have a ground. The real ground is an existence dissolved into the concept of power; the existent that has no ground does not exist, its being is a hollow phenomenon, a semblance.

Materialism's fundamental error now shows itself in this, that it makes the matter "existing from eternity" into an existent without a ground. The atoms, the absolute existences upon whose separation and combination all finite, material formations rest, then stand to actual things in their finitude as the groundless to the grounded! A more unnatural separation of ground and existence cannot be given. Materialism has a sharp eye for reality as object-power. It begins with a language of power, points to the fact, takes the existent as something inviolably given; that is sound and right. But to overlook the concept of power and to deny the essence of existence, in order to build upon an existent without a ground, is unsound and false.

Objective idealism holds to the grounding, in that it holds to the concept; but its fundamental fault is that it overlooks object-power and effaces the difference between reflexion in knowing and reflexion in power. What this misstep signifies can be seen, in matter, in atoms and mass. Existing mass has the atoms as its foundation; but the existence of the atoms has in turn the presupposed mass as its foundation. To form a foundation, an existent must go to the ground. The existent is represented; but the ground and the foundation are thought. That the atoms exist means that they are posited in representation, while the mass understood along with them is posited in thought; if, on the other hand, the mass is in representation, the atoms being understood along with it, then it is the mass that exists. The perishing of the atoms in the mass and the dissolution of the mass into atoms are imaginary movements that go on within thought and representation; it is a reflexion which, however many conceptual determinations it takes up, never gets beyond a represented existence with a thought ground, never reaches actuality.

In the reflexion of power the start is expressly made from actuality, insofar as the

start is made in the existent with object-power; the ground is a concept of power, and the foundational determinations are real moments of dissolved existence. The transition from existence to ground, and from ground to existence, is an essential change, an objective act of power, an event of nature. The mass exists, and the atoms exist; and yet here there are not at one and the same time two sorts of existent or two layers of existence. If existence falls in the mass, then the atoms have ceased to exist; they are taken up, as moments of the foundation, into the existence of the mass. But when the atoms perished, something actually took place: they approached one another, gave up their independence in cohesion, and became moments in the mass. If existence falls in the atoms, then the mass is dissolved: it has perished. But when the mass perished, something actually took place: it dissolved itself and in the dissolution lost its independence. That the mass lost its independence is the ground of the atoms' acquiring independence and coming to exist. The existence of the atoms has its real ground in the perishing of the mass just as surely as the existence of the mass has its real ground in the perishing of the atoms. In this reflexion of power the one is not an unconditioned first and the other a derivative; rather both existences, that of the atoms and that of the mass, are equally original and equally derived. Nor, then, is perishing annihilation; the existent that perished in another existent comes to exist again when the existence resting upon its perishing perishes.

The reflexion of the concept of power and of object-power which furnishes the general foundation of mechanical nature is everywhere recognizable by an abrupt conversion of existing members into moments, and of moments into existing members. The concept of power's reflected unity with object-power is expressed in the harshest opposition. The existent is no concept; it is object, actual fact. The concept does not exist; it is reflected in dissolved existence as ground and foundation. But the unity is in turn just as mighty as the opposition: the existent has the ground in itself: the object is the material expression of its realized concept. From the side of reflexion the concept is what overreaches; from the side of the object it is fact, and fact first and last. The infinite reciprocal determination of reflexion and fact, ground and existence, is the form of actuality for *power in another*. But power in another presupposes self-power; the concepts dwelling within the facts must be conceived by an overreaching sole power: *the mechanical power of nature is borne by the omnipotence of Spirit*.

In what sense existing matter must be said to be the first mechanical presentation of power in another can be seen from the properties which physics has particularly ascribed to it as matter. Matter is *sensible*, not merely for us, but in itself. To think matter is to transform it into a being of thought and to deny its existence. If matter is for Spirit only a being of thought, then the whole of nature is a semblance. But the truth is that matter's essence is an essence of power; its concept is a concept of knowing converted into power, its existence an object-power, a resistance for thought, an immediate object, for Spirit too. To be Spirit's immediate object, the real other of intuition, a limit for thought, is to be sensible. Matter *fills space*. Limited matter occupies its place and offers

resistance to other matter that would occupy the same place; this power of resistance is otherness-power, power to exclude another from another. Matter is *divisible*. Divisibility rests upon otherness. The mass is one thing, the atoms another. The unity of the mass is no self-unity, but a unity of another with another; the mass subsists by its other, for it consists of atoms. Were the atoms absolutely to subsist by themselves, it could still not possibly be as mathematical points; it would have to be as small masses. But what holds of the larger masses holds of the smaller, the smallest, the very smallest: the mass subsists in another by another. Each single atom has the same subsistence as all the others. The reality of one atom is the reality of all; but the reality of all is the mass: the atoms subsist in the mass, and so in another. Matter is *indifferent to rest and motion*; this indifference is quite a characteristic sign of otherness. That matter is subject to the law of inertia is also expressed by saying that it is without will, which in other words is to say that it is without self-determination. When it is at rest it cannot determine itself to motion, but must be determined by another; when it is in motion it cannot determine itself to rest, but must be determined by another. This doctrine of matter's selfless independence (inertia) has decisive significance for the knowledge both of matter's essence and of the laws of motion.

None the less the doctrine of immanence, as speculative philosophy of nature, will still, in the teeth of empirical physics and rational mechanics, maintain the mystical representation of a selfhood dwelling within matter, a silent longing for liberation, a hidden capacity to set itself in motion. Matter, says Hegel, is removal in space. It offers resistance, repels itself from itself: that is *repulsion*, whereby it posits its reality and fills space. But the single members repelled from one another are all one. That it is the same one which, by repelling itself from itself, cancels the removal of the single members — in that consists *attraction*. Both together constitute, as gravity, the concept of matter. The unity of gravity is only “an ought, a longing, the unhappiest striving to which matter is eternally condemned; for the unity does not come to itself, it is not attained.” Matter can, however, console itself that “it is not so stupid” — as the physicists — “as to hold one and many apart from one another.” It is only in the finite sphere, in the circle of “the selfless bodies,” that matter suffers from inertia; in its “free concept,” in the “celestial sphere,” matter has selfhood and moves itself.

In thus stepping forth polemically against physics, the doctrine of immanence at the same time comes into a dubious position with respect to the doctrine of Spirit. To make good the lack of Spirit's conceiving self-power, and to let nature conceive its own concepts, personifications are brought in: as a substitute for the excluded Spirit, felicities of wit are introduced. “Just as time is the simple formal soul of nature, and space, according to Newton, God's sensorium, so motion is the concept of the true world-soul. We are accustomed to regard it as predicate, as state, but it is in actuality the self, the subject as subject, the abiding in all vanishing.” Just as motion is the essential self, so matter is, in the mechanics of the infinite, naturally the actual self. The fact that matter

moves matter to infinity is explained, by a logical paraphrase, to the effect that matter accordingly *moves itself*. By such explanations the concept of nature's development is mystified from the ground up.

b) Force. Matter moves and is moved; the one mass moves the other: matter has the ground and cause of motion in itself, but under the form of otherness. Only when it is clearly seen that being able to move and to let itself be moved necessarily belongs to its space-filling reality is the concept of matter exhausted. It is not the abstract determination of the form of motion, but actual motion, that is treated of here. We must look about us for a fundamental phenomenon from which it can show itself most clearly what the moving thing properly is.

From a finite mechanical point of view one has pointed to *impact*. In the contact of two colliding bodies the mutual resistance occasions a struggle over the place, which seems to furnish sufficient ground for change of place, and so for motion. The ground is so much the more natural in that it manifestly lies in the reality with which every mass occupies its place. Immediate contact (ἄπτεσθαι) is accordingly posited by Aristotle too as the condition on which the one matter can act movingly upon the other. But that impact cannot be the moving thing itself is evident. For impact presupposes that the two colliding bodies must either both have moved in opposite directions, or, if the direction is the same, that either the one must have moved with greater velocity than the other, or the one have moved and the other been at rest. It is seen then from all sides that impact cannot be what properly moves; it is itself, after all, a result of motion.

To avoid the presupposed motion, one might try to place the moving thing in a pressure and make *pressure* a fundamental phenomenon. Pressure and counter-pressure now express straightforwardly the power of resistance identical with matter's reality; motion enters as a necessary consequence, whose presupposition is that the equilibrium between pressure and counter-pressure is cancelled. Motion begins with a striving toward motion; such a striving really lies in pressure, and so far the explanation has a ground; but it is an insufficient ground: a something is here presupposed which is not yet grounded. The phenomenon of pressure can be construed partly so that both bodies, each from its own side, strive to press into one another while yet mutually resisting one another; partly so that only the one is pressing forward, the other merely resisting. In both cases the striving coincides with the pressing forward; and the question becomes whence this pressing forward then derives. Pressing forward presupposes approach: *the mutual approach of two separated masses* must accordingly be the fundamental phenomenon sought.

Two masses separated in space, *A* and *B*, will, when extraneous hindrances are thought away, approach one another, and so come into motion; but what is the ground? Hegel appeals to gravity and places gravity in a common seeking after a common centre lying outside the masses. "The centre," he says, "is not to be taken as material; for the material is precisely this, to have its centre *outside itself*. Not the centre, but a striving

after it, is immanent in matter. Gravity is, so to speak, the confession matter makes of the nothingness of its outward being for itself, of its dependence, its contradiction.” But to ascribe to the masses such an inner lack, a yearning toward an ideal existing outside them, is poetic enough. but not scientific; just as it also expressly conflicts with the hard otherness, the selfless independence, to explain the mutual approach of the masses by an obscure sympathy and to invent a — naturally unconscious — longing in *A* for *B*, answered by a — naturally unconscious — longing in *B* for *A*. Strict mechanics has another construal. *A*, it is said, attracts *B* because *B* attracts *A*: what attracts in both masses, indeed in every atom of both masses, is *force*.

Force is, mechanically understood, the cause of motion; but — physics adds — an unknown cause. We shall now examine more closely how matters really stand with “the unknown cause we call force.” When it is said that we do not know what force is, it is tacitly understood that we do after all know well enough what matter is. But if matter really is a known quantity, then it is impossible that force, in its peculiar connexion with matter, should be wholly unknown to us. Force, the non-sensible, answers to matter, the sensible, as the reflexion of power answers to immediate object-power: matter is the outer, force the inner. But an inner can never show itself straightforwardly; that it shows itself means that it utters itself; it reflects itself in its outer and must be known by its utterances. A proposition such as this — we do not know what force is in itself, we know only its utterances, its effects — accordingly states nothing else than what lies in the nature of reflexion, namely that the inner must necessarily be known by its outer. The abstract logical essence reflects itself in a semblance of semblance, but the reflecting of power is an efficacy, its phenomena are facts: as the categorical expression of the reflexion of power, force is not a “hypostatized thought” in human imagination, but a fact reflected in its effects. Force as inner is nothing absolute in itself; the inner is a possibility which first becomes actuality by uttering itself and working. The proposition cited above cannot then, in the objective sense, be regarded as any sure designation of the absolute limit of human knowledge, but rather as a subjective indication of unclarity about the essence of thinking, of reflexion, and especially of the reflexion of power.

How unclear and hovering the usual representation of force is comes to light from the way in which its relation to matter is more closely determined. On the one side one speaks as though matter were one thing and force something quite other. It is then not the mass *A* that by attraction sets *B* in motion; no, it is the force, the force of attraction present in *A*, whereby the motion of *B* is caused, and conversely: it is not *B*, but the force in *B*, that moves *A*. The masses are moved, but the forces move: matter is *passive*, force *active*. On the other side one speaks again as though force were only a property of matter. Then it really is the mass *A* that in its capacity as mass attracts *B*, and conversely. The latter way of seeing stands in manifest conflict with the former. By making force a property of matter and, as regards universal attraction, a fundamental property at that, one lets the concept of force merge without more ado into that of matter. On the first

way of seeing there would here be two essences, but on the second there is only one essence. As a property of matter, force comes into a most questionable position with respect to matter's other properties, notably inertia. By inertia matter can only be moved, not move; by force it can only move, not be moved. It must then seem as though matter were a self-contradicting reality. As what fills space, matter has besides a number of properties in common with space, not to speak of a quantity of immediate properties of sensibility which do not well admit of being reconciled with force as force. For one will surely be as little able to speak of a broad force, a round force, a four-cornered force as of a thin or thick or grey force.

The difficulties cited fall away when we hold to the reciprocal determination of the reflexion of power and immediate object-power set out above. Matter is in its own way reflected, namely in the mutual grounding of atoms and masses; force is in its own way immediate, namely in its utterances. And yet the difference is conspicuous; for with matter immediacy, but with force reflexion, is what is decisive. The reflexion of power is something more than a logical mirroring of the one in the other; it is an actual determining, an efficacy in which it shows itself what the one can do over the other. It is precisely such a capability that mechanically utters itself in motion. The motion of *B* reflects the force in *A*, in that the motion of *A* reflects the force in *B*. The mutual motion is a phenomenon of the essence of the reflexion of power: the reflexion that overreaches in essence proves that the universal in the relation of the members governs the separate members. Physics is so far within its rights when it apprehends force, the active, as an essentiality for itself, in distinction from inert matter, the passive. But in this physics does not after all overlook that the two essences are at bottom side-determinations of the same essence. The unity of the reflexion of power with object-power is expressly brought out by force being apprehended as matter's own reflected determinateness, and so as its property. As matter's pervasive property, force is something other than an immediate quality or constitution; it is a categorical expression, not for matter's sensible reality, but for its reflected essence, its indwelling energy, its inner. Herewith the contradiction between force as a reflected property and matter's other, immediate properties disappears. The immediate properties belong to *the extensive*, but the determinations attaching to force belong to *the intensive*, for they belong to reflexion — be it noted: reflexion in power. “One can,” it is said of the centre of gravity in mechanical natural science, “represent things to oneself as though the whole weight of the body were united in the centre of gravity, and all the remaining parts had no weight.” This representation is more than a representation; it is an actual truth. The weight distributed over the parts, weight in the extensive, is transformed in the reflexion of power into a unity of weight, a weight in the intensive: the centre of gravity is the intensive point of unity.^{*)}

The validity of the above remains to be tested more closely, so far as a final construal of the otherwise so enigmatic phenomenon of attraction is concerned. The masses *A*

^{*)}Cf. *Mathematik og Dialektik*, pp. 129–131.

and *B*, which at a distance each fill their own space, seem to have nothing whatever to do with one another; and yet they have to do with one another, for they approach one another. It is a strange opposition of essence and existence, reflexion of power and object-power, that lets itself be seen in the phenomenon; existence says one thing, essence another. The solution of the riddle lies in the insight that material actuality has, through essence and existence, uttered two opposed but equally irrefusable demands, and that the phenomenon of attraction is precisely the fulfilment of both demands. The first demand is that the material members must have immediate independence; this independence is made intuitable in the sensible existence of the masses. Existence answers to place. Each mass occupies its place, and it may easily appear to the beholder as though space-filling reality were the complete expression of matter's actuality. That, however, is a deception of sense; what is thus seen in immediate existence is only half an actuality. Existence conceals something in the existing members that can come into view only in the form of reflexion, as essence. The second demand is that the essence must reflect itself in the relation of the members, in the mutual relation of the masses. Terms of a relation without a relation is just as abortive a thought as a relation without terms. If the members are actual, the relations must be so too; their actuality is in their activity, and this activity is the reality of the relation. When power is reflected in a relation of actual members, the essence shows itself as force. The distance-relation of the masses becomes, through force, an active and so an actual relation. In the masses matter's existence is limited, restricted to the place; the masses are as actual members moments in actuality. As essence, by contrast, force goes from mass to mass out beyond the limit, and so out beyond the place, and the moment of actuality is seen in the mutual change of place. That force goes through space, that masses act at a distance, is a mechanically characteristic expression for that reciprocal determination of the reflexion of power and object-power which makes actuality complete. The apparent contradiction, that a mass has force to move another mass but no force to move itself, has with this at once found its solution. Matter's actuality is an independence in the form of otherness, a selfless independence. Selflessness is dependence; material actuality is a dependent independence. The dependence is matter's inertia, the independence its force. Inertia is that matter is moved by another; force is that it moves another. In masses that attract one another, inertia and force are united on both sides: force is the complement of inertia, inertia is receptivity for force. Thus independence and dependence too are united on both sides. The mass *A* has independence, it sets *B*, from its own place, in dependence; but *B* also has independence and sets *A*, from its own place, in dependence. The relativity is that both masses are, in their independence, dependent. Here no outward observer is needed to hold the two places together in representation and determine their distance; for so pervasive is the relativity that it is in actuality the masses that mutually determine one another's places: matter's relativity is, in the motion, actual by means of force.

c) The reflected unity of matter and force: Spirit and matter. The development carried

thus far has the ideality of power, reflected power, as its foundation; its significance stands and falls with the answer to this question: *how is the reflexion of power possible?* That gravitating masses come forward as holders of power, exercising in proportion to their magnitude a certain dominion over one another, is only a figurative turn of speech so long as the possibility of the fact is not seen into. It is, as already shown, the reciprocal determination of members and moments that matters. Outwardly, in the form of parcelling out and dividedness, the moments are actual masses; their reflexion is relation, and the relation reciprocal action. But in the concept, inwardly, the actual masses are moments of the same unity, reflexes of one and the same conceiving. To conceive is to master. It is then clear that the omnipotent conceiving which unites all separate concepts of power in the unity of the universal concept must go on in an all-conceiving self-power. Only an infinite self-power, a knowing self that objectifies in power, avails to give the system of objective concepts actuality in the existence of the masses and to raise itself above the patchwork. The reflexion of power in another has its ground in reflecting self-power; herewith matter's absolute dependence upon Spirit is known in the concept.

The history of science teaches that all attempts in the philosophy of nature to clear up the relation between matter and motion, matter and force, have hitherto foundered upon a total misapprehension of the relation between Spirit and matter. Matter's unity with force is, namely, reflected in such a way that, besides being a logical unity in essence, it is at the same time an actual unity in fact. The reflexion in which the double-sided unity is grounded is itself double-sided; it is a reflexion of the essence of matter and of force; but it is at the same time a reflexion of existing matter and of force's utterance. If the unity be posited in essence, as when one says: matter and force are at bottom the same, then it is broken in existence; for existing matter is not force, and acting force is not matter. If the unity be posited in existence, as when one says: force is material (the atomistic conception) or: matter is a product of forces (the dynamical conception), then the difference in essence comes into view in the reflexion; for matter is an outer, force an inner. If the unity is to be known in a double reflexion of essence in essence, of fact in fact, then it is possible only by self-power and power in another, essence-power in ideality and object-power in reality, furnishing extremes so strong, so hard, that Spirit's omnipotence can show itself as absolutely overreaching sole power.

Without selfhood no reflecting, and conversely; without a reflecting self no reflexion; without reflexion no relation; without reflexion in power no active relation, no cause and effect, no reciprocal action. In the form of conceiving the reflexion of power is subjective; in the form of the concept, of the conceived, it is objective; in a reflexion of subjective and objective reflexion the objectification is completed.

α) Subjective reflexion of power. Self-power is not power's empty, merely formal likeness to itself without inner difference and opposition, as when infinite power is assumed without more ado to be existent and is determined by a tautological proposition:

power = power, $A = A$. No, self-power must expressly posit its own likeness and thereby make it self-likeness. In actuality this can happen only by the unity of power showing itself different from itself, doubling itself through another and confirming itself by opposition to another. Only that confirmation in which the same essence is in the result its own presupposition and in the presupposition its own result can in truth be called *self*-confirmation, a confirmation of the *self*.

The greater the independence the other has, the mightier is the self thereby confirming itself, for the self reflects itself subjectively by mastering its other: the stronger the resistance, the greater the power by which it is cancelled. From this it is seen that self-power cannot be apprehended as an immediate existent, an absolute substance indifferent to everything else. Self-power's being is efficacy, and this efficacy an infinite reflecting of itself in another, of another in itself. The inner motion in which it eternally subsists is a circular motion. Absolute self-power does not leave itself because it goes out from itself; it does not reach its other as an outwardly given, for it has its other in itself; it is not changed, for it masters its other; the motion is no unrest, for the self is in its motion at home with itself, its motion is rest; it rests in itself upon itself: *absolute self-power is Spirit*.

For Spirit as Spirit the whole material world is a nothing and yet an actual something. Both determinations are reconciled in the fundamental determination that the material world is Spirit's other. As other for Spirit, the material is a moment for the omnipotent self-reflecting, but nothing, nothing at all in itself: only Spirit, the one, is; all else is nothing. As other for Spirit, the material is actual; it is, after all, object-power, the inner power of resistance belonging to self-confirmation, the measure of Spirit's omnipotence. Were matter a semblance and the world a dream, then Spirit itself were a dream, then there were no reality, either inward or outward, no actuality, either ideal or real.

β) Objective reflexion of power. The immediate and yet reflected unity of matter and force is a unity of power. But the objective unity of power is at the same time a broken power, power in another, power of multiplicity; upon this rests its peculiar reflexion. The reflexion is recognizable in gravity. Physics apprehends gravity from three points of view: as a *property* of matter, as a particular *force*, and as a *power* spanning the universe. As property, gravity is reflected in the independence of the masses; as force, in their mutual reciprocal action; as world-power, in the great laws. It is from all three sides the same essence, the same power, the same substantiality; each side of the power reflects the two others in such a way that it must itself be said to be the whole power. If we have before us the actual masses, then we have in their common property force, in force attraction, and in attraction gravity as universal power. If the attracting forces are given, then we have with them at once the masses, for the masses are, in this light, nothing other than the existence of the forces, and force and mass again moments of gravity as power. If we have, finally, gravity as power, then the existence of the masses with the corresponding distribution of the forces is precisely its own reality, an issuing of the

particular in which it subsists as the universal. From all sides, then, objective power is reflected.

But objective power cannot reflect itself by itself; only self-power can do that. Let us look again at the two masses. The moving *A* is cause, the motion of *B* effect. The cause, which is reality in *A*, is ideality in *B*, for the ideality of the cause is to be — reflected — in its effect. Now since *A* is removed from *B*, it has as cause its effect outside itself, is as reality removed from its ideality. And yet the ideality is in the cause, and the cause is in *A*. *A* cannot then possibly be removed from what it itself is; as surely as it has the cause in itself, it must necessarily also have the ideality in itself. This contradiction — that each mass, namely, has its ideality in itself by having it outside itself — is a contradiction in the objective reflexion, which plainly proves that it cannot bear itself, but must be borne by Spirit. From Spirit, in and by the inspiriting, the reality of the mass receives an addition in which the ideal unites itself objectively and essentially with the real; this ideal-real addition is force. By means of force, matter in *A* is according to possibility more than matter, it is cause. That the activity sets in, that the cause becomes actual and gets its effect, is objective reflexion in power. If objective power were not reflected in Spirit, then gravity would not be the universal power, then there would be no universal powers in nature, then the laws would be impossible. Objective, universal power is not an abstraction from masses and forces, but their own substantial essentiality, their indwelling foundation. Objective power reflects itself — by being reflected; it is in actuality objective, but — by being objectified.

γ) Objectification in power. The concepts of nature have a double-sided subsistence, subsistence in actual things and subsistence in the self's conceiving. The latter conditions the former; the concepts could not give themselves an imprint in existing objects if they were not themselves stamped by an infinite conceiving. It is not, after all, sensible natural things that in the objective sense determine the concepts; on the contrary, it is the concepts of nature that non-sensibly determine sensible things. To be the determining essence of the thing, the concept must necessarily have determinateness in itself, must be determined in advance to furnish determination. The determinate concept of mass does not first come to be by means of existing masses, nor the concept of force by forces already existing. If masses and forces had an existence independent of concepts, then things could at once dispense with all concepts; there would be no reason at all in nature. If the coming-to-be of the concepts coincided immediately with that of things, it would be altogether impossible to say whence the determining came. It could not then be the concept that determined the thing, for the concept was precisely to be determined by the thing; but that neither could it be the thing determinate in itself that determined the concept goes without saying, since what determines a thing is just its concept. It stands fast, then, that the concepts of nature must have in them a determinateness independent of the existence of things and decisive for things themselves. Whence do the concepts now get this inner determinateness, indispensable for all outward existence? From a

conceiving. To determine concepts is to conceive; to present the concepts, determinate in themselves and for that reason determining, in existing and existence is not to *conceive*, it is to *work*. Nature presents concepts, forms concepts out, but it does not conceive, it works. Spirit's ideal working is conceiving, nature's rendering of what is conceived is not conceiving, it is real working, material activity, the unfolding of the determinations of the concept in sensibly actual facts.

The opposition of Spirit's conceiving and nature's working is grounded in the reflexion of subjective and objective reflexion of power, an all-sided expression for objectification in power. In the subjective reflexion of power, the ideal objectification, the concepts of nature have being as spiritual determinations of power, decisive for all material existing. But under the form of ideality, in self-power's conceiving, the concepts do not themselves have existence; their being, a being in knowing, is the infinite reciprocal shining of the determinations of power in one another; the single and the particular reflect themselves here in the clear, free universal: power in another is a moment for self-power. From the side of existence, and so from the side of nature, the emphasis falls by contrast upon power in another, upon object-power, upon reality, and the real objectification deposits concepts in corresponding objects. The objects are things, a swarm of single members, atoms and masses. The universal that determines as concept here merges immediately into the single, is bound in power by single members, by outward objects. So it is said in the language of physics: only the single members, the fundamental parts with the things composed of them, have existence in nature; the universal does not exist. And yet it is precisely the universal that comes to existence; it is, after all, objectified in power precisely through the issuing of single members. The single members work, take on through their reciprocal action the stamp of the particular, and fall due to the universal; then it shows itself that the single and the particular are subject to the universal. Objectively reflected power, the universal power of nature, of objectivity, comes to existence in the forces of things and the laws under which the forces work.

C.

Forces and Laws.

In objective reflexion universal power is both existence itself and the essence of existence; but the difference between essence and existence is just as mighty as the unity. Power in another, which is one with itself in essence, is different from itself in existence; it subsists with this difference as object-power in the masses of the members and as acting forces in the reciprocal action of the members; but it also has subsistence in the universal form of essentiality, as reposing, all-determining power, namely in the laws.

The utterances of forces are facts, but the laws look like thoughts, mighty thoughts, thoughts of nature. To know the acting forces and the conditions of their activity is to know the laws; and yet the forces are not laws, and the laws not forces. To come to an insight into the universal unity-in-difference of forces and laws, we shall now investigate, from a mechanical point of view, the laws' *origin, fulfilment and mutual connexion*.

a) *Represented forces: the origin of the laws.* "The understanding," says Kant, "does not borrow its (a priori) laws from nature, but prescribes them to it." This proposition points to the subjective origin of the laws and is justified by countless examples from analytical mechanics. If two equally great forces act upon a material point in opposite directions, they must cancel one another, and the point remain unmoved; that is a law. That this law has objective validity is just as evident as that its validity is independent of all experience. The law's universal validity lies in the necessity with which the posited presupposition yields its consequent: the law is, in logical form, a hypothetical judgment, in mathematical form an equation. No weight attaches to whether the assumed case occurs in outward nature or not, for the law is the nature of the matter; it has its nature in itself. It is the thinking self that sets up the presupposition; it is the thinking self that draws the consequence; it is the thinking self that gives itself the law and assures itself of its objectivity. There is neither in heaven nor on earth any outward power that avails to overturn the law given in the train of thought or to make it doubtful; the only holder of power able to cancel the law is the legislating subjectivity. Subjectivity cannot, indeed, contradict the law without contradicting itself; but it can cancel it by cancelling the presupposition, it can alter it by altering the presupposition.

That a law must alter when its presupposition is altered is itself a law. If the two forces acting in opposite directions are unequal, the point must come into motion, and the difference of the forces will then be the moving force; but if the two forces form any angle whatever with one another, it will at first glance look as though we had got a new law. The necessity with which the law holds is, namely, always bound to a presupposition of one or more conditions, and consequently is not an absolute but a hypothetical necessity. By introducing now fewer, now more conditions into the presupposition, and letting the conditions vary, the legislating subjectivity can let the one law proceed from the other, can in this way lay out and develop systems of laws. We see it in the parallelogram of forces. One should think that here a law of its own was set up for a special case of its own; but the thinker soon discovers that the apparently special case is precisely the general case, for two forces acting at one point must always form an angle with one another.*)

Motion along the diagonal of the parallelogram is, essentially regarded, a composite

*) Let the two forces be P and Q and their resultant R . For the parallelogram of forces the equation then holds: $R = \sqrt{P^2 + Q^2 + 2PQ \cos(P, Q)}$. If now the angle $(P, Q) = 0^\circ$, then $\cos = 1$, and $R = P + Q$; if the angle $(P, Q) = 180^\circ$, then $\cos = -1$, and $R = P - Q$. The law of the parallelogram is a fundamental law for the composition and resolution of mechanical forces, from which an infinite system of particular laws lets itself be developed.

motion, for the force by which it is determined is a co-working of two component forces; but just as the difference between simple and composite force is only relative, so too the difference between simple and composite motion is relative. The resultant marked by the diagonal can in its continuation itself become a component force, whose angle with another corresponding one yields a new resultant in a new direction, so that the track of the motion becomes, with the finite alteration, a broken line. If the alteration happens continuously, so that the component forces (the composants) are incessantly converted into a resultant, and the resultant again into a component force, then a curved line comes forth.

The true concept of motion in the curved line, whose law mechanics exactly determines, rests upon a resolution of two layers of misunderstanding. The ancients supposed that it belonged to the peculiar essence of certain bodies to move in curves, that notably the celestial bodies, in accordance with their higher nature, moved in circular orbits. That was a misunderstanding. By resolving this misunderstanding physics became an exact science. But to make composite motion clear, one had to resort to parcelling out. Under the parcelling the curved line becomes a broken line: let the fragments be as small as they will, they remain fragments nevertheless. Hereby a new misunderstanding came forth. The method of parcelling has to such a degree given offence to the more recent philosophy of nature that speculation has openly broken with mechanical physics and sought the truth in a modified conception of the ancient representation. Hegel is so angry at the parcelling that he not only forbids physics — what he concedes to logic in ample measure — any use of abstraction of its own, but goes so far as to want the representation of forces wholly rooted out. “Newton,” he says, “assures us that a leaden ball *in coelos abiret et motu abeundi pergeret in infinitum*, if (quite so: if) one could only impart to it the requisite velocity. Such a separating of outward from essential motion belongs neither to experience nor to the concept, but to abstracting reflexion alone.” By introducing “the representation of forces,” Newton is supposed, for all his reservations, to have done considerable damage to the true knowledge of nature. These are speculative misunderstandings. These misunderstandings resolve themselves and disappear when the relation between continuity, parcelling and infinite discreteness is sufficiently cleared up.

If a body be thrown upward in a slanting direction, then the force answering to the impulse can be regarded as a composition of two forces, of which the one would carry the body vertically upward, along the Y-axis, the other drive it horizontally forward, along the X-axis. If now every other action be thought away, the body will go along the diagonal of the rectangle which the coordinates x and y form with one another, and the path s traversed in the time t will be equal to $\sqrt{x^2 + y^2}$. The angle α formed by s and x will be constant, and its tangent $\frac{y}{x}$ of course likewise. In such a case it is then indifferent whether one brings out the continuity or the parcelling or the infinity of the discreteness. Every finite piece of s is a continuum, and the arbitrarily posited limits do no damage;

but s can also be resolved into elements answering to elements of x and y :

$$ds = \sqrt{dx^2 + dy^2}; \quad \frac{y}{x} = \frac{dy}{dx}.$$

It is otherwise when the body is, during the motion, represented as subjected to a vertically downward action; here an essential difference between parcelling and continuity shows itself.

If the action be represented as setting in at certain intervals, then s will separate into finite pieces, s_1, s_2, s_3 and so forth; the angle α will alter each time, since each piece of s gets an inclination of its own to the axis of abscissas, and $\frac{y}{x}$ will obtain different values. Every piece is a diagonal in a parallelogram of forces, one side of which is a continuation of the previous resultant, and the other side of which is formed by the supervening action: the body's path becomes a broken line whose parts can be measured and exactly determined.

If the partial action be exchanged for a continuous one, such as gravity g , then s alters its direction continuously, and the path becomes a curved line. Straightforwardly in the form of continuity such a path cannot be determined, for the infinity of the alteration cannot be determined for itself. Straightforwardly in the form of parcelling the path cannot be determined either, for the parcelling is a finite discreteness which breaks the continuity. Only that reciprocal determination of finite and infinite which is conditioned by infinite discreteness — whereby an operation with elements can become just as exact as a calculation with finite magnitudes — can resolve the parcelling into a finite determining of the infinite.

The equation: $y = f(x) = \operatorname{tg} \alpha \cdot x - \frac{g}{2a^2 \cos^2 \alpha} x^2$ shows how the smallest alteration in x continuously brings with it a corresponding alteration in y ; from which again follows a continuous alteration in s . The manner in which y depends upon x expresses the law. We have in the law the universal essence determining for the infinity of the phenomena — the infinite quantity of single cases. But in this infinity we have at the same time finitude. In an essential manner finitude is determined by the determinate function; in the outward form of the appearances, laid out for division and parcelling, the infinity of the finite comes into view in the swarm of single values. The transition from finite to infinite and from infinite to finite is determined by the mutual relation of the elements. Every element of the curve coincides with a corresponding element of the curve's tangent; how, i.e., by what law this tangent alters its direction in the infinite, is seen from the equation: $\frac{dy}{dx} = \operatorname{tg} \alpha - \frac{g}{a^2 \cos^2 \alpha} x = \varphi(x)$.

What holds of the law holds also of the forces. The resolution of forces is no straightforward parcelling; it is an analysis under which each component force is brought out for itself by an abstraction necessary for science. From the side of essence every one of the component forces is a moment of the resultant. Where a parcelling really takes place, it is nature's own parcelling, existence itself, that in the form of otherness makes the

one force independent of the other. Nor is the composition of forces a simple addition; the force acting in the resultant is the unity of the component forces, a unity in which their independence is wholly cancelled. Here there is not, as Hegel nevertheless asserts, the least ground for entering, on the concept's behalf, any objection whatever against the analyses of physics.

But is it now, strictly speaking, the laws that determine the forces, or might one not perhaps reverse the relation and say that it is really the forces that determine the laws? A superficial conception of the manner in which rational mechanics presents its problems seems at first to speak for the latter. An example. A certain velocity is imparted to a body in any direction whatever, only not vertical, whereupon it is left to the action of gravity: of what character will the path be? The answer is that the path, if the motion be assumed to go on in a vacuum, must be a parabola. The law of the parabola is accordingly the law of the motion. The represented forces and the posited conditions seem here to be the first, and the law the second, derived from them; and so: the forces are determined by representation, the law by thinking, and what is thought depends upon what is represented. How one-sided this way of seeing is comes to light from the relation between representation and thinking. The merely representing consciousness does not know the law at all. It cannot even, without the aid of thought, set up what is represented as a presupposition; for the presupposition is only in the reflexion of its consequent, but the consequent is only for thought. A determining of the forces by mere representation is an act of imagination, a fleeting, preliminary determining that loses all significance when thought is wanting. The determining representation is a function of thought; it is sent on ahead by thought itself and stands from the beginning in thinking's service.

From the standpoint of mechanics the matter easily takes on an appearance as though thought's task were exclusively to discover the laws, as though the laws were in and by themselves objective without needing objectification. The objectification would accordingly be merely for subjectivity's sake, not for the laws'. It is an illusion which strict science will instantly reject once it becomes aware of it. If the law-discovering subjectivity were not itself legislating, it could not discover the laws at all; it could not judge of the conditions, not distinguish the possible from the impossible, not determine the hypothetically necessary, not assure itself of the a priori validity of the alleged laws. And what becomes of the objective laws when they are separated from the legislating, objectifying subjectivity? Dead, meaningless formulas! A law of equilibrium for two points connected by a line and acted upon by the forces P_i and P_m , whose angles with the X-axis are α_i and α_m , the unknown force in the connecting line being R , its angle with the X-axis δ , and φ_i and φ_m arbitrary angles, is preformed, for a system of several

mutually connected points, by the equations:

$$\begin{aligned} P_i \cos(\alpha_i - \varphi_i) + R \cos(\delta - \varphi_i) + \dots &= 0 \\ P_m \cos(\alpha_m - \varphi_m) + R \cos(\delta - \varphi_m) + \dots &= 0 \end{aligned}$$

But as little as these equations would have significance in themselves if no one understood them, just as little would the law preformed by the equations have in itself the least validity if there were no subjectivity, human or divine, that could think the law, discover the law with self-legislating assurance, and objectifying maintain the law's objectivity.

b) Actual forces: the fulfilment of the laws. From the subjective standpoint, so far as it is merely a question of represented forces, the agreement between the divine and the human reason is pervasive; for the laws of nature are laws of thought, and the laws of thought at once both human and divine. But although all laws of nature are laws of thought, the laws of thought are in their free ideality not yet laws of nature. Apprehended for themselves, the thought laws have only an ideal objectivity, articulated by the representations; in nature itself, by contrast, the laws have a material actuality answering to the forces; this actuality is their fulfilment.

From the objective standpoint, so far as one looks expressly to natural actuality, the dualism, the difference in principle between divine and human objectification, comes into view again. From the divine side all facts must surely be derivable from the determining laws; but from the human side the actual laws must be drawn out of given facts.

The laws uttered in factual propositions about nature as in a divine language of power cannot, however peculiar they may be, be thought to stand in conflict with the general laws of reason; but it makes an essential difference whether the laws are treated as abstractly universal determinations of form, and so *mathematically*, or are shown in intuitable unity with the content of actuality in which they have their fulfilment, and so *physically*.

“Just as mathematics,” says H. C. Ørsted, “must treat the objects of nature in its own way, so also must physics. The ideal of the former is to accomplish everything without borrowing anything from experience, and when, owing to the impossibility of wholly attaining the ideal, it does make a loan from experience, this must be something as general, as little individualized, as possible. It belongs to its artistic perfection to take the most circuitous ways rather than to take up the alien stuff of experience into itself. Quite otherwise does it stand with the science of nature. Nature is its scientific property; it does not borrow from it, but draws from it as from its proper source. Its ideal is to present the objects to intuition” — and so the laws in their fulfilment — “and with a view to this it works up forms and magnitudes.”

The scientific significance accorded to physics as a science of experience is naturally dependent upon the manner in which the essence of experience is conceived as a whole. True experience rests upon an even co-working of sensation and thinking; but

this co-working can be perverted, according as it is either sensation that overwhelms thinking, or the reverse. If sensation is one-sidedly predominant, then the gaze fastens exclusively upon the facts in their singularity. Wading in heaps of accumulated facts, the observing consciousness gets so busy sorting, combining, adding and subtracting the many singularities that it can hardly recognize the thought in the universal representations it has itself abstracted. If a priori thinking is one-sidedly predominant, then the facts are involuntarily reduced to fleeting, transient appearances of the universal. Without entering upon any exhaustive investigation of the material in its singularities, the observer, tempted by one or another favourite hypothesis, will all too easily spin out a theory and give it out for a truth grounded on experience. Against both aberrations physics, the true science of experience, must wage a constant struggle.

The fundamental scientific concept upon which the science of experience builds is the concept *induction*.

That the induction must become incomplete when the observations are defective goes without saying. But that an outward heaping together of merely sensible observations, even if they be multiplied on an ever so enormous scale, can likewise give no true knowledge of nature, is then also evident.

One of the most thorough proofs of a superficial conception of induction's logical validity we have in Stuart Mill's inductive logic. "We learn," in this philosopher, "from experience that there is in nature an invariable order of succession, and that every fact is always preceded by another fact. What we call cause is the invariable antecedent, what we call effect is the invariable consequent. The concept of necessity, of which so much is made, here finds its explanation. Necessary is that which will occur, whatever we may assume with respect to all other things. Necessity is present when the antecedent is sufficient and complete, when it contains all the conditions sought, and no new condition is required. Unconditionally to be about to follow — that is the whole concept of cause and effect." So far as such a manner of explanation can be supposed to have something dazzling about it, this lies in the unembarrassment with which thought as thought is in one and the same breath invoked and dismissed. Seen in the light of thought, it is incontestably true that the relation between cause and effect encloses necessity, that the consequent must unconditionally occur when all the conditions are present. But thought is not allowed to count as thought; it is not thought's a priori assurance, but the testimony of the sensations, of the repeated sensations, that here, in the name of experience, gets to signify something. That a sum of sensible observations, however ingeniously they be sifted and weighed, cannot possibly secure us an "unconditionally to be about to follow," is evident to such a degree that it has not even been able to escape the author's own logical attention. Stuart Mill "is convinced that a man accustomed to abstraction and analysis" can very well "conceive that in certain places — for instance in one of those firmaments into which astronomy now divides the universe — events might follow one another by pure chance without any fixed law, and there is nothing,

either in our experience or in our mental constitution, that gives us sufficient ground to believe that this nowhere takes place.” According to such a logic the laws do not belong to the essence of things, but are in certain regions of nature accidentally pasted on the outside of things, much as one pastes a device on a bottle. If Hume rose from his grave, he would soon convince Mill that he has no need whatever to trouble himself with the long journey to “one of those firmaments” in order to get the laws away and necessity dissolved; that it happens far more easily when he stays where he is and bethinks himself of what he has himself said.

There is reason in nature; this central thought, which Ørsted has so beautifully uttered and maintained in our literature, gives induction another and deeper scientific significance than the Millian explanation; here one starts from the premiss that the laws lie in the essence of things. This central thought must have stood clear before Newton when the idea of universal gravitation dawned upon him. The eye that saw in the falling apple an utterance of the force that also draws the moon down toward the earth belongs not to sensation but to thinking. It was thought’s intimation of a universal essentiality indwelling in all bodies that gave the impulse to the great discovery. But with the thinking, sensation too came into its right. The universal essentiality indicated in one fact had again to be more closely determined by facts. Only so could the thought law confirm itself as a law of nature, a universally valid law of experience; in other words: the law of gravitation had to be sensed and seen in its fulfilment.

The reciprocal action between empirical physics and rational mechanics shows itself here in the fairest light. For the observations to begin and experience to be introduced, the problem had to be put into an equation; but the equation that was to give the observations significance had itself to be built upon observations.

That the moon’s fall must stand to the apple’s as the force that draws the moon to the force that draws the apple goes without saying; but in what relation do these forces stand to one another? The force that draws the apple is known from the law of fall — $15\frac{5}{8}$ feet in the first second; the force that draws the moon — the moon’s fall in 1 second — can, since the moon’s mean distance and mean period of revolution are known magnitudes, likewise be determined. If the moon’s distance be denoted from the earth’s centre by R , and the angle answering to the arc (in one second of time) by Θ , then the piece denoting the moon’s fall toward the earth is $\rho = 2R \sin^2 \frac{1}{2}\Theta$. The relation of the forces is accordingly determined by the quotient $\frac{2R \sin^2 \frac{1}{2}\Theta}{15\frac{5}{8}}$; for R to be determined, the earth’s radius r must also be known. Here is a circle of presupposed observations, and yet it cannot without special preparation be enriched by new observations. It lay near, indeed, to seek the ground of the difference in attraction in the difference of distance, but with this general surmise no experience could yet be made; only with the assumption that the attracting forces stand in inverse ratio to the squares of the distances was the

equation determined:

$$\frac{2R \sin^2 \frac{1}{2} \Theta}{15 \frac{5}{8}} = \frac{\frac{\mu}{R^2}}{\frac{\mu}{r^2}} = \frac{r^2}{R^2}.$$

The equation thus determined was a question put to actuality in so exact a form that the fact itself necessarily had to answer either yes or no.

The first answer that Newton received to his question was, as is well known, an unfavourable one. Since actual nature could not err, he drew, disheartened, the conclusion that the error lay in his thought; that, curiously enough, was his real error. In the matter itself his original thought was as faultless as nature. The error lay in the value then current for the earth's radius. When this error was afterwards corrected by a new survey, Newton found his thought confirmed; and only then had the turning point arrived at which the law found for a particular case could, by an induction answering to continued observations, be extended to a universal world-law.

The Newtonian law has in its time furnished a magnificent example of how the fulfilment of a law can, by means of the supervening conditions, alter a law's appearance so that it becomes almost unrecognizable. In occupying himself with the perturbations, particularly with the peculiar alterations to which the moon's orbit is subject (the motion of the apogee), Clairaut came to raise doubts about the universal validity of the Newtonian law. He remarks, on the occasion of a dispute with Buffon, who would defend the law with a priori grounds, that the ways in which the Newtonian law had hitherto been supported were so encumbered with exceptions and piecemeal taking into account of various influences that a law different from $\frac{a}{x^2}$ could give just as good an agreement between the results of calculation and of observation. In solving "the problem of three bodies" Clairaut came provisionally to the result that the expression for the law would have to be, not $\frac{a}{x^2}$, but $\frac{a}{x^2} + \frac{b}{x^4}$. Further investigations led, however, to the result that the final solution of the problem precisely confirmed Newton's formula. The reciprocal action between the three bodies — the central body, the perturbed and the perturbing body — goes on, namely, in each and all according to the same law, so that the varying conditions bring with them only modifications: the particular is in this example a modification of the universal unchanged in itself.*)

But the conditions of actuality under which a law is fulfilled may also be of such a character that they introduce one or more new determinations of law, heterogeneous with the given universal form, and thereby effect an essential alteration in the law. One may think of the alteration that goes on with the law of fall when the fall happens on an inclined plane and the modifying circumstances are taken into consideration; of the alteration that happens with the law of pendulum oscillations, according as one

*)That an alteration of the fundamental formula may nevertheless perhaps be possible must be conceded, since distinguished astronomers, such as Bessel, still harbour some doubt.

considers either “the simple” or “the compound” pendulum^{*)}; of the difference between the parabola and “the ballistic curve,” etc. The deviation is in the last case particularly so pervasive that in actuality a wholly new law shows itself.

The fulfilment of a law happens, then, in essentially different ways, according as the supervening conditions either merely alter the phenomenal expression, or bring with them a transformation of the law, or even change it in such a way that it is taken up into another law and disappears in it. It stands with the laws as with the forces. How the particular laws, in mutual reciprocal action, partly confirm, partly alter, partly mutually cancel one another can first be fully elucidated when we consider the corresponding forces from the standpoint of totality.

c) *Totality of forces: the realm of laws.* The laws of nature are unalterable; this fundamental proposition, on which physics stands fast, and without which it would not be the strict, exalted science it is, seems, when it is uttered in all unconditionality, to strike down at one blow everything set out above about alterations in forces and laws. And yet the philosophy of nature knows itself, on this point, in full agreement with physics. Physics has, namely, a right to set up its proposition — that the laws of nature are unalterable — only because logic has a right to set up its own: that the laws of thought are unalterable. When, then, logic proves that the unalterability of the laws of thought consists precisely in their alterability, it has *eo ipso* shown that the same must hold of the laws of nature. The unalterable cannot be held fast for itself in finite separation from the alterable, any more than essence can be set up finitely over against semblance and phenomenon. Essence for itself is semblance and nothing, when semblance and phenomenon are thought away. Essence is itself involved in the alterations; its unalterable subsistence shows itself precisely in this, that through all its alterations it preserves itself unaltered. The unity of unalterable and alterable does not efface the difference; no, it confirms the difference: here is essential difference, because here is essential unity.

The forces of nature are unalterable; and yet they are altered, in that they work and enter into reciprocal action: in their essence they are unalterable, but in their utterances they are highly alterable. So too with the laws: they are in their essence unalterable, but in their fulfilment alterable. When the proposition about the unalterability of the laws of nature is not understood in this way, but is proclaimed outright as a dogma, an article of faith in which essence and phenomenon, possibility and actuality run out into one, then

^{*)}At equal angle of displacement and force of gravity, the times of oscillation of two simple pendulums stand to one another as the square roots of the pendulums' lengths. With the compound pendulum the law cited holds, by contrast, only for the pendulum's reduced length, i.e., for the distance from the point of suspension to the centre of oscillation. In determining the centre of oscillation the moment of inertia is introduced. If $m, m', m'' \dots$ are the parts of the mass, and their distance from the point of suspension $x, x', x'' \dots$, then the moment of inertia is $mx^2 + m'x'^2 + m''x''^2 + \dots$. If the whole mass be called M , and the distance of the centre of gravity from the point of suspension a , then the reduced length $l = \frac{mx^2 + m'x'^2 + m''x''^2 + \dots}{Ma}$ is the designation of a law of its own. That the laws fused together in the fulfilment do not contradict one another goes without saying.

protest must be made. In spite of the catholic semblance of infallibility with which the dogma is preached, sound sense will easily smell a rat and ask: what does accidentality mean? what becomes of freedom?

That necessity and accidentality are everywhere united in the great whole of nature, that the necessary in the world of facts is accidental and the accidental necessary, has its ground in this, that the unalterable subsists in alterations. Every actual law must have actual conditions and show itself determining for active forces. Let a law of nature be as universal as it will, it can still never be fulfilled in the merely universal; a universal fulfilment is an abstraction: the actuality of the fulfilment is in the actual cases. The cases all fall under the law; but they do not for that reason lose their accidentality, do not cease to be cases. Were there no bodies that could fall, the law of fall would not be fulfilled. If there were no other bodies in celestial space, the force with which the earth's body acts upon other bodies could not utter itself, could not become actual. In essence and possibility the law of fall is the same unalterable law, whether in given cases it is fulfilled or not fulfilled; but in actuality it is not the same, for when it is not fulfilled it is unactual. In essence and possibility the earth's force of attraction is unalterably the same, whether it works or does not work. But actuality introduces an alteration, for only the working force is actual; the force that does not work is only possible.

It will now easily be seen how law and case enter into a many-sided reciprocal relation. That a stone falls to the earth is, with respect to the single stone, only a single case, but it expresses at the same time the necessity with which all stones in all like cases must fall. Bodies are, as regards gravity, under the same law because they are under the same necessity, a necessity that expresses their indwelling, universal essence. But, curiously enough, there are also laws which permit the accidentality in each single case to withdraw itself from the universal necessity.^{*)} There are, then, in the world of cases two essentially different circles of laws, *laws of essence* and *laws of accident*. In the first circle accidentality is necessary, in the second necessity is accidental. The two circles form, in their sharply stamped opposition, two extremes which must not be confused. But the extremes do not stand over against one another separated by a yawning chasm; innumerable determinations of transition form a web in which the threads of necessity are, through manifold crossings, as it were woven into the motley carpet of the accidentalities.

There is a pervasive connexion in the whole of nature, no leap, no interruption, no newly created matter, no interpolation of new forces, no essentially new laws: everywhere events engage in one another as links in a closed chain; the new that is formed is formed out of the old, out of what already was in primeval time. The Hellenes, says

^{*)}For because one man marries at a certain age, it is not necessary that others marry at the same age; because one man hangs himself, it is not at all necessary that others hang themselves too; because one man in distraction posts a letter without an address, it is not for that reason necessary that others must do likewise. And yet the calculus of probability shows that large groups of such cases, from whatever sphere of existence they be drawn, let themselves be brought under corresponding laws.

Anaxagoras, wrongly assume becoming and passing away, for no thing becomes or passes away, but things are composed out of subsisting things, and thus they would more correctly call becoming composition and passing away separation. What Anaxagoras divined, mechanical natural science has from its standpoint sufficiently endorsed. Just as there is in the universe a given system of “eternal forces,” so there is also a system of “eternal laws” closed in itself; this system is *the realm of laws*.

To have discovered and conceived nature’s infinite unity of connexion, and thereby opened to itself a magnificent view into the realm of laws, is “modern science’s” pride. The philosophy of nature acknowledges what is true and warranted in this. But in the hymns that ring out in honour of deified necessity, unresolved dissonances are not seldom heard, jarring tones which recall that modern science, out of excessive joy over necessity, is on the way to denying freedom.

Can freedom, one asks, really consist with necessity? This question presents, as regards the knowledge of nature, two principal sides, according as one reflects rather upon accidentality or upon the pervasive unity of connexion.

Just as all material actuality has a moment of accidentality, so freedom’s ideal actuality has a moment of arbitrariness. What accidentality is in the outward, the factually given, arbitrariness is in the inward, what originally determines. If necessity can be reconciled with the accidental outwardly, then it must surely also be reconcilable with the arbitrary inwardly. But arbitrariness in nature? The mere representation of this is an offence to physics. The offence will pass off, however, once the physicist comes to himself. It is, after all, precisely physics itself which, without in any way having to do with the arbitrary, supplies philosophy with the most important presuppositions and most reliable points of support for the recognition of the moment of arbitrariness.

Physics has broken with speculation in order to hold unconditionally to the fact. That is a decisive turning point in the history of science. To hold to the fact is to take it as it is, without reasoning about why it is now just so, whether it might not perhaps just as well have been otherwise. The fact is because it is; but what only is because it is, in its given necessity, immediately to be apprehended as something accidental. Observation sees in the many peculiar facts nothing but peculiar cases; whence, then, do the given cases have their original determinateness? From other cases. And the other cases? From other cases. And these again? From other cases. The endless reasoning that incessantly exchanges case for case drowns in accidentality without getting so far that it can even so much as glimpse the grounding necessity.

Only by holding to reason and starting from the a priori presupposition that nature’s order, grounded in itself, rests upon an all-embracing ground-unity that determines everything from within, can the observing investigator discover necessity. Every particular case must, after all, have its particular ground: as many cases outwardly, so many particular grounds inwardly. But the separation of the ground-unity into particular grounds cannot possibly be accidental; to make what grounds accidental is to annihilate

all grounding. The ground-unity must therefore itself determine itself from within to a manifold of different grounds, connected in their difference; the ground-unity must, for the sake of the connexion, necessarily be a determining self-unity.

If the observer will now, from this standpoint, take aim at accidentality, he shall assuredly discover arbitrariness. So as not to be dazzled by extraneous considerations, one may choose such general, uncompounded facts as: the force of cohesion, the force of gravity, magnetic forces, electrical attractions, etc. None of these forces lets itself be derived from the others; nor has physics any prospect of being able to deduce the corresponding properties of things either from their other properties or from the general essence of matter. The difference of the fundamental forces must have a determining ground. But what is the determining ground of accidentality? It is arbitrariness. As what is determined is, so must what determines be. The greater the emphasis that falls upon the determinate force in its singularity and immediate accidentality, the greater the emphasis that must fall, from the side of the grounding, upon the exclusive utterance of will, the act of arbitrariness by which it has been determined. Accidentality says that the given force is because it is; arbitrariness says that it is because it shall be. If the gaze fastens upon the single facts in their singularity, then everything inwardly is arbitrary, just as everything outwardly is accidental. What accidentality is for the system of facts, arbitrariness is for the system of grounds.

Rational mechanics is from first to last systematically conditioned by arbitrary and accidental presuppositions, and yet an articulation of necessity so sure and firm in thought that it truly does honour to human reason. Without a reciprocal determination of accidentality and arbitrariness it is impossible to set oneself any mechanical problem whatever. Let the mathematician choose whatever problem he will, he sees at once that he must either posit something himself, and so begin arbitrarily, or presuppose something as given, and so begin accidentally.^{*)} There is, in the circle of conditions, the same relation between the necessary and the arbitrary as between the necessary and the accidental: the consequence is equally necessary whether the presupposition is arbitrary

^{*)}How arbitrariness and accidentality stand in principle to the necessity of connexion can, as regards exact analysis, be seen for instance from *interpolation*. A number of scattered points, lying on a certain determinate curve, is given. Whether the given curve answers to an actual fact and so far falls under the accidental, or is expressly posited by a free self-determination and belongs under the arbitrary, is indifferent. The only thing that can be known from the beginning, because it lies straightforwardly in the necessity of connexion, is that the ordinate y and the abscissa x must for every interval answer exactly to one another; $F(x, y) = 0$ or $y = f(x)$. The problem to be solved by the interpolation calculus is then the following: from the *given* values of a function, answering to *given* values of the independent variable, the argument, to calculate the value of the function answering to any value whatever of the argument lying between the given ones.

If it can now be presupposed that the curve answering to the given function has a continuous and even course, then the problem will be soluble by an approximation which is the greater, the closer the given values of the argument lie to one another; the interpolation formula becoming:

$$y = a_0 + a_1x + a_2x^2 + \dots a_nx^n. \quad (1).$$

To determine the $(n + 1)$ constants entering into this, $(n + 1)$ corresponding values of the argument and

or accidental.

Just as the moment of accidentality merges into the unity of connexion, so the moment of arbitrariness merges into self-unity. The reciprocal determination of self-unity and unity of connexion shows itself in the solution of any mechanical problem whatever.

Chapter Two.

the function must be given. If these are respectively:

$$x_0, x_1, x_2, \dots, x_n \text{ and } y_0, y_1, y_2 \dots y_n,$$

one can write down $(n + 1)$ equations of the form:

$$y_r = a_0 + a_1 x_r + a_2 x_r^2 + \dots a_n x_r^n.$$

By the solution of these $(n + 1)$ equations of the first degree the constants in equation (1) can be determined. Geometrically regarded, the problem can accordingly be solved by drawing through the given points a parabola of the n^{th} degree.

The solution of the $(n + 1)$ equations of the first degree is avoided by bringing equation (1) into the form: $y = X_0(x-x_1)(x-x_2)\dots(x-x_n) + X_1(x-x_0)(x-x_2)\dots(x-x_n) + \dots X_n(x-x_0)(x-x_1)\dots(x-x_{n-1})$, in which every term contains all the factors $(x - x_0)$ etc. but one, the first term namely lacking the first factor, the second the second $\dots$ the n^{th} the n^{th} . The constants $X_0, X_1, \dots X_n$ are determined from the corresponding given values of x and y ; thus X_r from $y_r = X_r(x_r - x_0)(x_r - x_1) \dots (x_r - x_{r-1})(x_r - x_{r+1}) \dots (x_r - x_n)$. The interpolation formula sought (Lagrange's) accordingly becomes:

$$\begin{aligned} y = & + \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} y_0 \\ & + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} y_1 + \dots \\ & \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_n)}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} y_n. \end{aligned}$$

This so richly articulated necessity of connexion can, as said, have arbitrariness just as well as accidentality for its foundation.

If we now let the given curve with the even, continuous course be an image of given nature, then it will be possible to see how speculation and empiricism, each from its own side, are tempted by the necessity of connexion to apprehend this actual world, this present order of things, as the only possible one. Speculation holds to the universal, stops at $y = f(x)$, and for want of mathematical insight cannot get further. Empiricism starts from the particular, the given single members. The more scattered the observed single members are, the looser the whole connexion of nature seems to be. But as the sum of the observations increases, the single members move closer to one another: the countless possible curves approach coinciding with a single actual one. In his restless labour at finding out and calculating the new intermediate values, the empiricist gets an ever stronger impression that this order of nature, this curve with the even course, must be the only possible one, not merely under the given presuppositions but in unconditional universality. To lift the illusion, however, the empiricist need only go back to pure mathematics and bethink himself of the general foundation of interpolations. The *constants* in equation (1) must for every *different* curve be *different*; but an infinite quantity of different curves, answering to an infinite quantity of different series of $(n + 1)$ corresponding constants, are after all possible, and just so are infinitely many different worlds possible.

Mechanical Causes.

A.

Mechanical Ideality.

That natural necessity unites in itself the arbitrary and the accidental as cancelled determinations has its ground in this, that the mechanical cause is at once to be apprehended both as concept and as object. As object the cause is real, as concept it is ideal. To make the ideality of the cause recognizable, we shall now show in particular how matters stand in mechanics with *force*, *condition* and *cause*, and upon what it essentially rests that the unity of the cause masters the outward manifold and takes up into itself a plurality of *heterogeneous moments*.

a) *Force and cause*. When mechanics makes “what we call force” an “unknown” cause, it thereby indicates both a difference and a unity of force and cause. The difference is that the unknown lies in the force, not in the cause. That the concept of cause has validity, that every motion of matter must necessarily have its cause, cannot be doubted; in the contrary case rational mechanics would have to cease being a science. But one is in doubt about what the active element in the cause really is, how it is best designated, and so makes a kind of apology by saying: “we call it force.” It has been shown in our foregoing that this apology is, from the standpoint of essence, quite superfluous, that the concept force is in its kind just as trustworthy a concept of nature as the concept cause. And yet the apology has an acceptable ground when we look to actuality. Precisely by the cause being determined as force, the concept goes out beyond itself; it is not exchanged for another concept, but is set over into activity and expressed in fact. The active element is a concept and yet more than a concept; it is force. It is precisely through force that the mechanical concept of cause passes over into actuality and becomes a working cause.

How conceiving and working stand to one another in this respect can be found out by seeking the origin of the peculiar designations in which mechanics — though it is otherwise indifferent to the question of the forces’ “physical origin” — holds fast a qualitative difference among the forces themselves, between accelerating force, moving force and living force. It is in actuality the same force — the active element is force, and force the active element — and yet, on account of the determinations of content with which the concept of cause is enriched in a progressive development, qualitatively different forces; it is in essence the same cause, and yet, on account of the identity of the active forces, different causes.

α) *Accelerating force*. As the cause of motion, force can either work suddenly, e.g. in impact, and then gets the name of instantaneous force, or continuously alter the velocity and is then called — positively or negatively — accelerating. In the concept itself the difference is vanishing. The instantaneous force requires a finite time, however

small, to effect a finite velocity, and is so far essentially accelerating; conversely, the accelerating force can be resolved into infinitely many impacts and is in each impact to be regarded as instantaneous. Essential for the concept of cause is, by contrast, the peculiar circumstance that force in and for itself as force is not the cause of the acceleration, that it is so only by means of inertia. If we think inertia away, then force would, in order to accelerate or retard a motion continuously, itself have either to be increased or diminished continuously; but in that way the cause of the acceleration would come to lie not in the force, but in what caused the alteration in the force. If the cause of the alteration of force be again regarded as a force, then this force must again have a cause of alteration, and so on to infinity.

Only by working in unity with inertia is force accelerating. In acceleration we have, then, the sought unity of reflecting and working, a categorical expression for the mechanical ideality of the cause. Analysis shows exactly how matters stand with this unity. A constant force k works upon a material point; in the relative instant dt , $kdt = dv$. The material point receives in each instant an infinitely small but constant increment of velocity; but what brings it about that the velocity, which in the first instant is only dv , becomes in the next instant $2dv$, in the next again $3dv$ and so forth, so that in the n^{th} instant it has grown to $ndv = v$, for $n = \infty$, is precisely inertia. It is inertia that takes up, and it is inertia that preserves every increment of velocity, and thereby makes it possible for the velocity to grow with the time, to be a function of the time: $v = f(t) = kt$. From this it is seen that the passive is just as much determining for the cause as the active. The active is passive, for the force's action upon its point of application is from first to last dependent upon the inertia that receives and preserves the action; the passive is active, for inertia is on its side just as co-operative in the acceleration as force. So pervasive is the reflexion: the accelerating cause is the reflected, active unity of the active and the passive.

In the form of reflexion the cause is concept; but the reflexion here spoken of is the facts' own determining reflexion, a reflexion in power. Every reflex in logical conceiving has on the side of actuality its antilogically existential moment in the conceiving power; infinite reflecting has activity for its actual expression: point for point this reflecting goes together into one with a corresponding working. In nature itself the cause is concept and yet more than concept. The concept does not work; it is the force reflected in inertia, the physical cause, that works. But the physical cause is determined to activity by its concept; everything mechanics can elucidate about force and inertia must in every respect borrow its validity from the concept of cause. If we take away the active element in the cause, then there is only a logical concept, no actual cause; but if we take away the concept of cause, then force, inertia, the active element, and whatever else one will call it, are only idle representations; there is then no cause at all, either actual or unactual.

β) Moving force. Since it is precisely the moving cause that is called force, it might at first glance seem superfluous to introduce a category of its own and set moving force

alongside accelerating force. The accelerating force is moving, and the moving force accelerating; upon what, then, does the difference rest? Upon a particular development of the concept of cause. In determining acceleration and the accelerating cause, the material element in the matter was tacitly understood, but not expressly brought out. True enough, force issues from matter, inertia is in matter, the reflexion of inertia and force has matter for its foundation; but the material moment first comes into its right when the concept of mass enters into the concept of cause, and the quantity of matter becomes decisive for the cause. If we choose for example gravity at the earth's surface as accelerating force g , then, as regards the acceleration, it is altogether indifferent whether g works upon a small or upon a large mass; every smallest part of the body will — in empty space — fall with the same velocity, whether the parts be connected or separated: $g dm$ and $\int_0^m g dm = mg$ give in this respect one and the same result.

It is otherwise with the moving force, which is cause on account of the mass. If the mass M is greater than the mass m , then the moving force in Mg has undeniably a greater strength than in mg ; but while between Mg and mg there is only a quantitative difference, since they both belong under the same concept of cause, there is by contrast a qualitative difference between g and mg , for here the concept of cause is different. For the moving force as cause it is not, strictly speaking, necessary that it cause an actual motion. The cause is indeed not without effect, but the effect may be cancelled by a counter-effect, a pressure, and the cause has its reality none the less. So it is with weight. Weight is a product of mass and acceleration without preceding motion: $Mg = P$. The weight P is a reflected unity of the extensive in the mass and the intensive in gravity, an immediate and yet reflected unity of reality and concept; that is weight's mechanical ideality as cause. Mass in its immediate being is no concept; it may easily appear to a superficial observer as though mass might well be cause without entering into the concept of cause, but that is a contradiction. From weight it is seen that the concept of mass is an essential concept of nature, for from weight it is seen that mass is in itself only what it is, in its reflexion with gravity, for another. If gravity alters its intensity, then the mass gets another weight and, as the concept uttered in the equation $M = \frac{P}{g}$ shows, another reality. The mechanical equations are in this respect more than formulas with an equals sign, they are judgments. The equation $M = \frac{P}{g}$ can indeed, in the abstract quantitative sense, be resolved into a tautology: $M = \frac{Mg}{g} = M$, but, apprehended as a judgment, it means something quite other. That Mg is expressed by P means, namely, that the factors M and g must not be held fast in their product as heterogeneous units, but apprehended as moments in the real concept P , the concrete concept that gives existing mass its peculiar value as working cause.

As cause, the moving force is a mass-cause. If now the resistance to the effect be cancelled, then motion sets in, time gets its significance, and the moving force displays itself in the product of mass and velocity, i.e., in the *quantity of motion*: again an alteration in the concept of cause. If we call the moving force mk , the time t and the velocity v ,

then the equation $mkt = mv$, and so $mk = \frac{mv}{t}$, shows that any constant force whatever can be measured by the quantity of motion it brings forth in a unit of time. That the moving force mk , working in the time t , brings forth mv , means then that mkt is cause, mv effect. Insofar as the quantity of motion gained is expressly posited as effect, the working cause has taken the determination of time up into its concept and has become a function of time.

Here now arises the question: is the concept of working cause completely expressed in that concept of time and force which has the quantity of motion for its measure? Cartesius answers yes, but Leibnitz no. To appraise a force, one must know exactly what it can accomplish. It is, says Leibnitz, not enough that the effect be proportional to the cause; the complete effect must be perfectly equal to its cause. To lift a body A of 1 pound to a height of 4 ells, the same force is required as to lift a body B of 4 pounds to a height of 1 ell; it is precisely the force the two bodies attain by falling, each from its own height. But the velocity A attains by the fall stands to that B attains as 2 to 1. The quantity of motion in A will then be 1×2 , while that in B is 4×1 . The quantity of motion in A is only half as great as the quantity of motion in B, although the force is the same; consequently force cannot be measured by the quantity of motion. Abbé de Conti objected that the law set up by Cartesius required regard to simultaneous forces, or to motions which two bodies attain in the same time; but — asks Leibnitz — “when one sees a body of a determinate magnitude move with a determinate velocity, should one not be able to appraise its force without knowing how long it has taken to attain this velocity? And when two equal bodies have the same velocity, but the one has obtained it by a sudden impact, the other by falling for a longer time, will one then say that their forces are different? That would be the same as to say that a man was richer because he had taken longer to win his money.”

γ) Living force. The ground why the quantity of motion cannot be the completely uttered effect of the moving force, of the moving cause, lies in this, that the opposition of cause and effect is as yet unactual. The equation $mkt = mV = Mv$ states that the moving force mk can in the time t awaken a quantity of motion in m which is mV , and in M which is Mv . If we now ask how great the cause is, we are referred to the effect. If we ask how great, then, the effect is, we are referred to the cause. The cause is equal to the effect; that is a merely formal judgment, a tautological proposition, so long as it is not yet elucidated wherein this greatness actually consists. The question is whether the moving cause in the form of mV can really accomplish the same as in the form of Mv ; for if it cannot, then by assuming two different forms it has come under two different concepts and has thus become two different causes.

To test the working cause it is necessary to see it in activity. The activity cannot then be stopped by an immediate resistance; the resistance must be overcome, and the activity each time determined by the manner in which it overcomes its resistance. To this corresponds the distinction Leibnitz has introduced between *dead* and *the living*

force. Among dead, elementary forces he reckons those whose effect is immediately cancelled by a counter-effect; the living force, by contrast, is “always connected with actual motion.” During the motion the resistance gives way, while the force is gradually expended.

If $mkt = mV$ be posited, the activity in the first element of time will be $mkdt = mdV$. What is now accomplished in the running time? The force moves the burden forward in space, but in dragging the burden the velocity decreases, and the decreasing velocity is again an expression of the decreasing force. How long the force can hold out must then be seen from the length of way through which it avails to carry its burden. The equation $mdV = mkdt$ must accordingly be transformed into $mdV = mk\frac{ds}{V}$, so that it is known from the space what is accomplished in the time. The old saying that time is equally long but not equally useful here finds a peculiar confirmation; dt is constant, but V varies, from which it follows that ds also varies: $ds > ds_1 > ds_2 \dots$, since $V > V_1 > V_2 \dots$. From $mdV = mk\frac{ds}{V}$ follows $mVdV = mkds$; by integrating this equation we obtain $mV^2 = 2mks$ and with it the real opposition of cause and effect

But when to the quantity of motion mV there answers a living force mV^2 , then to the quantity of motion Mv there answers a living force Mv^2 . Since now mV^2 , by the presupposition $mV = Mv$, has another value than Mv^2 , it is seen from this that the force in mV , when it is set to work, accomplishes something other than the force in Mv . The living force, which is measured by the work, is accordingly qualitatively different from the force moving in time, which is measured by the quantity of motion. Here there is no leap, either in nature or in thought; it is force, the active element in the cause, that in the course of the development of the concept takes up and gathers step by step the one-sidedly emerging determinations and thereby reaches its complete expression in the form of living force.

b) Condition and cause. When the determinations of existence belonging to the concept of cause are held fast in representation as existing, each for itself, they become conditioning, and the working cause resolves itself into a sum of conditions. But it lies in the nature of the condition to point out beyond itself to a conditioned and to other conditions; as the conditions reflect themselves in one another mutually, they lose their apparent independence, and the unity of cause, undivided in itself, comes into view again as the unconditioned. This unconditioned, which accordingly does not fall outside its conditions but goes together into one with them, is the conditioned cause.

α) The condition. The cause is a concept, but a concept of actuality, and a concept of actuality always has reflexes of actual things in its content. The reflexes of things necessary for actuality are, on causality's behalf, *the condition* and *the conditioned*. Not even force can in this connexion be held fast against the conditions, for force for itself can merely be represented; what is represented, “we call force,” sinks to being a condition. The condition is a presupposition which the representing consciousness in the train of thought partly puts in, partly finds in actuality, a demand for an independent

which for the dependent's sake must be satisfied. The condition, the independent, encloses the conditioned, the dependent, and yet they stand over against one another as member against member. The reflexion of condition is a reflexion of represented members, an outward reflexion hovering between essence and existence, concept and actuality, characteristic of mechanical activity. Thus force and inertia are conditions for the acceleration; they are represented each for itself, and yet point incessantly to one another. Mass and acceleration are again conditions for weight; the same mass can be subjected to another acceleration, and the same acceleration applied to another mass, and so far the conditions must be separated; and yet they cannot be separated, for the altered product of mass and acceleration alters the weight. If time be drawn into the circle of conditions, then weight comes into motion, shows itself as moving force, and comes to be determined as quantity of motion. If, finally, the condition be added that the given quantity of motion shall be applied to overcoming a resistance successively, and be measured by the space the resistance traverses, then we have the living force, though not as cause; for now the cause is dissolved, it is altered into a sum of conditions.

From the categorical peculiarity of the condition it follows that several particular conditions can be enclosed in one fundamental condition, and this again resolved into a quantity of single conditions. Rational mechanics is laid out so as to show how resolution and composition happen systematically. For forces acting upon a point to be in equilibrium, their resultant must be zero; this is the universal condition of equilibrium, its fundamental condition. But what a swarm of particular conditions and varying single conditions can enter into this!

If two forces act in parallel directions toward the same side upon two points connected by an inflexible straight line, then one can determine the magnitude and position of their resultant, and thereby of a force opposed to it which holds them in equilibrium. One sees at once how the condition set up comprehends several conditions: that there are two forces, that they are parallel, connected, etc. If the forces be called P and Q , the resultant R , then we have the conditioned: $R = P + Q$. If it be further made a condition that A , the point of application of P , and B , the point of application of Q , shall have from an arbitrary point O in the prolongation of the connecting line the distances a and b , then the distance between the arbitrary point O and the resultant's point of application C , a distance we may denote by c , is expressly conditioned thereby. The composition of the conditioned, the relation between Rc , Pa and Qb , answers exactly to the composite condition, and this conditioned can by a turn in the reflexion again be set up as a condition. For since $Rc = Pa + Qb$, the equilibrium between the three forces P , Q and R is precisely conditioned by R being applied at the point C and in the direction opposite to P and Q .

Progress in analysis happens on the whole in such a way that what in a preceding connexion was the conditioned passes over in a following one into being a condition, and that new ground is continually won by partly generalizing, partly specifying a given

fundamental condition. To the above circle of conditions belong further the equations $Q = R - P$, $b = \frac{Rc - Pa}{Q}$; if the conditions now be specified so that $R = P$, but not $c = a$, then $Q = 0$, and $b = \infty$. From this it is seen that two equal parallel forces acting in opposite directions (R and P) cannot possibly be held in equilibrium by a single force (Q), for an infinitely small force ($Q = 0$) at an infinitely great distance ($b = \infty$) are conditions that cannot be brought about. Thus we are led by the way of the conditions from the doctrine of parallel forces to the doctrine of couples — an example in little which shows what it means in the large that the way of the conditions is the way of mechanics, that exact science with its equations of condition and its final equations has ground to resolve the cause into conditions. But philosophy of nature is not mechanics; it goes the reverse way and resolves the conditions into

β) The cause. The conditions are not to be effaced, and yet they are to disappear. What is characteristic of the condition is that in its represented independence it is so dependent. The hypothetical judgments — if A is, then B is; if an aberration of the fixed stars takes place, the earth must move about the sun; if B is, then C is; if the earth moves, the fixed stars must have parallax — show us the dependence of the self-subsistent, the dependence of the independent, when the judgments cited furnish premisses for corresponding inferences. One infers: A is, therefore B is; the fixed stars display an aberration, therefore the earth must move, although we cannot immediately sense it; B is, therefore C is; the earth moves, therefore the fixed stars must have parallax, even if on account of its smallness we are not able to measure it. In both cases the inference is by *modus ponens*, and the inference is a proof of the conditioned's dependence upon its condition. But in actuality the relation can also be reversed; it is precisely the condition that is dependent upon its conditioned — which is expressly seen when we infer by *modus tollens*: B is not, therefore A is not; C is not, therefore B is not. Which of the two modes of inference has actual validity in a given case must of course depend upon facts and be determined by observations; but that both have logical validity proves that the condition, dependent as it is, stands and falls with its conditioned.

Where a fundamental condition is resolved into particular conditions, the same dependence shows itself. Motion in a straight line has another fundamental condition than motion in a curved line. If we analyse the latter, then the condition for the straight line will repeat itself here and deposit single conditions, for motion along the tangent gives a straight line, and so does motion in a direction forming an angle with the tangent. If a fixed centre be introduced as a particular condition, then the position of the two directions — along the tangent and toward the centre — will likewise come under particular conditions and be again conditioning. If the one direction is constantly perpendicular to the other, the curve will be a circle and the velocity constant; if, by contrast, the position varies so that the two directions form different angles, now a right, now an acute, now an obtuse one, the velocity will vary, and the curve cannot be a circle.

It will now be possible to see from this how the conditions cancel one another and

disappear in an unconditioned. None of the single conditions cited is particularly fulfilled; the one incessantly prevents the fulfilment of the other. Motion along the tangent is at each instant prevented by the motion toward the centre, and this again by the former. Motion along the tangent disappears into two conditioned results; the one is a flight from the centre — for representation the so-called centrifugal force — which is cancelled by a corresponding striving, the normal force; the other is the advance in the orbit, the only result that is not cancelled.

The unity thus conditioned, the motion in the orbit, now depends upon a corresponding unity of the conditions which does not let itself be resolved, since it contains the conditions not as a sum but as cancelled conditions. This actual unity of the cancelled conditions is the unconditioned, the matter in itself, the original matter, *the cause*. But when the unconditioned is determined as cause, then it is thereby more than a conditioned, it is something caused, an effect.

That there is another fundamental relation between cause and effect than between condition and conditioned is easy to see. If the conditioned is not, then neither is the condition, for the conditioned is a content of the condition, represented as a member for itself; the condition with its conditioned belongs solely to outward reflexion. But the cause embraces actuality, and thereby both inward and outward reflexion. The cause has a being in possibility, in inward reflexion, even when it does not work, just as force has a being as an inner, even when it does not utter itself. A condition must supervene for a force to be able to utter itself, for force and condition are coordinate categories. But the cause is above them both, for it is in them both. The cause does not scrape by, borrowing here a force, there a condition; the cause itself owns the forces, it masters the conditions; its essence is power, the unconditioned of the conditions.

γ) *The conditioned cause.* As the pervasive unity of the conditions the cause is an unconditioned; but since this unity is itself determined by the conditions taken up into it, the cause determined in itself is conditioned. If one reflects upon the difference between cause and condition, then the same cause can, by taking up different conditions, have different effects; if one reflects upon the unity of condition and cause, then the differently conditioned cause becomes different causes, and each effect answers exactly to its cause. That this double-sided reflexion is by no means idle, but on the contrary decisive for mechanical ideality, we learn from rational mechanics.

The cause of rest is the equilibrium of forces; the cause of motion is the cancelling of the equilibrium: it is at bottom the same cause and yet in actuality a different cause. If we reflect upon the difference, then statics and dynamics have each its own circle of problems and fall so far apart from one another; if we reflect upon the unity, then the static problem becomes dynamic, and the dynamic static. The reciprocal determination is grounded on two corresponding principles, the principle of virtual velocities and D'Alembert's principle of the equilibrium of the lost forces.

Everywhere that there is talk of a principle, one must necessarily, however the matter

is otherwise to be understood, think of an unconditioned. The general equation of condition for equilibrium makes in this respect no exception; for it does indeed receive such an extension that it enters into all particular conditions and finds application to any system whatever of material points; but precisely the manner in which the particular conditions go together into unity gives the equation the stamp of an unconditioned. The fundamental thought uttered in the characteristic equation of equilibrium, the formula of virtual velocities,^{*)} — a fundamental thought which comprehends all conditions in the conditioned unity of the determinate cause, the unconditioned — can with full right be called a principle.

According to this principle equilibrium is, to be sure, tested upon a motion which is nevertheless no motion, for a virtual velocity is a possibility that does not become actuality. But this possibility, the merely represented displacement of the points, serves meanwhile to let motion shine through in the equilibrium, to let the equilibrium show itself as cancelled motion, and to make the static problem dynamic.

In the formula for the principle of virtual velocities one has a sum of virtual moments in which each term is a product of the force P_k and the virtual velocity δp_k projected upon the direction of the force p_k , and so $P_1 \delta p_1 + P_2 \delta p_2 + P_3 \delta p_3 \dots$. What is peculiar about this sum is that it indicates at every instant the total effect of all the forces, however different its value may be. What for statics is therefore the universal — namely that the sum of the virtual moments shall expressly be zero — can then, with respect to dynamics, be regarded as a special case, a conditioned cause which goes together with another condition of cause and forms another, more comprehensive unconditioned, a principle which at once determines the cause both to equilibrium and to motion — D'Alembert's principle.^{*)} Let a system of material points m_k , acted upon by the forces P_k , be given.

^{*)}By a point's virtual velocity is understood any infinitely small displacement of the point that can be thought to take place, without regard to the acting forces, solely in accordance with the conditions to which the point is subjected." If the force be called P , the projection of the virtual velocity upon the direction of the force δp , and the reciprocal determination of the universal and the single — the logical and the antilogical — be indicated by an appended index k , then the equation of condition for the equilibrium of a single point is $\Sigma P_k \delta p_k = 0$. The forces are reckoned positive, but the projections positive or negative, according as they fall either along the direction of the force or along its opposite prolongation. The formula holding for a system of points, in which all conditions are taken up, can be expressed by

$$P_1 \delta p_1 + P_2 \delta p_2 \dots + P_n \delta p_n + \lambda_1 \delta u_1 + \lambda_2 \delta u_2 \dots \lambda_r \delta u_r = 0.$$

The terms $P_k \delta p_k$ are called the virtual moments of the forces, the terms $\lambda_s \delta u_s$ the virtual moments of the resistance, and the formula presents the principle. The sum of the virtual moments $P_1 \delta p_1 + P_2 \delta p_2 + P_3 \delta p_3 + \dots$ indicates at every instant the total effect of all the forces. That this sum must be zero at the instant the position of equilibrium is passed is a straightforward consequence of the concept of equilibrium, since the forces then neither promote nor counteract the motion.

^{*)}Let a system of points m_k be determined by rectangular coordinates and acted upon by the forces P_k , whose components along the axes are X_k, Y_k, Z_k . The actual motion of m_k in the time element dt can be assumed to be the same as if this point were free and acted upon by the final velocity v_k and by an accelerating force Q_k which in magnitude and direction is different from P_k . If now every point m_k be assumed subjected, besides the given force P_k , also to the force Q_k and to another equal and opposite to it, namely Q'_k , then the given system of forces is not thereby altered. The whole system moves as though the

Since now the forces with their conditions fall together, according to our foregoing, in the cause, we have then in this system one single conditioned cause, the cause A . But analysis shows that a part of the forces P_k is cancelled by the reactions lying in the system, while another part is applied exclusively to the motion. The cause A accordingly yields two mutually different effects, of which the one, B , is seen in the system's maintenance, the other, C , in its motion. Here is the same cause and yet two different causes. If we reflect upon P_k , then it is the same cause; for it is P_k that is decomposed, so that a part of P_k , namely Q_k , yields the motion, while Q_k , which is equal and opposite to Q_k , forms with P_k a resultant p_k that is lost to the motion; P_k is accordingly decomposed into p_k and Q_k . If we reflect upon the opposition between p_k and Q_k , then we have two different causes, namely in p_k the cause of the system's maintenance (A') and in Q_k the cause of the motion (A''). How intimately the causes thus separated engage in one another is seen from this, that an alteration in the one brings with it point for point an alteration in the other. During the system's motion Q_k varies, and consequently p_k too. The sum of the virtual moments of the lost forces and of the resistances must of course be zero at every given instant; but as p_k and Q_k are determined as functions of time, and with them the cause A' answering to the effect B , so Q_k , with the cause A'' answering to the effect C , is determined at every instant, and conversely; the problem is at once both static and dynamic, because the cause conditioned by the system is itself both a cause of equilibrium and a cause of motion.

c) The unity of the moments of cause. When in a mechanical phenomenon one is expressly to look to what is immediately active, one names force and, insofar as this is in matter, also mass. If, by contrast, one will particularly bring out everything which, without itself being active in and for itself, gets an influence upon the activity and thereby becomes a co-operating element, then one speaks of conditions. If now, as we have seen, the cause has all forces and conditions in its power, then it is intelligible from this that the mechanical cause takes up and unites highly heterogeneous moments. In its essence

points were free and subjected to the force Q_k alone. That part of the forces which goes to overcoming pressures and tensions in the system itself gets no influence upon the motion and must in this respect be regarded as lost forces. The two forces P_k and Q'_k acting on m_k have a resultant p_k ; this resultant is *the lost force*, which by composition with Q_k gives P_k as resultant. If now the components of

$$Q'_k \text{ be denoted by } -\frac{d^2 x_k}{dt^2}, -\frac{d^2 y_k}{dt^2}, -\frac{d^2 z_k}{dt^2},$$

$u_1 = c_1, u_2 = c_2 \dots$ indicating equations of condition between the coordinates, then one has the characteristic equation:

$$\begin{aligned} \Sigma m_k \left[\left(X_k - \frac{d^2 x_k}{dt^2} \right) \delta x_k + \left(Y_k - \frac{d^2 y_k}{dt^2} \right) \delta y_k \right. \\ \left. + \left(Z_k - \frac{d^2 z_k}{dt^2} \right) \delta z_k \right] + \lambda_1 \delta u_1 + \lambda_2 \delta u_2 \dots \lambda_r \delta u_r = 0, \end{aligned}$$

which shows how the problems of dynamics are reduced to problems of statics. Cf. C. Ramus, *Anal. Mech.*, p. 210.

the cause is concept; in this lies that all moments of cause drawn from actuality, whether they be otherwise called forces or conditions, must be *ideal*. But in its working the cause is existent; in this lies that the active moments of cause must all be *real*. In the naturally determined fundamental relation between the ideal and real moments of the mechanical cause the unity-in-difference of ideal and real cause must accordingly be clearly uttered.

α) The ideality of the moments. Just as forces and conditions merge into the causes thereby determined, so the causes merge into the existing bodies; bodies are from this standpoint to be regarded as the only causes working in space and time: to decompose bodies into corporeal elements is to decompose causes into moments of cause.

It is instructive to see how rational mechanics, though its gaze is fixed most immediately upon forces, conditions and laws, precisely for the laws' sake transforms corporeal causes into concepts of cause and, without rightly knowing it, sets a whole world of mechanical causes forth in the light of ideality as a system of embodied concepts of cause.

To penetrate properly into the corporeal relations of cause, the mathematician begins by resolving the body into its elements: $M = \int dm$. But he does not stop at the infinitely empty explanation that every body consists of infinitely many small parts; he takes note of the manner of limitation; he takes aim at the concept of corporeality and is thereby put in a position to determine punctually both a body's inner structure and its outer form, its figure. If ρ be the density, V the volume, then $dm = \rho dv$ and $M = \int \rho dv$; if the body is of unequal density, and so ρ variable, then it is a matter for him of determining the ρ answering to every element of volume as a function of that element's coordinates. But how does he get the function determined? By presupposing determinate corporeal relations in the body in question, and so by starting from the concept of the represented body. He does not have in advance a given body to which he adjusts the concept — the representation of this is from this standpoint only a disturbing fancy; no, he has by means of the intellectual intuition the body completely in his power; he determines it in virtue of his concept and derives the law from the concept. Whether he says "if the body is of unequal density" or says "a body which is to be of unequal density" is, with respect to the concept of body, one and the same. It is precisely what is rational in rational mechanics that its functions and laws do not lean upon empirical inductions, but are developed out of concepts. The corporeal elements are not objects of experience; they are reflexes of representation which the conceiving understanding receives from intuition, ideal moments in images of corporeal reality.

In indeterminate universality the one dm is like the other, an element that can again be resolved into elements; but in virtue of the concept the one dm can become altogether heterogeneous with the other. An element of a sphere and an element of a parallelepiped can, as regards the infinite vanishing in infinite smallness, certainly pass for homogeneous; whether the infinitely small lines of limitation be apprehended as elements of straight lines, $dx dy dz$, answering to a rectangular coordinate system, or whether by the use of

polar coordinates curvilinear elements be taken up into the limitation, $r^2 \sin \theta d\omega d\theta dr$, makes in this respect no difference. But when the finite is to be brought forth out of the infinite, when an integration, $\int \int \int dx dy dz = \int \int \int r^2 \sin \theta d\omega d\theta dr$, is to be carried out, partly so that the body becomes a parallelepiped, partly so that it becomes a sphere, then it is recognized that the two bodies have heterogeneous elements: the moments of the concept are different because the concepts themselves are different.

Just as every seed has in it the disposition to a plant of a determinate species, answering to a determinate genus, so too every element of body determined by its concept has in it the disposition, under the integration, to grow a peculiar growth and become a determinate body. That the mass of a spherical shell $dm = d\frac{4}{3}\pi r^3 = 4\pi r^2 dr$, when integrated between the limits $r = 0$ and $r = a$, must give a sphere $= \frac{4}{3}\pi a^3$, is just as necessary as that the seed of a rose must, when it comes to growth and development, bring forth a rose. The categories *genus*, *species* and *individual* have just as much validity in mechanics as in botany.

That a homogeneous sphere must, on the presupposition that attraction stands in inverse ratio to the square of the distance, attract an external point as though the whole mass of the sphere were gathered at the centre — this actual relation of cause is not determined by the observation of any outward fact, but solely by insight into the essence of the sphere and of attraction. Even when empirical determinations enter into the calculation, it is still always a system of a priori corporeal concepts of cause that furnishes the rational foundation. Physics has weighed the planets, by empirical observations no doubt, but on the scales of ideality.

β) The reality of the moments. One should now suppose that the analysing mathematician would have to be fully assured of his concepts of body having physical validity, since he is fully assured of the physical validity of the laws he finds; but no! About the laws' validity for nature he harbours no doubt; they confirm themselves for him through all the modifications which their fulfilment under particular conditions brings with it in actuality. But with the relation between the concepts of body and actual bodies it is another matter; here the mathematical physicist is inclined to regard the fundamental determinations necessary for the analysis as arbitrary designations, subjective ways of representing, which have nothing to do with bodies in nature, and consequently nothing to do with the corporeal causes themselves either. To rely on the laws and yet doubt the concepts from which the laws are derived is assuredly a most singular inconsistency; but whence does it come? From an unclear conception of the reality of the moments.

An element such as dm does not exist in nature: *un infiniment petit est une grandeur moindre que toute grandeur donnée de la même nature (Poisson)*; it is the problem of division that here presses forward.^{*)} It goes with the element of body dm as with the element of space ds and with the element of time dt ; the mathematical elements do not exist and yet do exist. They do not exist as particular subsisting parts, but they are

^{*)}Cf. *Mathematik og Dialektik*, Copenhagen 1859; the problem of division, pp. 116–131.

moments in actual existence. The material body is just as surely an integral of material dm as the mathematical body is of mathematical dm ; the actual event is just as surely an integral of moments of event as the time in which it goes on is an integral of moments of time. The question does not turn at all upon the relation between the absolutely simple and what is composed of it, for the element is just as little in the real as in the ideal sense anything absolutely simple. The fundamental question is: how can that which is without magnitude bring forth magnitude? how can that which is itself without extension bring forth something in actuality extended?

To derive the real from the ideal, Leibnitz introduced the monads and called them metaphysical points: *il n'y a que les points metaphysiques ou de substance, qui soient exacts et réels, et sans eux il n'y auroit rien de réel, puisque sans les véritables unités il n'y auroit point de multitude*. But out of the monadic points, the soul-like unities, he was still not able to deduce corporeal actuality; as regards matter, it remained with "obscure, confused representations." In his exposition of "the amphiboly of the concepts of reflexion" Kant showed very rightly that there is a difference between matter as a concept of the understanding built upon "the power of representation" and the natural phenomenon matter, which is an object of sensation and whose apprehension is conditioned by the a priori form of our intuition of sense. From the standpoint of intellectual philosophy matter is only a complex of representations, a phenomenon of the representing being, a semblance of immediacy belonging to the intellectual; for critical philosophy the chasm between the subjectively ideal and the objectively real is so inconceivably great that matter can neither be brought forth by representation nor have anything to do with "the thing in itself." Both ways of seeing are one-sided. The actual unity of the ideal and the real is a unity in opposition, a unity of power. The ideal in dm does not bring forth the material, any more than the reverse; but both mutually opposed fundamental determinations are brought forth in and by the objective, universal power which in its doubling is both ideal and real.

γ) Ideal-real cause. To determine still further how matters stand in the mechanical sense with the concept of cause and the working cause, we choose, with a view to the foregoing, a determinate example: the formation of a sphere.

By starting from the prismatic element

$$r^2 \sin \alpha \, d\alpha \, d\theta \, dr$$

and carrying out the integrations $\int_0^\pi \int_0^{2\pi} \int_0^r r^2 \sin \alpha \, d\alpha \, d\theta \, dr$, one sees how the one element arises out of the other, and the whole sphere out of the system of the elements.^{*)} The sphere's concept in unity with the productive intuition is the sphere's ideal cause. But what is thus produced is likewise only something ideal, only the sphere's form, not the actual sphere. The sum of conditions of form to which the integrations answer is

^{*)}Cf. *Forelæsninger over Phil. Propæd.* 1861–62, pp. 31–33.

arbitrarily chosen with particular regard to known functions, and stands in no exclusively necessary connexion with that sum of material constituents and outward conditions in which the formation of an actual sphere must have its real cause. The moments of the concept and the elements of actuality fall into two circles separated from one another: here is a yawning chasm between the ideal and the real.

This chasm is, in the matter itself, filled out by the divine concept of power, whose energy embraces the ideal and the real evenly. But a human insight into the divine concept is attained only on human terms; that is why the example is chosen.

From the ideal side the mechanical essence of the concept of cause can be made clear by means of the required integrations. Out of pure nothing nothing can be formed. Every formation must begin with something; the something with which the beginning is made is itself a beginning formation, an element of formation. The element's possibility lies in conceiving intuiting; its definitive determinateness is posited arbitrarily as the first presupposition of the formation: the prismatic element with its three infinitely small dimensions is only a general condition of formation. What can now be formed on such a condition? According to the formula set up, the successive formations are particularly determined by the particular concept. If we integrate first with respect to r , then an elementary pyramid comes forth, an element of one finite and two infinitely small dimensions; if we integrate next with respect to θ , then a conical shell comes forth, an element with two finite and one infinitely small dimension; if we integrate finally with respect to α , we obtain the form sought, the spherical form with three finite dimensions. Thus, according to the governing concept which is the ideal cause of the sphere's formation, the prismatic element will condition the pyramidal element, the pyramidal element the conical shell, and the conical shell the sphere.^{*)}

This series of conditions shows what the mechanical concept, apart from finite operations of the understanding, is in itself as a concept of cause. It is not abstract logical universal forms, but punctual conditions, that make up the concept's content. What is singly seen as arbitrary becomes necessary in the connexion. No single condition is anything for itself, but all of them are moments for one another mutually. Outside the concept the inward, ideal conditions are meaningless representations; only in being penetrated and inspirited by the governing concept do they have, in the unity of conceiving, each its peculiar significance.

To the conditions in the ideal there answer the conditions in the real point for point, when the concept is a concept of power. Power's dialectic is, as we have seen, this double-sidedness, that the moments are existing members, and the existing members moments. Outside the concept of power the outward, real conditions are meaningless representations. Every condition must necessarily be a condition for something, but the

^{*)}By carrying out the integrations in six different orders, new elements will indeed come in and the course of the conditions be modified, but since the governing concept is the same, these modifications are inessential. Only when the concept of the sphere is exchanged for another concept must the elements be employed in another manner, and the formation becomes another.

something whose condition it is cannot be immediately determined by another condition. For the conditions to be able to be determined by one another, they must, in the matter itself, be reflected in one another; the matter in which they are mutually reflected is the cause, the concept in power, the ideal-real cause.

If we now return with these presuppositions to our original question — *how can that which is itself without extension bring forth something in actuality extended?* — a punctual answer can be given. The greater the mass of the sphere, the greater the force with which it attracts. If the sphere be resolved into its material elements, then the force is resolved into nothing but corresponding moments: each moment of force is just as dependent upon its peculiarly determined element of mass as the collected force is upon the whole mass. That the moments of force can be integrated into a finite force is conditioned by the elements of mass being integrable into a finite mass. The integration presupposes that in each element of mass there is a disposition to actual extension, immediate reality in the extensive, just as original as the disposition in each moment of force to actual strength, reflected reality in the intensive. The ideal-real cause both of matter and of force is power, Spirit's power, the divine object-power.

B.

Mechanical Reality.

In mechanical reality the cause is to be apprehended as immediate, natural object-power. It then comes to look as though the concept cause were only a name, for the object steps into the cause's place: the investigation answering to this moves about *power and cause*. Further, object-power is power in another. It then comes to look as though the existing causes fell outside one another in all accidentality: the investigation moves about *the parcelling out of causes*. Finally, the outward manifold of causes presupposes a foundation for many-sided reciprocal action, so that the members in their mutual connexion come forward as caused causes, and the question of reality becomes how matters originally stand with the infinite *series of causes*.

a) Power and cause. An exactly determined separation of what belongs to the object, the outwardly actual, and what the contemplating consciousness itself brings with it to the apprehension of the object, is not so easy as it might seem at first glance. Declarations such as this — “there are neither bodies nor causes, but only systems of facts, groups of present or possible motions and of present or possible thoughts” — show an endeavour to fix the separation, but at the same time furnish proofs of the superficiality with which an immature reflexion can fancy itself raised above more penetrating investigations. If one says: the object *stone* is not a body; *the object*, the subject in the judgment, is a *fact* lying outside consciousness, but *body*, the judgment's predicate, is a word, a universal concept

falling within consciousness — then we have the contradiction. In the one judgment it is the word body that is predicate; but in the other it is the word fact. If the predicate body is not to concern the object as object, but solely the predicating consciousness, then the same must necessarily hold of the predicate fact. If one will insist that it is exclusively the fact that is object, then it becomes still worse, for in the judgment “the fact is object” the object is itself predicate and enters into consciousness. If one says: the fact is not cause, for the cause is a concept, but the fact is only fact — then it is manifest that one is playing with words. In the one case one lets the word point out beyond itself to what is not a word but an existing thing; in the other case one lets the thing coincide with the word and makes the thing a word. To end this play on words it is necessary to bethink oneself of the relation between the logical and the antilogical.^{*)}

In the antilogical sense there are no thoughts or concepts whatever in nature: everything is actuality, everything is object, thing, fact, in the smallest as in the greatest. The words object, thing, fact will not be understood as words; they point out beyond themselves, they speak to the one who senses, they say to the beholder: open your eyes, come and see! But in the logical sense the fact is image, representation, concept. The investigating consciousness finds in the whole of nature nothing but images, nothing but representations, thoughts and concepts; and that for the good reason that consciousness, however busy it may be discovering objects, things and facts, meets in everything its own apprehensions and cannot possibly at any point leap out beyond itself. The earth attracts the moon. In the antilogical sense the earth is earth, the moon moon, and the motion a fact that is observed; but the cause is not observed as cause, the effect not as effect, for concepts are not objects of immediate observation. In the logical sense, by contrast, the earth is cause and the moon’s motion effect, for the actual relation between earth and moon is impossible without a concept of relation belonging to omnipotent conceiving.

The relation of experience between object and consciousness is well suited to clear up the relation in the matter between power and cause. The object’s reality is its power; power is known by reality, and conversely. That reality is power is brought to consciousness through the immediate impression of experience. If the impression could not with certainty be assumed to issue from the object as effect from cause, then an experience of the object would be impossible; one could not, after all, know that outside consciousness there really was any object. In the relation of experience the object is cause and its apprehension effect. The object is cause because it is power, and it is for consciousness power because it is the cause of the impression. Just so the object’s apprehension, its image and corresponding concept, is an effect determined by the impression: the effect is determined by being referred to the cause. To doubt the cause and rely on experience is accordingly meaningless, since the relation of experience is a relation of cause and effect. In the knowledge of the real cause thought goes together

^{*)}Cf. *Grundideernes Logik I*, pp. 146–166; *Grundideernes Logik i kort Begreb*, p. 28 ff.

with sensation, the logical with the antilogical. From the logical side the object is cause, from the antilogical side the cause is object: the language of experience is, in the style of fact, a language of power that speaks for the cause's reality.

But whence does the observer now know that the relation between bodies themselves is also an actual relation of cause? Solely from this, that the fundamental relation of experience is itself a relation of cause. That the sense organs are acted upon by the sensible object is an unmistakable proof from experience that body acts upon body, that, in other words, the relation of cause has corporeal validity. And yet — as follows straightforwardly from the extremes in the essence of power — the opposition of the logical and the antilogical is so abrupt that an observer can be fully assured of the concept of cause being applicable as a concept of experience and still harbour doubt about its a priori worth as a trustworthy expression for the essence of things.

The mechanical phenomenon in which the relation of cause utters itself as a relation of power in a quite immediate and intuitable manner is impact. For centuries impact was acknowledged as an actual cause before human knowing got so far as to be able to fix the laws of its effect or, what is the same, to get the still indeterminate concept of cause exactly determined. From the six propositions Cartesius set up it is seen that he misses the problem, in that he directs himself more by his own self-made concept than by the facts. His fundamental thought, that the quantity of motion in the whole corporeal world must always remain equal to itself, misleads him, namely, into the assumption that two motions opposite in direction do not cancel one another as positive and negative magnitudes, but that bodies colliding in this way must go back, each with the same velocity it had before the impact — *si duo illa corpora essent plane æqualia et æque velociter moverentur, B a dextra versus sinistram, C a sinistra versus dextram, cum sibi mutuo occurrerent, reflecterentur et postea pergerent moveri: B versus dextram, C versus sinistram, nulla parte suæ celeritatis amissa*. The formula holding for inelastic bodies, $x = \frac{MH \div mh}{M + m}$, which for the mass $M = m$ and the velocity $H = h$ gives the velocity after impact $x = 0$, now shows that Cartesius has not become aware at all of the difference between elastic and inelastic bodies.

Mistakes of this nature indicate that the determining power lies not in the human concept but in the fact. The investigator's knowledge is conditioned by trials; he must experiment. And yet the fact, immediately regarded, can teach him nothing; it can be resolved into a manifold of facts, but each of the parts is like the whole — a fact, no concept. To learn anything from the fact he must himself think and judge, and so bring concepts with him. But when it is the investigator who must add the concepts out of his own consciousness, the special ones as well as the universal, then all his facts are, as regards the conceiving, only occasions, not working causes: then he learns only occasionally from the fact what he originally learns from himself. So after all the determining power comes to lie in the concept.

The elastic body, for example, is in antilogical determinateness a fact, but elasticity

is a concept. How does the observer now get from fact to concept and from concept to fact? If B and C, he says, are elastic, they will in striking against one another be compressed to a certain maximum. In this maximum of compression they are, if only for an instant, inelastic, and the common velocity comes, as for inelastic bodies, to be expressed by $x = \frac{MH \pm mh}{M + m}$. B has accordingly lost the quantity of motion $M(H \div x)$, and C the quantity of motion $m(h \div x)$. But now it lies in the essence of elastic bodies that after the compression they strive to return to their original form; effects similar to those that took place during the compression then also take place during the expansion, whereby B and C again lose the same quantity of motion as before. The whole loss in quantity of motion is accordingly, for B's part, $2M(H \div x)$ and for C's $2m(h \div x)$. If the observer could give a punctual account of how much in this explanation he has from the fact and how much from his own concept, a problem of great significance for the science of experience would be solved.

The explanation given is built upon the observation of facts; and yet it goes far out beyond the observed facts, for it is built upon concepts and judgments. Every fact is antilogically a *this*; only this, the single, can each time be observed; the universal cannot be immediately observed. *This* body can be observed, but a body in general, or the universal in the body, cannot be observed; *this* compression and *this* elasticity can be observed, but elasticity in general, or the universal in the tension, cannot be observed. None the less the observer may rely unconditionally upon his concept as determined by the fact. That there are found in nature neither absolutely elastic nor absolutely inelastic bodies never holds him back from laying the concepts elasticity and inelasticity at the foundation of a mechanical analysis and, in spite of all particular conditions, letting the consequences of the concept hold good for the universal essence of the facts. Led by his concept he observes even in the fact itself not only the visible but also the invisible.

How it goes when two bodies, be they more or less elastic, momentarily alter one another's form during the collision cannot be seen with sensible eyes, and yet the observer sees mathematically what happens. Before the impact the body B with the mass M and the velocity H had a living force MH^2 ; after the impact, that is, at the maximum of compression, it has a living force Mx^2 . The body has accordingly lost in living force $M(H^2 \div x^2)$, and why? Because in compressing the body C it has performed a work. If the pressure answering to the penetration s be called P, then $P = f(s)$, and $M(H^2 \div x^2) = 2 \int_{s_2}^{s_1} P ds$. It lies in the nature of the case that the interval $s_1 - s_2$ must be greater when the impression is made, under otherwise equal conditions, in a softer than in a harder body. If in both cases the left side of the equation is to be the same, and so $M(H^2 \div x^2)$ to have the same value, then the right side, regarded as a sum of $P ds$, must contain, in the midst of infinity, a greater number of addends for the softer than for the harder. The sum is namely equally great in both cases, for, as the equation shows, it depends only on the impact's initial and final velocity; but the single addends must, in equal sums, necessarily

be greater where there is a smaller number; in other words: the pressure with the harder body is greater than with the softer.

When the investigator in virtue of observations goes in this way out beyond what admits of being observed, then he argues not from facts but from concepts. By laying concepts at the foundation of the analysis he is put in a position to derive the laws for elastic bodies from the laws for inelastic ones, and to see the necessity that the latter must always lose in living force upon collision, while the former must not.

But there still remains the answer to this question: what entitles the observer thus to exchange the single that he sees in the fact for the universal that he does not see? He abstracts, it is said, the concept from the fact; we must then take note of what such an abstraction essentially presupposes. The single is not the universal; but if the single in the fact could not present the universal, then it would be impossible to abstract the universal from the single. The object is not a concept; but if no concept were uttered in the given object, the investigator would deceive himself in supposing that he could abstract a concept from the object: to abstract a concept from something in which there is no concept is a contradiction; to discover and pursue consequences of the thoughts uttered in natural things, when there are no thoughts in the things, is a contradiction; to make the thing concept and the concept thing is a contradiction. As an object of scientific experience the thing is at once the concept's effect and its cause. It is effect; for the divine concept of power is its ideal cause. It is cause, namely cause of experience in reciprocal action with the observing consciousness, which thereby comes to a concept of the thing, and by the thing's concept to a concept of the reciprocal action of things, and so to a concept of the cause's reality. As the object's concept power is cause; as the concept's object the cause is power.

b) The parcelling out of causes. The real cause lets itself be parcelled out; the whole of outward nature is a magnificent example of the parcelling out of causes. Now if every real cause is an embodied concept, then it should surely seem that the embodied concepts must have the same fate as the bodies; and so: when corporeal causes are parcelled out, the concepts of cause must be parcelled out; and when corporeal causes are pieced together, the concepts must be pieced together. But concepts can neither be parcelled out nor pieced together; the concept is as concept an indissoluble totality of determinations belonging together: how, then, does it stand with the mechanical parcelling?

The gravity indwelling in all bodies works as cause; the concept of gravity is a concept of power pervading the universe; and yet here is a division. The planets in space have in actuality each its own share of universal gravity, the earth its own and the moon its own. But divisibility in the real is precisely grounded upon indivisibility in the ideal. If the concept of gravity, and with it gravity's mighty essence itself, were divided and by the division had gone to pieces, then the piece of gravity dwelling in the earth would find itself at a mean distance of 51,032 Danish miles from the piece dwelling in the moon; but with this the bond of gravity would be broken: gravity could not be gravity. The

secret of the cause is precisely this, that the real can be divided while the ideal is and remains an indivisible power. In the real sense the earth is cause, and the moon is cause, and the two real causes really find themselves at a distance of 51,032 miles from one another; but in the ideal sense gravity is cause, and gravity's ideal bond cannot burst. Between the two causes, ideal for earth and moon, there is no distance: they are, each for itself, the whole cause, and yet moments of one cause.

That the indivisibility of the ideal can consist with a parcelling out of the real has its ground in this, that ideal indivisibility itself consists in a fundamental division, but a fundamental division is no parcelling out.^{*)} The same concepts of power can be individualized in nature in different ways, just as the same laws can be fulfilled in countless ways. It must constantly be held fast that the laws of nature in all facts, just as much as the facts themselves, rest upon the concepts of nature. Physics does indeed discover manifold laws whose validity it secures by ordinary induction, without yet being able to ground them upon a clear insight into the nature of the real causes; but induction itself would lack a true scientific ground if the investigator did not dare presuppose that it is everywhere concepts — if hidden, as yet undiscovered concepts — that furnish the foundation of the laws. The laws for the effects of electricity cannot indeed yet be derived from the concept of electricity in the same penetrating manner as the laws for the collision of elastic bodies can be derived from the concept of elasticity; but there is surely no physicist who doubts on that account that the fundamental relation between concepts and laws must constantly be the same.

How the laws stand to the concepts of cause, and these to the real causes, can already be seen from the so-called mechanical powers. Of such simple machines as the lever, the inclined plane, etc., one may certainly say that each of them is a piece of cause, an arbitrarily formed implement; but that the whole concept of cause is just as much realized in the lever as in the inclined plane, that every one of these implements presents a whole concept for itself, is evident. That these restricted concepts, in all their one-sidedness, are not fragments but moments of the great mechanical concept of cause embracing the universe follows from this, that the laws of equilibrium and motion individualized in these implements are laws of nature, universal world-laws.

The law as law is one side of the concept and itself the whole concept, namely in the form of universality, while the corporeal causes in their mutual difference present the concept in the particular, and each body for itself the antilogically existing single concept. From the side of reality the single concept is, however, to be seen not as concept but as matter, as body. The observer does not notice that bodies are in their essence concepts just as much as the law is, and that the law is for that reason sensible in its fulfilment just as much as bodies are. He senses bodies and abstracts the law; the law thus abstracted therefore becomes for him not a concept but a categorical form for an unexplained necessity. The one-sidedness of the standpoint may, however, be said to

^{*)}*Grundideernes Logik II*, pp. 347–458.

be justified by the peculiar character of the science of experience; it does not lie in the investigator's interest to develop the concepts of cause as real concepts of nature: he holds to facts and laws.

But what has the philosophy of immanence accomplished? Hegel inveighs against the physicists and complains of their parcellings; but the physicists are surely excused when nature's "being, as it is, does not answer to its concept." How should the investigators of nature be able to hold to "the Idea" when nature cannot even do so, but must in the Hegelian sense be regarded as "the Idea's falling away from itself"? In its endeavour to set nature, fallen into "outwardness," on its feet again by means of "the concept" — or rather, to make it conceivable how "the Idea" works its way up step by step out of its degradation — speculative philosophy of nature has in a sorry manner had to overstrain logical necessity.

As surely as it is "the absolute Idea" that begins its absolutely necessary journey upward from below, the philosophy of nature must quite rightly furnish the proof that the stepwise ascent happens with immanent necessity, and that this present world-order is the only conceivable one in which the Idea could come to outward, actual self-revelation. If such a proof had really been furnished or could be furnished, then we should certainly be bound to acknowledge the proof and bow to the necessity; we should, as H. C. Ørsted says, have to "be ashamed in our inmost self if we caught ourselves wanting a truth other than the actual one." But when such a proof has neither been furnished nor can ever be furnished, then we offend against the actual truth in bowing to a fixed idea.

It is true that everything that happens must have its cause and so far its necessity; but it is untrue that any relative actuality whatever should be absolutely necessary. In abstract universality necessity is an empty thought; to give the thought a content we must give it a form and take it up into a determinate concept. Only by the concept can anything necessary come out of necessity. Necessary is that whose opposite is impossible. Conceptual necessity everywhere marks the limit of the possible and the impossible, but never reaches the limit of the actual.

The limit of necessity for an actual inclined plane, for example, is a limit of its possibility. All determinations of logical necessity concern actuality only so far as they lie in possibility. It is necessary that every inclined plane must have a base, a height and a length; why? Because an inclined plane without height, length and base is an impossibility. But within possibility there can be an infinity of actual inclined planes, different in their actuality; to the actual differences necessity is indifferent: in actuality it is just as content with the one specimen as with the other. The same holds of the laws. "The tendency to fall on the inclined plane must stand to the pressure against it as the inclined plane's height to its base." That is a law, for it is necessary; its opposite is impossible. But this necessity decides not the least thing concerning the actual height, the actual base, the actual pressure. Conceptual necessity is hard and strict against the impossible, but supple, compliant, extremely liberal toward the actual. Speculation

may widen its horizon as much as it will; its logical necessity never reaches out beyond possibility.

It might perhaps appear at first glance to one or another that the doctrine of this world's absolute necessity had a reliable foundation in rational mechanics; but not to speak of the fact that mechanics, in its presentation of the laws for the motion of the heavenly bodies, lays the Newtonian law at the foundation without an a priori proof of that law's exclusive, absolute necessity, it goes on the whole with necessity for the world's planets as it goes in particular with necessity for the inclined plane. If we represent a planet with the mass m moving about a central body with the mass M , and decompose the accelerating forces along three mutually perpendicular axes, so that the coordinates for M become X , Y and Z and for m x , y and z : then the distance r between M and m will be $r = \sqrt{(x \div X)^2 + (y \div Y)^2 + (z \div Z)^2}$; that is necessary, for its opposite is impossible. If we reflect solely upon the X -axis and call the angle r forms with this axis α , so that $\cos \alpha = \frac{x \div X}{r}$: then the attraction the central body exerts upon the planet, and so the latter's acceleration, will be $\frac{d^2 x}{dt^2} = \div \frac{M(x \div X)}{r^3}$, and the attraction the planet exerts upon the central body, and so the latter's acceleration, will be $\frac{d^2 X}{dt^2} = \frac{m(x \div X)}{r^3}$; that is necessary, for its opposite is impossible. If we let the central body be immovable, then its acceleration must be added negatively to the planet's, and the equation for the latter's motion becomes $\frac{d^2(x \div X)}{dt^2} + \frac{(M+m)(x \div X)}{r^3} = 0$; that is necessary, for its opposite is impossible. That under such analyses images of actual planets, an actual central body, actual planets and actual distances may float before the mathematician signifies nothing with respect to actuality itself and the actual parcelling. Where actual motions take place it is certainly necessary that the general laws must be fulfilled; but from this logical necessity to the absolute necessity of an antilogically determinate planetary system there is an infinite leap.

The absolute Idea, one says, has all relative moments, all essential conditions for its realization in itself. The whole outwardly cast finitude is accordingly a parcelling out in which the absolute must be absolutely omnipresent. The more one widens the concept of the whole, the more conditions one gradually takes up into the general presupposition of the analyses, the more one will be able, by the crossing ways of inner necessity, to approach the outwardly actual. We have in this way of seeing one of the many immanent ambiguities with which "modern" philosophy has become so overloaded that only a reference to exact determinations of form can be supposed likely to exert a purifying effect. Let us then add to the presuppositions given this further one, that besides the planet m other planets m' , m'' , etc., move about the central body. What does this now lead to? On the last presupposition the sum $\frac{d^2(x \div X)}{dt^2} + \frac{(M+m)(x \div X)}{r^3}$ or, when $x \div X$, m 's abscissa with respect to M 's centre, is denoted by ξ , the sum $\frac{d^2 \xi}{dt^2} + \frac{(M+m)\xi}{r^3}$ cannot indeed be zero; here a perturbing function R comes forth, and the equation becomes $\frac{d^2 \xi}{dt^2} + \frac{(M+m)\xi}{r^3} = \left(\frac{dR}{d\xi} \right)$. Such an equation certainly gives the expression of necessity a more articulated form; but the limit is only, in possibility, a limit for the impossible. Even if we were, guided by

the method of “the variation of the arbitrary constants,” to wander through the whole primeval forest of equations, series and formulas that lies behind the indications above, we should still not reach the antilogically determinate facts, should not get the material actuality of the parcelling pressed into absolute necessity.

c) *Series of causes.* The parcelling out of the mechanical causes must necessarily have a connexion of things of its own for its foundation. From the side of universality the connexion is realized in gravity, which under the form of particularity varies with the corporeal states. Physics sees, in the resistance bodies offer to the separation of their parts, a force of cohesion different from the force of gravity, and surely rightly.*)

But however little the force of cohesion may be regarded as a mere offshoot of the force of gravity, it will yet be intelligible that the two heterogeneous forces must be referred, not indeed to one fundamental force, but still to one fundamental concept, the comprehending concept of power whose inner articulation is the ideal cause of all forces. Just as the force of gravity, as has been shown in its place, rests upon reflexion in power, so too does the force of cohesion. The particular concepts in this reflexion are *whole* and *parts*. That the body is a whole real in parcelling out is synonymous with its consisting of parts. These parts are in the mechanical sense dependently independent members. Their dependence is an expression of the real concept’s power in the given whole; but their independence shows the whole’s being pieced together, for the real concept is power in the whole only by being particularly power in the parts: the expression of cause reflected in existence for the dependence of the independent parts is the force of cohesion. From the forms of state resting upon particular relations between “force of expansion” and “force of cohesion” it is seen how the properties of the parts reflect themselves in the whole, and from this again how matters stand, on the foundation of the parcelling, with the concepts of body and existing bodies.

It is a fundamental proposition in mechanics that action and reaction must answer exactly to one another. The proposition holds for gravity; for the attraction a mass exerts upon a point is always equal to that which the point exerts upon the mass. A falling grain of sand attracts the whole earth with a force equal to that with which the whole earth attracts the grain of sand — that the acceleration is highly different is quite another matter. — The proposition holds for quantities of motion. If a constant force k act for a certain time upon the mass M , then the acceleration is $\frac{k}{M}$, and the quantity of motion in a certain determinate direction $Mv = kt$. But the force acting upon the mass M must

*)The attraction of cohesion holds, namely, only for the small parts among themselves; if this attraction were an utterance of the force of gravity, it would, on account of the smallness of the parts and their great mutual nearness, have to become extraordinarily slight. According to the law of gravity $T = \frac{\mu}{r^2}$ — T being the attraction, r the distance and μ the intensity — the intensity for a spherical mass with radius r and density ϱ will be $\mu = \frac{4}{3}\pi\varrho r^3$. If now the sphere shrink to an atom, the intensity for the atom will be $\varepsilon = \frac{4}{3}\pi\varrho(dr)^3$ and consequently $\frac{\varepsilon}{(dr)^2} = \frac{4}{3}\pi\varrho dr$, a vanishing magnitude. The objection that the small parts are not infinitely small signifies nothing in this connexion; for the formula shows that the force of cohesion must become the weaker, the more intimate the joining is — something which is both at variance with experience and in itself a contradiction.

have issued from another mass, e.g. M' , and the mass M acted upon by k then reacts upon M' with a force which is also k ; only that the activity goes in the opposite direction, and the quantity of motion becomes $M'v' = \div kt$. From this follows $Mv + M'v' = 0$, an equation showing that the sum of the quantities of motion for both bodies continues unaltered. — The proposition holds for living force. It cannot indeed be said that the work given off by one body is always received by another, for the activity can assume different forms; heat, for example, can give off work, and work again be stored up. But if the activity be regarded under all forms as the same function of force, and be measured by the same measure, then this function too will remain unaltered in passing from body to body: action and reaction answer to one another.

The fundamental proposition teaches us that forces always come forward two by two, parallel, equal and opposite: pressure answers to counter-pressure, impact to counter-impact, A's attraction of B to B's attraction of A. The mechanical cause is, in other words, never any immediate single cause, but always a reciprocal action of two causes. That every outward cause is, essentially regarded, a reciprocal action of two outward causes spreads a peculiar light over all mechanical relations of cause.

To begin with gravity, the most comprehensive of all outward causes: it is not merely outward in the relation between the masses upon which it works, but it is in itself outward. The mass m attracts m' insofar as m' attracts m ; that the attraction is equally great from both sides proves that the cause in both masses is equally original and equally derived; both causes are in their activity *caused*: they cause one another mutually. What holds of two members holds of all; here an infinite series of caused causes forms itself; no member marks the first cause. The contradiction that gravity is everywhere equally original and equally derived is resolved in nature in such a way that gravity, which presupposes itself to infinity, comes outside itself and concentrates its effects in other causes, different from it: *pressure, impact, quantity of motion*.

Where the distance between the heavy masses is cancelled, gravity works as *pressure*. According to the fundamental proposition of the equality of pressure and counter-pressure, a body's weight is to be determined by gravity; and yet pressure as cause is different from gravity. By gravity the parts of mass strive to unite in one point, the centre of gravity, but pressure counteracts this striving and is, though itself an effect, nevertheless cause — cause, namely, of the parts keeping outside one another, so that masses fill space. Only what is pressed presses; pressure as cause is not merely outward with respect to its object; it is in itself an outwardly, infinitely caused cause. Series of causes formed of pressure and counter-pressure cannot possibly have any first member in which pressure as pressure was original cause. The contradiction that every particular pressure in its immediacy is a derived, caused cause resolves itself into a transformation of pressure. By this transformation it becomes a moment in another cause, *impact*. Impact presupposes motion, and motion must have its cause.

Where a distance between bodies is given, gravity may indeed cancel the distance

and become the cause of the impact; but gravity, which is not originally the cause of the distance, cannot be the original cause of the impact. Just as pressure is caused by a striking together, the distances of bodies would accordingly have to be caused by a striking apart; impact would then from this standpoint be apprehended as the concrete and so far original cause, having gravity for its universal and pressure for its particular moment. Older mechanical philosophy of nature (Cartesius) has not let references be wanting to a first, original impact, a fundamental impact, from which all motions and world-phenomena were to be explained. But that the representation of a first original impact must resolve itself into a conceptless concept is clear from the fundamental proposition: only from impact comes impact. But when only what has previously been struck can strike, then every impact is a cause caused by an impact. The infinite series of causes in which impact unbrokenly presupposes impact cannot then possibly begin with a first impact. If a beginning be made with a first impact, i.e., with an impact whose cause is not an impact, then the beginning of the series falls outside the series: the beginning is then made with a cause which does indeed contain the possibility of impact, but is nevertheless different from impact; the beginning is made with a *quantity of motion*.

So the quantity of motion would then be the cause rich in content which, by taking gravity, pressure and impact up into the circle of its real moments, could in spite of the parcelling be the principle of motions and come forward as fundamental cause; but this too is impossible. Quantities of motion can, says physics, even without regard to the unalterability of force, be propagated from body to body without altering magnitude or direction, so that what the one

body receives, the other loses. For a world of bodies the quantities of motion in every direction will accordingly be unalterable magnitudes, whatever forces work between the bodies or between the corporeal parts themselves; but in the form of quantity of motion too the cause is outward to itself. The cause of an Mv is in a $\div M'v'$; the quantity of motion that communicates itself must everywhere have been communicated: the infinite series of causes is a series of nothing but communicated, *caused* causes. Nor does the relation of causality alter its nature by the quantity of motion pointing out beyond itself, being taken up into a still more concrete cause and converted into *living force*.

A body's increase in living force is, it is said, equal to twice the work received, and a body's loss in living force equal to twice the work given off; the equation $M(v^2 \div c^2) = \pm 2 MGs$ shows both: the conversion of work into living force and of living force into work can be continued to infinity. With the co-operation of the different solid, liquid and gaseous states in which bodies find themselves, the outward causes can be transformed in countless ways and enter into manifoldly entangled reciprocal actions; but that these outward causes everywhere, in the great whole as in the smallest parts, necessarily presuppose a ground-cause and yet keep the ground-cause hidden, is and remains the mark of character of all mechanical reality.

C.

The Ground-Cause: the First Mover.

According to mechanical ideality the working causes are reflected in corresponding concepts of cause, and these holders of power in causality then show themselves to be the ideal causes determining from within. According to mechanical reality, by contrast, the concepts of cause are, by means of object-power, hidden under the outward material existence in which they have their actuality; the ground-cause has disappeared, and the beholder of the infinite series looks about him in vain for a first mover. If it is to be possible to discover the ground-cause in the series of motions, then the difference between the two heterogeneous motions — the *ideal* in the concept and the *real* in the fact — must be expressly shown. It is under the ideal motion that Spirit, the self-power that causes everything, determines the particular concepts of power to be the material causes of object-power. But the ideal motion presupposes precisely what it itself determines, namely the real. Not immediately, but through actual causes in nature and real motions, is self-power *the original cause of the whole mechanical parcelling out*. The parcelling rests as little upon an absolute necessity as upon a blind accidentality: the ground-cause itself, omnipotent freedom, is *the first mover*.

a) *Ideal and real motion*. The doctrine of immanence has done its utmost to make ideal and real motion look like one and the same thing. In Hegel's logic the motion of the concept is to pass for the motion of the matter; in his philosophy of nature the motion of the matter is in return to pass for the motion of the concept. One need only bethink oneself of the fundamental contradiction lying in this to come to certainty about how hollow the absolute is upon which "immanence" must go on building its "pantheistic," "naturalistic," "atheistic" castles in the air.

If it is to hold in earnest that the logical Idea has an eternal being resting in itself and independent of nature, then there must necessarily be presupposed a thinking, reflecting and conceiving exalted above space, time and matter; for without thinking, reflecting and conceiving there is not a single logical determination, far less an absolute logical Idea. In declaring for the logical Idea's originality and holding fast its supernatural being, the philosophy of the absolute goes out beyond its own vaunted immanence and makes a questionable loan from transcendence. From this standpoint the ideal motion in the concept is then also essentially distinct from the real motion of nature's development, and yet the two motions are supposed to be one.

To evade the contradiction in the consequence just drawn, one has recourse to another explanation. That the absolute is in its essence logical, and the concept its original form, must not then be understood as though the Idea of logic had really thought through its categorical determinations of content before it altered itself, struck over into the real and began afresh as the Idea of nature. The absolute's eternal being is on the contrary an original natural being; its immanent determinations of essence then transform themselves as at a stroke of magic; they are not in the absolute in an absolute manner; their existence

comes to be apprehended as the sensible, material forms of nature, of outwardness, inadequate to the Idea: the motion is real. For the absolute nevertheless to be able to remain absolute in the midst of the dispersion, the thoughtless must itself be thinking. It is the universal, the all-thinking, the great thinking neuter, upon which immanence rests. This universal is indeed a form lost in things, bound to nature, which can first be freed into conscious thinking in the human spirit; but the unfree state of nature makes no breach in the universal's infinite thinking; consciousness is quite superfluous; conscious thinking is only a lower kind of thinking. Absolute thinking is a thinking without consciousness, just as absolute knowing is "an unconscious knowing," a clarity without light, a conceiving without insight, an understanding without understanding. On these terms it is not at all unreasonable that time can conceive space, and space time, and that the matter brought forth by the mutual conceiving of space and time can again conceive and move itself: the real motion is ideal, for the ideal is real.

What is true in all this is that the concepts of nature are indwelling in natural things. It is Hegel's immortal merit to have developed Spinoza's barren doctrine of immanence into so fruitful an all-sidedness that "the concept" is seen to hold a world-content. Hegel's philosophy of nature deserves in this respect just as full appreciation as his logic. But the fundamental error is that immanence, however much is said of the "overreaching" — and so transcendent — concept, everywhere dissolves and denies the original presupposition upon which it itself rests: immanence is impossible without transcendence. The concepts of nature have a double medium; they are indwelling in things: actual nature is their real medium; they proceed from infinite conceiving: the all-knowing, self-powerful Spirit is their ideal medium. The unity of the two media is a unity in opposition which grounds the essential difference between ideal and real motion.

α) Ideal motion. For the mechanical cause to be concept, it must be conceived. How the conceiving goes on in human knowing has been shown above. Thinking and representation are functions of one another mutually. The cause cannot be thought without a something representing the active element, namely force, and the active element cannot be thought without a something upon which it works, namely matter, whose inertia is determined precisely by its receiving and preserving the action. Force and inertia reflect one another mutually: the phenomenon answering to this in time and space is acceleration, and the cause represented in the concept is the accelerating force. In acceleration the mass, limited matter, is understood along with it but not yet taken up into the concept itself. The concept of cause demands that acceleration and mass expressly enter into the reflected unity of the moving force, the more determinate expression of the cause conceived in time, whose measure is quantity of motion. But quantity of motion in exact form is a one-sided measure. The relation of causality demands that the effect be opposed to the cause, namely in such a way that the cause reflected in its effect answers exactly to what is effected. The completely determinate effect is work; the cause is set to work and assumes the form of living force. However many determinations

representation may have brought to thinking in the course of this development of the concept, it is nevertheless evident that none of these determinations of representation is drawn in under the cause from without, either arbitrarily or accidentally, but that they all develop from within out of the mechanical cause's own concept. If the beginning be made in the abstract with acceleration, then the motion cannot stop before the concrete is reached in living force; if the beginning be made in the concrete with living force, then analysis will discover in it space, time, quantity of motion, and in the quantity of motion acceleration: that is ideal motion.

And yet the development of the concept is not yet completed. The mechanical cause is by character a cause of otherness; its activity is in the outward; the concept of outwardness belongs originally to its essence. The conceiving reflexion is determined by reflexes of outward things, and what causes spreads itself into a circle of conditions. But the independence of the conditions is only apparent; the conditions, cancelling one another, disappear as moments in the cause's unity, the unconditioned. By taking the conditions up in the form of unconditionedness the cause is conditioned in itself, and in such a way that it can develop its own difference from within out of its conceptual content: one cause can, on the foundation of the conditions, become several, and several again one. It is no real but an ideal motion that is seen in these transitions. The concept of cause is as yet only a concept; but the conceiving shows that the ideality of the moments reflects itself point for point in a corresponding reality.

As a concept of nature the cause has its ideal medium in the all-conceiving Spirit; to the infinite conceiving there must then also answer an ideal motion. As unreasonable, indeed insane, as it would be to represent the absolute Spirit after the likeness of the human one — observing, deliberating, abstracting, calculating, differentiating, integrating — just as unreasonable, indeed insane, is it to want to think a cause without a concept, a concept without a development of the concept, a development of the concept without inner, free, ideal motion. The metaphysics of immanence, with its selfless thinking neuter, ought at least to be able to learn this much from the analysis of infinity that lies at the foundation of rational mechanics: that it should at last come to consciousness of how matters absolutely must stand, within conceiving itself, be it human or divine, with the articulation of the infinite by the finite.

β) Real motion. Real motion stands in sharp opposition to ideal motion, because the existing cause stands in sharp opposition to the concept of cause. A body's motion from place to place has, outwardly regarded, no likeness to a motion of the concept. It is a language of power that dictates the opposition; but this language of power is the inborn natural language of the real concepts. The concepts' transition from the ideal into the real medium is no finite natural-historical occurrence; it is a fundamental motion, an infinite event belonging equally to time and eternity. The transition does not happen in such a way that the real concepts, untouched by all realities, are preserved in self-power's eternity so that object-power may afterwards supervene in time and fill the

empty forms with stuff-like content. Self-power and object-power are extremes of the same omnipotence in an eternal-temporal unity of doubling. It has been made clear by analyses that the same differential, dm , which is ideal in the body's concept of nature, is real in the existing body — that, in other words, object-power, which posits existence, is infinitely penetrated and inwardly moved by the concept of existence. But precisely by expressing its concept the object comes to conceal the concept; the fact veils the ideality, and the intellectual semblance is so faint that the observer can hardly discern whether it is the thing's concept or his own that he has to do with. This opposition of the ideal and the real, stemming from the moment dm , is objectivity's opposition in power; but the observer objectifies in powerlessness.

The real concepts' transition from the ideal into the real medium is grounded in an omnipotent objectification. The objectification is double, inward and outward. Insofar as a determinate system of world-concepts is thought through in self-power, the motion is logical and the act of objectification ideal; insofar as the same system of concepts is translated into object-power, the motion is — now *here*, now *there* — antilogical, and the act of objectification real: the real motion has the ideal for its presupposition.

γ) Reciprocal determination of ideal and real motion. For real concepts to be formed, an outward must unbrokenly reflect itself in its inward; reflexes of space and time, matter and force with all the determinations of outwardness belonging to them, belong essentially to the formation of the concept. There is no thinking in space, for space is no form for abstract thinking; and yet when material actuality is to be conceived, a gleam of spatial relations must necessarily fall into the ideal medium of conceiving. The moments of the real concepts are dissolved representations, and the representations dissolved images; there is in every image an intuiting, however fleeting, of space. Matter is no thought, but thought's object, its resistance; the material in matter is not thought, it is conceived as thought's other in a reflexion of intuiting and thinking: the image of sense, whose significance is that thought's other has existence outside thought, brings the reflex of the material with it. There are accordingly in the ideal motion of the real concepts indications everywhere of real motion.

Real motion now has what the ideal presupposes: an actual matter in actual masses, an actual space with determinate places, an actual time within given limits. The two heterogeneous motions cannot, to be sure, be parallel; the intellectual way from concept to concept is incomparable with the outward way from place to place; but that real motion is impossible without ideal motion is the pulsing central thought of the philosophy of nature.

Ideal motion belongs to eternity; real motion goes on in time. The great world-system of mechanical concepts of cause does not come to be in time; it is eternally finished. Infinite conceiving is simultaneous, and the succession inseparable from ideal motion is only the indication of time reflected in eternity itself. The real causes in which the concepts are realized belong by contrast to time as time, and the motions go on with

different velocity in space. In the eternal sense all motions are determined in advance, for the ideal must be completed in itself before the real can come to be. Motion along the diagonal of the parallelogram of velocities must be completed in the concept, for real motion can only in an imitative manner express what is originally in the concept. The laws, self-power's thoughts, which are eternally thought and have been thought, are determined by *ideal*, but the fulfilment of the laws by *real* motion.

b) The cause of the parcelling out. In the reciprocal determination of ideal and real motion lies this, that the mechanical distribution of masses and forces in space must be referred to a ground-cause working through outward causes. From the ideal side the ground-cause is to be sought in the all-determining self-power, from the real side by contrast in object-power, the power of existence answering to outwardness. If now the emphasis falls particularly upon self-power, then the fundamental determinations point to nothing but acts of will, and the necessity uttered in the connexion rests upon *arbitrariness*. If conversely the emphasis falls exclusively upon object-power, then the division proceeds from particularly existing causes, and the necessity of connexion rests upon *accidentality*. Only by absolute self-power being known to have all power in another as the reality of its unconditioned supremacy can the moment of arbitrariness from inner and the moment of accidentality from outer causality be reconciled with necessity; and the plan of wisdom in the given world-order then presents itself as a work of divine *freedom*.

α) Arbitrariness. Every original determination, be it otherwise of whatever kind, is a determination of the self, a self-determination. If the self-determinations be apprehended in part, singly, they take on the stamp of arbitrariness. Yet it is easily seen that a plurality of determinations of arbitrariness must either fall apart from one another, contradict one another and lose themselves in the meaningless, or by mutual agreement presuppose one another, lie in a plan and form a unity of connexion.

By means of the connexion the determinations of arbitrariness come under necessity: the arbitrary becomes necessary, and the necessary arbitrary — in the infinite reflexion. But in the finite reflexion it looks as though arbitrariness and necessity fell apart from one another, as though the arbitrary belonged to one part and the necessary to another part of the plan.

Let it lie, for example, in the plan of omnipotence that two planets in world-space shall stand to one another in such a way that the one furnishes a fixed centre about which the other moves at a certain distance: then certainly, when it is omnipotence that absolutely wills it, each of the planets can be arbitrarily posited for itself, the one be given a motion along the tangent, inertia be added, and a law of gravity be fixed so as to preserve the relation; here in this parcelling out there are no fewer than five omnipotent acts of arbitrariness. But now comes necessity.

The acts of arbitrariness presuppose one another mutually in the plan itself. Rational mechanics maintains with full right that to the solution of such a problem there belong

equations of condition,^{*)} which must inviolably be fulfilled, that the principles of the conservation of areas and of living forces have, under the presupposition given, a necessity from which not even omnipotence itself can withdraw.

He who wills the end must will the means; that means, in the language of infinity, that the end is willed in the means and the means in the end. The conceiving self-power, which in unclouded clarity sees the whole determined by the parts and conversely, chooses the conditioned for the conditions' sake just as much as the conditions for the conditioned's sake. At no point in the chosen connexion does the arbitrary fall outside the necessary. In infinite conceiving there can accordingly be no thought of arbitrariness either weakening or breaking the laws of necessity; but there may well be a danger here that the real causes should from this standpoint become empty phenomena of power belonging to the ideal ones, that the abstract ideality of the determinations of arbitrariness should undermine natural reality, that all-determining self-power should in such a one-sidedness devour object-power.

β) Accidentality. The real cause coincides with object-power in corporeal existence. If one now looks away from the question of ideal causality and takes things as they show themselves, then the laws are laws of nature, and the conditioned necessity suffers no damage. Indeed it might almost seem as though the reciprocal action between the natural causes furnished a surer guarantee of lawful necessity than a formal reflexion of divine acts of arbitrariness. To keep to our example: the two bodies will now take their place and attract one another not by divine order, but in virtue of their own material nature. Every side-thought that it might perhaps one day please absolute self-power to transform matter, impart new properties to it, introduce new laws, a new connexion with a new necessity — all such representations of arbitrariness fall away, and the factually given necessity is as though built of solid masonry.

By this conception something is certainly won; the cause's mechanical reality is maintained, and nature so far come into its right. But those who suppose that necessity

^{*)}If the forces be resolved along three perpendicular axes into X, Y and Z, and the running coordinates be denoted by x, y and z, then the equation $X : Y : Z = x : y : z$ states a condition that must necessarily be fulfilled if the resultant of the forces is to pass through the origin of the coordinates and have in it a fixed centre. Since $\frac{d^2x}{dt^2} = X$, $\frac{d^2y}{dt^2} = Y$ and $\frac{d^2z}{dt^2} = Z$, the equations $\frac{xdy - ydx}{dt} = a$, $\frac{zdx - xdz}{dt} = b$ and $\frac{ydz - zdy}{dt} = c$ show that all the constants a, b and c cannot possibly be zero if the moved body is not to plunge straight toward the centre, but its radius vector to describe an area. From the equations it further follows with necessity that the areas projected upon the planes XY, XZ and YZ must stand to one another as the times employed, and that the orbit must be a plane curve, since the equation derived from the presupposition, $ax + by + cz = 0$, is precisely the equation of a plane passing through the origin of the coordinates. Finally, since from the equations set up it can likewise be derived that $\frac{dx^2 + dy^2 + dz^2}{dt^2} = \frac{ds^2}{dt^2} = 2F(x, y, z) + C$ (s denoting a piece of the orbit, and $F(x, y, z)$ the integral of $Xdx + Ydy + Zdz$), and that the velocity $\frac{ds}{dt}$ — as surely as $Xdx + Ydy + Zdz$ is a complete differential containing neither the time nor the velocity — depends solely upon the coordinates, it follows from this that the living forces are maintained, that the moved body has the same velocity every time it comes to the same place in the orbit, and that the motion is accordingly continued to infinity in the same orbit.

itself, rational necessity, would be better secured upon the real than upon the ideal foundation are remarkably mistaken. Of the concept of necessity it does indeed hold that, in the character of a selfhood-form, it has the conditions as moments in its essence; but of mechanically real necessity precisely the opposite holds; as an otherness-form it has the conditions outside itself: the existing members by which it is conditioned everywhere present a side of existence that lies under accidentality. We learn it from our example. The two planets, of which the one moves about the other, must indeed be acknowledged to stand in a real relation of necessity and to presuppose one another in outward reflexion; but the one is nevertheless not the cause of the other: their being together is, in the form of the parcelling out, a pure accidentality. If one now tries to lift the accidentality by carrying both masses back to a common original mass, then the accidentality is there again; it is in the mass's existence. If one lifts the accidentality of this existence by letting the originally rotating mass come forth as a necessary result of atoms scattered in space, then the accidentality is there again; it is in the atoms' existence.^{*)}

γ) *Freedom.* Necessity in and for itself is a powerless being, a brooding night in the womb of possibilities, too heavy and formless to help itself forward. To win shape and come to standing, necessity must be supported from within by arbitrariness, from without by accidentality. It is Spirit, self-power in eternal reciprocal action with object-power, that breaks the seal of the possibilities and brings dark necessity to the light.^{**)} Every complete actuality, every mechanical, relatively closed world-whole must originally be determined from within as a totality of concepts satisfying the demand of selfhood. This is the plan, the arbitrarily formed but content-rich, comprehending and pervasive universal concept in which the particular concepts reflect and determine one another with rational necessity; this is the ideal cause of the parcelling out.

And yet here there is as yet no parcelling out. Actual parcelling points everywhere, in the whole as in the single case, to real causes. With the real causes the facts come forth, with the facts accidentality; accidentality is woven into necessity, and outwardly conditioned necessity, restless and yet at rest, reposes upon accidental presuppositions.

^{*)}In his work *Über die Natur der Cometen, Beiträge zur Geschichte und Theorie der Erkenntniss*, 2^{te} Auflage, Leipzig 1872, J. C. F. Zöllner — under the heading *Ueber die Endlichkeit der Materie im unendlichen Raume* — has by reasoning and calculation sought to show that, since the assumption of an act of creation is an *arbitrary* limitation of the series of causality, it is — on the presupposition that the elements of matter are subject to the Newtonian and the Mariotte law — necessary to assume a finite quantity of matter, existing from eternity, in infinite space. — However interesting these investigations are in their details, and however willingly we concede that physics too may profit from “starting out from universal concepts,” that the progress of knowledge “shall not be hampered by inherited prejudices,” it still stands fast that a physical series of causes cannot possibly begin with an outwardly given, a first member that is not caused. To avoid “arbitrariness” one throws oneself into the arms of accidentality; finite matter in infinite space comes, from the Zöllnerian standpoint, to be taken as the first, uncaused cause, the great accident in eternally groundless accidentality.

^{**)}On the concept of necessity, cf. *Grundideernes Logik* II, pp. 253–275; *Grundid. Logik i kort Begreb*, pp. 78–83.

If now there could be discovered in the whole universe a real cause so free of presuppositions, so factual, that in its given existence it fell outside every determining concept of cause, then we should have here a genuine representative of facticity's unconditioned power, eternally blind accidentality. But so long as such a discovery has not yet been made, we can freely let "modern" science's "free thought" play at necessity with blind accidentality, while we inviolably hold fast the fundamental proposition that every fact, as an outward cause, must be determined from within by its concept of cause, its ideal cause.

Upon the original unity of ideal and real cause all knowledge of nature must be grounded. Philosophical criticism has plainly shown that matter, as little as its properties and forces, admits of having natural existence imparted to it by any power outside nature. This is a decisive thought for the philosophy of nature; but the thought is misconstrued and distorted when it is used to defend nature's unconditioned autonomy, its absolutely necessary self-grounding. The truth is that all development of nature rests upon natural presuppositions: nature begins with nature, however far back we go. But if this truth is not to be perverted, it must necessarily be understood along with it that the causes in nature are real, and that every real cause is in essence determined by an ideal one. The cause of the parcelling out is as little to be referred to an absolute necessity as to facticity's accidentality; the natural cause is in object-power — an all-embracing system of outward causes.

The system of natural causes is a world-plan uttering itself in a categorical language of power. Arbitrariness, accidentality, necessity are one-sided concepts. Accidentality is cancelled when it is seen that the accidental is because it *shall* be; and so: accidentality is cancelled when it is reflected in arbitrariness. Arbitrariness is cancelled when it is seen that the arbitrary is because it *must* be, that the closed system is necessary for the plan; and so: arbitrariness is cancelled when it is reflected in necessity. Necessity is cancelled when, in the unity of connexion of the members, it is reflected as the form for absolute causality's content. Absolute causality, ideal in self-power, real in object-power, is overreaching self-determination, is freedom; and so: necessity is cancelled when it is reflected in freedom.

c) *The first mover.* The relation between the motion of thought and motion in space has been developed in a peculiar manner by Aristotle. Motion in nature is transition from possibility to actuality, and conversely; motion presupposes two members, a moving and a moved, an actual being in the form and a potential one in the matter: motion is eternal, as are form and matter. If motion had a beginning, then prior to this beginning there must either have been a moving and a moved, or not. If they have not been before, then they must have come to be, and a motion has then preceded the first motion. If they have been before, then it is unthinkable that they should not have brought forth motion; in the contrary case an activity must have set in whereby they came to move, and we should then in this case too have a motion prior to all motion. Just as motion

in a finite manner presupposes itself to infinity, so in the absolute sense it presupposes a first mover that is not moved (τὸ πρῶτον κινεῖν ἀκίνητον ὄν); this first mover is the Godhead. Everywhere in the order of things the moving is itself a moved; but what is moved is altered and cannot work any constant, uniform motion. The first mover must be unalterable; but unalterable it can be only by excluding, according to its essence, the possibility of being otherwise; and to exclude the possibility of being otherwise is to exclude matter. Matterless, incorporeal, indivisible, outside space, without motion, passivity or alteration (ἀπαθὲς καὶ ἀναλλοίωτον), the first mover is the perfect form, eternal actuality, pure energy — in a word: God.

With this metaphysical concluding thought Aristotle remains standing on the border between two mutually irreconcilable world-views, pantheism and theism.

In a theistic direction Aristotle maintains Spirit's absolute originality and lays a decisive weight upon the opposition between God and the world. The concept of the Godhead coincides, in perfect thinking, with the perfect good. As the eternal, perfect good the Godhead moves everything in being desired by everything; for everything in nature is laid out toward the good as toward a highest common end (πρὸς μὲν γὰρ ἔν ἅπαντα συντέτακται). As the good perfect in itself the Godhead can move everything without itself being moved; for that which is desired and thought moves without itself being moved. What *shows itself* beautiful is desired; what *is* beautiful is willed. We desire the beautiful because it shows itself, but it does not show itself because we desire it. The end moves because it is loved, and what is moved by the end moves in turn the rest. Since the divine consists in eternally completed energy, it consists in eternal living, for life is energy (ὥστε ζωὴ καὶ αἰὼν συνεχῆς καὶ αἰδῖος ὑπάρχει τῷ θεῷ). This life is blessed; such a joy is granted to us only in brief instants; but all joy (ἡδονή) is a draught of the divine principle.

That Aristotelian theism nevertheless stands on weak feet, criticism has no difficulty in showing. A Godhead that does nothing but eternally think the thinking of thinking (αὐτὸν ἄρα νοεῖ . . . καὶ ἔστιν ἡ νόησις νοήσεως νόησις), and which, so as not to stain its perfection with anything imperfect, cannot possibly have a thought to spare for anything in the world, is an empty, natureless being whose absolutely necessary content of thought resolves itself into pure tautologies.

Where the working of the self-subsistent God upon the world is treated, the turns of phrase are ambiguous. There was, it is said, neither eternal chaos nor eternal night when motion began; but the same has always been, either in constant alternation or in another way, as surely as actuality is prior to possibility. If now the same is always to work in constant alternation, then there must be something that always works in the same manner; if there is to be arising and passing away, then another must incessantly work otherwise and again otherwise. The first mover, which is always like itself, thus comes into a questionable intercourse with what works otherwise and is always unlike itself; and yet the eternal thinking of thinking was supposed to be untouched by everything

else. What is said of the need and desire with which things move toward the perfect one is only a teleological, figurative expression for the negativity indwelling in everything finite, which shows that the relative is a moment for the absolute. In a far more immediate manner thinkers such as Cartesius and Gassendi have made God the first mover. However much these two philosophers otherwise disagree about the causes in nature, they both start from a dogmatic separation between the absolute principle, the first cause of everything, the Creator, and the secondary causes, among which is reckoned “everything which besides God has a force to effect and accomplish something in the world.”

The cause of motion, says Cartesius, “is twofold; partly a universal and original one, which causes all the motion there is in the world at all; partly a particular one, which brings it about that the single parts of matter obtain motions they did not have before. The universal cause is God, who created matter both with motion and with rest, and who moreover by his natural co-operation maintains all the motion and rest in matter that he laid into it from the first.”

What is characteristic of this way of seeing is that, starting from an abrupt opposition of the supernatural and the natural, it is laid out so as to explain nature’s coming-to-be by a finite act of creation, and so by an act that is not natural. First matter is created; the Creator implants in it certain properties, a certain quantity of motion, etc. With a created matter, with created forces and properties, created nature is then to begin working in a natural manner. Such a beginning is, on the presupposition set up, impossible. As the supernatural effect of a supernatural cause, created matter with its properties and forces gets at the beginning a supernatural existence of its own; its first motion is accordingly supernatural: whence are the natural causes with the natural motions now to come? They cannot lie in matter’s essence, for matter’s created being is supernatural; just as little can they be ascribed to the forces, since the forces are themselves likewise supernatural. As the cause is, so must the effect be; cause and effect must, in spite of the separation in the outward, absolutely have the same essence. That created matter with its properties and forces is wisely distributed by the Creator and becomes, by the supernatural act of distribution, “secondary causes,” will then mean that the secondary causes are not natural either.

The dogmatic assurance that a supernatural beginning must necessarily be presupposed here, though we human beings are not able to derive the natural from the supernatural, is only a convenient evasion for theologians, but signifies nothing in science. One of two things! Either nature as nature has a supernatural beginning, and then it must necessarily continue as it began: all so-called natural causes, effects and reciprocal actions must in being and working be supernatural; or, if there really are natural causes and naturally necessary motions, then the beginning must answer to the continuation: nature, natural in being and working, must have a natural beginning.

It is an unclarity resting upon the concept of omnipotence that from all sides breeds the misunderstandings. While Aristotle apprehends the absolute in the form of thinking

as a metaphysical self-being without actual omnipotence, the absolute is in Gassendi and Cartesius represented as an omnipotent self, but without object-power, without power in another; while Aristotle gets a first mover which in actuality can move nothing and must alongside this let nature be nature and move itself on the foundation of a motion smuggled in under a negative form, Gassendi and Cartesius make absolutely ideal self-power into an immediate power of reality and presuppose an omnipotence which must consistently devour the nature it itself brings forth.

The fundamental contradictions find, as we have seen, their solution in power's original self-doubling. The mechanical causes in nature belong to object-power and move with *necessity*; but the ideal causality, the inner determining of the natural causes, belongs to self-power, the first mover, which moves with *freedom*.

Chapter Three.

The Mechanics of the Heavens.

A.

The Structure of the World.

There is in the world nothing objective except what is objectified: space and time, matter and its forces, everything in outward nature, in the realm of power, has objectivity – not immediately of itself, but from Spirit's omnipotence. From the standpoint of finitude it does indeed hold that visible things are in their outward actuality independent of the eye by which they are seen; but if the question is of knowledge's truth, then the law of objectification teaches that the validity which can belong to the object as an existing object is dependent upon the validity which, according to scientific insight, can belong to the object's objectification. The objectivity of the world-structure is a magnificent and eloquent example of this. The starry heaven is indeed an object of immediate contemplation, and it might so far seem as though everyone who looks up toward the high vault must involuntarily see in what style this building is raised. And yet it is an illusion; the unsurveyable building with its many rooms does not stand open to naively curious spectators. With earthly buildings it is the actual erection that costs labour; the contemplation comes of itself. To get the heavenly building raised costs no labour; that work is completed; but to see into the building and, if only approximately, to acquaint oneself with the plan in its peculiar construction is conditioned by labour. It has cost the human spirit millennia of labour to get the world-structure objectified in all the main features answering to actuality; this ideal work of building is of infinitely greater

worth than all the material toil expended on Egypt's pyramids. We follow the progress of the objectification from *the stage of illusion* through *reflexion* to *the final testing*.

a) *The illusion: the Ptolemaic world-structure.* Antiquity's consciousness of nature is essentially mythical. What is brought to the senses from without and must be referred to outward nature melts involuntarily together with what comes from within, from the ground of the mind, under a mood-laden co-operation of feeling, fantasy and thinking, and transforms the outward; how it comes about that the world-picture and the actual world, in spite of their mutual difference, meet and fuse in consciousness itself — this secret of objectification is not there for a naive contemplation. The sense of sight is objective, but by reason of an illusion; in the apprehension of the visible heaven the illusion rules so much the more as the earthly standpoint is fixed of itself, and none of the other senses reaches with its testimony out into infinite space. It is then a matter of course that the first picture of the world-structure must be illusory. But the illusion is not a simple deception. The revolution of the celestial sphere about the earth, the sun's rising and setting, the moon's particular motion and changing appearance — all such things have from an earthly station their truth just as surely as the separation of light and darkness, morning and evening, day and night.

The Ptolemaic world-structure rests upon ancient foundation stones; the first material was gathered at a time when a sustained beholding of the phenomena in the heaven was less a concern of science than of religion, of popular faith. The seven planets — the moon, Mercury, Venus, the sun, etc. — were of course not regarded as lifeless masses of rolled-together matter, but as high powers in radiant splendour, as gods. In this illusion Spirit's unity with nature was held fast. The gods wandering across the heaven did not merely, by their measured motions, give men a standard for calculating civil time; they were themselves the rulers of time. In order to survey events and govern earthly affairs these lords of time divided time among them; it was then no accident that each of them got his own day in the week.^{*)}

If the illusion is the naturally necessary form under which the human spirit, at the stage of naive intuition, avails with the gods' assistance to see a reflected splendour of its objective essence, then it cannot surprise us that antiquity was not only involuntarily dazzled by the illusion, but we understand that it also, in an arbitrary and yet instinctive

^{*)}By arranging the planets in the following series — Saturn, Jupiter, Mars, the sun, Venus, Mercury and the moon — and starting from the presupposition that each of them has indeed its hours in the day, but in such a way that the one that reigns in the first hour is overlord and gives the day its name, one will understand — in the sense of the illusion — how the seven days of the week can be subjected to the seven planet-gods. If the first hour of the day begins, for example, with Saturn, then we have a Saturday, *Dies Saturni*. By sharing the dominion with the six co-regents Saturn will further come, in his own person, to rule the eighth, the fifteenth and the twenty-second hour; then Jupiter will get the twenty-third, Mars the twenty-fourth, and the regency in the first hour of the following day falls to the sun, so that Sunday becomes the sun's day, *Dies Solis*. As it went with Saturn, so it now goes with the sun; when it has last ruled in the twenty-second hour, Venus rules in the twenty-third, Mercury in the twenty-fourth, so that the first hour of the following day falls to the moon and becomes a Monday, a moon's day, *Dies Lunæ*; and so forth.

manner, half consciously, half unconsciously, held the illusion fast as the truest form for its richest and deepest content. Thus the constellations are by no means formed by merely arbitrary grouping with arbitrary names; and just as little can it be supposed that the ancients really fancied they saw the heroes of the past alive among serpents and monsters in the heaven. The arbitrary and the involuntary melt together in a need of the mind for exalted symbolism, for an illusory form for Spirit's self-objectification in nature. The constellations are, as J. L. Heiberg has so aptly expressed it, "true pictures which antiquity has hung in the great gallery where all generations shall behold them. Here it rightly shows itself that the starry heaven, far from being foreign to the earth, is a reflex of earthly destinies, earthly thoughts and dreams."

The ground-plan for the Ptolemaic world-structure was supplied by Greek philosophy; it is a plan that lies open to the illusion from first to last. The fantasy filled with the world-idea, the beautifully rounded cosmic whole, knew how to form the stuff-like content to thought's liking so ingeniously that the phenomena of the illusion received a most profound rational grounding.

In mythically ideal images Plato's *Timaeus* describes how the god, the highest master-builder, arranged everything from the beginning for the best, in forming the world-soul into that intermediate being in which reason and sensibility, idea and corporeality alone admit of being united. Everything corporeal is moved by another; only the soul, self-motion's own force (ἡ δυναμένη αὐτὴ αὐτὴν κινεῖν κίνησις), moves itself and everything. Even before the corporeal elements were formed, the god mixed the indivisible substance, like to itself, and the corporeally divisible in such a way that a third being came forth, namely the substance of soul, in which the self-like and the other are joined together. This whole was now divided, according to the fundamental numbers of the harmonic and astronomical system, in such a way that the circles of the fixed-star heaven and of the planets came forth by a division of length of its own; for all relations of number and measure are contained in the soul, its essence is nothing but number and harmony. Plato lets the seven parts of the world-soul stand to one another as the numbers 1, 2, 3, 2², 3², 2³, 3³; upon this division rests the planets' distance from the world's centre; the sun is 2 times, Venus 3 times, Mercury 4 times, Mars 8 times, Jupiter 9 times and Saturn 27 times as far from the earth as the moon. As the soul, in the circular motion, returns to itself, it says through its whole being everywhere to itself what it is that it touches in its revolution; whether it is something divisible or indivisible, with what it is of one kind, from what it is different, and how it in general stands in its becoming and being. This speech propagates itself soundlessly in the self-moving and begets knowledge. From the circle of the other, the ecliptic, proceed right representations and opinions; from the fixed-star circle, the equator, the circle of the self-like, stem rational knowledge and understanding knowing. The illusorily profound central thought is that the harmoniously moved world-whole is a living, ensouled being.

Aristotle takes a step forward. He disdains the mythical clothing; he subjects Plato's

doctrine of the world-soul to a strict criticism and holds to actuality. One should now suppose that the Aristotelian building, freed from the dazzling light-mists of myth and illusion, would have to stand fast on the ground of reason, and the actual outlines keep themselves in the clear day of science. And yet the whole objectification is a complete illusion. How little human reason is able, by a few general observations, to penetrate so objectively into nature that it can in virtue of its own world-idea construct the actual world — of this Aristotelian physics is a grand example. Everything from first to last has this philosopher reasoned through, grounded, defined, demonstrated; the system, worked through with enduring acuteness, makes the impression of a comprehensive, irrefutable necessity of thought; but the necessity rests upon an illusion.

How Aristotle has thought the first mover has been shown in the preceding chapter; here it shall now be touched upon briefly how he apprehended the motions in the corporeal world. All motion in space is either rectilinear or circular or composed of both. Rectilinear motion is, according to the nature of the elements, again either *toward* or *away from* the centre. The four elements move along the straight line — earth, which is absolutely heavy, toward the centre; fire, which is absolutely light, away from the centre; water and air form transitions; but circular motion belongs to the aether. Since now every element has its natural motion and its natural place, the fundamental shape of the world-structure is determined with necessity. Immovable, in the middle of the whole, rests the spherical earth. It is impossible for the earth to move; it conflicts with its nature. The earthly can move only insofar as it moves toward the middle; but the terrestrial globe is the middle; every body comes to rest when it has reached the place to which its motion carries it: everything earthly accordingly has its rest in the middle. If it stands fast that the earth with its sphere of air and fire is spherical, it follows that the heaven surrounding the earth must also be spherical. But it is seen besides on other grounds. The heaven must, namely, as the most perfect, have the most perfect form; but the most perfect corporeal form is the sphere. As the measure of all motions the heaven must move with the greatest velocity, and so follow the shortest way; but the shortest way from the same to the same is the circle. All heavenly bodies move daily from east to west; but seven among them move besides, in shorter or longer periods, from west to east. Since it is now settled that the earth rests, either the whole heaven must move, or only the stars, or both the heaven and the stars. But that both the heaven and the stars should move is unthinkable; for how should it come about that the stars kept pace in velocity, as of themselves, with the velocity in their own circles? That the stars should move and their circles rest is also unthinkable; it would have to be an inexplicable accident if a star's motion should answer exactly to the size of the circle. There remains, then, only to assume that the circles move while the stars fixed upon them rest (τοῦς μὲν κύκλους κενεῖσθαι τὰ δὲ ἄστρα ἡρμεῖν). The famous astronomer Eudoxus of Cnidus had already sketched a system with 27 spheres, of which 26 fell to the planets. To these Callippus added a further 7 spheres, so that sun and moon each got two, and Mercury,

Venus and Mars one each. From the manner in which Aristotle takes up and grounds the intricate theories of spheres it becomes quite evident that he, far from cancelling the illusion, on the contrary works it so deeply into the thinking consciousness that nearly two millennia had to pass before science got it purged out from the ground up.

On an illusory foundation an understanding objectification can meanwhile go on upon the prudent condition that the observer will not undertake to give grounds for everything, but holds to facts and by empirical induction sets about finding out the laws of motion. Herewith scientific astronomy begins.

The understanding objectification now runs through three principal stages. The first thing necessary is to draw circles and mark off points, whereby it becomes possible to orient oneself on the celestial sphere. To this answer the familiar great circles — the horizon with zenith and nadir, the equator with the poles of the world, the ecliptic — in short, what belongs, from different points of view, to indicating a star's coordinates, that is, determining its spherical or angular place. Although such points, poles and circles have no material actuality, they have nevertheless an undeniable validity for intuition and thinking; they have an essential objectivity.

As the sum of the first conditions of objectification is gradually widened, and continued observation of the phenomena of motion leads to ever more exact and more special results, it will dawn upon consciousness that the world-harmony, the wonderful connexion of the whole, is far more intricate than one had represented it from the beginning. Here a contradiction sets in between the original presupposition of the earth as the centre of all circles, of the heavenly bodies' even, uniform motion, etc., and the manifest, at least apparent irregularities which the observations display; the endeavour to solve this great problem, indicated by the contradiction, is characteristic of the next stage. If the sun's orbit is a circle and the motion even, how can it then come about that the seasons are not of equal length, that the sun takes a very different time to traverse each of the ecliptic's four quadrants? To explain the phenomenon Hipparchus set up two hypotheses which, however — as Ptolemy afterwards saw — are different only in form, not in actuality. According to one of these hypotheses the sun describes with even velocity a circle whose centre lies outside the earth; according to the other the sun is ascribed an even motion on a circle — the epicycle — whose centre itself moves on a circle — the deferent circle — which has the earth for its centre. Hipparchus is the real founder of the epicycle theory. Finally — and this is, as regards method, the last, completing step of the objectification — the whole theory of the planets is brought to a relative conclusion by the continued application and extension of the epicycle hypothesis. Precession, the retrograde motion of the equinoctial point, was shown by Hipparchus; the unevenness in the moon's orbit which afterwards got the name of evection was discovered by Ptolemy, whose *Μεγάλη σύνταξις* set the garland upon the ancient edifice of doctrine. In illusory objectivity, with a few Arabic repairs, the building stood unshaken for fourteen hundred years.

b) Reflexion: the Copernican world-structure. Fantasy can breathe in thinking, set

thoughts in motion and reflect; conversely, thinking can build upon an intuition and, when the intuition is hovering, fantasize. Illusion and reflexion form in and for themselves no sharp opposition. There can, as we have just seen, be much reflexion brought to bear in developing and grounding an illusory theory, and likewise there can, as we shall see in what follows, be illusions of various kinds that creep into an otherwise thoroughly reflected fundamental intuition answering to actuality. But when the talk is of what on the whole characterizes a standpoint of objectification, the difference becomes recognizable.

There is a deep-going difference between a view of the world whose starting point belongs unconsciously and straightforwardly to the illusion, and an investigation of the connexion of the phenomena which begins by subjecting the foundation of the view, the first, general presupposition, to a strict scientific testing. In this sense we can, as regards the doctrine of the world-structure, rightly say that Copernicus stands in the history of astronomy as a heroic figure, insofar as he, with an opened eye for a higher world-harmony, is the first who had the superiority of genius and the intellectual courage to break with the power of habit, place himself on the standpoint of reflexion and combat the illusion. "The authors," he says, "are usually so agreed that the earth rests in the middle of the world that they consider it meaningless, indeed even laughable, to assume the opposite. But the question is not yet settled and cannot be dismissed with contempt."^{*)}

To choose one's standpoint in the sun, when all the adherents of tradition are agreed on keeping to the earth, must in many eyes look like a whim, a foolhardy notion of a high-flying fantasy. But Copernicus will not fantasize; he will reflect in earnest. It is the actual relation between earth and heaven — a relation that only a carried-through reflexion, not an immediate judgment, can appraise — that he will have investigated from the ground up.^{**)} It is the difference between apparent and actual motion, and so the objectivity of motion, upon whose determination it turns.^{***)}

That the heaven's daily motion from east to west can be explained as a deceptive rendering of the earth's actual motion from west to east, Copernicus is so far from wanting to give out as any new thought that he on the contrary recalls that among the ancients, for instance, the Pythagoreans Heraclides and Ecphantus and Nicetas of Syracuse^{*)} were, according to Cicero, of this opinion. But, he adds, when this is conceded, then another no lesser doubt follows about the earth's place, although it

^{*)}De Revolutionibus orbium coelestium. Norimbergæ MDXLIII. Lib. I. Cap. V.

^{**)}Qvam ob causam ante omnia puto necessarium, ut diligenter animadvertamus, qvæ sit ad cœlum terræ habitudo, ne dum excelsissima scrutari volumus, qvæ nobis proxima sunt, ignoremus, ac eodem errore qvæ telluris sunt attribuamus cælestibus. De Revol. Lib. I. Cap. IV.

^{***)}Omnis enim, qvæ videtur secundum locum mutatio, aut est propter spectatæ rei motum, aut videntis, aut certe disparem utriusque mutationem. Nam inter mota æqualiter ad eadem non percipitur motus, inter rem visam dico et videntem. Terra autem est, unde cœlestis ille circuitus aspicitur, et visui reproducitur nostro. Si igitur motus aliquis terræ deputetur, ipse in universis, qvæ extrinsecus sunt, idem apparebit, sed ad partem oppositam, tanquam prætereuntibus, qualis est revolutio cotidiana imprimis. Lib. I. cap. V.

^{*)}Should no doubt be Hicetas.

is now assumed by almost everyone that the earth is the middle of the world. From this doubt reflexion then begins in such a way that the rejected way of seeing is set against the authorized one, and the grounding happens by reasoning against reasoning. Ptolemy finds it unreasonable that the earth should move. By the rapid rotation the connexion of things would have to be dissolved, all earthly bonds burst — indeed, what is quite a laughable representation, the heaven itself would have to fall down. But, asks Copernicus, why does one not rather apply this to the world, whose motion must be so much the more rapid as the world is greater than the earth? Has the heaven perhaps become so infinitely great because it is driven away from the middle with indescribable velocity (*ineffabili motus vehementia*)? In truth, if this reasoning holds good, the heaven's greatness must be infinitely increasing; the velocity must bring forth the greatness and the greatness again the velocity, without measure or limit. But according to the physical axiom that what is infinite can neither be overstepped nor moved, the heaven must necessarily stand still (*stabit necessario cælum*). But when the heaven stands still, it must be the earth that moves. If the earth were the world's motionless middle, then it would have to be the common centre for all the orbits; but that it is not the fixed common centre comes to light both from the manifest unevenness in the planets' motion and from their changeable distances from the earth. Since there now demonstrably exist several centres, there is every ground to look about for another world-centre, another centre of gravity.^{*)} All the planets' testimonies (*tot tantaque errantium syderum testimonia*) agree (*consentiunt*) that the earth moves forward in space. Copernicus ascribes to it a threefold motion: a *daily* one about its axis, an *annual* one about the sun through the ecliptic, and — superfluously^{**)} — yet another motion, whereby the axis's inclination remains constantly parallel with itself (*tertius declinationis motus*).

We see the ancient philosophers starting from the fundamental consideration that bodies moving with equal velocity must seem to move the more slowly in proportion as they are at a greater distance — that the moon, with the smallest orbit, therefore has the shortest period of revolution, and Saturn, with the greatest orbit, the longest. But about Venus and Mercury opinions are divided. Whether one places them with Plato above the sun or with Ptolemy below, the matter is equally confused. As a ground why Venus and Mercury ought to be placed below the sun, it has been urged that otherwise there would be an all too enormous space between the sun and the moon. That is a most unreliable ground; for precisely the measures of distance upon which the assumption is to be grounded show that the epicycle of Venus, if the earth is to be the deferent's

^{*)}Pluribus ergo existentibus centris, de centro quoque mundi non temere quis dubitabit, an videlicet fuerit istud gravitatis terrenæ, an aliud. Equidem existimo, gravitatem non aliud esse, quam appetentiam quendam naturalem partibus inditam a divina providentia opificis universorum, ut in unitatem integritatemque suam sese conferant in formam globi coeuntes. Qvam affectionem credibile est etiam Soli, Lunæ, cæterisque errantium fulgoribus inesse, ut ejus efficacia in ea qua se repræsentant rotunditate permaneant, quæ nihilominus multis modis efficiunt circuitus. Lib. I. cap. IX.

^{**)}As Rothmann had already noticed.

centre, encloses a space so great that this empty space could more than hold the earth, the air, the aether, the moon and Mercury together. Another argument is still unluckier. Ptolemy considers it fitting that the sun should get a middle place between the planets which move away from it on all sides (*omnifariam*) and the planets which do not; but how false the proof is can be seen from the moon, which in this case would have had to have its place above the sun. Accordingly, Copernicus concludes, either the earth is not the centre, or the order in which one arranges the planets is without significance.^{*)}

To change the geocentric standpoint into a heliocentric one is possible only for one who has become quite clearly conscious of the difference between appearance and actuality. From the beholder's actual standpoint all motions are apparent; only from an unactual, reflected-to standpoint are the motions actual; it is a reversal which only a superior reflexion has been able, at first hand, to grasp and carry through. Whether the Pythagorean Philolaus^{**)} or Aristarchus of Samos were of the opinion that the earth was a planet makes in this connexion no difference; for what it turns upon is an astronomical-geometrical reflexion which, with due insight into the manifold of entangled relations, avails to carry the appearances back to actuality and then again, starting from actuality, to get the apparent exactly explained and grounded on all sides in all its details. It is this problem that Copernicus, in relation to the astronomical knowledge of his age, completely solved. An exalted central thought led him, an idea about which his objectifying fantasy was from first to last at one with his reflexion: the idea of an "admirable symmetry of the world" (*admiranda mundi symmetria*) and "a harmonious connexion between the motions and the size of the orbits." The fixed-star heaven, "the highest of all," is immovable, for it encloses everything; but the planets Saturn, Jupiter, Mars, the earth with its moon, Venus and Mercury all move about the common centre, the sun. "Who could in this beautiful temple set this lamp in another or better place than there, whence it can at once irradiate everything? Some call it, quite aptly, the world's lantern, others its soul, others its governor; Trismegistus a visible god, Sophocles's all-seeing Electra. Yes, thus the sun sits as on a royal throne, governing the whole family of encircling heavenly bodies."^{*)}

Copernicus has not enriched astronomy with any new art of observation or with the discovery of new particulars; his discovery belongs not so much to the facts as to reflexion; it is a secret of objectification that he has discovered; by this discovery he has transformed not the world, but our consciousness of the world: he has, for our knowing, remade heaven and earth.

c) *The discussion: three hypotheses.* The Ptolemaic system was dogmatic; dogmatism, with its naive reliance on the objective validity of human objectification, is on good terms

^{*)}Oportebit igitur, vel terram non esse centrum, ad quod ordo syderum orbinmqve referatur: aut certe rationem ordinis non esse, nec apparere cur magis Saturno quam Jovi seu alii cuius superior debeatur locus. Lib. I. cap. X.

^{**)}Cf. De Revol. Lib. I. cap. V.

^{*)}De Revol. Lib. I. cap. X.

with illusion. About particulars doubts might arise, but the foundation as a whole stood fast, for it rested upon immediate intuition. The Copernican system is thetic in the form of reflexion, and so far dogmatic too, though in a way of its own. The senses' immediate testimony is indeed in each and all subordinated to the understanding's testing; but the standpoint reflexion has chosen must for the time being hold to the dogma of harmony. Even though Copernicus now and then expresses himself with caution, as though it were enough for him if the reader merely found his view probable, or at any rate not unreasonable, it is nevertheless clear from the whole impression that he himself considers it irrefutable; the world-structure he has raised is for him not one of several possible ones, but the only possible one, because it is the only reasonable one.

Now it lies in the nature of the case that two mutually contradictory theories, both coming forward with equal title, since both of them, each in its own way, satisfy the observations in about equal degree, must make one another problematic. But when the foundations are problematic, then either one of the theories must be laid down as an article of faith, or both must remain standing as hypotheses.

The unnamed author of the preface to the work *De Revolutionibus* has expressed himself in this respect in a quite characteristic manner. Since it is impossible, he says, to discover the true causes, the astronomer has only to devise and invent such pre-suppositions or hypotheses as can give him a convenient and satisfactory foundation for the calculations. In several turns of phrase it is then put before the reader most emphatically that the Copernican theory neither is nor will be anything but a plain and simple hypothesis. It is surely more than considerations of prudence that stamps this presentation; it is honest conviction.^{*)}

The two hypotheses stood meanwhile on so strained a footing with one another that the adherents of the one showed themselves irreconcilable toward the adherents of the other; it might then, for the calming of minds, be quite to the purpose to attempt a mediation by setting up a third hypothesis, a hypothesis which removed what was offensive in the new one and in part made good the imperfections of the old: this third hypothesis was set up by Tycho Brahe. The old standpoint of illusion is maintained, insofar as the earth continues to be the immovable centre: about the earth move the moon and the sun; but the sun — that is the concession made to the new theory — is the centre for all the planets. One can from astronomy's present standpoint apprehend the Tychonic hypothesis as a modification of the Copernican, arising from the intersection point of the coordinate axes being transferred to the earth's centre; but one can with just

^{*)}Est enim Astronomi proprium, historiam motuum cælestium diligenti et artificiosa observatione colligere. Deinde causas earundem, seu hypotheses, cum veras assequi nulla ratione possit, qualesumque excogitare et confingere, quibus suppositis, iidem motus ex Geometriæ principiis, tam in futurum quam in præteritum recte possint calculari.

Neque enim necesse est, eas hypotheses esse veras, immo ne verisimiles quidem, sed sufficit hoc unum, si calculum observationibus congruentem exhibeant . . . Sinamus igitur et has novas hypotheses, inter veteres, nihilo verisimiliores, innotescere . . . Neque quisquam, quodad Hypotheses attinet, quicquam certi ab Astronomia expectet, cum ipsa nihil tale præstare queat.

as much ground regard it as a special case of the Ptolemaic, namely the one obtained by taking the sun's orbit as the common deferent circle for all the planets. On this hypothesis one then knows the relation between all the planets' mean distances, but the value of the distance in radii of the sun's orbit is, like the relation between the moon's and the sun's distance, still unknown.

With the three hypotheses the world-structure's objectivity has been lost; actuality itself has gone missing for consciousness. But that the actual connexion between earth and heaven should not admit of being known is an intolerable representation, equally unnatural for scientific research and for naive contemplation. If the reflecting understanding has, in an ingenious, combined play of hypotheses, squandered its original dowry, the certainty of knowledge's first immediate reality, then neither does it find peace until it has been sated anew with actual content and has conquered back the lost objectivity. Doubt and groping led to comprehensive *discussion*; contradictory considerations crossed one another; the points of view from which they proceeded are characteristic of the scientific condition of the time.

The *ecclesiastically dogmatic* point of view was at bottom subjective, but, as usual, mistook itself and fancied that it was in all infallibility objective. It was some time before the theologians became properly aware of the danger with which the Copernican theory threatened Protestant as well as Catholic orthodoxy; but hardly had they got their eyes open before they, from both churches, defied the danger and went into the fight with contempt for death; history gives them the testimony that in this matter "important for salvation" they conscientiously used every occasion on which they could in a fitting manner come to prostitute themselves in a properly scientific way. What Copernicus foresaw in the dedication to Pope Paul III — that "empty word-pourers (*ματαιόλογοι*), destitute of all mathematical knowledge, would arrogate to themselves a judgment upon the work by wilfully distorting one or another passage of Holy Scripture (*propter aliquem locum scripturæ male ad suum propositum detortum*)" — was amply fulfilled.

The *empirically philosophical point of view*, represented by Baco, was undeniably better suited to a prudent discussion of the contending hypotheses, grounded on the method of induction; but, remarkably enough, Baco had, in open contradiction with his own theory of induction, so thoroughly settled himself in a certain conception of nature's fundamental qualities and of the *spiritus* indwelling in bodies that he not only held to the immovable earth — without, however, either settling down with Ptolemy or accepting Tycho Brahe's modification — but was not even able to enter upon exact investigations. He sees in Copernicus a man who takes up notions of every kind and introduces them into nature whenever they can in any way be brought into agreement with his calculations. By such an appraisal Baco has, however, characterized not Copernicus but, as regards the astronomical problems, rather himself; for an assertion in *Thema Coeli* such as this — that the planets' so-called retrograde motion from east to west does not on the whole exist at all, but is an empty semblance arising only from the fixed vault of heaven advancing

further on the one side, so that the planet on the other side seems to be left behind — must surely be reckoned among notions of the kind no astronomer finds it worth the trouble to refute. By boldly breaking with scholastic formalism, laying bare the barrenness of syllogistic, maintaining induction and pointing to experience, Baco gave the thinking of his age a mighty push forward. But the fundamental relation between the empirical and the philosophical method he was unable to see through. For a penetrating analysis of the peculiar reciprocal action of empirical and a priori thinking he lacks on the one side insight into the mathematical-geometrical, on the other side an eye for the critical-dialectical. He has the general propositions in the form of wit; but, unable to let the idea itself penetratingly dissolve and confirm the peculiar facts of concrete actuality, he remains standing at the merely witty and does not reach the spirit: Baconian empiricism is only a somewhat improved edition of the Aristotelian.^{*)} Those who in the name of the philosophy of experience still consider Baco the non-plus-ultra will on closer testing assuredly make the discovery that there is, for the principle itself, too little experience in his philosophy and too little philosophy in his experience.

So there remains only a third point of view which could, and then really did, acquire a true scientific significance in this discussion, namely the *exactly empirical*. Its great representatives are Gilbert and Galilei. It is not the final result but the way to it, not the verdict upon a hypothesis — whether it ought to be accepted, rejected or doubted — but the grounding of the verdict, upon which everything in the discussion first and last turns. As regards doubts and misgivings, Gilbert stands close to Baco; but in grounding reflexion they are widely separated. Gilbert does not speculate, he investigates; when he cannot convince himself that the earth has an annual motion, but wavers constantly between Tycho and Copernicus, it is not the ideas but the facts that give him trouble. In his work *De Magnete* he does indeed assume that the earth turns about its axis,^{*)} but

^{*)}In vehement polemic against the Copernican hypothesis Baco adduces: that Copernicus burdens the earth with a threefold motion and tears the sun loose from the planets, with which it is nevertheless so nearly akin; that his system introduces too much that is immovable into nature, in that it deprives the sun and the fixed stars — and so precisely the bodies that shine and radiate most clearly of all — of motion, etc.; these arguments are in scientific worth altogether Aristotelian. The truly significant central thought that the form is the matter's inmost essence, *ipsissima res, natura naturans, fons emanationis*, is Aristotelian too. Baco does indeed make the form-giving element in bodies, the principle of motion and alteration, the spiritus indwelling in every body, into a material substance, and will on no account have this material form-being confused with an Aristotelian energy; but by this unclear conception of the nature of bodies he is so far from raising himself in the concept above Aristotle that he on the contrary sinks down to a kind of witty dalliance with magic and sorcery. That Baco, with all his zeal for a unity of experience and thinking, has not been able to reconcile the empirically investigating science of nature with the philosophy of nature proper is explicable from this, that he has not been able, for his time — as little as the English to this day — to become determinately and clearly conscious of the opposition between the two qualitatively different sciences. In Baco the interest in the empirical is preponderantly philosophical; therefore he has no sense for mechanics and looks down loftily upon mathematics; in Newton the interest rests decisively on the mathematical-mechanical, but for all that Newton considers his physically exact principles to be principles of the philosophy of nature.

^{*)}Guilielmi Gilberti Colcestrensis, Medici Londinensis. *De Magnete Magneticisque corporibus et de Magno magnete tellure, Physiologia nova*. Londini MDC. — *De terrestris globi diurna revolutione*

sets the doctrine of this in an unclear connexion with the doctrine of the magnet and of electrical attraction (*Globus telluris per se electrice congregatur et cohæret*), in assuming that the electrical striving aims at a binding heaping-together of matter (*motus electricus est motus coacervationis materiæ*). Although his proofs in this respect are unsatisfactory, there is nevertheless something that shows he stands on a scientific foundation in quite another sense than Baco. He does not decide by any speculative explanation what nature can and what it cannot do, what suits its *Ingenium* and what does not suit it:^{*)} he takes the facts as they show themselves to him; he can be mistaken in determining the premisses, but in this he is not mistaken, that the grounding of physical phenomena must be derived exclusively from physically determinate data;^{**)} he holds fast to the exactly empirical method even in his missteps.

The weightiest contributions to the great trial were made, however, by Galilei. Not content with letting one of the speakers in his Dialogue^{**)} conduct a defence of the Copernican hypothesis — which afterwards drew upon him the persecution of the Inquisition — he aims higher, he arms the eye and widens the horizon; with the telescope he himself constructed he makes astonishing discoveries in celestial space. With Galilei the exactly empirical method makes its triumphal entry into the history of science.^{*)} Of all the senses sight is the most objective; with the widened sight, the widening thinking gets a widened foundation; so actuality is overtaken, and the objectivity that had fled is called back again. The mountains of the moon, the phases of Venus and above all the moons of Jupiter^{**)*)} belong to a cycle of facts which, by the new and surprising in what they offered to the sharpened sight, necessarily had to make a strong and magnificent impression upon the age, shake confidence in the old way of representing, open the eye to the Copernican world-structure and awaken thought-rich intimations of a hitherto

magnetica. Lib. VI. cap. III. Terram circulariter moveri. cap. IV. Terræ motum negantium rationes et earum confutatio. cap. V.

^{*)}Theologi etiam curiosi, mysteria divina ultra humanum sensum posita, per magnetem et succinum illustrent, ut vani metaphysici, cum inutilia phantasmata fundunt docentque, magnetem habent tanquam Delphicum gladium, exemplum semper ad omnia accommodandum. De Magnete. Lib. II cap. II.

^{**)*)}The last chapter — Lib. VI cap. IX. — De præcessionis æquinoctiorum et obliquitatis Zodiaci anomalia closes in a quite characteristic manner with the following words: Quo magis hæc omnia de inæquali motu tam præcessionis quam obliquitatis incerta et incognita sunt. Quare neque nos illius causas aliquas naturales proferre et certo statuere possumus. Quare etiam et nos magneticis rationibus et experimentis hic finem et periodum imponimus.

^{***)}Dialogo di Galileo Galilei, dove ne' Congressi di quattro giornate si discorre de' due massimi sistemi Tolemaico e Copernicano. Firense per Landini 1632.

^{*)}It is of course of Galilei as the representative of his age, not of Galilei as an isolated investigator, that this can be said. That the telescope was first invented in Holland, and that it was the news of the invention made in Holland that gave Galilei occasion to construct his telescope, is well known.

^{**)*)}Jupiter's moons, the first secondary planets of all found by the telescope, were, as it seems, discovered almost simultaneously by Simon Marius in Ansbach on 29 December 1609 and by Galilei in Padua on 7 January 1610, without these men standing in the least connexion with one another. With respect to the publication of this discovery, on the other hand, Galilei with his *Nuntius Sidereus* (1610) forestalled Simon Marius's *Mundus Jovialis* (1614).^{*)} A. v. Humboldt, *Kosmos*, translated by C. A. Schumacher, second volume, Copenhagen 1849, pp. 289–90.

unknown mechanics of the heavens.

B.

The Solar System and the Universe.

Both at the stage of illusion and at that of reflexion there shows itself a peculiar relation between the planetary spheres and the fixed-star heaven; but it is impossible to form a reasonable concept of the universe so long as the representation of the solar system is still unclear. The immediate task accordingly becomes: by sharpened observation and corresponding calculation to bring about the most exact possible agreement between the orbits and the laws, and thereby to secure *the objectivity of the formal foundation*. Next, when the phenomena of motion are so far cleared up that they show themselves as law-determined effects of determinate causes, the causes can be investigated, and the causal connexion of the facts lead to an assurance of *the objectivity of the real foundation*. Finally, when the foundation, the real as well as the formal, is secured on all sides, the method's two elements — the art of observation and mathematical analysis — will mutually drive one another forward, and the objectification of the mechanical world-structure, in its details as in its great outlines, complete the picture of *an outwardly objective universe, infinite in finitude*.

a) *The objectivity of the formal foundation: the Keplerian laws.* One essential hindrance to a real and true insight into the plan and structure of the solar system still remained to be removed: it was the epicycle theory. However resolutely independent Copernicus was where it was a question of making the sun the centre, he was nevertheless still dependent upon the old presupposition of the necessity of circular orbits. "We must," he says,^{*)} "assume that the celestial motions are circular, for their inequalities follow a determinate law and return in unalterable periods, which they can do only when they go on in circles." We shall now see, from the standpoint of the philosophy of nature, that the epicycle method must, by the nature of the case, make a final solution of the great problem of objectification impossible, noting particularly wherein what is confusing for the objectification really consists.

What dazzles and confuses cannot be placed in this, that the observer is tempted to take the many circles, great and small, for so many actual orbits and to ascribe to each of them a material existence; for the unreasonableness of such a way of representing is too conspicuous to be able to dazzle the knowledgeable. "Even if dogmatic philosophers such as Aristotle have regarded the celestial spheres as really subsisting, solid bodies, Ptolemy nevertheless speaks of them only as of merely imaginary things, and already from the proof he gives for the identity of an epicycle with an eccentric circle it clearly

^{*)}De Revol. Lib. I Cap. IV.

follows that he has not held the epicycle to be anything but a geometrical conception by which he attempted to present the apparent motions of the heaven in agreement with the observations.” (Whewell.) The mystical reveries of the Middle Ages and the Reformation period about solid heavens and celestial spheres were most sharply distinguished by the astronomers from the mathematically geometrical epicycle theory.^{*)}

Nor can what misleads the objectification be sought in this, that the epicycle theory, by piling circle upon circle, had at last to become too intricate; for in the first place a part of the superfluous hypotheses did gradually admit of being reduced. One had begun by laying the deferent circles and first epicycles in the plane of the ecliptic, and had therefore to explain the latitudes by introducing a motion on small circles perpendicular to the ecliptic. Copernicus, however, helped himself by assuming certain librations or oscillations of the epicycles; and Tycho Brahe avoided the perpendicular circles by giving the planetary orbits a slight inclination to the ecliptic. Next, and this is what is decisive, the growing complications hung most closely together with the multiplication of the observations and the imperfection of the mathematical methods at the time. If the intricate^{*)} were in and for itself misleading, then the infinite series of the newer analysis would be misleading in quite another degree, for in comparison with these the complications of the epicycle theory may surely be said to be very simple.

No: what misleads the objectifying consciousness lies in this, that the alternating hypotheses make it impossible to distinguish objectively between actual and feigned orbits. That the sun’s orbit, apprehended abstractly and geometrically, must be the same whether one assumes that the sun with even velocity describes a circle whose centre lies outside the earth, or ascribes to the sun an even motion on a circle whose centre moves evenly on a circle having the earth for its centre, is incontestable.^{*)} But although the mathematical result is the same, the mechanical connexion of the matter is nevertheless highly different. On the first hypothesis there is here in the objective sense only one

^{*)}... “in the middle of the sixteenth century, when the learned polymath Girolamo Fracastoro’s theory of the 77 homocentric spheres met with so much approval, and when later the opponents of Copernicus sought out every means for the maintenance of the Ptolemaic system, the representation — favoured especially by the Church Fathers — of the existence of solid spheres, circles and epicycles was still very widespread. Tycho Brahe expressly prides himself on the merit of being, by his considerations of the orbits of comets, the first to have proved the impossibility of solid spheres and shattered their artificial scaffolding.” (A. Humboldt.)

^{*)}When the intricate arises from clumsiness in setting and treating a problem, an improved method will be able to remove all difficulties to an astonishing degree. Thus the higher mathematics now treats with the greatest ease a quantity of problems whose solution is difficult, cumbersome, indeed impossible, if it is to be done solely by the means the elementary mathematics offers. It is another matter when the intricate lies in the problem’s own nature; for in this case the increasing entanglement of the problems will stand in direct relation to the progress of the science. Entanglement presupposes development. In this sense it must be said that the problems in the newer astronomy are far more intricate than in the older.

^{*)}If, on the latter hypothesis, the intersection point of the coordinate axes be laid in the earth’s centre, while the axis of ordinates coincides with the line of apsides, and the ordinate for the deferent circle be denoted by y , and that for the sun on the epicycle by y^1 ; then $x^2 + y^2 = r^2$ is the equation for the deferent circle, and $y = y^1 - \varrho$; whereby one obtains $x^2 + (y^1 - \varrho)^2 = r^2$, which is precisely the equation for the eccentric circle answering to the first hypothesis, the eccentricity being $= \varrho$.

orbit; on the second there are, by contrast, two. In the first case one will for the time being be able to explain the orbit by pointing to the nature indwelling in all heavenly bodies of moving in circles; the only metaphysical difficulty that remains must then turn upon the ground why the earth has not become the actual centre of the sun's orbit. In the second case it is really not the sun but the sun's epicycle that moves about the earth. Here there must then in the objective sense be a double ground for the double motion: a particular ground for the motion of the epicycle's centre about the earth's centre, and alongside this a particular ground for the epicycle's motion with the sun or — what for the grounding gives a new problem — for the sun's motion about the epicycle's centre. If two alternating hypotheses, however simple each of them may be in itself, can already give the connexion of the matter so subjective and doubtful an appearance, then it follows of itself that along this way one comes with every step ever further from the objective. With the orbit of Mars, for example, the mechanical connexion and its grounding is still more entangled, since account must be taken of the sun's orbit lying between the earth and Mars. If one lets Mars move on an epicycle whose centre moves evenly, with a period of revolution of 687 days, on a deferent circle about the earth in such a way that Mars's place on the epicycle is determined by the sun's place, the epicycle radius to the planet being constantly parallel with the line connecting earth and sun: then the objective element in the orbit of Mars is already doubtful in itself, and becomes still more doubtful in that the objective character of the sun's orbit is doubtful. For motion in circular orbits speculative reason could still think a ground; but for eccentric orbits and motions of epicycle centres not even the most ingenious speculation has been able to find out anything that could resemble a ground in nature. Grounds without causes can be thought, but causes without grounds cannot be thought; if the phenomena are so entangled that one cannot glimpse a determining ground, how should it then be possible to discover a working cause? If the laws of motion are to be known, the epicycle theory must go: here is the turning point at which Kepler's genius breaks through.

To follow the discoverer of the great world-laws on his thorny way with its bewildering windings — *per errores ad veritatem, per ardua ad astra!* — is not the task of the philosophy of nature; an indication of the movement must suffice for a presentation of the course of the objectification.

In the year 1600 Kepler attached himself to Tycho Brahe, whom he found in Prague eagerly occupied with a theory of the planet Mars answering to the observations he himself had made. It was these new and improved observations that led Kepler onto the new path and put him in a position to publish, nine years later, his epoch-making work *De motibus stellæ Martis*. Had it been possible by immediate observations to determine an arbitrary number of places in the orbit of Mars about the sun, then the problem of finding out the orbit's character would undeniably have been considerably simplified; but the problem was entangled to such a degree that the relations in the objective had to pass through a quantity of testings in subjective reflexion before the solution could be

attained. The reciprocal determinations of the determinate and the indeterminate with the known and the unknown are manifold. There must indeed in the objective sense be a system of points lying in Mars's heliocentric orbit; but these points had first to be determined in relation to the earth's orbit. First the period of revolution had to be found by means of the oppositions^{*)}; next, one and the same place at which Mars had twice been in its orbit^{**)}, had to be seen in relation to the two corresponding places for the earth in its orbit: only by two lines of direction from two known stations toward the same point did it become possible to find out the planet's distance from the sun and determine the completely heliocentric place in the orbit of Mars. Only when one point had been found could the determination be repeated for an arbitrary number of points. When Kepler now, with these points, set about establishing by calculation the nature of the line, he found that the orbit of Mars could not, as had hitherto been assumed, be any circle, but that it must have an oval form. By assuming the line to be an ellipse he found, not indeed complete agreement with the observations, but nevertheless a significant approximation. Why only an approximation? Dazzled by the theory of the circle^{*)}, Kepler had started from the presupposition that the earth's orbit, whose eccentricity is comparatively insignificant, had to be a circle; but when this error was corrected, the so-called *first law* was also discovered: *the planetary orbits are ellipses, in one focus of which the sun is situated.*

The polar equation for the ellipse shows that the *radius vector* is a function of its angle with the major axis^{**)}; this leads in turn to a closer determination of the motion in the orbit. Hitherto it had counted as an astronomical dogma that the planet must absolutely move with even angular velocity — in other words, traverse equal angles in equal times. In honour of the dogma one invented the *punctum æqvans* with corresponding epicyclic artifices. Kepler at length saw the unreasonableness of this. There was indeed something even in the motion; but it was not the angles traversed, it was rather the elliptical sectors described by the *radius vector*^{***)}. Hereby the so-called *second law* is determined: *the areas which the radii vectores describe stand to one another as the times employed in the motion.*

Like Copernicus, Kepler too has a magnificent conception and a living assurance

^{*)}At the instant of opposition the planet's heliocentric longitude is the same as the geocentric; the geocentric longitudes of planet and sun differ by 180 degrees from one another. The Tychonic series of observations of Mars covers a period of 20 years and gives 12 oppositions.

^{**)}If one chooses, among the geocentrically angular places a series of observations can give, two which are separated by exactly an interval of 687 or more exactly of 686.98 days, then one knows that Mars has both times been at the same point of its orbit.

^{*)}Primus meus error fuit, viam planetæ perfectum esse circulum, tanto nocentior temporis fur, quanto erat ab auctoritate omnium philosophorum instructor et metaphycicæ in specie convenientior. De motibus stellæ Martis. Pars III. cap. XI.

^{**) $r = f(v) = \frac{a(1 - e^2)}{1 + e \cos v}$.}

^{***)}From the law of the areas it follows that the planet's angular velocity at every point of its orbit stands in inverse ratio to the square of the distance. Hereby a new means is won for determining the planet's distances from the sun, when one of these distances is assumed as given.

of an objectively subsisting world-harmony. There was in this respect no thought so bold that this wonderfully cosmic genius did not dare take it up. In his *Mysterium Cosmographicum* (1596) he expresses himself about the relations of the planetary orbits in the following manner: “The earth’s orbit is a circle. If one describes about the sphere to which the earth’s orbit answers as great circle a dodecahedron, then the circumscribed sphere gives the orbit of Mars. If one describes about this orbit a tetrahedron, then the corresponding circle presents the orbit of Jupiter. If one describes about Jupiter’s orbit a cube, then the circle answering to it will give the orbit of Saturn,” and so forth. These Pythagorean-Platonic speculations found at length a fruitful scientific clarification, insofar as they underwent a transformation and after 27 years’ exertions led to the discovery of the *third law*, which says that *the squares of the different planets’ periods of revolution stand to one another as the cubes of their mean distances from the sun*.

These laws, whose objective validity no one can draw into doubt without at the same time drawing the whole validity of the science of experience into doubt, furnish the formal foundation for a scientific objectification not only of the solar system but of the universe. That the formal element in these laws’ objectivity points out beyond itself with a promise of the real is manifest. Instead of empty geometrical centres, as in the epicycle theory, the elliptical orbits have a full corporeal centre, an actual central body in the focus; instead of the imaginary circles of subjective reflexion we find here objective relations between the central body and the peripheral bodies moving in the orbits. As attention fastens upon these fundamental relations between the bodies themselves, the question of moving causes gets sure points of support and a new life that promises a new and rich yield: formal knowledge has prepared the real; abstractly empirical mechanics is on the point of becoming rational physics.

b) The objectivity of the real foundation: gravitation. The first objections of any significance against the Copernican system stemmed partly from unclarity about the distance between the planetary system and the fixed-star heaven, in that one demanded that the Copernicans should point out the motions which the fixed stars, on the new doctrine, must necessarily have^{*)}, and partly, and chiefly, from ignorance of mechanics. If the earth moved from west toward east, then a stone falling from a high tower must necessarily — so it was objected — fall in a westerly direction, whereas the fact is that it falls in an easterly one: inertia was not yet known. By Galilei’s discovery of the law of fall the reciprocal determination of inertia and attraction was shown in exact form, and mechanical physics founded. Meanwhile, as regards the mechanics of the world, several introductory steps and groping attempts still had to be made before Newton came to utter the fundamental propositions, work the fragmentary into a rational whole, and mark the new epoch by a work such as *Philosophiæ naturalis Principia mathematica*.

^{*)}Remarkably enough, the question of the fixed stars’ distance was afterwards, when the law of light’s propagation had been discovered, to lead to one of the strongest proofs for the Copernican system, through Bradley’s discovery of aberration.

It is surely more the work's title than its content that has in more recent times given Hegel and the Hegelian school^{*)} occasion for objections against the Newtonian principles as unclear as they are unwarranted. If one holds strictly to the designation philosophy of nature (*philosophia naturalis*), and takes the word as a category in the Hegelian sense, then objections can undeniably be made against the manner in which Newton has defined and set up his fundamental propositions; but such a conception rests upon a colossal misunderstanding of the Newtonian standpoint. To avoid unnecessary wrangling about obscure words, Newton declares straightforwardly that the expressions he has chosen — inertia, attraction and several others — are not to be taken in a physical but only in a mathematical sense.^{**)} From this speculative philosophy has taken occasion to show how ungrounded his separation between the physical and the mathematical is in itself, and how little he has been able to hold the separation consistently. But how foreign this whole conception of physics and mathematics is to Newton's train of thought one will easily be able to see on a little reflexion. Newton guards against one's taking his expressions physically, that is, metaphysically, and the meaning is that he will not at all enter upon speculative investigations concerning the inner How of the physical activities^{*)}; but to tear the physical loose from the mathematical could surely not possibly occur to the man who from first to last is laid out so as to make clear how the quantitative in the effect points point for point to the qualitative in the cause.

Mechanics is originally a unity of mathematics and physics; space and time, matter and force are in equal degree physical and mathematical factors. To develop the mathematical Newton must lay weight upon the physical, and conversely. Mechanics expressly demands that the relation between the two be apprehended in a double-sided way, partly from the standpoint of cause and effect, partly from the standpoint of law and phenomenon. As regards law and phenomenon, it is intelligible that Newton can exchange categories such as attraction and impact for one another, for what attraction is in the form of continuity, impact is in the form of discreteness. As regards cause and effect, there is by contrast an essential difference between attraction and impact. But that, mechanically understood, there is no conflict between the first and the last conception is seen straightforwardly from this, that a body is acted upon *from without* just as much when it is attracted as when it is struck. The contradiction between "*vis attractiva*" physically apprehended and the law of inertia, to which Schaller appeals, does not in actuality take place. What Newton brings out — that the force of attraction cannot

^{*)}Jul. Schaller. Geschichte der Naturphilosophie. 1ster Theil. Leipzig 1841. S. 353–400.

^{**)}Voces autem attractionis, impulsus, vel propensionis cujuscunque in centrum, indifferenter et pro se mutuo promiscue usurpo; has vires non physice, sed mathematice tantum considerando. Princ.Mathem. Amstælodami MDCCXXIII. pag. 5.

^{*)}Vocem attractionis hic generaliter usurpo pro corporum conatu quocunque accedendi ad invicem; sive conatus iste fiat ab actione corporum vel se mutuo petentium vel per spiritus emissos se invicem agitantium, sive is ab actione ætheris aut aëris mediæ cujuscunque seu corporei seu incorporei oriatur corpora innatantia in se invicem utcunque impellentis ... I. l. p. 172, confer. p. 483. The fundamental thought of the whole standpoint can hardly be uttered more plainly.

have a physical significance for the very reason that the centres from which it issues are only mathematical points — must be understood from the connexion in this way: that the mechanical theory which makes a beginning by concentrating the attracting causes in mathematical points thereby gives notice that it will not investigate the inner unity of essence of matter and force; but from this it can in no way be derived either that the physical is excluded, or that the mathematical in attraction should cease because the centre is laid into a corporeal mass.

(The relation between Kepler and Newton is, at the stage of mechanics, a relation between objectification in empirical reflexion and objectification in pervasive conceiving. The Keplerian reflexion abstracts the laws from the cases before it and secures them by empirical induction; the Newtonian method, by contrast, aims at knowing the causes from their effects, determining the particular on the foundation of the universal, and conceiving the laws in their fulfilment. That is, on the way of objectification, a great advance.

Kepler speaks only of the areas described in the planetary orbits; but Newton proves that the law of the areas must hold for any motion whatever of a material point about a fixed centre, since it is in general determining for the objective concept of central motion.

Kepler holds to the factual — that the planetary orbits are ellipses — but, bound to the fact, he is not able to state the cause; from Newton's principles, by contrast, the cause can be seen in the effect, insofar as it can be seen that not merely the ellipse but the conic section generally is a naturally necessary consequence, an express consequence of cause, of attraction's peculiar law: $T = \frac{\mu}{r^2}$, and that conversely this law is a necessary consequence of motion in a conic section.^{*)} Kepler confines himself to setting up the three laws alongside one another without any deduction whatever; but on Newton's principles the one law can, if one starts from the universal and introduces conditions, be

^{*)}J. Schaller remarks that Newton has not been able to "prove generally" that the curve must be a conic section when the attraction stands in inverse ratio to the square of the distance. It must in this connexion be expressly brought out that what essentially matters is not the provisional manner of proof, but what admits of being derived from the fundamental thought itself. Everyone who makes himself acquainted with the fundamental propositions of rational mechanics will have ample occasion to convince himself of the universal validity of the Newtonian principles. For central motion in general — r being the radius vector, c an arbitrary constant, θ the angle r forms with a fixed axis, and ϕ the accelerating force — one finds (cf. Duhamel, *Cours de Mécanique*, Paris 1846, II, p. 3) the formula:

$$\phi = \frac{c^2}{r^2} \left(\frac{1}{r} + \frac{d^2 \frac{1}{r}}{d\theta^2} \right)$$

If now the orbit is a conic section, the angle between r and the polar axis θ , that between the major axis and the polar axis ω , and the equation of the orbit

$$\frac{1}{r} = \frac{1 + e \cos(\theta - \omega)}{\frac{p}{2}},$$

deduced from the other.^{*)} That the deduction must constantly be tied to given conditions recalls that the proofs, as regards the physical, are neither merely a priori nor merely empirical, but that they must, by the nature of the case, rest upon an exactly articulated reciprocal determination of experience and thinking.)

It has been asserted from the standpoint of “the concept” that “the more universally attraction is apprehended, the more the mathematical proof loses in significance and

then it follows from this that:

$$\frac{d^2 \frac{1}{r}}{d\theta^2} = \div \frac{e \cos(\theta - \omega)}{\frac{p}{2}},$$

which, in connexion with the equation of the orbit, gives:

$$\frac{1}{r} + \frac{d^2 \frac{1}{r}}{d\theta^2} = \frac{1}{\frac{p}{2}}$$

and consequently:

$$\varphi = \frac{c^2}{\frac{p}{2}} \cdot \frac{1}{r^2} = \frac{k}{r^2}$$

From this it is seen that the attraction stands in inverse ratio to the square of the distance when the orbit is a conic section and the central force issues from the focus.

^{*)}If the force of attraction be put $\varphi = \frac{\mu}{r^2}$, then the equations in the plane (xy) become $\frac{d^2 x}{dt^2} = \div \frac{\mu x}{r^3}$, $\frac{d^2 y}{dt^2} = \div \frac{\mu y}{r^3}$ (I). If one introduces polar coordinates, so that $x = r \cos \theta$, $y = r \sin \theta$, and the area described by the radius vector $\frac{1}{2} c dt = \frac{1}{2} r^2 d\theta$, then by multiplying the equations (I) by $c dt = r^2 d\theta$ one gets the equations $c d \frac{dx}{dt} = \div \mu \cos \theta d\theta$ (α) and $c d \frac{dy}{dt} = \div \mu \sin \theta d\theta$ (β). If these equations be integrated so that the constant for (α) becomes $\div \gamma \sin \omega$, and for (β) $\gamma \cos \omega$; if the first be then multiplied by $y = r \sin \theta$ and the second by $x = r \cos \theta$ (remembering that $x dy - y dx = c dt$ and $\sin^2 \theta + \cos^2 \theta = 1$), one obtains:

$$r = \frac{\frac{c}{\mu}}{1 + \frac{\gamma}{\mu} \cos(\theta \div \omega)}$$

that is, *the orbit is a conic section with the centre of attraction in the focus.*

If the equation $r^2 d\theta = c dt$ be integrated so that $\theta = 0$ when $t = 0$, then $t = \frac{\int_0^\theta r^2 d\theta}{c}$, which put in words means that *the areas described by the radius vector are proportional to the times in which they are described.*

If, finally, the special case be chosen in which the conic section is an ellipse, and the integral $ct = \int r^2 d\theta$ be taken from 0 to 2π , then this becomes twice the ellipse's area, $2ab\pi = 2a^2 \sqrt{1 \div e^2} \pi$, and $T = \frac{2a^2 \sqrt{1 \div e^2} \pi}{c}$ the time in which the whole orbit is traversed. If a , e and T are known, c can be determined, since $c = \sqrt{a(1 \div e^2)\mu}$, whereby one obtains $t = \frac{\int r^2 d\theta}{\sqrt{a(1 \div e^2)\mu}}$, and consequently $T = \frac{2a^2 \sqrt{1 \div e^2} \pi}{\sqrt{a(1 \div e^2)\mu}}$; so

that $\frac{T^2}{a^3} = \frac{4\pi^2}{\mu} = K$. That the ratio between the square of the period of revolution and the third power of the major axis is a constant magnitude shows that this ratio holds for different material points moving in ellipses about a common focus; in other words: *the squares of the periods of revolution stand to one another as the cubes of the semi-major axes, or mean distances.*

On these conditions the Keplerian laws can be deduced from Newton's principles.

shows itself insufficient to strike the essence of the phenomenon.”^{*)} The opposite of this is precisely what is true. The history of science teaches that insight into the essence of the phenomena has advanced steadily with the knowledge of the laws, and the knowledge of the laws with the mathematically mechanical determination of the objective relations, from which alone the causes and their conditions admit of being found out.

The doctrine of gravitation, the doctrine of the planets’ mutual attraction, is grounded evenly upon concept and fact, and is in equal degree a priori and empirical. It can be shown from experience and fixed by induction how the masses act upon one another; but the mutual action that admits of being observed is in turn grounded by an activity that does not admit of being immediately observed, but must be determined by thinking – namely the reciprocal action whereby the smallest parts in the one mass attract the parts in the other. The attraction of the parts is explained from that of the masses, and that of the masses again from that of the parts. The mutual attraction between two masses is proportional to their product, while the force with which the one mass is accelerated in its motion about the other is proportional to the sum of the masses.^{*)} When the masses, the actual planets, are introduced in place of material points, the Keplerian laws do indeed undergo manifold modifications; but these modifications are so far from weakening their validity that they are on the contrary proofs of their actual fulfilment.

If the planets were perfectly spherical, formed of homogeneous concentric layers, they would act upon one another as though each of them had its whole mass compressed into its centre; but since they deviate somewhat in form from the sphere and, what is more essential, stand in relations of attraction not merely to the sun but to one another, the reciprocal action becomes infinitely entangled, and the perturbations, the periodic as well as the secular, make in this respect great demands^{**)} upon the observing and calculating science. The perturbations are not, however, what the word one-sidedly suggests, disturbances of something actual; on the contrary! they belong essentially to the character of mechanical totality; they are phenomena of many-sided reciprocal action; it is precisely the perturbations that give the actual the stamp of outward material objectivity. For a scientific objectification of the world-structure it is not sufficient to consider the single planets in the single orbits, each for itself, or to stop at a general representation of the various supervening actions to which a planet in its orbit may

^{*)}Schaller, Anfr. Skr. p. 367.

^{*)}If M be the sun’s and m the planet’s mass, and the distance r , then the mutual attraction is to be expressed by $\frac{f \cdot Mm}{r^2}$; the accelerating force for the planet by $\frac{f \cdot M}{r^2}$, for the sun by $\frac{f \cdot m}{r^2}$; for the planet’s *relative* motion about the sun the driving force must by contrast become $\frac{f(M+m)}{r^2}$. By f is denoted the mutual attraction of two units of mass, reduced to simple points and placed at a distance that gives a unit of length.

^{**)}The formula for the moon’s longitude presented by Delaunay in *connaissance des temps* for 1865 takes up no fewer than 45 printed pages, and yet this formula takes account only of the perturbations due to the sun.

be subject. The concept of actuality must be developed so exactly and completely that the reciprocal determination of the universal and the single, the logical and the antilogical the laws and their modifications in the particular cases, stands clear before thought.^{*)} Only from the standpoint of objectification does it become fully evident what it means that realistic sciences such as mechanics and physical astronomy stand in Spirit's service. It is something far higher than a petty craving to get the manifold phenomena of motion punctually calculated that drives the investigations forward; it is the investigating Spirit that will have actuality theoretically in its power; it is the thinking subjectivity that desires the objective and works restlessly with the conditions of objectification. A magnificent example, showing what science can attain by building on Newton's foundation, the present age has admired in Laplace's *Mécanique céleste*.

c) *The universe's mechanical objectivity: a system of systems.* When Hephaestus was of old cast out of Olympus's window he had to fall for 9 days and 9 nights before he reached Lemnos. A fall from such a height may well awaken a shudder; but when one

^{*)}If one denotes the sun's mass by M , its coordinates by X, Y and Z , the planets' masses by $m, m', m'' \dots$, their coordinates by $x, y, z, x', y', z', x'', y'', z'' \dots$, and if one further lets the sun's distances from the planets be $r, r', r'' \dots$, and the perturbed planet's distances from the perturbing ones $s', s'' \dots$, then the equations for the motion of the single planet, m , about the sun as a fixed centre will be, with respect to the X -axis:

$$\frac{d^2(x - X)}{dt^2} + \frac{(M + m)(x - X)}{r^3} = 0;$$

but if the planet m is perturbed by $m', m'' \dots$, the equation for the X -axis will be:

$$\begin{aligned} \frac{d^2(x - X)}{dt^2} + \frac{(M + m)(x - X)}{r^3} &= \frac{m'(x' - x)}{s'^3} + \frac{m''(x'' - x)}{s''^3} + \\ &\dots - \frac{m'(x' - X)}{r'^3} - \frac{m''(x'' - X)}{r''^3} - \dots \quad (I) \end{aligned}$$

and the corresponding ones for the Y - and Z -axis have the same form. That we have in such equations a system of objective relations is certain; but by what is the objectivity secured? By the objectification! And by what is the objectification secured? By an articulating unity of observation and thinking. It is thinking that sees that when the law for the motion of the single planet, m , is $\frac{d^2x}{dt^2} = -\frac{M(x - X)}{r^3}$ etc., and for the sun's motion $\frac{d^2X}{dt^2} = \frac{m(x - X)}{r^3}$ etc., then the law for the perturbed planet, m , must become:

$$\frac{d^2x}{dt^2} = -\frac{M(x - X)}{r^3} + \frac{m'(x' - x)}{s'^3} + \frac{m''(x'' - x)}{s''^3} + \dots$$

and for the sun:

$$\frac{d^2X}{dt^2} = \frac{m(x - X)}{r^3} + \frac{m'(x' - X)}{r'^3} + \frac{m''(x'' - X)}{r''^3} + \dots \quad (II)$$

On the left side the concept of (perturbing) force is determined in exact logical universality; on the right side it is shown how the concept expresses itself in existing cases. From the equations (II) for the absolute motions the equations (I) for the relative ones are then derived. The objectifying reflexion articulates the logical, the universal, by making the law clear in its different forms and formulas; but the language of signs also shows that the general formulas are only empty symbols so long as the moments of actuality, with number and measure in corresponding antilogical determinations, are still wanting. The facts are objective, but the analyses subjective. What do we learn from this? That no science, not even the most objective, can take the subjective away and keep the objective — that, in other words, nothing in the world is objective except what has been objectified.

considers that it is the way from the highest heaven down to the earth that had to be traversed on this journey, one will not be able to say that the length is exaggerated. In the Greeks' view of the world limitation is so essential a moment that Aristotle can even in a manner be said to make the Godhead itself the limit of the heaven (τὸ πέρας τοῦ οὐρανοῦ). Observing astronomy had indeed to come upon the question of the distance-relations of the heavenly bodies from many sides, but the observations were imperfect, and the Ptolemaic epicycle theory had to content itself with an imagined standard. Tycho Brahe himself still assumed the sun's parallax to be so great that the earth became a considerable magnitude in the universe. Only after the discovery of the telescope and the introduction of the new methods of observation did science get so far as to obtain measure and standard. Not only did the radius of the earth's orbit become a known magnitude; even the fixed stars' infinitely slight, generally vanishing parallax led to insight into immeasurable measures of distance. When Ole Rømer, by means of the eclipses of Jupiter's moons, had discovered the velocity of light, thought went with the ray through celestial space; the immeasurable was measured, the incomprehensible understood. Fantasy gave up, but insight triumphed. To traverse the distance from the sun to the earth, 20 million miles, light needs 8 minutes and 13 seconds; to traverse the distance from the 61st star in the Swan, whose parallax Bessel has determined, light needs $9\frac{1}{4}$ years; such a length of way is incomprehensible for intuition, and yet it is exactly apprehended by the understanding — it is, after all, measured. But when one must further conclude "from the character of the telescope" that the light from the most distant stars in the Milky Way has been 2 or 3000 years on the way — indeed, that there are stars we see by pre-Adamite light — then the immeasurable makes even thought giddy.

For the classical intuition of fantasy the world was limited; for the modern — we may say: romantic — reflexion the universe is unlimited. Each of these ways of seeing is one-sided. If the actual universe really extends as far as space, then it must be unlimited like space. But the unlimited is something unactual; everything actual is limited: insofar as the universe is actual, it is limited. The apparent contradiction that the universe is both limited and unlimited is, however, as already shown in the doctrine of space, no insoluble antinomy. Astronomy itself has indicated the solution of the riddle. "If one compares," says Humboldt (Kosmos I, p. 68), "world-space with one of our planet's island-rich seas, then one may think of matter as distributed in groups: now as irresolvable nebulae of various ages, condensed about one or several nuclei, now as clotted together in star-clusters or isolated sporades. Our star-cluster, the *world-island* to which we belong, forms a lens-shaped, flattened layer separated on all sides, whose major axis is estimated at seven or eight hundred, its minor by contrast at a hundred and fifty Sirius-distances." We are thereby led to apprehend the universe not merely as an unsurveyable manifold of worlds and world-islands, but as *a system of systems*.

The *physical* double stars, which by mutual attraction describe orbits about one

another according to the Keplerian laws, contain a confirmation that Newton's principles hold good outside our solar system too. Now if attraction is a property indwelling in all matter, then it follows that not merely the planets in each single system must attract one another, but that the systems must also act upon one another mutually and show themselves as members which, by the infinite roundabout way of mediation, can be regarded as realizations of a single, all-embracing fundamental system. This way of seeing is further confirmed by there being nothing absolutely at rest in world-space; everything is in motion, not merely planets and comets but the sun itself and the suns: there are, properly speaking, no fixed stars. The question presses forward: is there here a common centre for all these motions? Corporeal this centre surely cannot be. As is well known, the moon does not move about the earth's centre, but earth and moon about a common centre; and the same holds of the sun and the planets. When, then, our whole solar system has a translatory motion, this does not exactly indicate that there is a single fixed body about which the motion happens; it rather presupposes that the system is acted upon by another, or rather by other systems, so that the common centre of gravity is determined by the systems' mutual attraction. In the universality of the concept the matter is accordingly to be apprehended thus: the universe is limited, for every actual system has its limit; the universe is unlimited, for outside the one actual system there is another actual one, and outside this again another: every system is a whole for itself, and yet a part, a member of a more comprehensive, still more composite system: the actual limit is everywhere and is everywhere overstepped. The finite does not end here or there, any more than the infinite begins here or there; the infinite is not the ceasing of the finite, does not mean that the finite ceases, but on the contrary that it repeats itself incessantly; *the finite is everywhere, because the infinite is everywhere*: this is, in short, *the universe's infinity*.

How must the objectification of a universe infinite in finitude now necessarily go on? As the object is, the objectification must be, and conversely. That the infinite is in actuality finite, and the finite infinite, comes to light most clearly of all from the newer science's art of observation and analytical methods. The articulating reciprocal determination of finite and infinite, which is in equal degree characteristic of observation and of calculation, points to the doctrine of *infinite approximation*.

Although with the Keplerian laws corporeal centres were introduced, the centres of gravity in the doctrine of gravitation do not always have material existence. They are determined by the co-working of given forces, and are thereby distinguished from the empty designations of place in the epicycle theory. But besides such ideal realities the newer astronomy has a quantity of feigned magnitudes, especially mean magnitudes — mean sun, mean anomaly, mean motion, mean time and so forth. To see what an infinite approximation signifies in this respect it will be sufficient to fasten attention upon a single problem, for example the so-called Keplerian one: the equation of the centre.

It is something peculiar about the mathematically empirical sciences — and astron-

omy is a mathematically empirical science — that they demand the exact and yet must incessantly struggle with the approximate. The stricter the method, the more plainly it becomes evident that there is a disagreement between the actual and its apprehension, between the object and its objectification, which never quite admits of being lifted. The old astronomy stopped in all naivety at the approximate. On the hypothesis that the sun with even velocity describes a circle whose centre lies outside the earth, it was then a rather simple problem to find the equation of the centre.^{*)} If we start, by contrast, from the Keplerian laws and carry the analysis through in a peculiar direction, then we get a sine series with multiplied arcs which is indeed strongly convergent, but nevertheless

^{*)}Let the diameter passing through the earth be the line of apsides, its end points P and A, perigee and apogee; let the eccentricity, the distance from the earth J to the circle's centre C, be denoted by ε ; the angle which the lines from J and C to the sun S form with one another, by E; the angle PCS, the mean anomaly, by nt : one then has, for determining the equation of the centre,

$$\frac{\sin E}{\sin(nt + E)} = \frac{JC}{CS} = \varepsilon$$

or, in all approximation, $E = \varepsilon \sin nt$.

infinite:**))

$$E = 2e \sin nt + \frac{5}{4} e^2 \sin 2nt + \frac{1}{4} e^3 \left(\frac{13}{3} \sin 3nt - \sin nt \right) + \dots$$

Here we have, in the form of calculation, an example of infinite approximation.

If we fasten attention upon the task of observation, then the same shows itself from another side. No single observation is exact; one can repeat an observation as often as one will, one commits an error every time, though not the same one. It is not clumsiness on the observer's side, not an accidental teasing by fate's caprice; it lies in the nature of the conditions, in the objective necessity of accidentality — it is a law. How characteristic that it is precisely the exact science that has become aware that there is a law of the errors, a *law of error*. The calculus of probability, with the law of error and the method

**)The ellipse's major axis is the line of apsides, and the eccentricity e . The earth, not the sun, is assumed to be in the focus. E is the angular distance between the mean sun, a feigned sun, and the actual sun; m the mean anomaly. If we compare the expression from the previous solution, $E = e \sin nt$, with the first term of the infinite series opposite, it is seen that the hypothesis of the eccentric circle, in that $E = 2e$, gives the equation of the centre a value agreeing with the assumption of the elliptical orbit, so long, be it noted, as one is content with the approximate. But if the approximate is to be lifted, and the demand for the exact satisfied, then the limit of finitude must be overstepped within finitude, to infinity. That is, as regards the analyses, an enormous step. What transformations the mathematical method has had to undergo before it became possible to get from the first term in the series opposite to all the following ones — of that one gets a collected impression by casting a glance at the elegant solution of Kepler's problem to be found in Duhamel (*Cours de Mécanique* II, pp. 63–69). As a general foundation, reference is made to Lagrange's formula $z = x + \alpha f(z)$, from which one gets:

$$z = x + \alpha f(x) + \frac{\alpha^2}{1.2} \frac{d.(f(x))^2}{dx} + \dots;$$

from this, with respect to the formulas for the problem:

$$u = nt + e \sin u;$$

$$r = a(1 - e \cos u), \quad \operatorname{tg} \frac{1}{2} \theta = \sqrt{\frac{1+e}{1-e}} \operatorname{tg} \frac{1}{2} u,$$

one then obtains the equation:

$$u = x + e \sin x + \frac{e^2}{1.2} \frac{d. \sin^2 x}{dx} + \dots \frac{e^m}{1.2 \dots m} \frac{\sin^m x}{dx^{m-1}} + \dots$$

To carry out the differentiation, however, one must know in advance what is given in equations such as:

$$(2 \sin z)^m \cos \frac{m\pi}{2} = \cos mz - \frac{m}{1} \cos(m-2)z + \frac{m(m-1)}{1.2} \cos(m-4)z \dots$$

$$(2 \sin z)^m \sin \frac{m\pi}{2} = \sin mz - \frac{m}{1} \sin(m-2)z + \frac{m(m-1)}{1.2} \sin(m-4)z \dots$$

(Ramus, *Algeb. og Functionsl.* p. 202.) and so forth, before one (Duhamel, p. 69) reaches the definitive form of the series in question. For the sake of the objective relations the analysing thinking must sink itself in operating, combining reflexions. The thoughts in the objective must be thought *subjectively* in order to be thoughts; an unconscious objectification of thoughts is just as meaningless a turn of speech as a shining darkness, a seeing blindness, an unconscious knowing.

of least squares, is a telling testimony to the persistent struggle which mathematically empirical knowing must wage in all directions against the approximate by means of infinite approximation.

Astronomy's objectification of the world-structure culminates in approximation, and a culmination of that kind is infinitely dialectical. From the one side it looks as though the actual, the objective in itself, withdrew itself from the observer's measure to infinity; for if exact, definitive determinateness could really be attained, then the approximation would cease of itself. From the other side actuality with its determinate limits comes constantly nearer, for if there were a finite fixed limit to the approximation, then it would not be infinite. The exact cannot be attained by a restricted number of single calculations and carefully verified observations; for the one series of observations points back to the other, and of course the one calculation to the other as well; but in this lies also that observations and calculations can be corrected and secured by one another mutually. The more exact the various series of observations become, the more means the analysing investigator obtains for correcting them further and approaching the exact; the residue of indeterminateness that still remains does not disturb him by any foggy approximateness, for now he knows exactly between what limits it falls, and under what conditions it can be regarded as vanishing^{*)}.

Observing science must have instruments and set store by their perfecting; telescopes, refractors, universal instruments and so forth play an important part in the exact investigations. It is necessary for the observer

to know his instrument and understand how to use it. For the sake of the objectification the philosophy of nature has yet to draw attention to one indispensable instrument — in truth a universal instrument — whose share in determining the objectification is otherwise so easily overlooked, and that is consciousness itself. Outside consciousness there is indeed a presupposed actuality, but no knowing of its objectivity. Not merely in the determination of the objective, but in the essence of objectivity — be it the objectivity of the earth or of the sun, of the Milky Way or of the comets — the objectifying consciousness is an essential factor. However gigantic the steps exactly empirical science has taken with the objectification of the actual world-structure, however much it has in theory widened its ideal power by continued conquests from the immeasurable realm of the starry heaven: the building itself it has not conquered — the actuality of nature, the objective in itself, is in small as in great a barrier to consciousness, a barrier which at every step reminds it that, infinitely far from being the first objectifying agent, which objectifies in power, it must be content with its lot: to objectify in powerlessness.

C.

^{*)}Cf. *Mathematik og Dialektik*, pp. 34–56.

The Origin of the Solar System.

As an infinite system of worlds the universe is itself without a finite beginning. The becoming, arising and passing away of the single worlds takes place in the fundamental system; but this neither is nor becomes. To investigate what science at its present stage avails to elucidate about the formation of our solar system, we bring out: from the standpoint of the facts the physical-astronomical hypothesis of *the primal nebula*, and from the standpoint of “the concept” the system’s self-formation in virtue of *the immanent Idea*. By placing the two sharply opposed views under a common illumination of objectifying omnipotence, we can then convince ourselves that both are insufficient, while they nevertheless, each from its own side, furnish essential contributions to the clearing up of the matter, when the question is of the system’s origin and the fundamental relation between *nature and Spirit* is fully known.

a) *The primal nebula: the physical-astronomical hypothesis.* The ground why no particular weight can be laid upon the supernatural presuppositions of creation and creating omnipotence, with which the older physicists and philosophers, from Gassendi and Cartesius to Newton and Leibnitz, willingly equip the foundation of their view of nature, has been treated in the preceding chapter. The only yield which the doctrine of gravitation might be supposed to afford the supernaturalistic view would in any case consist in this, that a more deeply penetrating insight into the mechanics of the world-structure gave the beholder renewed occasion to trace and admire the divine wisdom. That the Creator must be as great a mathematician as physicist followed from this standpoint plainly enough from the whole plan. What good would it have been that the Almighty had “implanted” inertia and attraction in matter, if he had forgotten to distribute the masses, to place sun, planets and secondary planets each in its determinate position, and to calculate the first impulse so that the orbits could become ellipses? Nor is this all. There is, namely, not only a mutual attraction of the sun and each single planet, but so manifoldly crossing a reciprocal action among the planets themselves that the perturbations thereby caused manifestly threaten the system with destruction. This the Creator has foreseen and from the beginning taken his precautions against. He has indeed not wished to make the planets perfectly spherical, just as on the whole he has not shunned the intricate, so that his mechanical wisdom might be properly revealed to the investigator of nature. But he has seen to it that the orbits got only a slight inclination, that the periods of revolution became incommensurable: $\frac{T}{t} = \frac{A^{\frac{3}{2}}}{a^{\frac{3}{2}}}$, and, what is more, that the sun got so great a mass that it more than outweighs all the rest^{*)}. The last especially is of extraordinary significance; for the one planet cannot now act disturbingly upon the other without the sun in majestic superiority, so to speak, placing itself between them and keeping order. The more one thus sinks oneself in the mathematical analyses,

^{*)}The sun may, even after Neptune’s discovery, be assumed to be about 720 times greater than all the masses together that move about it.

in the manifold of infinite series through which a Laplace has developed the laws of perturbation, the more the admiration of the world-creator's divine mechanics grows.

Amid such edifying considerations it must then be highly surprising to hear Laplace himself exclaim: *Philosophe, montre moi la main qui a jeté les Planètes sur le tangent de leur orbite!* What does this challenge mean? Is the mathematician then, for all his wisdom, unable to discover the Creator's hand and know his wisdom?

There is a fundamental proposition which says that *everything in nature must go on naturally*. By departing from this fundamental proposition natural science would make itself guilty of a contradiction as dangerous to life as the peasant's in the tree when he sawed through the branch on which he himself was sitting. But that everything in nature goes on naturally means, in other words, that all causes working in nature must without exception be natural. There must then just as necessarily be a natural cause for the distribution of the planets in the solar system as for their motions this very day; just as much a natural cause for the preponderance of the sun's mass over the rest as for the difference between sidereal and solar time. It is not an arbitrary, presumptuous craving out beyond the limit of what can be known, but a scientific need for a coherent account of given facts, that lies at the ground of the world-hypothesis by which Kant first and in detail^{*)} and afterwards Laplace in all brevity^{**)} have sought to explain to themselves the natural coming-to-be of the planetary system.

Physical astronomy presupposes that matter with its forces is "eternal" like space, and then sets about conceiving, out of the first existent, the becoming and stepwise formation which the present must in the course of the ages have passed through. The first existent is to be apprehended as a state in which everything existing can be assumed to have found itself from the beginning. By a magnificent reduction we are then carried back to a matter evenly spread in space and as yet indifferent, an extraordinarily thin but enormous mass of vapour whose diameter is estimated by some at 12,000 billion miles.

The formation now begins with the parts attracting one another, whereby condensation, rotation and heat come forth. The mass of vapour becomes spherical, raised at the equator; the rotation increases, and the centrifugal force grows: all of it natural! In the raised belt of the spherical mass there will then gradually form rings, which loosen, are separated off and roll themselves together again; the rolled-together masses move about their axes and about the main mass: here we have a first sketch of the system, in all naturalness. What happened with the great mass can repeat itself with the smaller ones; some among these can again deposit one or more rings, and these rings be separated off and become moons. Saturn's ring, which in actuality consists of three rings plainly separated from one another, is an element arrested in its development, from which one may learn how it has gone with the formations on the whole. Since the sun's mass is

^{*)}Allgemeine Naturgeschichte und Theorie des Himmels. Kant's Werke. Achter Band. Leipzig 1838. S. 224 flgd.

^{**)Exposition du Système du Monde. A Paris. L'an IV de la republique francaise. Tome second. S. 293 flgd.}

precisely the remaining main mass, we easily see the natural ground of its preponderance; only concerning the closer character of the planetary orbits does a doubt remain which the theory, if it is to be accepted, must be able to lift. It must indeed in general be conceded that we no longer need a supernatural cause to explain the first impulse along the tangent, since the force of rotation is already in the rotating mass; further, it follows from the law of attraction that all the orbits must become conic sections. But of the three conic sections only one, the ellipse, is suited to give a closed system of motions; the other two would by contrast make the system meaningless, since the planets, moving either in a parabolic or in a hyperbolic branch, would gradually remove themselves from the sun without ever returning. If one will not set it down to chance that there is reason in the solar system, then it undeniably seems as though precisely the particular initial velocity on which the elliptical orbits are conditioned must presuppose a particular arrangement of wisdom, and so after all point to an invisible hand, an arbitrary intervention by a supernatural cause.

The defenders of the hypothesis find it by no means difficult to meet such misgivings. To prevent the orbits from becoming parabolas, no divine intervention was in truth needed. Among infinitely many possible cases there is for the parabola only a single favourable one^{*)}: the parabola is, on the presuppositions contained in the hypothesis itself, the very least probable form; but that the very least probable does not happen is quite natural. As regards the hyperbola, this curve has, when one considers the matter in all generality, certainly a probability in its favour, just as much as the ellipse; but when the closer circumstances are brought into account, then it is in the first place a question whether the force of rotation in the rotating mass has ever been great enough to give an initial velocity answering to the hyperbola. But even if the possibility of this be conceded, what then? Then the consequence will in any case only be that some planets must in primeval times have left the system, and that we have only those left which got a smaller initial velocity.

If one goes into detail with the hypothesis's application, then there are indeed not a few facts — all motions from west to east; the axial rotations answering to the sun's rotation and to the directions in the orbits; small inclinations; slight eccentricity; the outer planets' more rapid rotation and slower orbital velocity^{*)} and so forth — which in

^{*)}If the radius vector be denoted by r , the semi-major axis by a , the intensity by μ and the velocity by v , then the formula common to all three conic sections is:

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right).$$

Since now a is positive, negative or ∞ according as the conic section is an ellipse, parabola or hyperbola, it is seen from this that it must depend upon the velocity (the initial velocity) which of the three conic sections is to come forth. For the ellipse $v^2 < \mu \frac{2}{r}$, for the hyperbola $v^2 > \mu \frac{2}{r}$, for the parabola $v^2 = \mu \frac{2}{r}$. Of the infinitely many possible cases only a single one will accordingly be favourable to the parabola, while the number of favourable cases is infinitely great both for hyperbola and for ellipse.

^{*)}Spectral analysis too seems to speak in the hypothesis's favour. We will not, however, omit to adduce

broad outline speak for its validity; but there are nevertheless also several phenomena which will not readily accommodate themselves to it. Everything does not fall out so neatly as Kant assumes.

“The telescopic planets Vesta, Juno, Ceres, Pallas (and the new Astraea), with their mutually intertwined, strongly inclined and more eccentric orbits, have,” says Humboldt ^{**}), “been considered as a zone dividing our planetary system into two spatial sections, indeed as a kind of middle group within it. On this view the *inner* planetary group (Mercury, Venus, the earth and Mars) presents, in comparison with the *outer* (Jupiter, Saturn and Uranus), several striking contrasts. The inner planets, nearer the sun, are of middling size, denser, fairly alike and slowly rotating (in about a 24-hour period), less flattened, and, with one exception, namely the earth, altogether moonless. The *outer* planets, further from the sun, are considerably larger, five times less dense, have twice as rapid an axial rotation and are therefore strongly flattened as well as moon-rich, in the ratio of 17 to 1, if it is confirmed that Uranus really has 6 satellites.^{*)} But” — he adds — “this general consideration of certain properties characteristic of whole groups does not admit of being applied with equal right to the single planets in each of the groups; nor to any relation between the single circling heavenly bodies’ *distance from the central body*, or to their absolute size, their density, period of rotation, eccentricity, inclination of orbits and axis ^{**}).

The hypothesis may on the whole be acceptable, even if it does not succeed in accounting completely for all the details and removing all misgivings. What is peculiar in the co-working of conditions and forces under which those first formations went on loses itself in the darkness of primeval time, inaccessible to special investigations. The hypothesis has probability in its favour so long as no facts can be pointed out that stand in manifest conflict with its assumption. But suppose even that irrefutable objections

the following, though, as it appears to us, somewhat weak objections by a distinguished French geognost: Mais l’hypothèse émise par Laplace, il ne la présentait lui-même qu’avec „defiance“ . . . ses conjectures n’expliquent point la forme elliptique des orbites planétaires, l’inclinaison de leur axe; elles paraissent en outre démenties par le mouvement rétrograde des satellites d’Uranus . . . *Enfin la découverte de l’analyse spectrale, qui sera l’éternel titre de gloire de M. M. Kirchhoff et Bunsen, autorise à croire que la composition chimique du soleil diffère très-notablement de celle des planètes de son système, car cet astre ne contient, du moins dans ses couches extérieures, ni silice, ni étain, ni plomb, ni mercure, ni argent, ni or (!).* La Terre par Élisée Reclus. I. Paris 1870. pag. 19.

^{*)}Kosmos. I. S. 71 flgd.

^{*)}In the most recent time only 4 moons have been expressly observed.

^{**)}As an example that we know no mechanical law of nature which, “like the beautiful law binding the squares of the periods of revolution to the third powers of the major axes of the orbits, makes the planetary bodies’ six elements just named, or the form of their orbits, dependent either upon one another or upon the mean distances,” it is adduced that Mars, further from the sun, is smaller than both the earth and Venus; that Saturn is smaller than Jupiter and yet much larger than Uranus; that Uranus itself seems to be denser than Saturn; that we, as regards the inner group, find on both sides of the earth both Venus and Mars less dense than the earth itself. The time of rotation (about the axis) does indeed on the whole decrease with the distance from the sun; but it is nevertheless greater with Mars than with the earth, greater with Saturn than with Jupiter. Saturn stands midway between Jupiter, whose axis of rotation is almost perpendicular, and Uranus, whose axis by contrast almost coincides with the orbit; and so forth.

led at length to a straightforward rejection of the hypothesis; would it then follow that natural science were compelled to resort to supernatural presuppositions and make a loan from theology? By no means! If the hypothesis cannot be defended, then one of two things happens: either the question of the solar system's formation is held in abeyance as unanswerable, or a new and more satisfactory hypothesis forms itself. In both cases science will with equally unshakeable certainty hold fast its fundamental proposition, the irrefusable fundamental proposition that everything in nature must go on naturally. This is what is decisive; the manner of explanation is not what is decisive. Even if the solar system's stepwise coming-to-be, by transitions from state to state out of a primal state, could be established by a theory as trustworthy as the doctrine of gravitation, a theory of a series of alterations of state would still be no explanation of the origin of the whole; but such a theory would assuredly, by clarifying our view of the inner concatenation of the natural causes, furnish a great real contribution to the determination of *the concept: nature*.

c) *The system's self-formation: the immanent Idea.* While physics with its hypothesis holds exclusively to phenomena and conditioned causes, the speculative doctrine of immanence goes the other way, taking decisive aim at the unconditioned: *the Idea*. The Hegelian deduction of the solar system is in this respect remarkable.

Nature is more than a swarm of facts; it is the Idea itself in the outwardly existent, or in the form of being-other. It lies in the divine Idea to open itself up, in order to be subjectivity and Spirit. From this standpoint the solar system must be apprehended; for its "absolute mechanics" is the form in which the Idea of nature has realized itself. The abstract forms, space and time, the self-conceiving Idea has filled with matter, whose essence is gravity; for gravity consists precisely in this, that matter, seeking its point of rest outside itself, moves from place to place and by place and motion cancels the contradiction of time and space. The motion is threefold: α) the mechanical, imparted from without, is uniform; β) the half-conditioned, half-free motion, namely the motion of fall, shows that a body's separation from its gravity is still accidental, while the motion nevertheless already belongs to gravity itself; γ) the unconditionally free motion, "the great mechanics of the heavens," is seen in gravitation.

"*Gravitation* is the true and determinate concept of material corporeality, realized into Idea. Universal corporeality divides itself essentially into particular bodies and joins itself together, as the phenomenal being of motion, into the moment of singularity or subjectivity, which is herewith immediately a system of several bodies."^{*)} — There can be no well-grounded question whence these bodies come; they are real moments of the Idea's being: they exist, for in them the Idea has existence. The independent bodies of which the solar system consists refer themselves essentially to one another, are heavy, but hold themselves in this relation and set their unity in an other outside themselves. Thus the plurality is no longer indeterminate, as with the stars, but the difference is

^{*)}Vorlesungen über die Naturphilosophie. Berlin 1842. S. 94 flgd.

posited; and the determinateness rests solely on *absolutely universal centrality* and on *particular centrality*. From these two determinations follow the forms of motion in which matter's concept is filled out. The motion falls to the relative central bodies, which are in themselves the universal determinateness of place: and yet their place is at the same time indeterminate, insofar as it has its centre in an other; this indeterminateness must then likewise have being, while the place determinate in and for itself is only one. It is therefore indifferent to the particular central bodies at what place they are; and this comes into view in their seeking their centre, i.e., leaving their place and setting themselves at another place. The third is this: taken immediately, these bodies might be removed equally far from their centre; were they so, then they would not be removed from one another. If they moreover all moved in the same orbit, then they would not be different from one another at all: they would then be one and the same, the one only a repetition of the other, and their difference of course only an empty word. The fourth is this, that as they alter their place at different distances from one another, they return into themselves through a curve; for only thereby do they present their independence against the central body — as also their unity with the centre by moving about it in the same curve. As independent against the central body they keep to their place and no longer fall into the centre. The double-sidedness of the relation is that the particular bodies at once posit themselves a central body and are posited by the central body: the centre is meaningless without periphery, and the periphery without the centre.

On this foundation the Keplerian laws are to be deduced from the pure Idea. As regards first the orbit's form, the circle is only an orbit for plainly uniform motion; but in free motion space and time must hold according to their difference, and hereby the abstract, simply uniform velocity is cancelled. Thinkable, as one says, it may well be that a uniformly increasing and decreasing motion can happen in a circle. But this thinkability or possibility is only an indeterminate, superficial representation. The circle is the line returning into itself in which all the radii are equal; that is, it is completely determined by the radius, or: the radius is the whole determinateness. But in free motion, where determinations of space and time in their *difference* enter into a qualitative relation to one another, this relation must in the spatial itself necessarily come forward as a *difference* within the same, which expressly demands *two* determinations. The demand is that different arcs be traversed in the same time. If the arcs are to be different, then the radii must be so too: from the accelerating motion follows difference in the radii. Thereby the form of the orbit returning into itself becomes essentially an *ellipse*; — the first Keplerian law.

The abstract determinateness which is the circle's distinguishing mark shows itself also in this, that the arc or angle enclosed by two radii is a magnitude *independent* of them, an empirical one. But in the motion determined by the concept, the distance from the centre and the arc traversed in a certain time must be comprehended in *one* determination, must make up *one whole* (the concept's moments are not accidental to

one another); then there comes forth a spatial determination of two dimensions, *the sector*. The arc is in this way essentially a function of the radius vector, and, as unequal in equal times, carries the inequality of the radii with it. That the spatial determination becomes, by means of time, a determination of area hangs together with what holds of fall: that the same determinateness must occur the one time as time in the root, the other time as space in the square^{*)} Only that here the quadratic element in space is, by the line of motion's return into itself, limited to the sector. These are, as one sees, the general principles on which the second Keplerian law — that in equal times equal sectors are cut off — speculatively rests.

This law, however, concerns only the arc's relation to the radius vector; and time is moreover the abstract unity in which the different sectors are compared, because it is what determines as unity. But the further relation is a relation of time, not as unity but as quantum in general, as period of revolution, to the orbit's size or — what is the same — to the distance from the centre. As root and square we saw time and space stand to one another in *fall*, the half-free motion, which is indeed from the one side determined by the concept, but from the other side nevertheless determined outwardly. But in absolute motion, in the sphere of *the free measures*, every determinateness gets its totality. As root, time is a merely empirical magnitude, and in a qualitative respect only abstract unity. As a moment in the developed totality, by contrast, it is at the same time a unity determined thereby, a totality for itself, produces itself and refers itself therein to itself. As what is in itself dimensionless, time in its production comes only to a formal identity with itself, to the *square*; space, by contrast, as the positive outside-one-another, comes to the concept's dimension, to the *cube*. This is the third Keplerian law, the relation of the cubes of the distances to the squares of the times; — “a law which is so great precisely because it is so simple and presents the matter's own reason.”

As regards the bodies themselves, they have, for their determination of their different nature, the principal sides that make up the moments of their concept. One of them is the universal centre for the abstract relation to itself. Against this extreme stand the *immediate*, centreless *singulars*, existing outside themselves, likewise in the form of independent corporeality. The particular bodies stand under the determination of being-outside-self as well as of being-in-self; they are centres for themselves and yet refer themselves to that first as to their essential unity. The central body presents the abstractly rotatory motion. The centre is to be a point; but being a body, it is at the same time extended and consists of what seeks. This dependent matter, which the central body

^{*)}Running time is negative unity, while space spreads itself out as something existent. Space is measured by time in such a way that the units of space give the numerator, the chosen unit of time the denominator (time in the root). Now as the units of space, under the accelerated velocity of the motion of fall, reflect themselves in the unit of time, there comes forth a number which is the unit of time's own product, and so the square of time, a measure for space. This is the most intelligible paraphrase we are able to give of the enigmatic words of which Hegel (Anfr. Skr. p. 88) says *Diess ist der Beweis des Gesetzes des Falls ans dem Begriffe der Sache*.

has in itself, requires that it must rotate about itself. For the dependent points, removed besides from the centre, have no fixed determinate place — they are only falling matter and thus determined only in one direction. The remaining determinateness is wanting; every point must therefore occupy all the places it can occupy. What is determined in and for itself is only the centre; the remaining outside-one-another is indifferent, for hereby only the place's distance is determined, not the place itself. This accidentality of the determination comes to existence in matter's altering its place, and is expressed by the sun's rotating-in-itself about its centre. This axial motion is no change of place^{*)}; for all the points keep the same place with respect to one another. The whole is a reposing motion. — The *dependent* bodies, which at the same time have a free existence and do not make up connected parts of the extension of a body endowed with a centre, also have rotation, but not about themselves, for they have no centre. They rotate accordingly about a centre that belongs to another individual body, from which they have been expelled. Their place is in general this or that; and this accidentality of the determinate place they then express by their rotation. But their motion is a sluggish and rigid motion about the central body, since they constantly remain in the same position with respect to it, as for example the moon in its relation to the earth. The dependent heavenly bodies form that side of the concept which is called particularity; therein lies that with their difference they fall apart from one another, for in nature particularity exists as twoness, not as in Spirit in the form of *one*. First, then, the moment is posited that the reposing motion becomes a restless motion, a sphere of excursion or striving out beyond immediate being toward a beyond. This moment of being-outside-self is itself the substance's moment as a mass, a sphere; for every moment here obtains its own being. This sphere is the *cometary*, and expresses a vortex, a constant standing on the point of dissolving and dispersing into the infinite and empty. The extreme of excursion consists necessarily in approaching the central body's subjectivity once and then yielding to the repulsion. But this being-other has immediately its opposite in itself, an opposite which is: to seek its centre, its rest — the cancelled future, the past as moment; this is the *lunar* sphere, which points not to an excursion from immediate being but to what has come to be, or to being-for-self, to the self. The cometary sphere is therefore referred only to the immediately axis-turning body, the lunar by contrast to the new centre reflected into itself, to the planet. — The *planetary* sphere is finally the one that expresses relation both to itself and to another; it is axis-turning motion and yet has its centre outside itself. The planet has its centre in itself; but it is only a relative one; it does not have its absolute centre in itself, and is so far also dependent. The planet has both determinations in itself and presents them both as change of place. As independent it shows itself in this, that its parts themselves alter place with respect to the position they have to the straight line connecting the absolute and the relative centre; this grounds the planet's rotatory

^{*)}That, according to "the mechanics of the heavens," there can hardly be any axial rotation without a translatory motion, may easily be overlooked when one looks only to the "absolute Idea."

motion. The planet is, as the third, the syllogism in which we have the whole. The solar, planetary, lunar, cometary nature! this fourfoldness of heavenly bodies forms the completed system of rational corporeality. The planetary nature is the totality, the unity of the oppositions, while the others present only its isolated moments; the planetary nature is the most perfect, also with respect to the motion, which here alone comes into consideration.

The fundamental weakness in this deduction is conspicuous. By one-sidedly fastening attention upon the Idea and ideal causality, the thinker comes at every decisive turning point to overlook the actual conditions and mechanical causes. Here one argues from grounds; that is the old Aristotelian manner. But as little as causes in nature admit of being known without grounds, just as little do grounds in nature admit of being known without causes: *vere scire est per causas scire* — that is a fruitful word of Baco's.

None the less everyone who understands how to read will get an impression that a magnificent endeavour with a great thought stirs and breathes in this deduction, an endeavour aiming at cancelling the patchwork, carrying the phenomenal determinations back to the Idea's unity and beholding the universe in the light of reason. That nature is Idea means, in other words, that it is not a piecing together of countless fragments, but one infinite essence connected in itself with one ground; that all single facts, all particular phenomena and laws, are side-determinations of the one fundamental essence, real moments in the one content-rich concept.

c) Nature and Spirit. Physics and the philosophy of immanence are agreed that nature is a whole connected in itself, and the world-system an active revelation of eternal reason; but from neither side is there offered us any clear or penetrating account of what is really to be understood by the word *nature*.

Physics simply refers the question away from itself; it treats of everything that is in nature, but what nature itself is it does not treat of. When it goes back to the primal nebula, it understands this in the sense that the primal nebula was originally in nature, that there was nature in the primal nebula; but that nature is in its essence a primal nebula, or that the primal nebula is nature itself, it by no means asserts. The philosophy of immanence takes the question up and answers it resolutely with the declaration that nature is the absolute, the Idea itself in sensible outward being; but the answer is illusory.

The mysticism of the philosophy of immanence shows itself in its making, without more ado, the absolute into Idea and the Idea into the absolute. The Idea for itself as Idea is by no means the absolute. The pure Idea is the hollowest, most powerless, most ambiguous being, when it is to be held fast in its own absoluteness, separated from every corresponding medium. *No Idea without a medium!*

On the deduction given above it looks almost as though it were the planets themselves, in concrete corporeality, that out of logical interest in a system of three syllogisms agreed to meet — not at the same place but, for difference's sake, each at its own place — and came to an understanding about setting themselves a common centre, the sun; but since

simple planets must at bottom be supposed too stupid to see the concept's logical point, it cannot be the planets simply, but the planet-Idea, the Idea existing as planet in planets, that conceives the whole. Matter is indeed conceiving too, but in unclarity, in obscurity; the clear conceiving, the pure thinking, is in space and time. It is then not existing matter with properties and forces, but a dialectical process between speculative space and reflecting time, that has made the Keplerian laws necessary. In order that two such conceiving parties as time and space might in the concept's name each get its own, the difference between them had absolutely to come forth, and the planetary orbits to become ellipses; time had to become root, so that the quadratic element in space could be limited to a sector. But time would not be content with being root; it strove further forward, it wanted to be squared, wanted by the Idea's way to rise to being square. Only when time attained "the formal identity" of being square, while space advanced to the concept's dimension, to cube, did the solar system come into order: the solar, cometary, lunar, planetary nature merged into the Idea.

As essential content the Idea is common to nature and Spirit; but, it must be expressly noted, the Idea stands in another relation to Spirit than to nature; for Spirit it is a derivative, for it is determined and formed by an omnipotently conceiving will; for nature it is an original, for it determines and forms the natural whole. With this standpoint we now return to the cosmogonic hypothesis, to the primal nebula.

α) Origin and beginning. The primal nebula with which the beginning is made must itself have a beginning; for if it be assumed that the matter spread through space exists from eternity, then we have, as shown above, in this beginningless thing an existent without coming-to-be, a nebular phenomenon without essence, an existent without ground, a dreamt presupposition, a natureless, unnatural first. But with a natureless, unnatural first no formation of nature can begin. According to Kant, the whole infinite world-space is from the beginning to be filled with a matter whose properties and forces lie at the ground of all alterations. "This fundamental matter, an immediate consequence of the divine existence, must," he says, "at once be so rich, so complete, that the development of its combinations in the course of eternity can spread itself over a plan which encloses in itself everything that can be, which admits of no measure — in short, which is infinite."^{*)} The theological-supernaturalistic representation of a single miracle of creation is the scientific makeshift Kant still^{**)} has in reserve.

Now it might well seem as though a miracle lying an eternity back in time, a single general miracle which moreover happened with necessity and did the whole thing at one stroke, might after all be tolerated in science — just for a beginning; and yet the contradiction is easy to discover. Existing matter cannot by an immediate word of power spring out of self-power; matter belongs to nature as nature and must have a natural beginning. Matter's coming-to-be is object-power's first real possibility, and the issuing

^{*)}Allgem. Naturgeschichte. u. Theor. des Himmels. 7tes Hauptstück. S. 319.

^{**)}1755. Kritik der reinen Vernunft appeared only in 1781.

of power is determined solely by the real concept of the atoms. The beginning happens as the concept is realized. Of what character the matter thus coming to be will otherwise be, what properties its atoms and masses will assume from the beginning, depends solely upon the peculiarity of the concepts of power that determine from within; but their peculiarity depends in turn upon the peculiarly determinate plan, the content-rich disposition, the Idea.

The Idea is accordingly indwelling in nature; the development is in its beginning as in its continuation a conditionally necessary realization of determinate possibilities, of dispositions determining from within. Nature does not give itself its possibilities, but realizes the given possibilities; it does not impart dispositions to itself, but develops the imparted dispositions: *all development of nature is, essentially regarded, a development of imparted dispositions*. Hereby we are carried with scientific consistency from the concept *beginning* to the concept *origin*.

The beginning is natural, but the origin supernatural. Nature's imparted dispositions presuppose an imparting of dispositions; this imparting does not stem from anything whatever in nature; it is to be sought solely in what — not outwardly but inwardly — is above nature, and so in the supernatural itself, in Spirit. The Idea of nature's development could not possibly be determining for a self-developing natural whole if such a determining Idea did not originally have an inner determinateness in itself, if it did not in its essence contain a system of strict, inviolable fundamental determinations. Whence, then, has the Idea these determinations? A nature-Idea can as little determine itself as a nature-thought can think, or a nature-concept conceive itself. The all-thinking, all-conceiving power-essence that determines all Ideas is Spirit as Spirit, the omnipotent self-being. If it is then true that object-power does not stand finitely outside self-power; if it is true that all determinations of object in the doubling of power, through real concepts and dispositions, point to sole power's absolutely overreaching self-determination: then it is also true that this world is created, that the act of creation is Spirit's omnipotent act of will; then it is true that the whole of actual nature has a natural beginning, not *although*, but precisely *because* it has a supernatural origin.

β) Fundamental will and formative drive. Although physics is on the whole inclined to presuppose an eternally existing matter, it is nevertheless far from wanting to ascribe the same age to the many worlds in the universe; it assumes on the contrary that the different world-systems begin their formation at highly different points of time. "Just as in our forests," says Humboldt, "we see the same manner of formation simultaneously at all stages of vegetation, and from this consideration, from this coexistence, receive a visible impression of the onward-striding development of life, so we know also the different stages of star-formation in the great sea of worlds. The process of condensation, which Anaximenes and the whole Ionian school taught, seems here to go on as it were before our eyes."

To explain to himself the naturally necessary connexion between the many world-

formations setting in at different times, Kant assumes that the infinite whole must from the beginning have a centre of condensations and motions determined by the Creator's wisdom, a first material central point, and that formation then follows upon formation as the ordering motion spreads itself and gets the better of the confusion in the original chaos. There is in this way of seeing a double crookedness. In order to preserve the natural series of causes, Kant overlooks that to every peculiar world-formation there must answer a peculiar disposition, and so a peculiar act of creation. He further overlooks that an act of creation, scientifically understood, by no means interrupts but on the contrary makes possible a law-determined connexion among the world-causes themselves: the many worlds in celestial space make up a system of systems because all the world-Ideas make up moments of one and the same Idea of the universe, because all creating acts of will proceed from one divine will, an eternal fundamental will.

That the arrangement of the solar system cannot immediately derive from God's hand, but must be a product of mechanically working causes, Kant will prove in the following way. All the planets move in the same direction about the sun; this circumstance can be explained only from the mechanism out of which the motions have arisen. If God with his own hand had set the planets in motion, his choice — since there is here not the least motive why he should bind himself to one single determination — would have displayed more freedom in all manner of deviations and differences. Why do the planetary orbits fall almost in one and the same plane, and yet not quite? Why do they approach being circles and are yet slightly eccentric? In all such things we recognize nature's usual procedure; it strives after something determinate but does not quite reach it; it must employ various intermediate causes, but the intermediate causes make it bend a little aside, so that it never hits the matter to a nicety. If it were true, as a philosopher has said, that God constantly practises geometry, and if this could be clearly seen along all the ways of the laws of nature, then quite another perfection and geometrical exactness would be traced in the immediate works of the almighty word.

The kernel of truth in the Kantian reasoning is that actual formations of nature are due to working causes in nature, and that it is particularly the mechanical causes which in their reciprocal actions take up the outward cases with all their accidentalities. It is precisely by the accidental that Kant will recognize the natural; against this there is nothing to object. But the manner in which he in his "theory of the heavens" sets the fulfilment of the general laws against the plan itself in the legislating will betrays a one-sided reflexion. With a human plan and its execution it does indeed sometimes go so that supervening circumstances, incalculable accidents, unforeseeable hindrances can effect such thoroughgoing alterations in the execution that the realized plan does not at all, or at any rate only in part, answer to the master-builder's end. But a divine plan cannot possibly be exposed to such unforeseeable hindrances. There is in the solar system's disposition, in the primal nebula, an inner manifold of dispositions to dispositions, of possibilities, of conditions for conditions, of causes for causes. The

contraction is cause of condensation and rotation, the rotation again a cause of the parts' heaping together at the equator, and so forth. The cause of the earth's formation and its motion about the sun, the cause of the moon's formation and its motion about the earth, all such causes with their conditions, which develop themselves time after time, are from the beginning enclosed in the ground-cause of the disposition, so that the will which has willed the disposition and has fully known what it itself willed has in each and all willed the whole development. As the many outward causes with the many conditions, inward and outward alike, gradually enter into activity, the supervening circumstances do indeed co-operate, but nothing, nothing at all, is left to mere accidentality: the laws lay claim to the cases they meet, so that all of them without exception must punctually serve the execution of the plan.

The creating will uses the cases, wills the accidental, because it in earnest wills the natural; it wills a naturally necessary formation of the parts because with Spirit's freedom it wills a naturally necessary formation of the whole. The whole's development through millions upon millions of ages is in all its details a natural fulfilment of the creating fundamental will. Between *the fundamental will* within and the manifold of mechanical causes without lies *the formative drive*, spiritual and yet natural, an essential link between nature and Spirit.

γ) *Teleology and mechanism.* The old dispute about the relation between the mechanical causes and the so-called causes of purpose must now in principle admit of being resolved. What is the purpose of the solar system? A natural development of everything enclosed in its original disposition: the development of the disposition is the attainment of the end. The development happens in virtue of mechanical causes; but when the development is the goal and the mechanical causes the means, then there can in the matter itself be no conflict between the mechanical and the teleological. Only when the consideration is entangled in finite representations can the conflict arise. Either one drags "the purposes" down into the finite and offends by arbitrariness, or one holds the real causes fast in the parcelling out and offends by necessity.

To drag the purposes down into the finite is to bring out certain arrangements beneficial and salutary for life, especially for men, and to apprehend them so singly and so separated from the connexion of the whole that it looks as though the Creator had arbitrarily had the attainment of such results in view. It is a teleology so egoistic that it varies and becomes different according to the standpoint on which the beholder is placed. Why does the full moon stand higher in the heaven in winter than in summer? On the Northerners' teleology it must be the Creator's purpose that it shall light up the long winter nights. A pity that there are also long nights without a full moon! Why has the earth's axis got an inclination to its orbit? From the standpoint of an earth-teleology this must of course have happened with the all-wise purpose that the separation into hot, temperate and cold zones, with the alternations and differences thereby arising, should furnish an inexhaustible wealth of conditions for the development of organic life on the

earth. Unfortunately those who live under the line, and those who dwell about the poles, must in spite of the all-wise fatherly purpose feel themselves stepchildren.

To hold the causes fast in the parcelling out is to make conditioned necessity everything. To this answers the altogether perverse, though among several physicists not uncommon, representation that a natural phenomenon cannot possibly express a certain relative end when it demonstrably proceeds as a law-determined effect of mechanical causes. If the teleological consideration at the stage of finitude is one-sidedly subjective, then the merely mechanical one is assuredly just as one-sidedly objective. As an investigating science, physics has meanwhile — however finite its mechanism may be — the unmistakable advantage that its investigations are fruitful and its groundings scientific. Finite teleology, by contrast, whether it appeals to inner or to outer purposiveness, is and remains unscientific; of it holds Baco's word: *Causarum finalium inqvisitio sterilis est, et tanqvam virgo Deo consecrata nihil parit.*

From the standpoint of infinity, of principle, that is, of Spirit, the matter stands otherwise. The development of nature is first completed when it in each and all satisfies Spirit. It does indeed satisfy Spirit at every stage, but surely only insofar as every stage marks a step toward the completion. The question whether the solar system has its goal solely in its mechanical form, or is destined to furnish the foundation for the attainment of other and higher ends of nature, has scientific significance and coincides with the question of principle: what forms the development of nature requires if the whole conceptual content answering to the disposition is to be thoroughly, all-sidedly and completely realized. Herewith the transition from the abstractly mechanical to the elemental, more concrete nature is motivated.

Part Two.

Elemental Nature.

Chapter One.

The Elements.

A.

The Origin of the Elements.

On the usual way of representing it must seem almost contradictory to speak of the elements' origin: an element is precisely that which is without origin. All things can be resolved into their elements, but the elements themselves cannot be resolved; they cannot pass away, and therefore cannot arise either: they are eternal. By examining more closely the significance of *matter and element*, of *matter and form*, and then testing matter's essential share in *the substantiality of things*, we shall however obtain a new confirmation of what the doctrine of mechanical nature has already shown: that the elements, in order to have a natural existence, must have a supernatural origin — in virtue of Spirit.

a) *Matter and element.* Greek philosophy begins, as philosophy of nature, in all immediacy with a doctrine of matter. What is characteristic of this beginning is that matter and principle go together into one; the opposition of the real and the ideal, of nature and Spirit, does not yet exist at this stage of consciousness: it is pure immanence in speculative innocence. The contradictions hidden in the immanence then gradually emerge, now under this form, now under that; the represented phenomena and the thought concepts come into conflict with one another; the problems become constantly more difficult, and the discussions more entangled.

The general thought that bears the Ionians' representation of a primal matter is the fundamental thought of *transformation*. How the matter was more closely represented in itself — as water, as air, as fire, and so forth — is inessential; the transformation of matter itself is the main thing. The energy with which the thought of transformation is developed in Heraclitus already makes it clear that nothing material has eternal subsistence, that matter in its immediacy is a contradiction. Fire is for him not an existing matter by whose condensation or rarefaction other existing matters might come forth; that at any rate is only a sensible way of representing. What is decisive is that every thing is just as much a non-existent as an existent. The visible fire that can be extinguished has in its being a non-being; for how else should it be able to be extinguished? That a fire is extinguished means that it passes over from being to not being; and so with everything existing. But fire itself, which in the ceaseless becoming, streaming and changing of the things of sense is the true universal fundamental essence — this fire is not kindled and is not extinguished; it is a fire of thought whose unity is difference and whose difference is unity; it is the world-producer, the everliving fire which in itself kindles measures and extinguishes measures (ἀπτόμενον μέτρα καὶ ἀποσπεννύμενον μέτρα). That in this way — since every part of the whole is not merely opposed to other parts but is something in itself opposed — there must show itself strife, discord and tension everywhere in nature, goes without saying. The like becomes unlike; the unlike joins itself together. Strife is the father and lord of all things (πόλεμος πάντων μὲν πατήρ ἐστι πάντων δὲ βασιλεὺς); life is a mixed draught; the world's strings are strung, like those of the bow and the lyre.

Against Heraclitus's pervasive dissolution of everything existent stood the Eleatics'

sharp denial of all becoming, arising and passing away. If becoming were to admit of being thought, then it would have to be either the like coming from the like, or the unlike coming from its unlike. The first is impossible, for the like *is* — and does not become — the like. The second is also impossible. If something were to come forth from its unlike, it would have to come forth from an other in which it was not, and so from non-being, from nothing; but from nothing comes nothing.

Between the sharply opposed views Empedocles attempted a mediation. On the one side he makes concessions to the Eleatics and asserts with Parmenides that arising and passing away — that is, a qualitative alteration in the original matter — is unthinkable; on the other side he will nevertheless not close his eyes to the fact that things in the world, like the whole world, are subject to a constant change. Nothing then remains for him but to refer the phenomena of alternation to motions in space, to combinations and separations of unoriginated, imperishable and qualitatively unalterable substances or *elements*. Of such elements there must necessarily be several of unlike kind, if the manifold things with their different properties are to be derived from them. Thus arose the doctrine of the four elements — *earth, water, air and fire* — a foundation for the view of nature of antiquity and the Middle Ages which the newer physics has dissolved.

In another way the representation element was apprehended in atomism. To explain actual becoming, Leucippus and Democritus started from the opposition between the full (τὸ πλήρες) and the empty (τὸ κενόν), between the atom and space. Earth, water, air and fire consist of atoms; the atoms themselves are accordingly the real elements. In every atom matter is independent and solid (τὸ στερεόν). The atoms are without becoming; there is in them nothing non-existent: they are indivisible. No particular properties can be ascribed to them; they are the immediately, positively existent. The only thing by which they are distinguished among themselves is their figure, size, weight, arrangement and position — nothing but quantitative differences. Where a combination of atoms takes place, the empty is always present as interval. Out of the atoms things are composed. A thing arises when a complex of atoms forms itself; it passes away when the combination of atoms dissolves; it alters when the atoms' position and arrangement change; it grows when new atoms enter into the combination; it decreases when atoms are separated out without being replaced. As regards their constituents, all bodies are alike; as regards their properties, they may by contrast be highly unlike.

Without dwelling on details, which concern the individual ways of representing more than the fundamental thought as a whole, and belong particularly under the history of philosophy^{*)}, we can already see from what has been adduced how matters stand with immanence in these first attempts at a philosophy of nature. In Heraclitus the thought is pervasive and unfalsified. The everliving fire which is the reason of things is also the reason in the soul. The essence in the transformations is a being of sense only insofar as

^{*)}Cf. Zeller's clear and excellent presentation. *Die Philosophie der Griechen*. 1ster Theil. Tübingen. 1856.

it is a being of thought: the thinking element in nature and the thinking element in man is the same universal. While Parmenides, in order to keep the being of thought pure as the differenceless one, lets the world of sense for itself furnish an empty, deceptive spectacle of the non-existent, Heraclitus by contrast reconciles the non-existent with the existent and purifies them both in the fire of thinking, for the sensible furnishes only fuel for an infinite burning. In the transformation the material comes everywhere into view with the immediacy of a being, as earth, as air, as finite things; but that the finite things are only appearances of the infinite being of thought is seen from this, that non-being, the negative, is everywhere present. The existent is — in *becoming* — a non-existent: that is immediate immanence.

It is otherwise when matter gets absolute independence and becomes element. The element is no being of thought; it is precisely thought's hard resistance, a barrier impenetrable for thought. And yet thinking does not give up without more ado; it grasps the element in representation; the representation is determined by sensation; sensation by impressions of sense; impressions of sense by the immediately qualitative; on this presupposition the original can be laid into elemental qualities, and Empedocles show the necessity of four elements. To what degree representation hereby gets the upper hand of thinking comes to light immediately from this, that the existent and the non-existent are held fast alongside one another in such a way that both of them, each in its own manner, get an immediate being. It is an advance in the atomists that they would not begin with sensible qualities. They would conceive the qualities by deducing them from the atoms' combinations; they exerted themselves to the utmost to make the atomistic elements of matter into pure moments of thought. The atom is not something straightforwardly sensible; it is a categorical expression for what thought expressly demands, an absolute existent, an existent whose only quality is to exclude all non-being by having the empty, the non-existent, as its limit. But with all this the being of thought is nevertheless originally denied, and thought has become a stranger to itself. In the place of transformation there now comes the category of *alteration* to play the principal part: position, arrangement, separation, combination denote nothing but alterations in the outward. With such outwardness immanence is in danger. Everything, the non-sensible too, becomes sensible; and yet something thinkable must be presupposed. Democritus is as little able as Empedocles to derive thought from the element or to show any essential unity-in-difference of sensation and thinking. If the being of thought is to be maintained and immanence preserved, it must happen by another way.

b) Matter and form. The simplest expedient for a reconciliation between elements and the being of thought must, from a popular physical standpoint, seem to be the one indicated by Anaxagoras. Anaxagoras assumes, as regards the material, a chaos of unlike elements which are, each in itself, determined as uniform in quality (homoiomeries) parts of gold, parts of flesh, parts of bone and so forth, infinitely divisible, in infinite mixture; as regards the ideal, he presupposes an ordering understanding, Spirit (νοῦς),

all-knowing, almighty, the finest, purest, absolutely simple, indivisible being, which in all beings is and remains like itself. On the foundation of this dualism immanence can be preserved so far as it can be presupposed that Spirit is present in everything, moves and penetrates everything; that the mechanical and teleological causes complete one another in such a way that the actual formations go on naturally by the parts' separation and combination, while the all-ordering, all-moving Spirit organizes the whole and builds a world of it answering to the ends of reason and wisdom. But on closer consideration one will soon see that a dualism of this nature must dissolve into contradictions. The very assumption that outside Spirit, independently of it, there subsists from eternity a manifold of absolutely independent fundamental parts proves plainly enough that Spirit is not almighty, but powerless.

It is the same fundamental contradiction that still haunts half-philosophical brains when it is asserted in one and the same breath both that matter in its elements is eternal, without origin, and that there is nevertheless a governing reason and a rational order in nature. Reason is a being of thought. If reason is to govern, it must unconditionally have the elements in its power; but that is possible only on the presupposition that the elements themselves are absolutely dependent upon reason. If one assumes that the elements' eternal existence is reason's own eternal reality, that reason not merely exists in the elements but that the independent elements are precisely the existing reason: then one has either outright denied that what we call reason is a being of thought, or brought so unreasonable a dualism into reason itself that the rational ceases.

The Anaxagorean dualism must indeed from its side have helped to strengthen the consciousness of the spiritual being in which the knowledge of nature has its truth; but the explanation was, as Socrates complains, in actuality mechanical, and the dualism unthinkable. One had then to choose between one of two things: either to hold to the independent elements and give up the being of thought, or to hold fast the being of thought and apprehend matter as a kind of non-existent, a formless stuff, and deduce the elements by means of form.

With Plato's doctrine of Ideas the being of thought came, in the Greek sense, to full recognition. Without concepts full of essence there is neither any true being nor true knowing. The concepts, which at once express the truth of being and of knowing, are self-valid forms of self-valid content, enclose all actuality and subsist by themselves as Ideas. The sensible alters, the phenomena change, natural things arise and pass away; but what subsists in and for itself, the universal, the self-like in the manifold of comparisons (*ἐν ἐπὶ πολλῶν*) is eternal, unchangeable, the object of knowing, the existent itself, *the Idea*.

The Idea's unity is indeed also a plurality; the existent has in itself manifold determinations of non-being; but the differences hereby arising all lie in being's essence and agree with one another. The existent cannot possibly be the differenceless one. If we say: the one is, then we make a difference between unity and being; we have accordingly two:

the one and the existent. If we say: the existent is a whole, then it lies in the concept whole that it must have parts: the whole is a plurality of parts. If anyone will assert that the existent has no parts and consequently is not a whole either, then he makes the existent impossible. How should it be able to be? In itself! But then it must be different from itself; for there is after all a difference between that in which something is and the something that is therein. From these dialectical indications it will be evident how matters stand in Plato with the manifold of Ideas. Every Idea is a determinateness of being because it is a determinateness of thought, a determinate thought, and conversely. The Ideas must accordingly originally have been considered from two sides, namely: 1) as beings of thought, whose being is a necessary thinking presupposing an eternal thinker; and next: 2) as objectively existent, universal substances in the fundamentally existent. But an even development of the two principal sides Plato has not been able to give. The Ideas' relation to the Godhead, the subjective side of the being of thought, is touched upon only in the form of representation under mythical clothing, while the objective side, the Ideas' real being-for-self, is developed through the very finest dialectic.

Now if the true world is a world of Ideas, whence comes the world of sense? A something must necessarily be presupposed that can make a co-being of the sensible, the material, the becoming intelligible. But what is that something? It would have to be matter, stuff. But a positive expression for matter and stuff is not to be found in Plato. Matter cannot be thought; for what is in truth thought is the Idea. Matter, the stuff formless in itself, cannot be sensed either; for what is sensed is always only a determinate, formed stuff, not the universal, formless foundation of everything material (only a *τοιοῦτον*, not *τόδε*). The formless stuff, the mass (*ἐκμαγεῖον*), is accordingly to be apprehended by means of a certain spurious thinking (*μετ' ἀναιθησίας ἀπτόν λογισμῶ τινι νόθῳ*). Stuff in and for itself is a non-existent, which nevertheless in a sensible sense must become a co-existent. If stuff is to be more closely determined, it must happen by its opposition to the Idea. While the Idea is the eternally existent, stuff is the always becoming, the unlimited, which gets its limitation, its form, its stamp from the Idea. By a combination of the Ideas with the material there come forth the sensible things, the finite yet eternal nature, the whole outward world existing in space. There is then in nature not a mixture of two sorts of being, the Ideas' and stuff's, for the stuff in things is essenceless; it is the Ideas that give things reality: the whole of actuality is in the Idea.

That stuff, the non-existent, can, by means of the Idea, acquire significance in the sensible existence subject to becoming and alteration, rests then upon its being formed, upon its merging into the forms of Idea necessary for outward being — that is to say, into fundamental geometrical forms. From this standpoint Plato's peculiar deduction of the four elements will be intelligible.

As a corporeal world, it is said in the *Timaeus*, the world had to be visible and tangible. Visible it could not be without fire; tangible not without earth, the ground of everything solid. Between fire and earth there had to be, as a middle, a third that could

bind them together. Since now the fairest binding-together is a determinate ratio, proportion, this third must stand in a ratio to both. Had we merely to do with surfaces, then one intermediate term would be sufficient; but now, since bodies are in question, there must necessarily be two. We obtain then four elements which among themselves form a continuous proportion, so that fire stands to air as air to water, and air to water as water to earth.

How matters stand more closely with this proportion in an arithmetical sense is in the present connexion without interest; characteristic, by contrast, is that the elements are formed without existing stuff out of geometrical figures. All surfaces, it is said, consist of triangles; all triangles are derived from the right-angled, which are partly isosceles, partly scalene. Among the right-angled and scalene triangles the fairest is the one whose smaller leg is half as great as the hypotenuse. Of six such triangles an equilateral one arises; but of four isosceles triangles a square. Of the square the cube, the hexahedron, is formed; of equilateral triangles the tetrahedron, the octahedron and the icosahedron. Fire's fundamental parts are tetrahedra, air's octahedra, water's icosahedra and earth's hexahedra.

As stuff shrinks to a non-existent, and the elements are formed of pure limiting forms in space, atomism is dissolved: only stuffless discreteness, pure mathematical punctuality, remains. One may now indeed say that immanence is maintained, that sensible things are immanent in the Ideas, since the Ideas themselves are things' own reality. But whence the sensible comes; how it comes about that a non-existent can look like stuff, and bring it about that the eternal world of Ideas becomes an outward, sensible world tangible in becoming and alterations — that is from a Platonic standpoint inconceivable. The emphasis falls on matter and form.

The relation between matter and form (ὕλη and μορφή) plays a principal part in Aristotle, in the philosophy of nature as in the metaphysics. Matter is the foundation of all becoming; for becoming consists precisely in a matter's assuming a determinate form, passing over from being in one form to being in another. Matter and form stand as possibility to actuality, as the passive to the active, in a particular sense as accidentality and mechanical necessity to purposiveness. Only form as form is conceivable; formless matter is without concept and so far a non-existent. None the less a certain positive being must be ascribed to this non-existent, for matter resists form: the accidental and inessential in natural things stems from matter's resistance to form. The Platonic deduction of the elements is, according to Aristotle, still more mistaken than atomism. In the atoms there is at least a formed matter, and so a reality; but Plato's elements are hollow figures, as much at variance with mathematics as with physics. To compose bodies out of surfaces leads consistently to assuming indivisible lines, indeed to resolving magnitudes into points. As pure mathematical figures the elements cannot possibly fill world-space, and yet Plato will not concede any empty space. From a physical standpoint the attempt is altogether unreasonable. For how can the corporeal, which after all has

weight, consist of surfaces which have no weight? Whence, on this presupposition, should the single elements' specific gravity or lightness derive? — The assumption of solid, indivisible atoms is nevertheless also false. On the absolute indivisibility it might happen that the atoms' figure did not fit the space in which they were to be. Nor does one see any ground why there should be only invisibly small and not also large atoms. If the atoms are moved by something else, then they are subjected to an action, and the apathy ascribed to them is cancelled; if they move themselves: then either the moving element in them is different from the moved, and they are then not indivisible; or, if the moving is one with the moved, then there are in one and the same atom conflicting properties.

Far from placing things' arising and passing away in a heaping together and dispersing of indivisible atoms, whose quantitative differences are insufficient to explain the corporeal qualities, Aristotle assumes a pervasive alteration of matter effected by form, a real becoming in essential transitions. When water freezes, or when ice melts, it does not happen by a mere alteration in the parts' position, but by a re-forming of the parts themselves, in the course of which the properties change; when bones and flesh are formed in the body out of the nourishment, it does not happen by single parts being drawn out like bricks from a wall, but a new matter forms itself. It is the Heraclitean doctrine of transformation, developed into an articulated concept.

The conditions for the qualitative alteration lie, as with every motion, in the relation between the possible and the actual, more closely determined as a relation between the suffering and the working. The working and the suffering must be partly of like kind, partly opposed. The alteration consists in the working making the suffering like itself, a transformation which can again happen in several ways; for the mechanical causes in matter are only intermediate causes; the emphasis lies on motion. Nature itself is motion's inner ground; every motion has in itself a goal by which its direction and limit are determined: nature works teleologically. It is the peculiarity of motion downward and upward that forms the foundation for the separation of the four elements. Yet account must be taken not merely of the opposition between the heavy and the light; the bodies' felt properties also come into consideration. The elemental properties are warmth and cold, dryness and moisture, the first two active, the last two passive. By placing these properties together in pairs we obtain, with the deduction of two impossible combinations, four possible ones: warmth and dryness in *fire*, warmth and moisture in *air*, cold and moisture in *water*, cold and dryness in *earth*; and so again the four elements.

The elements pass over into one another; for everything becomes from an opposite to an opposite. The more complete the opposition, the more difficult the transition; the less complete, the easier. When water's cold is cancelled, air arises; for water and air have moisture in common and are incompletely opposed. Water and fire, by contrast, are completely opposed; the transition must therefore happen mediately, in that water first passes over into air and then, when the moisture is cancelled, over into fire.

Without laying weight on the fact that the construal of the phenomena conflicts at essential points with present-day physics, we shall now, from the standpoint of the philosophy of nature, examine in particular how far Aristotle has by his explanation of the elements' origin really been able to reconcile form with matter, maintain the being of thought and carry immanence through.

c) Matter and substance. The atomists hold fast the being of matter in unconditioned independence and exclude the being of thought; Plato maintains the being of thought by reducing matter to the semblance of a non-existent. Aristotle makes the being of thought into form and the complement of sensibility into a positive substratum, a matter that lets itself be formed. He acknowledges with a certain evenness both the ideal and the real; in him we should then expect to find what the newer speculation seeks so eagerly: the ideal-real principle that can ground the material world's immanent unity with objective reason. But precisely in Aristotle we learn how unreasonable and self-contradictory the whole speculative endeavour is when immanence lacks its transcendent presupposition, and the unity of nature and Idea is to be sought immediately and at first hand — not in Spirit, but in nature. There is in corporeal things a peculiar union of something sensible with something thinkable; upon this union rests every body's peculiar subsistence or substantiality. If we abstract from the thinkable, that is, from the being of thought, then we have only a fleeting, dreamlike appearance, but no substance, no actual body. If we abstract from the sensible, the material, then we have a general concept of body, but the existing substance, the actually determinate body, is gone. So it is incontestably certain that matter as matter must have just as decisive a significance for a body's substantiality as form. This fact puts Aristotle in an embarrassment — an embarrassment which has by no means yet been overcome, either by Schelling or Hegel or Feuerbach.

The universal essence — the genus, the species — has, according to Aristotle, no subsistence for itself; it subsists only in and through the single. Substance in the fullest sense, the true, actual essence, which is subject but never predicate, is a single being. Since now form as form is the universal, a predicate which only in an abstract, improper sense can become subject, it follows that form can only be one side of substance, not substance itself. If the ground, the *real* ground, of the single being lies not in form but in matter; if it is matter that gives things their particular subsistence by giving them individuality, then what is properly substantial must lie in matter. But that Aristotle will not concede. If matter were substance, then the substantial would fall absolutely outside knowing, for all knowing goes upon form. Substance must then be conceived in such a way that matter is a single being only according to its possibility, but that the single being's realization is due to form. And yet this assumption too must be rejected. That matter has the possibility of becoming a corporeal single being by means of form must necessarily be thought in such a way that the same matter which is the single being's possibility is also form's possibility. Matter accordingly does not get form as an other, a fully finished actuality supervening; on given conditions it develops its form, its actuality,

out of its own possibility: matter lets itself be formed because it is its nature to have form in itself, to form itself. It is a pure fancy that there should be a general substratum (ὑποκείμενον) falling outside thought because it fell outside form! The formless matter is — as may indeed be recognized from this, that in its formlessness it is apprehended as something general — itself a thought, an abstract, abortive thought. Formless matter is a designation for something untrue, for a non-existent and yet existent formable thing waiting for form. Formless matter is no possibility; it is on the contrary an impossibility, for it is a contradiction. The contradiction does not concern matter at all, but thought. Matter formless in itself is just as contradictory a thought as form formless in itself, or matter matterless in itself.

The manner in which Aristotle will deduce the elements, and in general everything he teaches about the properties of bodies, shows plainly how contradictory it is to make form without more ado into the thinkable and the material into the unthinkable. Where do such elemental properties as warmth and cold, dryness and moisture belong? to the thinkable or the unthinkable. Insofar as they are apprehended as forms, they must fall exclusively under knowing, and so under the thinkable. But according to the deduction they answer to feeling, and so to the material, the unthinkable. How should dryness let itself be transformed into a thought? It would have to be a dry thought. But thoughts are in actuality as little either dry or moist as they are, in the natural sense, either warm or cold.

Matter in all indeterminateness is something non-sensible; it is determinateness of form that makes matter sensible. In consequence of the Aristotelian conception one may then with just as much right assert that form is sensible and matter thinkable, as the reverse. One can let matter be a moment in form's concept and make form the whole actuality; but one can also let form be a moment in matter's concept and place the whole actuality in matter. This swinging between a one-sided idealism which in form holds to the concept, and a one-sided realism which seeks reality in matter, has not merely shown itself in antiquity but — a necessary consequence of the ambiguity of the doctrine of immanence — has repeated itself in a quite characteristic manner in more recent times. Hegel, who makes matter and stuff into categories as logical as form and substance, culminates in a conceptual idealism; Feuerbach, who out of offence at the natureless hollowness in conceptual idealism goes to the opposite extreme, sets form as a moment in matter's self-individualization and ends with a sensible this-here, an unsayable.*)

Uncertainty about what matter, element and form of matter are must necessarily coincide with a corresponding uncertainty about what nature is. That Aristotle, for all his penetrating eye for experience, his superior acuteness and admirable endurance in comprehensive thinking about nature, does not know what nature is — is not at one with

*)The echoes assure us that Hegel is one-sided because he overlooks the real, and Feuerbach one-sided because he overlooks the universally ideal; so one may become all-sided by setting the ideal immanent in the real and the real immanent in the ideal; so one has the ideal and the real in pure immanence — that is to say, in immanent talk.

himself about how he is really to determine nature's essence — is demonstrable from his own system. The transition from the metaphysics to the philosophy of nature he marks as a transition from the doctrine of the unmoved to the doctrine of the moved. In the circle of the motions and everything moved there present themselves immediately the corporeal single beings, consisting of matter and form. What now is matter? It is not known and cannot be known, for matter as matter is no object of knowing. What then is form? That is precisely what is known; and yet, when we look more closely, it is not known either. Form, it is said, is actuality. The perfect form is the perfect actuality, the highest energy: that is the Godhead. This form-being is in the absolute sense a being of thought; it is the eternal thinking that thinks only itself, the blessed reason that conceives itself, an actuality without any possibilities. But the forms of nature — what sort of essentialities are they? Are they the Godhead's thoughts? By no means! To assume this would be in equal degree at variance with the Godhead's essence and with the forms' nature. With the Godhead's essence! For the perfection in the divine thinking rests precisely on its thinking the perfect alone. To think the perfect it must exclude the thought of everything else; for nothing is perfect but the Godhead itself. With the forms' nature! For if the forms of nature were the Godhead's thoughts, then they would have a being in an other, namely in the thinking Godhead, and would be derived from an other, namely from the divine thinking. On this presupposition form could not possibly be things' substance. Substance is precisely that which neither is in an other nor can be derived from an other, but unconditionally is in itself. What a splitting of form's concept! The highest form is not merely thinkable by us but thinkable in itself: its thinking is being and its being thinking. The forms of nature, by contrast, are thinkable by us but in themselves unthinkable, for in the form of thought they are not substances.

Determinateness of form is then in itself unthinkable too. Natural things' properties and determinations lie in the moving and determining forms. Matter is determined by the form determinate in itself; but form itself is not determined; form is determinate. Has form then formed and determined itself in advance? By no means! Were the forms of nature to have formed themselves in advance before they could begin to form matter, then they must in this self-forming have passed through a becoming; they must already have had the possible, the passive, the material in them before they came into contact with matter; there must then again have been a form finished in advance which could give the forms of matter their form. Determinateness of form immediately presupposes itself: the whole system of forms working in the realm of nature comes to be taken as a given, inexplicable fact. This fact then has again the questionable feature that it suffers from insoluble contradictions. Insofar as the forms are apprehended in knowing, they are apprehended for themselves in abstracted universality and cannot be substances; insofar as they go together with matter, they are burdened by the material and undergo an essential alteration; they are determined in determining; they become passive by being active, just as matter, by means of its resistance, becomes active by being passive. By

continuing the analysis one can find contradiction upon contradiction. The secret is that the immediate immanence of matter and form rests upon a fundamental contradiction; but that a fundamental contradiction, wedded to dialectical reflexion, breeds strongly is a well-known matter.

The yield of the old doctrine of immanence is then the following: matter is not immediately to be apprehended as element;

it is in itself a dependent thing, which first acquires quality and qualitative subsistence when it is determined by a form answering to knowing. The elements have an origin; the demand is that their origin shall be shown, that the elements shall be deduced. The demand in itself is true; but it is not yet fulfilled.

B.

Elements and Natural Ground.

The hard opposition of element and being of thought has its original ground in the opposition of immediate object-power and overreaching self-power. Absolutely overreaching self-power conceives the content of the ground of nature as a totality of moments in one being of thought; upon this rests *the ideality of the elements*. In object-power the reflected moments come to existence as the immediate issuing of the fundamental differences, as independent members in outward actuality; in this consists *the reality of the elements*. In order to be grounded in itself, nature must be grounded by Spirit: the infinite conversion of ideality into reality, and conversely, goes on in *the ground-medium*.

a) *The ideality of the elements.* The being of thought, determined solely by human knowing, cannot possibly be apprehended as the real essence at work in the elements of nature. One of two things: either consciousness must halt before a barrier of impenetrable reality, bow to outwardly mighty nature and ascribe to its elements and forces an eternal, absolute independence; or thought must, in speculative forgetfulness of its powerlessness, declare matter — the unthinkable ground of all outward being — to be a non-existent, posit *its* essence as the essence of things, and make itself into a godhead. If the being of thought is to master the elements, then it must be a superhuman essence, resting in divine knowing, grounded by an omnipotent thinking.

The Ideas, it is said in Plato, are not things of thought (*νοήματα*) which merely have existence in the soul. Right! But when it is thence derived that the Ideas cannot be the Godhead's thoughts, the inference is false. The soul's thoughts, apprehended for themselves, are thoughts of powerlessness, abstract things of thought which have the realities outside them; but the Godhead's thoughts are thoughts of omnipotence with inner, essential reality. That the forms of Idea are necessary forms of thought the Platonic

dialectic proves; but forms of thought are thinking's own forms, forms determined by thinking itself, forms which only a thinking self can bring forth. If the Ideas' eternal being is not grounded in an eternal thinking, then there is no true knowing, and the Platonic dialectic is an empty play on words.

What does it really mean that thinking is indeed dependent upon the Idea, but that the Idea, what exists in and for itself, is independent of thinking? It means expressly that the Idea is in itself an unthinkable essence. An existent that precedes all thinking is an existent whose determinateness, far from resting upon any determining thought, on the contrary falls outside every determination of thought. For such an existent to become an object for thought, it must be represented; it cannot melt together with thought, but must be held fast in existing independence as thought's resistance, its impenetrable other. Thought may now quite well in each and all direct itself by this object, invisible in representation, and assure itself that it exactly apprehends all the determinations contained in the existent; but what has happened? The eternal determinations of being, independent of thinking, have been falsified; they have been transformed into something which they nevertheless are not in themselves, namely into nothing but determinations of thought. What is in thought is accordingly not the Idea itself — the determinations of Idea are not determinations of thought — but an imitation, an intellectual copy of an original existing outside; the Ideas' actual being falls outside their thought being; that is a hard dualism of thinking and being.

To form the world and bring the best possible order into the disorderly, the highest master-builder had, says the *Timaeus*, constantly to look to *the archetypes*, the Ideas, things' eternal *models*; hereby, it is claimed, the Ideas' substantiality is incontestably maintained. Right! But when one will thence derive that the archetypes to which the Godhead itself looks cannot possibly be products of the Godhead's own thinking, the inference is false. The long, unclear dispute between the Ideas and the divine understanding must surely at length admit of being brought to an end.

So long as human consciousness, be it in intoxicating speculation or fasting criticism, arrogates to itself an absolute knowing of the absolute, the mystical representation of "ein unvordenkliches Sein" will constantly crop up and bring disturbance into the much-praised unity of thinking and being. In the psychologically limited self, real being necessarily precedes every thinking, however speculative, turn it how one will; only in the absolute self is the finite succession in time cancelled. Divine being and divine thinking presuppose one another to infinity: being follows from thinking, and thinking from being, in Spirit's eternal substantial essence. To make being the absolutely first and let a thinking follow after is, with respect to the concept of the Godhead, just as thoughtless as to make thinking the first and derive being from it.

To prove that there is in eternal being something which divine thinking cannot alter, and by which it must accordingly direct itself, such general propositions have been adduced as: that two is less than three, that a triangle must have three angles, that the

whole must have parts, that an effect must have a cause, and so forth. But who does not at once see that such so-called “eternal truths” are necessary determinations of being only because they are necessary determinations of thought? If God could ever grow weary of thinking, then he would at the same time be weary of being.

On the Aristotelian definition that substance is what cannot be in an other, the divine thoughts certainly cannot be things’ substances; but from the definition an either–or follows. Either there are no corporeal substances at all – which Aristotle cannot concede, since he most emphatically insists that the corporeal single beings are the real substances. Or the corporeal substances must be absolute – which he cannot concede either, since it stands in express conflict with his doctrine of the elements’ transitions, of corporeal formations, combinations and dissolutions. Were the corporeal substance a single being in the sense that its being in itself made its being in another impossible, then such a single being would have to be an eternal, indissoluble atom: the corporeal substances would then have to be raised above all alterations. That things undergo alterations, perish and arise through a transformation of matter is a proof that their substantiality is not absolute but relative. The substances’ relativity makes it incontestable that their subsistence in themselves is a subsistence in another.

From a general concept of divine thoughts to a sensible representation of such determinate substances of matter as carbon, sulphur, gold and the like there is, outwardly regarded, undeniably a great leap; but when the matter is thought through, it will nevertheless show itself that the categories substance of matter and substantial being of thought at bottom presuppose one another.

The substantial being of thought is objectified in a thinking that avails. In the infinite thinking the mutually different determinations of thought must shine through one another: every particular concept is a mirror of thought reflecting the whole system of concepts in which the content full of essence is formed out into Idea. What now gives this being of thought substantiality? The thinking, knowing, all-availing self-power! That thought is power, and thinking an infinite availing, means that thinking is working and has in itself all the moments of actuality as reflexes in the overreaching energy of the infinite reflexion. But when all the moments of actuality without exception are enclosed in the substantial being of thought, it follows that no matter, no real moment, can come from without, but that all reality, all actuality, be it ever so material and outwardly tangible, must come from within, from the being of thought, the essential ground of the real world’s, outward nature’s, elements and substances of matter.

We have seen how the being of thought in and by itself presupposes the substances of matter; we shall now see how the substances of matter in their reality presuppose the being of thought.

How far chemistry’s so-called fundamental substances really are simple substances and mark the outermost limit of bodies’ possible resolution is a question of its own; at science’s present stage they represent the elements, and that is enough for us.

The elemental substances of matter have first the mark of ideality that their unaltered subsistence comes into view in nothing but alterations. When oxygen and hydrogen unite and form water, they alter one another mutually; they give up their independence; their particular properties are transformed and disappear in new properties: they perish as the water comes to existence. The substances of matter's being in themselves is a being in another; their immediate independence is a one-sided and transient state; to know their essence one must know their combinations. What they are in themselves, they are for one another mutually; they play a new part in every new combination, they become incessantly an other through another; in the stream of alterations there remains no trace or residue of absolutely immediate independence: for their modifications and alterations there are no outward, finite limits. And yet there are limits, essential, law-determined limits; for the elemental substances have in all this being-other each its own particular substantial essence; the one cannot be transformed into the other, oxygen not into hydrogen, hydrogen not into carbon. But that the elemental beings of matter keep themselves unaltered precisely by undergoing alterations proves that their fundamental being is an essence reflected into itself, a *substantial* being of thought.

The next mark of ideality is that the elemental substances of matter are, as regards qualities and properties, both independent of one another and yet essentially determined for one another mutually. If we cast a glance at the somewhat more than sixty fundamental substances, metals and metalloids, at present known, it is indeed seen that some among them stand nearer to one another, others further apart; but it also shows itself that every single one has its properties, its peculiar essence, for itself. There is — if we except the few which, although they are composite, nevertheless play a part as fundamental substances — no trace of the one elemental substance having borrowed anything from the other. From this standpoint the fundamental substances are accordingly independent of one another; their differences look as if they had come forth by a pure accident. Who would see, looking at carbon, that its peculiar existence in any way rests upon that of hydrogen, or the reverse? But the hydrocarbons teach us that there must be an inner, essential and quite peculiar kinship between the two substances; they fit one another — indeed, it looks for all the world as though they had been fitted to one another. But how is that to be understood?

It is a fact that the properties of the one substance, although they immediately belong to the substance in question, acquire significance only through the properties that just as immediately belong to another substance. This fact shows that the properties of the one fundamental substance must originally, independently of outward coincidence, be reflected in the properties of the others; in the formation of the one there is expressly a regard to the formation of the others: here is among them all a harmony predetermined in divine knowing (*harmonia præstabilita*). But this again proves that the elemental qualities are moments of one and the same fundamental essence, the substantial being of thought.

Finally — and this is ideality's fully developed mark — the whole sum of elemental substances has the character of a system of concepts carried through in all directions and answering originally to the content of the Idea of nature. The real concepts of the substances of matter present to consideration two principal sides, a finite and an infinite. From the side of finitude they are, as concepts of existence, bound, each of them, to its peculiar matter; from the side of infinity they are in the Idea raised above parcelling out into an eternally clear and transparent unity.

The inner real connexion between the existing substances we learn to know in chemistry from their affinities. The ground of the difference of the affinities, upon which the whole manifold of combinations and separations essentially rests, is a real ground, the ground as being of matter. But that this real ground must be ideally articulated, and so originally determined by a rational ground, can be expressly shown. Affinity is in its essence double-sided: chlorine has affinity to gold because gold has affinity to chlorine, and conversely. Oxygen has affinity to hydrogen because hydrogen has affinity to oxygen, and conversely. And why, then, has oxygen no immediate affinity to fluorine? Quite straightforwardly, because fluorine has no immediate affinity to oxygen. But in this outward referring from the one to the other there is no grounding at all. Affinity's formal, self-cancelling because—because demands a deeper-lying motive, a rational ground decisive for the essence.

Affinity presupposes that the concepts of matter in question must necessarily be side-concepts of one and the same comprehending total concept. Oxygen's and hydrogen's mutual affinity, in virtue of which they unite and form water, is the expression in existence of the natural truth that the real concepts of oxygen and hydrogen are, as side-concepts of water's real concept, comprehended in one another. What holds of two concepts of matter holds of them all: they are all concepts of the same conceiving. From the side of infinity the concepts of matter are concepts of Idea; in each single one of these concepts of Idea the Idea's whole content is present in a particular manner. The real ground is grounded in the rational, the being of matter in the being of thought; that is the ideality of the elements.

b) The reality of the elements. Matter is as little an existent in and for itself as the empty, abstract form. Matter and form are side-determinations of real power, object-power, which is originally opposed to self-power's ideality. In the ideal form of the concept of power the element is a reflected moment; in object-power's real form the moment is a formed matter, an existing element. Nothing is added to the moment from without in order that it may become an element; nor is anything drawn from the element in order that it may become a moment: moment and element are, as regards content, essentially the same. And yet they are opposed to one another — indeed so essentially opposed that the one is something thinkable, the other something in and for itself unthinkable.

Only a being of thought lets itself be thought; but an existing element — coal, gold,

sulphur — is no being of thought; as an existing object it is thought's actual resistance. For powerless thought the resistance is a hindrance, an outward reality which by sensation can be apprehended in image and representation, but not conceived. For omnipotent thought the object is the substantial satisfaction, the other with which thought is sated, the object in which the objectifying self recognizes its own power. That there is ideal power in the divine thinking is categorically expressed by there being in the elements a real power, an actual object-power. Spirit does not think the elements; it creates them; it masters them in form-giving concepts: to *conceive* is to *master*.

From concepts of element to existing elements human thinking can show no finite transition. It can be seen in science that the elements express concepts, and that the concepts are conceived by Spirit. If the concepts be taken away, then the elements are impossible; if the elements be taken away, then the concepts are unactual: the conversion of concept into element is an issuing of power. But the transition itself, the actual act of creation, cannot be deduced from the general presupposition lying in the doubling of power. The doubling of power expresses the eternal fundamental relation between the ideal being of the concepts and the real being of the elements in general; but the bringing forth of these, just *these* elements with *these* qualities, of *these* substances with *these* forces in *these* properties, is for every particular world-system a self-conditioning, unconditioned act of will, an act of creation which in the unity of the disposition determines the particular dispositions. The peculiar character of the fundamental substances must be found out by the way of experience; but the science of experience for itself does not have the task of uniting the representation of the elements' manifoldness with a scientific knowledge of the unity of the ground of nature. Mechanical analyses have, especially in this century by means of the voltaic pile, led to the discovery of a great quantity of previously unknown fundamental substances. The question is now this: will a future separation of substances hitherto regarded as fundamental substances end with an increase or a decrease in the number of the fundamental substances? By the isolation of potassium, sodium, norium, thorium, erbium, terbium and so forth, the number has undeniably become greater and greater. But there are then also indications that the turning point might one day arrive at which the results turned about, and the movement went from manifoldness toward unity. It has shown itself that saltpetre, for example, far from being what the Middle Ages assumed, a modified element, is a salt (nitrate of potash) and nitric acid a combination of nitrogen and oxygen ($\ddot{\text{N}}$); it has shown itself that cyanogen, though it behaves as a fundamental substance toward the other fundamental substances, is a combination of carbon and nitrogen ($\text{C}_2 \text{N}$), and ammonium, of which the same holds, a combination of nitrogen and hydrogen (N H_4). There are now those who assume that by eliminating one alleged fundamental substance after another one will at last be able to resolve them all into a very few, indeed possibly into a single one. Upon so magnificent a reduction there would then have to follow a corresponding deduction. Just as one can already now transform oxygen into ozone — indeed let the

transformed oxygen come forward both as ozone and as antozone — so one would then, when the ideal was reached, be able to let the many qualitatively different substances arise and form themselves out of one and the same common fundamental substance.

However highly we prize the analytical art, to which the honour of so many great and fruitful discoveries belongs, we may nevertheless never forget what investigating science constantly recalls: that merely empirical analyses, even if they be continued with ever so wonderful success, cannot possibly by themselves lead to a true insight into the essence of the ground of nature. If one stops at the representation of a manifold of eternally existing fundamental substances, then one denies the unity of the ground, indeed the ground itself. If the fundamental substances are eternal, then they exist without being grounded, and then atomistic separations and combinations step everywhere, in all formations, into the place of grounding; things are then pieced together but not grounded. Without an original ground nature becomes only an exceedingly fine mechanical patchwork, a natureless nature, a contradiction. If one ends in the representation of a common, differenceless, general fundamental substance and puts this existing matter in the ground's place, the contradiction is still more conspicuous. If substances of different kinds were to come forth from a first matter in which there were no differences, then the differences would have to arise out of nothing. If in the presupposed fundamental substance there really were a quantity of different constituents, then it would in actuality not be one but several substances; but when there are no differences at all in it, then there is also nothing from which determinations of difference can be derived, either immediately or mediately. The differences belong just as originally and necessarily to the presupposition as the unity. The content-rich unity of the determinations of difference, the real ground of the substances of unlike kind, is no fundamental substance, no object of experience. If the ground of nature is not to be known, then we remain in the contradiction; if the ground of nature is to be known and the elemental differences grounded, then the limit of empirical knowing must be overstepped.

Another side-representation accompanying the science of experience will likewise obscure the apprehension of the fundamental relation between the original and the derived. The simple substances, the elemental realities that cannot be resolved, show themselves to the observer as the constant, the original, while the composite substances represent the inconstant and derived. A double salt has then from this standpoint less independence than the particular salts into which it can be separated; these can subsist without it, but it not without them. For the same reason the salts have less independence than their acids and bases, and these in turn less independence than the elements themselves. This way of seeing may within the circle of the realities themselves be perfectly warranted.^{*)} But for the knowledge of the elements' relation to the ground of nature it is misleading. Knowledge of the ground reduces the relative difference between simple and composite substances to a derivative and holds fast the elemental cycle. In the

^{*)}Cf. Forelæsninger over „Philosophisk Propædeutik“ from the university year 1860–61, p. 138.

cycle the ceaseless alternation of arising and passing away holds as much for the simple substances as for the substances of matter composed of them. That the composite arises means that the elements perish; that the composite perishes means that the elements arise. Every substance is a derivative; nothing is at bottom original but the ground itself.

The superficial side-representation of the elements' unconditioned priority, which does not concern research, confuses concept and conceiving. It constantly comes to look as though the first, fixed, unalterable concepts of nature lay exclusively in the elements, while the particular, more content-rich concepts uttered in the formations of composite substances were only derived, accidental concepts of mixture dependent on outward circumstances. That is a fundamental error. The elements are, according to their concepts, expressly laid out toward particular formations. Only from the particular formations can it be seen what they really signify as elements. The concept of water, the concept of quartz, the concept of feldspar are just as original concepts of nature as the concepts oxygen, hydrogen, silicon, aluminium, potassium. It is precisely the more all-sided, more concrete concepts that contain the more one-sided in themselves; not the reverse.

Upon insight into the relation between more abstract and more concrete concepts rests all essential insight into the teleological. The science of experience is within its rights when it dismisses loose reasonings about lower and higher, about means and ends in nature's housekeeping; but concepts such as teleology of nature and reason in nature it cannot treat. Reason in nature and teleology rest upon dispositions in nature, dispositions in nature upon the ground of nature; but the question which the science of experience must by its character leave unanswered is precisely the fundamental question, the question of the ground.

c) The ground-medium. Power's fundamental doubling necessarily brings with it a doubling of the ground. In the self-medium the ground is being of thought, ground of reason; in the object-medium it is being of matter, real ground. The ground of reason has indeed its truth, the confirmation of its ideality, in the real ground, and this in turn the confirmation of its reality in the former; but the former is reflected in inwardness, the latter merges into existential immediacy: they form extremes. The ground-medium, the immanent intermediate being in which the opposition of ideality and reality ceaselessly arises and just as ceaselessly disappears, is the middle of the extremes.

Outward, material actuality is an extreme of positive immediacy, an antilogical reality that excludes all negation, all reflexion, and with that all ideality. Every element expresses its concept in so realistic a manner that the concept disappears in the expression. If we say: this piece of coal has a limit, a limit is negation, therefore it has in its being a negation — then we confuse what belongs to thought with what belongs to the object. The piece of coal's limit is the positive in its surface; the parts lying in the surface have just as immediate an existence as those found in the interior. The limit as negation belongs to thought: the surface is the body's ceasing, its non-being — that is a judgment

of the reflecting consciousness. The body's non-being does not concern the existing body: the element's negation falls outside the element.

The antilogical immediacy of the actual is found indicated in age-old problems. The Eleatics denied becoming because the existent must exclude the non-existent. The Megarians went further and denied possibility, because the possible is in actuality a contradiction, an existent which nevertheless is not. To *be able*, they said, is to *work*; what nature can do, it does; what it does not do, it cannot do (*ὅταν ἐνεργῇ μόνον δύνασθαι, ὅταν δὲ μὴ ἐνεργῇ οὐ δύνασθαι, φασιν*): only the actual *is*; the possible is not at all. Aristotle explains becoming and possibility by overlooking the point and breaking the edge off the problem. To explain becoming he refers to possibility; to explain possibility he refers to matter — the inexplicable! What is matter? It is possibility. What is possibility? It is matter. In more recent times Hegel has given a dialectical impetus toward the clearing up of real possibility. The immanent negation is the secret of all possibilities; that is the matter to a nicety. A pity that the immanence of the negation has, as regards natural things, itself remained a mystical secret.

For the elements to be able to work, there must be in every actual element countless possibilities. In an isolated state existing carbon must indeed express its concept; and yet cannot in actuality express its whole concept. Where carbon comes forward as diamond, only one side of the concept is realized, for the possibility is that it can also come forward as graphite; where it in actuality comes forward as graphite, the possibility is that it can also come forward as in anthracite. In carbon's combination with hydrogen the possibility of carbonic acid is excluded. That the whole concept cannot possibly be realized at once and in one place in one and the same actuality rests upon existence's being exclusive, upon the one form of actuality necessarily excluding the other. According to its concept pure carbon is *both* graphite *and* diamond, but in actuality it is *either* graphite *or* diamond. It is accordingly not, as Aristotle assumes, matter, but the concept, that is the possibility; when the form of the concept has become a form of matter, we have the actuality. Every element is always something other in actuality than it is in possibility; the actuality is each time only *this*, only one, but the possibilities are manifold. The concept's manifold possibilities have a negative being in the one positive actuality with which an element exists. This negative being of the possibilities is the actual's immanent negation, its ideality, its inner.

But according to what we have seen above, the actual in nature has no inner; the natural thing is, in antilogical immediacy, what it is, neither more nor less. So there is dispute about the relation between nature's inner and its outer; the dispute goes on both in verse and in prose. Against Haller's strict separation of the inner and the outer — *Ins Inn're der Natur dringt kein erschaffner Geist* — Goethe pledges his intuitive genius for the inner's immediate unity with the outer: *Natur hat weder Kern noch Schaale, Alles ist sie mit einem male*. Who now is right, Haller or Goethe? In reality there is everywhere a unity of the inner and the outer, for reality is that the inner merges into its outer. But

reality is finite; against reality's finitude stands ideality's infinity, and so there is after all a decisive opposition between the inner and the outer.

Ideality, the inner, answers to the being of thought, the ground of reason; in the ground of reason the concepts of element are concepts of Idea: there is in every existence in nature an infinite surplus of ideality. Reality in its immediacy answers to the ground of matter, the real ground; for existence's real ground merges, with the given conditions, immediately into the grounded existence.

For intuition and representation it then easily comes to look as though ground of reason and real ground furnished two foundations lying one upon the other, so that the real ground went together with actuality while the ground of reason was to be regarded as a kind of invisible container of the possibilities. The realization of a possibility would then come to be apprehended thus: that the possibility came by an inflow from the inner and, by intervening in the outer conditions, got power to alter what subsists, take actuality from what is given in advance and thereby give itself actuality. In this way every new possibility would accordingly come as a conqueror from the depth of the idealities, ally itself with a circle of conditions and tear actuality to itself. An unnatural machinery! One represents the elements as swimming in a sea of possibilities; and then there is between possibility and actuality no substantial unity. One might then just as well represent the elemental realities as being nothing but occasional causes, loose counters which an omnipotent will set up and moved at pleasure.

All activity in nature rests upon a substantial unity of possibility and actuality, but this unity rests in turn upon a pervasive opposition of ideal and real power. In the way of representation everyone knows well enough what is generally meant by an activity in nature. So one assumes that working and activity is something that needs no explanation, and explains without more ado the transition from possibility to actuality by activity, pointing particularly to causes, forces, conditions as the active element. One forgets that precisely that by which one explains is in high degree in need of an explanation. What use is it to appeal to "the active element" and say: the causes *work*, the forces *work*, the conditions *work*, when one is not able to give an account of what really lies in the concept: to *work*?

Reality for itself, the antilogically determined actuality, cannot work, for it lacks possibility. Two pieces of wood, for example, cannot be rubbed against one another without the one having, as regards the impression, to furnish possibility for the other's actuality; in being mutually both possible and actual for one another, they work, and heat is brought forth. Oxygen and hydrogen can be mixed without giving up their immediate actuality; no new production is then effected. But if an electric spark passes through the mixture, then the possibility is unlocked: the elements work and bring forth water. Possibility for itself, pure ideality, cannot work either, for it lacks actuality. The possibility lying in the concept of water cannot realize itself; it must have the oxygen and the hydrogen for actuality in order to be able to work. Working is under all forms

an infinite conversion of ideal into real, of possibility into actuality, and conversely.

This conversion presupposes that the ideal is power. If ideality were not power, it could not possibly have the moment of actuality in itself and work out its own reality. All working is, seen from possibility's side, a categorical expression of the ideal's mastering the real. But the conversion also presupposes that the real is from its side power and masters the ideal. If immediate reality did not have ideality, possibility, as a hidden moment, an indwelling negation, in itself, then it would be powerless over against possibility; it could then not take part in the activity, it could not work at all. From actuality's side all working is a categorical expression of the real's mastering its ideal.

Possibility is an availing (*δύναμις*, *potentia*); its essence is an ideal power. Actuality, antilogically determined, is a being's independence, is resistance, object; its essence is a real power. Actuality is an issuing of power; possibility is ideality's surplus. Water is according to possibility ice, and yet here is a qualitative difference. The transition from water to ice is an abrupt, decisive conversion; the degrees of cold with their preparatory intermediate determinations lose themselves in the disjunction: either water or ice. So long as the water is actual, the ice is only possible; the new form of state, in which the ice is actual and the water possible, rests upon an issuing of power. The issuing of power may otherwise be of highly different kinds; but it is always recognizable by the qualitative conversion. Real possibility requires conditions. The conditions can enter one by one; so long as a single condition is wanting, there is here only a possibility and a number of conditions; but with the entrance of the last condition everything is altered; the possibility is no longer possibility, the conditions not conditions; the whole is a new actuality — it is an issuing of real power. The conditions' transformation into moments in the ground is the essence of the issuing. The real power masters the conditions, for it transforms them; and yet this is only the one side of the matter. The opposite is that real power is dependent upon the conditions, inasmuch as it is dependent upon possibility and upon ideal power. Upon the restless, yet at bottom quiet, unity of real and ideal power rests all working and activity. The elements combine, and the composite substances of matter come forth by an issuing of real power — on the ground of power's ideality. The composite is separated, mighty ideality unlocks its possibilities, and the elements return — in power's real issuing.

The ground-medium, the universal essence in which the ground of reason is one with the real ground, reflects in its inner what is parcelled out in the outer, and unites in the outer what is different in the inner. The ground-medium expresses the unity-in-difference of nature and Spirit; in the ground-medium the natural is spiritual and the spiritual natural. That an existing element can carry all its possibilities, its immanent negations, in immediate existence is possible only because the real power, whose issuing determines its restricted actuality, is in the ground-medium one with the inner ideal power that carries ideality's surplus.

So here, then, is an immanence, an original identity of being of thought and being

of matter. Yes! But the unity is grounded upon an opposition, the immanence upon a transcendence; the ground of nature itself is a derived ground. Nature is grounded by Spirit.

B.

Elements and Bodies.

The natural difference between element and body rests upon an immediate separation of independent and dependent, a movable determination of limit which comes particularly into view when the concept *corporeality* is investigated. That the elemental concepts of independence are concepts of power makes itself known in facts by this, that the determinations of concept, although their inner essential connexion must necessarily be presupposed, do not admit of being derived from one another in a rational manner; the factual connexion is known *from the corporeal properties*. Bodies' independence is a material substantiality, a selfless independence, which in elemental nature has its measure and its standard in *the individuality of bodies*.

a) *Corporeality*. From the foregoing it is seen that the word element can be taken in very different senses; now it is the fundamental atoms, now the masses of the simple substances, now again the universally corporeal forms of state and activities that are meant. Every element is a body, every smallest part of matter fills a space; but not every body is an element. In determining the element one has originally started from the premiss that it must be the absolutely simple, the independent in itself; but simplicity and independence do not straightforwardly answer to one another. The atom, the abstractly simple, is a dependent thing; the atoms subsist in the mass and are known only from the relations of weight and measure in which the masses combine.^{*)} Even though the fundamental substances are in themselves chemically inseparable, they are by no means on that account absolutely independent; a glance at the elemental cycle shows that the simple has at bottom no greater independence than the composite. Hegel protests against chemistry's exclusive right to determine what an element is; in him the limit between independent and dependent is *logically* fluid. The peculiar way in which he has dialectically refreshed the doctrine of the four old elements is too characteristic to be passed over here.

Matter, it is said, has individuality; it has shown in the solar system that it can determine and form itself out of its own essence. What gravity is in the mechanical, light is for the elemental qualities. Light is the first qualified matter, reflected being-in-itself, nature's immediate, universal self. Light's outward existence is *the star*, as a moment in

^{*)}On chemical proportions, equivalents and atoms cf. Forelæsninger over Phil. Propæd. 1860–61, p. 200 ff.

a totality *the sun*. The sun is the existing light presupposed by the planets themselves. Light itself is no element, but the principle for the elements' development. For light has not merely darkness; as qualified matter in the form of ideality it has at the same time a plurality of qualified matters in the form of reality for its opposite. It is these realities, the light-being's immanent negations and moreover matters for themselves in outward being, that come forward as four elements, answering to the four natures uttered in the solar system: air to the solar, fire to the lunar, water to the cometary and earth to the planetary nature.

Air is the element of universality, not in the form of selfhood but immediately as being-other; air, light degraded to being for another, is heavy; light itself is without weight. Passive toward light, transparent, mechanically elastic, a creeping, corroding power, air is everywhere on the point of dissolving and volatilizing the individual.

The elements of particularity, of *opposition*, are fire and water.

In the moon immediate being-for-self has stiffened into a sulphurous being-for-self, a fire-principle without fire — the reader will perhaps best be able to think of it as a frozen fire — this stiffness (*Starrheit*) now passes over, in the region of the elements, into a restlessness existing for itself, into burning fire. Air is at bottom fire, as shows itself when it is compressed. In fire, then, air is expressly posited as what it is in the concept, as negative universality or self-referring negation. Fire is materialized time or selfhood (light identical with heat), the simply restless and consuming — a consuming of another which at the same time consumes itself and so passes over into neutrality.

The cometary element, the second member of the opposition, the neutral, is water. As the opposition sunk together in itself, water has no singularity existing for itself, no fixity or determinateness in itself. There is in this element a pervasive equilibrium of possible states; every mechanical determinateness posited in it is dissolved. The limit of form is received from another and sought in another (adhesion). Water is fluid, like air, but not elastically fluid, so as to expand itself on all sides. It is more earthly than air, seeks a centre of gravity, stands nearest to the individual and strives toward it, because it is essentially the concrete neutrality which is not yet *posited* as concrete.

The planetary, concrete element, earth, is first the still indeterminate *earthy* in general; more closely determined, it is the totality which takes the other elements up into itself and comprehends the differences into individual unity. It is the earth's destiny to appropriate the universal astral powers which as heavenly bodies have a semblance of independence, to digest them, bring them under the dominion of an individuality in which these giant limbs are reduced to moments. The first great act of digestion is the elemental process.

The physical elements, it is added, are not individualized in themselves but shapeless; they are moments of corporeality, but not subsisting bodies. The inorganically individual is for the concept the most obstinate, because matter is here specified into individuality; and yet the individuality is immediate, and as quality immediately identical with the

universal.

This, now, is Hegel's central thought; it is the last and most complete development of the doctrine of immanence that we have before us here. Of the Hegelian deduction holds what was brought out above on the occasion of the Aristotelian. The question is not in the first place how far the explanation given satisfies physics — here, with the exception of a few small errors, there is really as good as nothing that concerns physics — the fundamental question is whether such a method in the philosophy of nature has validity, and what worth can be accorded it in science.

Ueber die Natur philosophiren heisst die Natur schaffen — these words of Schelling, which the editor of Hegel's philosophy of nature has chosen as a motto^{*)}, are well suited to set the immanent standpoint in a blazing light. There is something oracular, something high-priestly, something loftily dogmatic-episcopal in the speculative self-certainty with which nature's secrets are here revealed. It looks undeniably as though Hegel had himself created the four elements in order to introduce them, if not into nature, then at any rate into the philosophy of nature.

It is in truth not a want either of scientific earnestness or of logical depth, but a misunderstanding rooted in the doctrine of immanence concerning the objective concept's self-conceiving, that lets itself be seen in this abstruse creation of elements. The apprehension of the relation of the concept of corporeality to existing corporeality is mistaken from the bottom up. Here is no ground-medium for rational-real necessity, for here are no qualitatively determinate extremes; here is, in other words, no trustworthy starting point, either in the a priori or in the empirical. Not in the a priori; for concepts such as air and fire, water and earth do not admit of being derived with logical necessity from pure thinking. The elements are not beings of thought but beings of matter. So determinations borrowed from sensation and from ordinary everyday experience are smuggled in. Not in the empirical; for it is after all supposed to look as though the smuggled-in determinations of sense were not due to experience, but were most strictly derived from the immanent concept. The logical concept repeats itself with subjective necessity in three principal forms: the universal, the particular and the singular; ergo it is to repeat itself with objective necessity in four elements. And why? Because nature, which cannot hold the concept fast, must let the moments of particularity fall apart from one another into two members and therefore needs two elements, fire and water, to express the particular. One gets the impression that the last addition derives not from any higher conceiving but from a plain everyday experience. With the application of the four natures in the solar system it looks still more questionable; here there is nothing whatever, sensible or thinkable, to hold to. That the moon is sulphur is about as scientific as that it is a green cheese. What a leap from the fire-principle in the moon to a burning fire on the earth! Worst of all, however, is the comet-water. Suppose the comets were

^{*)}G. W. F. Hegels Vorlesungen über die Naturphilosophie als der Encyclopädie der philosophischen Wissenschaften im Grundrisse zweiter Theil. Herausgegeben von D. C. L. Michelet. Berlin 1842.

waterless — and that might well be the case — then there is no water-principle at all. But when there is no water-principle in the comets, it must go either against “the immanent concept,” which then by the new theory of comets suffers a blow it never recovers from; or, if “the concept” is absolutely to be in the right, it must go against poor nature — for how shall nature be able to hold its water when there is no water-principle?

Hegel maintains that in such deductions it does not really turn upon the character of the outward, material existence; the main thing is that one can everywhere in the corporeal world see the concept. Paracelsus, he remarks, has said that all earthly bodies consist of four elements — of mercury, sulphur, salt and the virgin earth — just as one also had four cardinal virtues. Such things must not be understood according to the letter; the higher meaning is that real corporeality has four moments. — One learns from this, then, that on these presuppositions it does not help to make much of realism by appealing to the unity of the real with the ideal. When the ideal demands no more than that real corporeality shall have four moments, then one may just as well seek the moments in Paracelsus and Jacob Böhme as in Schelling and Hegel.

The limit of the independent and the dependent in the world of bodies is a movable but nevertheless law-determined *limit of existence*; the moments in the concept of corporeality are actual facts.

b) Corporeal properties. The relation between the concept of corporeality and the existing body is, as we have already seen from several sides, a relation of power. In the ground-medium the unity of concept and actuality is reflected as working and activity; but in the object-medium the concept is so immediately one with its existence that the object stands in the concept’s place. Of what nature a body is must be known from its origin; but the body’s character, the whole peculiarity with which it exists, must be known from its properties. The body with its properties is an immediate given; the given wills, in antilogical immediacy, to be seen and sensed from all sides in all ways just as it is given. This necessity of the object involuntarily makes an impression of power; but just as involuntarily object-power sets reflexion in motion again. It is impossible for the observer to sense without thinking: this is a body, this body is hard, is yellow, and so forth. So the body becomes logical subject (substratum) and the properties predicates; for experience happens only in thinking and judging. It now turns on what direction the reflexion takes. The reflecting consciousness can, as the history of science teaches, leap over the ground-medium and throw itself upon one of power’s extremes, either upon self-power in all arbitrariness or upon object-power in all accidentality. Scholastic theological metaphysics is an example of the first one-sidedness, empirical scepticism of the last.

The body’s existence is a word of power, a revelation of self-power’s will; that is scholasticism’s theological presupposition. How this presupposition was in its time applied is seen from the Catholic doctrine of transubstantiation. Bread and wine are transformed into body and blood and yet retain their sensibly corporeal properties. How

now is this possible? Here a distinction must be drawn — *distingvendum est* — between substance and accidents. The arbitrary self-power which has created substance out of nothing and imparted certain accidents to it can at pleasure let two substances of unlike kind share one and the same sum of accidents, or else re-create the substance but let its accidents remain unaltered. When a mouse eats the Most Reverend, what then does the mouse get? Bread or breadness (*panis* or *paneitas*)? God knows, answers Petrus Lombardus. An apt answer. In a natureless nature the bond between substance and accidents is so arbitrary that omnipotence itself immediately looses and binds it, according as the priest chants.

Empirical scepticism takes its starting point in the immediately given; the given's necessity is hidden in accidentality: object-power is antilogical in this and this and this. Knowing rests upon experience, but what is experienced is each time a single thing, and every single thing something accidental. We suppose we know that there is a substance called gold; but that this is a fancy can be seen from the way in which gold's properties are put together. Gold, it is said, is a yellow metal which can crystallize in octahedra; it is so malleable that it lets itself be rolled out into leaves of $\frac{1}{20000}$ of a line in thickness, and so forth — nothing but empirical marks. Were it to be shown that gold was a substance, and the substance gold the necessary unity of these properties (accidents), then it would have to be possible to show how each single property followed from the others and from the combination they form, just as with a triangle, for example, it can be shown that each single side depends upon the two others and upon the angle they enclose. But this is so far from being the case that on the contrary each single property is experienced by itself without the least indication of what it really has to do with the others. To be yellow, to be malleable, to crystallize in octahedra are in actuality three things so unlike in kind that no one readily thinks of representing them in indissoluble unity. That gold has of all the metals the least affinity to oxygen is adduced in chemistry as a universally valid proposition. But how does one prove this proposition? Grounded upon experience the proposition cannot be; for one can experience a *number of single* cases, but not *all* cases, still less *a universally valid* case. One appeals to induction; that is to say: one puts some cases for all cases; one might just as well put $1 = 2$, $2 = 3$, $3 = \infty$.

To avoid arbitrary determinations from self-power and disturbing accidentalities in object-power, the doctrine of immanence, the philosophy of the absolute with its absolute knowing, has briskly struck a line through both extremes; it will in virtue of its ideal-real principle articulate the content of rational necessity; but the articulation founders on the arbitrarily accidental determinateness of the corporeal properties.

The doctrine of immanence stands in conflict of principle with physics. Physics is indeed raised above the looseness of scepticism, for it maintains reason's right and knows the necessity in "the unalterable laws." But rational necessity is in nature real, and real necessity inseparable from accidentality. In every corporeal property the necessary has its immediate being in the accidental. The accidental is that gold is yellow, that it is

malleable, that it has a stronger affinity to chlorine than to oxygen. But this accidental is precisely existence's expression for an articulated necessity, a law-determined connexion: physics relies on induction because it relies on the law; it knows a substance in gold because it knows a factual necessity in the combination of the properties. That the necessity is factual, untransparent, physics utters by declaring that we know only things' properties, not their essence.

Against such an utterance the doctrine of immanence supposes it can set an explanation of this sort: that the thing's essence reflects itself in its properties, so that a knowledge of the properties must necessarily go together into one with a knowledge of the essence. Such an explanation is good enough in a logic; but how does it look when it is to be carried through in a physics? Let us, for example, choose the property of a body that it is *magnetic*, in order to see what the philosophy of immanence avails to elucidate about this property. Magnetism's essence is, according to Hegel, altogether transparent in its appearing (*Erscheinung*). What is essential is that the concept's different moments are not set identical in an outward manner, but that they set themselves identical. The brittle point opens itself up to separation in the concept; thus we have the poles. On the physical line, which has the difference of form in it, they are the two living ends, each of which is posited so that it is only in relation to its other and has no significance when the other is not. Both are one determination. "This appearing," he adds, "is *the whole theory* of magnetism. The physicists say that it is not yet known what magnetism is, whether it is a current, and so forth. All this belongs to the sort of metaphysics that is not acknowledged by the concept. Magnetism is nothing mysterious."

After such an utterance one should now expect that the essence cleared up in the concept would have to spread an almost divine light over the otherwise so enigmatic difference between magnetic and non-magnetic bodies; but the expectation is disappointed. "Upon what bodies magnetism comes to display itself is," it is said, "perfectly indifferent to philosophy. Chiefly it is found in iron, but also in nickel and cobalt." We learn from this declaration that actual bodies' actual properties stand in an altogether outward, accidental and indifferent relation to corporeality's essence; and yet the body's essence is immanent in its properties! The mark of accidentality is inseparable from the fact; it is facticity's immediate determinateness, its own antilogical moment. The doctrine of immanence does not dissolve accidentality, because it loftily overlooks the accidental; it does not crack actuality's hard nut, because it throws the facts away as useless shells.

That the philosophy of immanence cannot come to terms with object-power has in a deeper sense its ground in its having excluded self-power; it cannot digest the accidental because it cannot appropriate what is arbitrary in the concepts of nature; instead of nature's categories, fixed from within, it slips its own self-chosen concepts underneath, and becomes falsely arbitrary: its idealism is as hollow as its realism.

We are to go to school with nature, says the physicist; it is not we who are to teach nature something, but nature that is to teach us something. The physicist accordingly

acknowledges that in natural things he has to do with determinate, expressly prescribed thoughts and concepts. And in this he is right. The corporeal properties show that the concepts of nature have their actual expression in a categorical language of power. Gold is a metal, iron is a metal; but from the universal concept metal it is impossible to derive the particular metals' particular properties. Just as every metal has in outward being a stamp of accidentality, so it has in essence and concept the stamp of an omnipotent arbitrariness. Why is gold not magnetic just as much as iron? Indeed, why is there in the earth both gold and iron? To such questions no physicist will answer. Real concepts such as gold and iron are, in their categorically ideal determinateness, concepts of will. There is in every determination of thought a moment of arbitrariness, just as there is in every property a moment of accidentality. The moment of arbitrariness stands to *rational* necessity as the moment of accidentality to the *real*; the unity of both is *natural necessity*, a unity articulated in the ground-medium.

c) *Corporeal individuality*. That every body's peculiarity must display itself in its properties goes without saying; he who fully knows all a body's outward (physical) as well as inward (chemical) properties must in truth be said to know the body's essence. But from a human standpoint the knowledge will, whether one otherwise holds to the real or to the ideal principal side, always rest upon an approximation.

The science of experience is realistic; its separation between hard and soft, tough and brittle, elastic and inelastic bodies is grounded upon the representation of the mutual relation of the invisible parts: the forms of state are of essential significance for bodies' individuality. And yet the form of state can alter without the body's thereby losing its substantial peculiarity. Ice, water and vapour may indeed be said to be different things, since with each mode of aggregation there follows an alteration of properties; but the substance, the chemical character, is for water-vapour the same as for ice and water. The physical properties can alter, and the chemical remain unaltered. Alterations of individuality can go on without alteration of substance; but an alteration of substance cannot go on without altering the individuality.

Upon what, then, does an alteration of substance really rest? This question gets a different answer in physics from the one it gets in philosophy.

It lies in the nature of the case that the science of experience with its exact analyses must lay particular weight upon material discreteness. "In our times," says H. C. Ørsted^{*)}, "the investigations of the laws by which substances of unlike kind enter into those sensibly perfect unions which we call chemical have led to the necessary presupposition that the substances in these unions do not completely penetrate one another, as Kant and his adherents assume, but that the act of union stops at certain fundamental parts of bodies, which hang together with great intimacy without penetrating one another. Hereby we are led to a doctrine of secret fundamental parts in bodies which has so much likeness to the old atomism that occasion was hereby taken to give the chemical

^{*)}Naturl. mech. Deel. 1859. S. 14.

fundamental parts here in question the name of atoms.”

In the formation of new substances it depends of course upon the constituents’ character, quantity and position. A new body arises not merely by corresponding fundamental parts uniting in a certain proportion of quantity; bodies with a new individuality can also come forth from the same constituents in the same quantity. With these so-called *isomeric* bodies a difference must then, as regards the manner of composition, be made between *empirical* and *rational* formulas. Let a given product be, for example, $C_6 H_{12} O_4$, and so, on the empirical formula, consisting of 6 atoms of carbon, 12 parts by weight of hydrogen, 4 atoms of oxygen; what substance is it now? There is in this matter a possibility of two substances, a doubleness which finds its explanation in the empirical formula’s admitting of being separated into two rational formulas: $C_4 H_6 O_3 + C_2 H_6 O$, in which case the substance is acetate of methyl oxide, and $C_2 H_2 O_3 + C_4 H_{10} O$, in which case the substance is formate of ethyl oxide. The substantial difference rests upon a rearrangement of the parts, for which reason such isomeric formations have also particularly received the name of *metameric*. In a corresponding way the alteration of substance is in certain cases derived from the combinations containing the same substances in the same relative but not in the same absolute quantity, so that *polymeric* formations come forth. Thus $C_8 H_{16} O_4$ gives the same percentage composition as $C_4 H_8 O_2$, but since the quantity of atoms in each of the elements is different, the difference between the two products may be substantial.

The atoms’ different manner of position can of course as little be an object of immediate observation as their size, figure or weight; but what substantial significance must be ascribed to the manner in which the constituents are thought to be grouped can be taught us by the influence alone which the crystal form exercises upon the determination of the physical properties. The crystal forms are determined chiefly by the axes, the central line about which the whole formation is regularly laid out. That the axis really furnishes the essential point of support for the position — which must accordingly be different according as it happens parallel with the axis or perpendicular to it — confirms itself from many sides. The determinate directions along which the crystal cleaves stand in a relation to the axis; the same holds of the passage of light which, since it rests upon the vibrations of the parts of the aether, is precisely dependent upon the position of the atoms. Parallel with the axis tourmaline, for example, is almost opaque, whereas seen in a direction perpendicular to the axis it is transparent; dichroite, which is blue in the direction of the principal axis, is brownish yellow perpendicular to it. Something similar holds of heat. Crystals of the cubic system (the spheroidal system), which has three kinds of principal axes, conduct heat equally well in all directions; whereas calcspar and quartz, which belong to the rhombohedral system, which has only one axis, show a conductivity that is greatest parallel with the principal axis and least perpendicular to it. — From all this it is seen that the science of experience must by its character on the whole go in an atomistic direction.

Over against physics, which derives the composite bodies' substantial individuality from the peculiar grouping of unaltered fundamental parts, the speculative philosophy of immanence goes to the idealistic extreme of making the substances of matter into immediate self-beings and ascribing individual subjectivity to the elemental bodies. So Schelling in intuitive mysticism; so Hegel in overstrained logic.

"The corporeal properties, predicates of the self-individualizing matter, are real moments of an indwelling activity of form striving toward subjectivity. The movement of form springing from the principle of subjectivity runs through two stages: from gravity to sound, and from heat to the electric light. Gravity is qualified into *specific gravity*; its articulating moments — punctuality, linearity and superficiality — are particularly posited in qualified *cohesion*. Cohesion's ideality consists in the material parts, existing outside one another, being set in one another; the ceaseless alternation of the parts' being in one and being outside one another is the vibrating and trembling that lets itself be heard in *sound*. Sound for itself is individuality's self. The tone touches our inmost part because it is itself an inner, a subjective. To individuality there belong matter and form; sound is the total form making itself known in time — the whole individuality, which is nothing else than that this soul is now set in one with the material. The next is that the provisional, still imperfect individuality is dissolved in *heat*. By heat matter is restored to its original formlessness, its fluidity; it is its abstract homogeneity celebrating a triumph over the first specific determinatenesses. The advance consists in matter's continuity, existing only in essence, being expressly posited as the negation of negation, as activity, as existing dissolving. Out of this dissolving the individualizing activity of form begins afresh in a deeper sense. The immediate shape, formless in itself, is in its one extreme punctuality, in its other extreme the spherical fluidity — shape as inner shapelessness. The point must now pass over into the line, and form set itself on the line in polar extremes with an indifferent middle. This syllogism makes up the shape-forming principle of the giving of form — *magnetism*. As the magnetic activity deposits its product, the shape is determined as *crystal*. The individual bodies are now developed into physical totalities, selfish subjects whose particular predicates show themselves as physical properties in relation to the universal elements with which they enter into real processes. In relation to light, the selfhood identical with itself, the corporeal individualities utter their peculiar self-being in opacity, transparency and the many darkenings that give them *colours*; in relation to air the individuality is *smell*, in relation to water *taste*. The development of form of the still immediately independent individualities ends with an actual conflict, a physical tension of their particularity, their real selfishness, which is equalized in the electric light. It began with the bodies having light outside them; it ends with every corporeal individuality having its light, its own subjective essence, in itself."^{*)} — The arbitrariness in such a deduction is conspicuous.

Only by the doctrine of the ground-medium can the philosophy of nature be brought

^{*)}Hegels Vorlesungen über die Naturphil. S. 129–360.

into essentially true agreement with the science of experience without giving up its independence as a science of its own. In the ground-medium we have the principle for the full acknowledgment of experience's facts; for object-power is presupposed, and the concepts' issuing in power happens in masses and fundamental parts, so that these must naturally play the principal part in the composition and dissolution of the substances of matter, the corporeal individualities. In the ground-medium we have the principle for an actual articulation of the concepts of body; for self-power is presupposed: the articulating thinking goes together into one with an articulating will. The system of the concepts of power, in its infinite, peculiar connexion, is a system of will in which the concepts' transitions punctually furnish the transformations that in actuality ground the corporeal alterations.**)

What elements and bodies are for one another mutually, in the ideal as in the real sense, must be known from their mutual reciprocal actions, their law-determined activities; the law-determined activities are functions.

Chapter Two.

Elemental Functions.

In the elemental world of bodies the existing bodies are to be apprehended: partly in reflected being as moments for the corporeal whole, and so subject to the *universal*; partly as parts, subsisting members which under many-sided conflicts hold themselves against one another in the *particular*; partly, finally, as substantial products, peculiar productions of the ground of nature at work in *the individual*. Insofar as bodies stand, according to their particular character, in a relation to a qualitatively determinate, universally corporeal essentiality, the activities flowing from this show themselves as *dynamic* functions, answering to *light and heat*. Insofar as what is proper to the bodies themselves maintains itself under mutually straining oppositions, the reciprocal action passing through the dynamic universal essence will come into view in *articulating* functions, answering to *magnetism and electricity*. Insofar, finally, as the whole activity of form is so pervasive that the members become moments and the moments members of individual formations, we have the *body-forming* functions, answering to *chemism and chemical processes*.

A.

**)On „Forandringsproblemet i Philosophien“ cf. Forelæsn. over „Phil. Propædeut.“ 1860–61, pp. 143–177.

Dynamic Functions: Light and Heat.

The usual representation of the dynamic — that substances are transformed into forces, penetrate one another and the like — is unclear and without significance in science. The dynamic has the mechanical for its foundation and consists in the active parts' finite discreteness approaching the infinite to such a degree that it makes an impression of continuity.^{*)} When light and heat are referred to vibrations in the aether, the activity is mechanical; but what is peculiar here is that the parts of the aether and their motions melt together in representation in such a way that the single terms of the relation are not held fast for themselves but merge as moments in the activity. The closer difference is determined by the qualitative. For gravitation's abstractly mechanical relations of weight all matter is of one kind; here there are only the heavy masses and the empty space. The dynamic in light and heat is by contrast conditioned by a qualitative difference in matter itself, an essential opposition between ponderable and imponderable matter. To call the aether an ideal matter is however superficial, for the aether fills its space just as much as the weighable masses do. When it is said that the aether "penetrates transparent bodies," then the dynamic in this penetration is a reflex of the mechanical. Even if every atom of a body is "surrounded by a sphere of aether," it is nevertheless unthinkable that the atom of aether should penetrate the body's atom and punctually occupy the very same space as it. That the aether is without weight means only that on account of its extremely slight density it is unweighable; but it by no means proves that it is in itself without gravity. If its density be assumed to be 845,000 million times less than air's, so that a cubic mile of aether would contain no more mass than $1\frac{1}{3}$ pounds^{**)}, then it is intelligible from this that the aether may very well be unweighable without being withdrawn from the law of gravity. The ideal moment in the aether is that it immediately represents universal matter, a hypothetical foundation for pervasive dynamic activities. But the opposition between the ideal and the real does not lie in matter itself; it rests upon the peculiarity of the functions. The dynamic ideality is in *light* what the corresponding reality is in *heat*; the reciprocal determination of the ideal and the real is articulated in *the relation between light and heat*.

a) *Light*. To know what light is, we must begin by investigating the active element upon which its being rests. Here, then, two activities of unlike kind must be presupposed, one in the form of otherness and one in the form of selfhood, a making-visible and a seeing, a material and an ideal activity: light itself is the reflected unity of the two activities of unlike kind. This can from the standpoint of natural science be uttered in the following two propositions: 1) Light rests upon a trembling in the aether; when the aether is at rest, there is no light. 2) Light rests upon an activity of sight; where there is no seeing, there is no light. — Light's essence is accordingly at once both objective and subjective. But although the two activities cannot possibly be separated from one

^{*)}Cf. Grundideernes Logik I, S. 354–69.

^{**)}C. Holten. Lysets Naturlære. Kbhvn. 1861. S. 167.

another without light's disappearing, there is nevertheless nothing to prevent each of them from being considered for itself — only that in considering the one, one must constantly be conscious of understanding the other along with it. We treat first the objective essence and then the subjective, in order to reach a developed reflexion of light's real-ideal nature.

α) Light's objective essence. Seen from the side of nature, light must necessarily have a material foundation answering to its activity. By apprehending the cause insofar as it reflects itself in its effects, natural science seeks to make clear from all sides the activity conditioned by the foundation. What is not seen reflects itself in what is seen: light's objective essence can be determined only by its objectification. What is peculiar about the hypothesis of vibration is that the sensible element understood along with it disappears for observation; for what is observed is not the vibrating parts of the aether, it is phenomena which as effects are referred to vibrations as to invisible causes; what is fleeting in the representation transforms itself into a reflex of the motion: concepts and laws enter into reflexions of the cause.^{*)} The cause of the vibration divides itself originally into two causes, a moving and a living force, which are ceaselessly converted into one another, ceaselessly consume and bring forth one another. This working is a pervasive reflecting; that the cause works dynamically means: it is reflected in its mechanical moments.^{**)}

If the opposition of light and darkness were merely immediately qualitative — either light or darkness — then the concept of vibration in the form of universality would be satisfactory. But the very fact that the illumination is weakened with the spreading presupposes a law and demands a grounding. The grounding follows naturally from the concept's development, for the concept of vibration presupposes that the waves spread themselves as spherical surfaces about the luminous point, and that the aether is perfectly elastic. In two concentric spherical surfaces the sums of the living forces will accordingly be equal. Since now the spherical surfaces' mass grows with the distance, and the spherical surfaces stand in direct ratio to the squared radii, the living forces —

^{*)}It lies near to compare water-waves, air-waves and aether-waves with one another; but they are nevertheless very different. Cauchy has — in his work *Memoire sur la dispersion de la lumière* — treated the aether-waves with atomism for a foundation and derived correct formulas for the dispersion of colours. From this Fechner has concluded: ergo atomism is proved. But the same formulas can be derived with the doctrine of continuity for a foundation; one might then with the same (in)justice conclude: ergo continuity is proved. What can be shown on both presuppositions is that the mechanical here passes over into the dynamic.

^{**)That a part of the aether is in vibration means that it moves to and fro perpendicularly over a given straight line; in the line itself it has its position of rest. If the distance from the position of rest be denoted by $\pm y$ answering to the time t , then the maximum for y can be denoted by $+a$ on the one side and by $-a$ on the other side of the line. If the aether-part's mass be called m , $m\varphi$ the moving and mv^2 the living force, then the relation between φ and v^2 is the following: $\varphi = -ky = \frac{d^2y}{dt^2}$; $\frac{dy^2}{dt^2} = v^2 = k(a^2 - y^2)$. From this it is seen that when $y = a$, then φ is at its maximum but $v = 0$, that is, when the moving force is greatest, the living force has disappeared. When $y = 0$, and so the aether-part is in the position of rest, then $v^2 = ka^2$ but $\varphi = 0$, that is, when the living force is greatest, the moving force has disappeared.}

and so the strength of the light — must stand in inverse ratio to the squares of the radii, that is, of the distances.^{*)} But now the opposition between strength of light and colour? For the grounding of this opposition too the concept of vibration contains the necessary conditions, namely such particular moments as the amplitude, the distance from the position of rest in the running time, and the time of vibration.^{**)} The strength of the light rests upon the amplitude, the colour upon the time of vibration. From this it is seen that the amplitude can be different and the time of vibration the same, that is, the strength of light can be different and the colour the same; conversely the amplitude can be the same and the time of vibration different, that is, the strength of light can be the same and the colour different.^{*)} Finally, the concept has in it the possibility of taking up all the particular conditions from which the laws for the aether-waves' forward-advancing motions^{*)} and their mutual actions upon one another admit of being derived. It can be exactly calculated how two light-waves which propagate themselves simultaneously

^{*)}For a spherical surface with mass nm let the radius be r and the velocity of vibration v ; for a spherical surface with mass $n'm$ the radius r' and the velocity of vibration v' ; one will then have $nmv^2 = n'mv'^2$. Since now $\frac{n'}{n} = \frac{r_1^2}{r^2}$, it follows that $\frac{mv^2}{mv'^2} = \frac{r_1^2}{r^2}$.

^{**)From the equation $\frac{dy^2}{dt^2} = k(a^2 - y^2)$ one obtains $\sqrt{k} dt = \frac{dy}{\sqrt{a^2 - y^2}}$ and from this again, by integration, $\sqrt{k} t + C = \arcsin\left(\frac{y}{a}\right)$, and so $y = a \sin(\sqrt{k} t + C)$. To $t = 0$ answers $y = 0$, consequently $0 = a \sin C$, and $C = 0$ — in the first quadrant — 0 . One accordingly has $\sqrt{k} t = \arcsin\left(\frac{y}{a}\right)$. If the time of vibration be denoted by T , then $t = \frac{1}{4}T$ answers to $y = +a$, and so $\sqrt{k} \frac{1}{4}T = \arcsin(1) = \frac{\pi}{2}$, $\sqrt{k} = \frac{2\pi}{T}$, and the formula of vibration $y = a \sin \frac{2\pi t}{T}$, in which a is the amplitude.}

^{*)}If the velocity of the light-waves be denoted by V , the time of vibration by T , the wavelength by β and the number of vibrations by N , then one gets $T = \frac{\beta}{V}$ and $N = \frac{V}{\beta}$, from which it is seen that the time of vibration is in direct and the number of vibrations in inverse ratio to the wavelength. The red rays have the greatest, the violet the smallest time of vibration; when 500 billion vibrations in the second give red, then 733 billion give violet. The strength of the light, by contrast, has for its measure the square of the velocity with which the aether-part passes through the position of rest. From the formula $y = a \sin \frac{2\pi t}{T}$ there follows, namely, $\frac{dy}{dt} = \frac{2\pi a}{T} \cdot \cos \frac{2\pi t}{T} = \frac{2\pi}{T} \sqrt{a^2 - y^2} = v$. In the position of rest $y = 0$, the velocity $\frac{dy}{dt} = v = \pm \frac{2\pi a}{T}$, and consequently $v^2 = \frac{4\pi^2 a^2}{T^2}$. "Since, however, it does not turn upon an absolute determination, one may just as well use a^2 as the measure of the strength of light, so that two equally strong" (monochromatic) "rays of light which meet under conditions as favourable as possible will give an illumination 4 times as strong as one of them." C. Holten. *Lysets Naturlære*. S. 173.

^{*)}If a wave propagate itself from the point m , in which the vibration is denoted by $y = a \sin \frac{2\pi t}{T}$, to m' , the way being x and the time employed upon it θ , then one gets for the vibrations of the aether-part m' the formula $y = a \sin \frac{2\pi(t - \theta)}{T}$. If two waves are simultaneously to run through the same line, but from different starting points, so that the one traverses the way x , the other x' , before they arrive at m' , then — on the presupposition that the wavelength β is for both the same, but the amplitude for the one a , for the other a_1 — the vibrations of m' will make up the sum of the vibrations of both and be expressed by the formula:

$$y = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\beta} \right) + a_1 \sin 2\pi \left(\frac{t}{T} - \frac{x_1}{\beta} \right).$$

along the same line, but from different starting points, will meet, in which cases they will strengthen and in which cases they will cancel one another; from this it can be explained that light with light can give not merely strengthened light but also darkness.

From these indications it must surely be possible to judge in what sense the concept of vibration answers objectively to the concept of light and determines light's cause in nature in such a way that the whole system of particular causes of particular phenomena of interference, reflexion, refraction, polarization^{*)} and so forth, with all the modifications and shadings belonging under them, admits of being developed from one fundamental cause. In the reflected unity of the phenomena of light with the natural concept of cause we know light's objective essence.

β) Light's subjective essence. Without sight no light. Apprehended from this side, light accordingly becomes not a function of the vibrating aether, not a function of otherness but one of selfhood, a function of the seeing self. This, considered for itself, is light's subjective essence, its ideality. This ideality, though it is demonstrably subjective, the speculative philosophy of nature has none the less transformed into something objective in nature, and thereby given a new proof of what ambiguities of abstraction the doctrine of immanence must resort to when it seriously attempts to engage with actuality. Light, says Hegel, is matter's pure selfhood, over against which matter itself, even according to its immediate being, is the selfless, darkness. Light is the infinite oscillating in itself, the pure reflexion into itself, the same as what Spirit's I is in a higher form. Space, light and Spirit so far denote a stepwise development of form of the same essence. "I am the infinite space, self-consciousness's infinite likeness with itself, the abstraction of empty self-consciousness, the pure identity of me with me. I am only the identity in the relation of myself as subject to myself as object. With this identity light is parallel; of this it is a faithful image. It is only not I because it is not darkened and broken in and by itself, but is only the abstract shining."

By zealously insisting on light's objective ideality the doctrine of immanence has supposed that it was really fighting for the right of spirituality, and has on this account raised a vehement — one may well say fanatical — polemic against physics. It is especially Newton who has to suffer for it; Hegel will show in his experiments insipidity, incor-

If $\frac{t}{T} - \frac{x}{\beta}$ be denoted by $\frac{t_1}{T}$, $x' - x$ by e , and $a = a_1$ be assumed, then the calculation shows that $y = 2a \cos \frac{\pi e}{\beta} \sin \left(\frac{2\pi t_1}{T} - \frac{\pi e}{\beta} \right)$; from this it is seen that for an odd number of half wavelengths one gets $y = 0$, so that light with light gives darkness; for an even number, by contrast, $y = 2a \sin \frac{2\pi t_1}{T}$, so that light with light gives strengthened light.

^{*)}Reflection (tilbagekastning) and refraction (brydning) always happen according to the same laws as regards direction, but not as regards intensity. Two parallel mirrors A and B with an inclination of a little over 33°, so that the angles of incidence and emergence at both become about 57°, show that the rays cast back from A are either wholly or only partly or not at all cast back by B, according as the angle between the mirrors' planes is 0°, 90°, 180° or something in between. Hereby light's polarization is determined. Usually, it is said, *the vibrations perpendicular to the ray go on in all directions, but under certain conditions the vibrations are bound to a certain plane, and the light is then said to be polarized.*

rectness, indeed dishonesty. The coarsest representations are ascribed to the physicists: that they speak of bundles of rays, of four-cornered rays, that they — *horribile dictu* — regard light as something material and composite and let every single ray have its single colour in the white light, just as every single horn has its single tone in a Russian horn band, and so forth. With all this, however, what is really the main thing in physics' exact investigations is completely misunderstood, and that to great damage — not for physics, but for philosophy.

Physics does not teach us that the vibrating aether is in and for itself light; it makes the eye the mediating organ between the active element in nature and the activity of sight, and apprehends the cause of light as a law-determined reciprocal action of two causes of unlike kind, an objective one in the aether and a subjective one in the soul. That physics will not engage in a particular explanation of light's ideal essence is no proof that it denies light's ideality; it is only a proof that it neither is nor will be philosophy. Physics holds fast its standpoint, knows its problems and knows what it has to do as an investigating science. It cannot, as regards light's side in nature, content itself with general turns of speech about "oscillating," "reflecting," "manifesting"; it develops the doctrine of vibration into an exact system of functions and laws. In expressly presupposing the activity of sight, physics teaches neither that the strength of light straightforwardly *is* a square of the amplitude, nor that the time of vibration *is* the colour, so that 500 billion vibrations in the second are red and 733 billion violet; but it teaches that the different sensations of light which the eye receives are effects of different corresponding causes, and shows in this respect exactly that the strength of light is a function of the amplitude, and the colour a function of the time of vibration. Physics does not teach that the rays of light are material lines shot out in bundles from the luminous body; it teaches on the contrary that they are mathematical lines, radii, that is, lines of direction upon which the wave-surfaces stand perpendicular. Expressions such as cones of light, cylinders of light, bundles of rays and the like are partly geometrical determinations of form, partly figurative designations aiming at securing the representation and supporting the intuition. Optics divides and composes the rays of colour, not in a material sense, but in a manner corresponding to the way mechanics divides and composes forces.^{*)} Physics cannot, as regards the wave theory, possibly assume that white light is a sum of colours existing in it; for when all colours melt together in the white, then they are precisely by this melting together cancelled; the vibrations for white light must then form a common wave in which the particular colour-waves are taken up and contained. It assumes, by contrast — and with full right — that white light is the actuality in which all the colours have their possibility, that the different colours come forth when the light is broken and the common wave divides itself into particular waves with particular times of vibration.

Had the speculative philosophy of nature seen itself able to give as sufficient an

^{*)}This can among other things be seen from the fact that Cauchy, for example, introduces imaginary waves into the analysis. Cf. Holten, *Anf. Skrift*, S. 280.

account of the subjective functions in the activity of sight as physics has given of the objective ones in the activity of nature, then it would have stood otherwise with the apprehension of light's ideality than it now does.

To place light's ideality in "matter's pure selfhood" is meaningless, for matter has no selfhood at all; it is precisely matter's character to be the selfless other. There is as little any immediately and straightforwardly luminous matter as there is any seeing, self-intuiting matter. What is luminous in matter is an activity, but an activity in the form of otherness; therefore this luminous element can shine neither for itself nor for any matter whatever. This is the contradiction: that the luminous element in matter cannot shine unless it is taken up into an activity of sight, and so into a self-activity; only as the self-activity sets in does nature, so to speak, get the benefit of the activity in another, for now what is luminous can shine, now matter can be illuminated.

That the activity of sight, though conditioned and determined by mechanically dynamic causes, makes what is not in itself light into something luminous, makes functions of another into functions of its own, of selfhood, is in the form of ideality a categorical expression of light's subjective essence.

γ) *Light's real-ideal nature.* From the objective side light is a function of the vibrating aether; in this consists its reality. From the subjective side it is a function of the seeing self; in this consists its ideality. Light is accordingly an indissoluble unity, reflected in its essence, of real and ideal, of objective and subjective. None the less this indissoluble unity is at the stage of finitude nevertheless dissoluble; the objective activity of sight falls in actuality outside the subjective. Light shines, after all, whether we see it or not; the sun of day would shine over the earth even if all the eyes of animals and men were closed. So strong is the fundamental sensation of the self-validity in light's shining, its changing clarity and the colours of things, that we are tempted at every moment to transfer its ideality into outward nature: only when by conferring with the speculative philosophy of nature we have properly come to our senses does the temptation pass.

The contradiction that light in nature has no ideality, and that light's nature is nevertheless both real and ideal — this apparent contradiction, existing in the semblance as semblance — cannot be resolved from a merely finite standpoint; its resolution is in the fundamental relation of nature and Spirit. If Spirit, the Spirit almighty in itself, which grounds and bears the whole of nature, were merely a thinking and not at the same time an intuiting, seeing being, then the concept of natural light would in actuality be as contradictory as the concept of a round square. But Spirit is intuiting, seeing, objectifying self-power; in self-power's objectification light has an objective natural ideality independent of all finite seeing.

When, however, it is particularly brought out that natural light — the light here spoken of — is not a spiritual but a sensible light, an objection arises. Sensible light is, after all, only insofar as it can be sensed and seen with sensible eyes. If Spirit by its seeing and intuiting were to give light sensible ideality, then this seeing and intuiting

would have to be sensible, not spiritual; Spirit as Spirit would then have to have sensible eyes. We must, as regards objections of that kind, dismiss all loose representations of a loose imagination. Here it is not asked what and how many eyes Spirit has, or how a Spirit looks with sensible eyes; here it is asked how it is possible that light can have an ideality independent of finite seeing; and the answer runs: it is possible only in virtue of an infinite seeing; possible only because the whole of nature has its ideality from Spirit.

That nature's ideality and its sensible reality are not finitely put together but are grounded in an original unity can be known from the ground-medium. In the ground-medium the causes of unlike kind whose co-working is the distinguishing mark of all aesthetophysical functions^{*)} have their common source. From the unity in the ground-medium the separation issues in the object-medium; upon this rests all the parcelling out of finitude. Outwardly regarded, the eye and light stand in an accidental relation; they can meet accidentally and be separated accidentally, for the eye can be without light, and light without an eye. But in nature and essence eye and light are mutually presupposing concepts, real concepts of the same conceiving: light's concept is originally reflected in the eye's concept, just as the eye's concept — as its structure sufficiently shows — is reflected in light's concept: upon the double-sided reflexion of the concept of light in the aether and the concept of light in the eye rests light's real-ideal nature.

b) Heat. Of heat too it holds that the objective is determined subjectively; feeling is for heat what sight is for light. None the less a difference shows itself here; for in the doctrine of heat the outwardly immediate objectivity plays a far more prominent part than in the doctrine of light. In all phenomena of light the effect of light must unbrokenly be determined by sight; but the phenomena of heat cannot be determined in that way by feeling. Feeling is a most imperfect measure of heat; so the objective side — the alterations which heat effects in bodies, the expansion particularly — must be brought out from the beginning. The expansion lies at the foundation of the construction of the thermometer. By the thermometer the subjective in the apprehension is altered into something objective for sight, the objective sense steps into feeling's place: heat, though invisible, becomes visible. But although heat is real in another way than light, it will nevertheless be possible, in good agreement with physics, to show that it too has its ideality, and that it is precisely this ideality that grounds its real activity.

α) Heat's ideal essence. Physics speaks of "the unknown cause of what we call heat." How matters at bottom stand with the representation of known effects of an unknown cause has already been elucidated in the doctrine of force. As regards heat we have only to show how the concept of cause originally stands to the concept of the elemental function. In the form of fact, and so outside the concept, the function is pieced together out of finite presuppositions. Heat's cause, what warms, is represented as a something for itself, an unknown matter, while the effect — the alteration that goes on with the warmed body — is observed for itself and drawn in under what is known. The warming,

^{*)}Cf. Grundideernes Logik I, S. 369–87.

the function itself, then becomes from this standpoint both inexplicable and explicable: inexplicable insofar as it has an unknown cause; explicable insofar as the visible effects admit of being observed. It is otherwise when the function is apprehended according to its concept, and so from the side of essence. That heat essentially consists in an activity no one will surely deny. One may now provisionally form whatever representations of the material element in the cause one will; it is nevertheless incontestable that what warms is a cause which must expressly be determined and known from its effects. Were heat without more ado a property of a matter of heat, and a body's warming to consist solely in the warm matter's penetrating the body in smaller or greater quantity, then there would be no activity of heat at all: the warming itself would be a pure fancy. The matter of heat does not become warmed, for it is warm. Nor does the body become warmed, for the heat in the body does not really concern the body; it is on this presupposition not the body but the matter of heat in which the heat is, and the matter of heat is warm. If one will not stop at the empty, meaningless representation that the warm is a matter of heat and the matter of heat the warm, then one must necessarily concede that heat is activity, and that what is warmed takes part in the activity just as much as what warms. That there is something which warms rests upon there being something which receives heat, which lets itself be warmed. But to receive heat, to let oneself be warmed, is just as surely conditioned by an activity as is warming. The working and the suffering do not fall apart as two things: the working is suffering, and the suffering working. Here is no unknown cause in outward opposition to a known effect, for the cause is reflected in its effect and the effect in its cause: the elemental function is a function of dynamic reciprocal action. Heat as cause, heat as effect, and the warming answering to the reciprocal action are side-concepts of one and the same concept of nature, heat's real concept. This real concept is heat's ideality.

Considered for itself as a universal cause reflected in its effects, heat is not an extensive but an intensive magnitude. It rises and falls in continuous transitions, and there is as little any absolute limit to its falling as to its rising. The whole of heat is in every degree of heat, and yet heat is different from itself in its different degrees. It has according to its ideality all conditions enclosed in its real concept; it grows and decreases in itself, by itself, out of itself: there is nothing from without that comes either to give or to take away heat.

But, one objects, the bodies which are warmed do come from without! That is, as regards the essence, a great misunderstanding. Just as the cause must reflect itself in its effects, so in turn the many effects reflected upon the bodies must reflect themselves in the cause. What the bodies are for themselves in other respects, and whence they come, does not concern heat; only what they show themselves to be according to their share in the activity of heat concerns heat. The corporeal forms of state have no subsistence outside or alongside universal heat; they are on the contrary forms of transition for a determining activity of heat, phenomena of heat's essence, functions of determinate

degrees of heat. That the forms of state stand in heat's service and make up moments of heat's concept can be seen from the transitions. If a body passes over from solid to fluid, or from fluid to gaseous state, then heat is bound; if the transition is the reverse, then heat is set free. From this it is seen that heat comes to existence in the form of corporeal states, that every form of state expressly represents a certain determinate capital of heat. That heat is bound means that the capital is increased, because the state entering is itself the measure of the increased capital; that heat is set free means that a deduction is made from the capital, because the state entering is precisely the expression in existence of the diminished capital. If specific heat be made the measure, the same repeats itself: bodies are measured with one another in being measured with heat; but heat is, by means of the bodies, measured with itself, for the measuring is done with units of heat. From this standpoint it shows itself that the otherwise different concepts of body are all comprehended in heat's universal real concept; that is heat's ideal essence.

β) Heat's real essence. From the standpoint of ideality, working and activity merge into reflecting and conceiving; but the reflexion of power demands at the same time the reverse: that the actual conceiving be expressed in a real manner in working and activity. For reflexion and concept the active members are reflexes, moments, side-determinations; for the sensible, outward-going activity, the actuality visible in facts, the real concept's essential moments are existing, active members. Thus heat gets its reality in virtue of its real concept. Heat as universal cause must then, like light, have its material foundation and in it a particular subsistence for itself. Over against this the different bodies then come forward, each with its material independence, and come from without under the heat communicating itself in different measure, and communicate heat to one another in turn. The warming, which rests precisely upon the real opposition of heat and body being converted into a real unity, hereby gets the character of material activity. That an actual unity of opposition cannot possibly be apprehended as straightforwardly existent is easy to see. The immediately existent unity of opposition is, namely, a contradiction, since it is composed of two mutually excluding determinations. If the opposition be posited as existent, then the unity is excluded; if the unity be posited as existent, then the opposition is excluded: the resolution of the contradiction is activity. The activity, the dynamic function of warming, consists in a ceaseless conversion, a reciprocal action in the course of which the unity proceeds out of the opposition — so that heat becomes corporeal and bodies warm — and the opposition again out of the unity, so that there is nevertheless an actual difference between heat and bodies. Hereby heat's real essence is determined in general.

If this actual essence is now to be more closely investigated; if the investigation is to go into detail; if it is the activity's inner material *how* that is the object of the investigation — then physics has surely ground to speak of the “unknown” in heat's cause. The unknown, so understood, is to be discovered by the way of observation, through an approximation. It lies near to let heat have the same foundation as light; so

physics then assumes that heat, in analogy with light, must rest upon vibrations in the aether. The real concept to which the philosophy of nature has to hold is accordingly the concept *analogy*.

As a concept of fact, analogy everywhere shows in the likeness a recognizable unlikeness, but in the unlikeness again a recognizable likeness; for the likeness must in the foundation be what overreaches. As regards radiant heat, the likeness to light stands in the foreground; as regards the conduction of heat and a number of other phenomena, it seems by contrast as though the unlikeness were on the point of veiling the likeness. It is a well-known matter that heat can spread itself in rays like light; that the rays of heat are refracted, and refracted differently, like the rays of light; that they are cast back according to the same laws: the angle of emergence = the angle of incidence, and so forth. What the difference between transparent and opaque bodies is for light, the difference between diathermanous and athermanous bodies is — as Melloni has shown — for heat. In all this the likeness is striking. But heat is not light; so the unlikeness also comes into view, among other things in this, that dark bodies — bodies which do not radiate light — can radiate heat. So analogy takes the unlikeness up into the likeness, and the explanation is that the mere rays of heat, even though they can, as Tyndall has recently shown, pass through the eye's aqueous medium, nevertheless cannot act upon the optic nerve, because the waves are too broad and the vibrations too slow.

The likeness's confirmation through the unlikeness is seen in a striking manner, among other things, from the fact that a dark body — a piece of iron, for example — which during the heating sends out ever more and more refrangible rays, becomes luminous as it becomes glowing, first red-hot and at last white-hot.

It is otherwise when heat propagates itself not by radiation but by conduction, passing by immediate contact from the one body to the other and spreading itself through the whole mass. Here the unlikeness becomes so great that the analogy at first glance seems to disappear; for to the highly characteristic difference between good and bad conductors of heat there answers no such difference between good and bad conductors of light. But the unlikeness bends again under the likeness, and the explanation is that the waves of heat — the vibrations of the aether which bring forth heat in the body — at the same time set the body's own small parts in vibration. The peculiar reciprocal action between the vibrating parts of the aether and the body's vibrating small parts finds a confirmation in the doctrine of specific heat and atomic heat: the analogy is re-established.

The attempts made from several sides (Redtenbacher and Clausius) to form a quite clear representation of the relation between the vibrations of the aether and those of the small parts — and so of the inner how of the reciprocal action — have no scientific significance as hovering hypotheses, but would have significance if there could thereby one day be reached a presupposition that really grounded the laws of heat; for thereby a deep and secure insight into heat's real essence would be opened to us.

y) *Heat's ideal-real nature.* How intimate the natural connexion is between heat's

ideal and real essence can be seen from the laws. Heat penetrates and expands all bodies; this fundamental law is in equal degree decisive for the nature of bodies and for the nature of heat. What is realistic is that heat and bodies have each their own nature with which they enter into reciprocal action; what is idealistic is that the two natures are at bottom one nature, and that it is precisely the unity of the fundamental nature that makes the reciprocal action, and with it the fulfilment of the laws, possible. Both conceptions find equal confirmation in the doctrine of cohesion.

On the realistic conception the body's fundamental parts would, if heat did not work, only draw themselves together, for which reason every body becomes firm and stiffens in the cold; but the aether surrounding the fundamental parts works by its vibrations repulsively, removes the trembling parts from one another and effects an expansion. Cohesion accordingly rests upon a double nature at work in bodies: that of the parts of the aether and that of the fundamental parts. The realistically mechanical way of seeing, however, dissolves of itself into the idealistically dynamic. It is a fact that it is by no means always increased heat that causes a body's expansion, but that a diminution of heat can effect the same. Pure water at 4° expands on being warmed, but it also expands on being cooled. Now it would be inconsistent to assert that the cause of the water's expansion above 4° lay in the vibrating parts of the aether, while the cause of its expansion below 4° lay by contrast in the vibrating parts of the water. But when the parts of the aether and the corporeal fundamental parts have, in the dynamic reciprocal action, an equally essential share in the expansion, this means in other words that the nature of heat and the nature of water work as one fundamental nature. The unity of the two natures has its exact expression in the law; the law is essence, for it is concept.

If we consider the particular laws, the realistic separation returns: the double nature comes forward, but always on the foundation of unity, that is, of ideality. The law for the expansion of solid bodies is in its universality to be expressed thus: the coefficient of expansion of volume is three times as great as the coefficient of expansion of length. In this universal the unity of essence, and so the ideality, is uttered. The coefficient of expansion of length — the figure indicating how great a part of its length at 0° a body expands under an increase of heat from 0° to 100° — is however different for different bodies: platinum, glass, iron, copper and so forth. Does this difference now fall outside or inside heat's nature? In reality it falls outside, for the coefficients rest upon the peculiarity of the bodies and are so far no concern of heat's. And yet in ideality it falls inside; for the actual expansion must, according to the law, always presuppose an actual coefficient, and the one actual coefficient is in the determination of the activity of heat just as essential as the other: the difference of the coefficients is a necessary condition for particular utterances of heat's nature. — The law for melting has its universal expression in the two determinations: the unalterability of the melting point, and the binding of heat. What is realistic is seen at once from the position of the melting points at different heights — when ice melts at 0° , then hammered English iron melts at 1600° and quicksilver at

÷ 39° — but the different binding of heat proves in turn what is idealistic, namely that the corporeal differences concern heat as much as they concern bodies; for the bound heat is just as much the ground of the altered state as the alteration of state is the ground of the binding of heat: the nature of the bound heat is the state, and the nature of the state the bound heat. — When the degree of heat in a body whose weight is m and whose specific heat is c is increased or diminished by t degrees, the body must receive or give off a quantity of heat which can be expressed by the product mct . This law, which utters in the product that weight, specific heat and degree of heat are not merely factors of a mathematical multiplication but real moments in an actual concept of nature, then also shows — as on the whole do all the laws of heat — how intimately the nature of heat and the nature of bodies make up, through all real differences, one indivisible fundamental nature.

Heat without bodies and bodies without heat are unnatural things. The subjective moment in heat's essence does not indeed come into view so immediately as in light; but the real concept, the pervasive concept of power which makes the actual differences into moments for the active unity, is for heat just as subjective-objective as for light. Heat has, like light, its real conditions and sources in nature, because it has, like light, its ideal conditions and sources, its real concept, in self-power's infinite conceiving, in Spirit.

c) Light and heat. The dynamic functions of light and heat presuppose, as we have seen, a universal material foundation upon which the activity of light becomes real and the activity of heat ideal — yet so that ideality is what overreaches in light, reality in heat. There form themselves accordingly, upon the foundation, two systems of functions analogous to one another and yet of unlike kind.

That light with all its reality is idealistic is seen from its relation to sight. The whole outward world is a world of images of light because it is a world of images of sight. We suppose we see the actual, material objects themselves, but that is — as optics sufficiently proves — a fancy. The space with objects that we see in the room has as little immediate actuality as the space with objects that we see in the mirror. There is indeed an actual difference between the mirrored space and the space in the room, but this difference is not objectified by mere sight. In light for sight everything is semblance, reflected semblance, mutual semblance; the difference of the colours rests upon a difference of semblance: it is a pervasive illusion. But this illusion is no deception; it is on the contrary an ideality of things, whereby they show themselves as phenomena in visible forms and relations and therein display their essence.

But what now is the world of bodies in relation to heat? Heat governs the forms of state: all solid bodies can melt. One has indeed hitherto, it is said, not availed to melt coal, “even though several physicists have claimed to have found traces of melting on the edges of the diamonds they have experimented with. By analogy one must conclude that there is no perfectly unmeltable body, but that all would melt if only one could bring forth sufficiently high degrees of heat.” By analogy one must then also conclude

that all bodies can evaporate at sufficiently high degrees of heat. With respect to heat the whole corporeal world is, according to its material essence, to be apprehended as a world of vapours; the difference between these vapours is again, when heat is thought to overstep all known limits and to rise to a mythical heat, vanishing. Everything is a vapour, and matter a uniform, space-filling being; that is heat's judgment upon the corporeal world. Although there is nothing illusory in the activity of heat, the ideality is nevertheless recognizable in this, that bodies' immediate independence perishes in heat: heat is accordingly the ground of inwardness upon which their outward coming-forward, their peculiar being, generally rests.

The two systems of functions, light's and heat's, proceed from one fundamental system; in this fundamental system all the real concepts of the world of bodies are enclosed; the whole of nature is of one casting, a total concept grounded and borne by infinite conceiving.

But that bodies, in virtue of the comprehending total concept, nevertheless also have an independence of their own over against light and heat can be seen from their properties. Colour, for example, is just as much a property of certain bodies as of light. The body that casts back the red rays is red; the one that casts back the blue is blue. The relations, in this respect very entangled, rest as much upon the inner structure of bodies as upon the refrangibility of the rays of light.^{*)} As regards heat, bodies show by the difference between their melting points, binding of heat, specific heat, diathermancy, conduction of heat and so forth, that it is they which, by their different properties, bring something peculiar of their own into the universal activity of heat.

By being set in relation to light and heat, bodies are at the same time set in relation to one another. If rays of heat which have passed through a glass plate be let fall upon a plate of alum, they are almost entirely absorbed, whereas a plate of alum lets through very nearly all those rays of heat which have previously passed through a plate of citric acid. Rays of light which have passed through a green glass can, as is well known, easily pass through another green-coloured glass, but are absorbed when they fall upon a red glass.

Bodies' peculiarity is an individual that serves the universal and is mastered by the general in offering it resistance. Were bodies so dependent that light could really transform them into semblance, then pure, unbroken light without shadows and without colours would be as good as pure darkness. As phenomena of light, bodies are phenomena of their own essence in being phenomena of light's essence. If all bodies had one and the same melting point, so that heat could transform them all into vapour at the same time, then universal heat would be as uniform and as insignificant as the universal vapour in the universal space. It may then also be said that bodies, in being phenomena of heat's essence, are at bottom phenomena of their own essence.

If particular weight now be laid upon bodies' own peculiarity and upon the self-

^{*)}Cf. Holten, *Lysets Naturl.*, S. 190, 308 and elsewhere.

subsistence reflected in the universal, then the dynamic universal that dwells in the bodily individualities takes on new and peculiar forms, and the functions alter their character in consequence of what is particular.

B.

Articulating Functions: Magnetism and Electricity.

In relation to light and heat, bodies do indeed display a multitude of different properties, whereby they make good their mutual inequality and the corresponding self-subsistence; but the opposition of the individual and the universal is nevertheless equalised, in the merely dynamic functions, in such a way that the universal determinations retain a one-sided preponderance. Only when the dynamic functions pass over into becoming articulating functions, functions of polar oppositions, as in magnetism and electricity, do the bodily peculiarities begin to come forward in full light; but, remarkably enough, it is then also that what is proper to the bodily essence begins in earnest to show itself as an impenetrable riddle.

The universal in magnetism and electricity is neither so at one with the essence of ponderable matter that it can be grasped as a property in being, after the manner of gravity; nor has it so uniform a persistent being for itself that it can be referred to a uniformly represented substratum after the manner of light and heat. On the contrary! When one tries to hold fast the essence of electricity in particular as something universal for itself, be it in the aether or in some other unknown fluid, it goes together with the bodies concerned; and when one then will hold it fast in the several bodies, it frees itself and takes on the stamp of the universal.

The enigmatic is increased still further by the unrest and tension that hereby arises between representation and thought. The physicist is not content to observe the phenomena and abstract the laws; he feels a need to form for himself a representation of what the inwardly operative really is. But the inwardly operative is essence, and an essence does not let itself be represented. What is magnetism? A peculiar force in certain bodies! This explanation satisfies neither representation nor thought. Representation is not in a position to hold fast what is peculiar in the force, for no force lets itself be represented straight off. Nor can what is peculiar in the magnetic force be held fast by thought; thought demands something universally valid and necessary, but this force is precisely so peculiar that no one has any inkling whence it comes, why just iron ore and iron, in preference to so many other bodies, should be endowed with the magnetic force.

Just as untrustworthy are the representations that have been formed of electricity with thought's support. Dufay assumed that there were two kinds of electricity or electric fluids, whereas Franklin maintained that there was only one kind; and, as it is said, "the

last century's dispute between the Franklinists or Unitarians and the Dualists cannot yet be regarded as finally decided."

Not even by keeping to the laws can the observer secure for himself any clear representation of the common operative cause. By analogy with magnetism one assumes that the electricity e at one point must, at another point at the distance r , produce a distributing force $K = \mu \frac{e}{r^2}$. By experiment it can further be shown that two small bodies with the quantities of electricity e and e' must at the distance r repel one another with a force $F = \mu \frac{ee'}{r^2}$. If now μ be assumed identical in both expressions, one would suppose that the equation for the moving force F and the equation for the distributing force K must point to one and the same fundamental force, one and the same common cause; but an acute physicist does not let himself be dazzled by a merely formal likeness.*)

Over against everything enigmatic and uncertain in these hypotheses and representations, the philosophy of nature has to lay down two wholly certain fundamental propositions: 1) *all bodies, however peculiar they may be in properties and combinations, are realised power-concepts, so that every utterance of force is a natural utterance of concept;* 2) *the power-concepts' expression in existence makes itself specially known in the laws, for the universal's going together into unity with the individual is to be seen only in the fulfilment of the laws.* The relation between physics and the philosophy of nature will then, with regard to these two fundamental propositions, be determined thus: physics starts from the facts and discovers the laws; the philosophy of nature starts from the discovered laws and develops the concepts of nature.

The articulating functions thus acquire natural-philosophical significance in that they lead, step by step, to an ever deeper insight into the individualisation of the bodily fundamental relations. It will then be possible to see how the polar opposition shows itself first more outwardly and immediately in *magnetism*, next more inwardly reflected in *electricity*, and at last unfolds its doubly articulating reciprocal action in *electromagnetism*.

a) *Magnetism.* An attraction according to the law of gravity presupposes at least two members, two masses or parts of mass; nevertheless there is in this no proper articulation; the one mass is as the other, and every difference in properties, even in magnitude, is irrelevant to the essence of gravity taken universally. It is otherwise with the attraction between two magnets; each of them has its poles, the unlike poles attract, the like repel

*) "It has been the view hitherto prevailing that electric distribution . . . is due to the electricities' mutual attractions and repulsions, and that these latter pass over into the corresponding mechanical forces between the electrified bodies themselves, when the electricity is bound to the bodies by surrounding insulators. The first assumption is an analogy drawn from our acquaintance with bodies' mutual attractions and repulsions; but this is hardly fit to be carried through any further, partly because the distributing force at a point appears independently of the electricity present at that point, partly also because the effects of this force are not an acceleration of the electricity's motion at that point, but must be measured by the quantity of electricity flowing through. But even if this representation be nevertheless held fast, these forces cannot, because the electricity meets a resistance, be regarded without more ado as the cause of a mechanical pressure upon the body in a definite direction, for this "resistance" is nothing else than a greater specific conductive resistance". L. Lorenz. Lectures on Natural Science. Copenhagen 1870. pp. 192–193.

one another; attraction is conditioned by repulsion, and conversely; thus pole works against pole, member against member: these are member-forming functions. In the poles the one mass is not merely outside the other, but the otherness is accentuated twice over, the one pole is qualitatively opposed to the other; the south pole has its determinate other in the north pole, and just for that reason the attraction is mutual; the south pole, on the other hand, has only a formal other in another south pole; uniformity is at variance with the principle of articulation, and the repulsion is mutual. How originally the opposed members presuppose one another is seen in the fact that one can indeed procure magnets with many poles, but not a magnet that has only one pole. The polar opposition is grounded in qualified unity; the unity is a unity of opposition expressly determined by articulation.

Seen immediately, it looks as though the magnetic force sat in each of the magnet's poles, while in the so-called *space of equilibrium* there were no such force; but already Gilbert has shown that the articulation goes all the way through. By breaking a bar magnet across the middle one does not get two half magnets, each with its pole, but two whole ones, each with two poles. The part is a complete whole, and the division can be continued down to the smallest particulars: the articulated whole consists of nothing but articulated parts (small magnets, elementary magnets).

The thoroughgoing articulation opens us an insight into the connexion of the whole; for the reason why the operative poles form themselves toward the end-faces lies in this, that the small magnets in the space of equilibrium equalise and cancel one another's effects, whereby there arises a peculiar opposition, conditioned by the circumstances, between bound and free magnetism.*)

If one will now say that the magnetic functions do indeed presuppose an articulation but are not themselves articulating, then we must for the present content ourselves with referring to what physics calls "the distributing force". A piece of iron becomes, by contact with a magnetic pole, itself a magnet that can carry small pieces of iron. The distributing force is doubtless as yet only a superficial representation, in so far as it immediately presupposes what it really ought to explain, namely the polarity of the elementary magnets; none the less it does cast a peculiar light upon its own presupposition and discloses a secret of cohesion.

For here there shows itself a peculiar relation between the mobility of the small parts and the fixity of the mass, a relation more closely determined by the *coercive force*. Steel is magnetised with more difficulty than soft iron, but then holds the faster to its

*"If we conceive that in a steel bar all the small parts had become equally strongly magnetic under the influence of the same magnetising force, with their poles turned alike and in the bar's lengthwise direction, then the poles of the free magnetism would lie in the bar's end-faces, since the everywhere equally great magnetic forces within these would have to cancel one another's outward effects entirely. This magnetic state cannot, however, subsist, on account of the elementary magnets' distributing effects upon one another, whereby there arise alterations in the inner magnetism, which consist partly in a weakening, and this must then be greatest in the bar's longitudinal axis, partly in a strengthening, which becomes greatest at the middle of the bar." Lorenz. Work cited. pp. 144-45.

magnetism; the reason is that the coercive force is greater in steel than in iron. Changes of temperature, shocks, vibrations and the like weaken the coercive force, in that they alter the cohesion.

The distribution that a magnet produces, especially in soft iron, always works back strengtheningly upon the magnet's inner magnetism, but now strengtheningly, now weakeningly upon its free magnetism — again a modification of articulating activity.

Magnetism's mechanical effects recall in certain respects those of gravity. Two magnetic poles, in which the quantity of free magnetism is m and m^1 , will at a distance r act upon one another with a force K that can be determined by the equation $K = \frac{mm^1}{r^2}$; and yet, as indicated above, the magnetic attraction is qualitatively different from that of gravity; a steel needle, for example, does not become heavier because it is magnetised. If the influence of the repelling pole be reckoned in, then the whole effect of a magnetic bar will, on particularly given conditions, prove to stand in inverse ratio to the third power of the distance.*)

In that the magnetic needle points, as it is said, toward the north, it points at the same time to a magnetism that does not let itself be explained by immediately existing small magnets, namely to terrestrial magnetism. It is a leap from the particular over into the total. Totality-magnetism points from many sides to articulating activity of diverse kinds, a widely ramified system of articulating functions. The old notion of magnetic mountains in the neighbourhood of the earth's poles has long since been given up; the earth is magnetic because it is unceasingly being magnetised. The periodic changes, both the daily and the secular, to which the needle is subject, partly in respect of declination, partly of inclination, are, like the changes in the strength (intensity) of terrestrial magnetism, so many testimonies that what is constant in the magnetising activity is united in a peculiar way with what is variable. What is immediate in the phenomena has, in the essence itself, a thoroughgoing reflexion of power for its presupposition. Provisionally it looks as though the whole system of magnetic forces turned within the earth from east to west.

A closer elucidation of the possible ground of the magnetic phenomena can, in relation to the present standpoint of physics, first be given when magnetism is brought into connexion with another and still more deeply intervening system of articulating functions.

b) Electricity. The analogy between magnetism and electricity is striking on account of the polar opposition; but as regards the inwardly operative, the likeness must rather seem to belong to the phenomenal than to the essence. For magnetism is, as "force" and "property", bound to the magnetic body, whereas electricity "can pass freely over from

*) Hansteen. Untersuchungen über den Magnetismus der Erde. Christiania 1819. — On Gauss's formula $K = \frac{mm^1}{r^p}$ cf. Lorenz. Wk. cit. pp. 152–56. A. Wüllner. Lehrbuch der Experimentalphysik. 2ter Band. Leipzig 1866. Wirkung magnetischer Massen auf einander aus der Ferne. pp. 540–58; Versuche von Gauss. pp. 558–64.

the one body to the other”, and in this passage shows itself analogous to heat. As regards conduction there is so remarkable a connexion between electricity and heat, that for the metals as well as for their alloys the same ratios have been found between their conductive power for heat as between their electric conductive power. If heat had poles, it would lie near at hand to conceive electricity as a polarised heat; but since no fact points to there being a positive heat-pole that attracted a negative one and repelled another positive one, since by bound and liberated heat something quite other is understood than by bound and free electricity, since heat, in other words, is only quantitatively, not qualitatively different in itself, it is seen from this that there must be a fundamental difference between electricity and heat.

It is characteristic to see how heat and electricity, however inwardly they may be united, yet in an essential sense each play their own part in the world of bodies. Out from its invisible, universal ground heat moves through the manifoldly different bodies, dissolves the differences, so far as it is able, into the uniform, effaces the bodily individualities and sets its highest in transforming the whole into vapour. In opposition to this, electricity must be said to be laid out precisely upon asserting the differences by force of something common, upon articulating what is proper and making the bodily individualities good. Heat can indeed be converted into electricity, and electricity into heat; but this by no means signifies that electricity and heat are one and the same; it is, on the contrary, an intimation that the difference is preserved in the unity. To the development of the unity of difference there corresponds the activity of form. When the common is posited, form demands the particular; and when the particular is posited, it demands the common again. Thus heat and electricity work, each in its own way, in form's service. And yet it is only the extremes of form's movement, not the substantial activity of form itself, that we as yet have before us. Electricity has indeed for itself a universally pervading essentiality after the manner of heat; for that glass-electricity is called positive and resin-electricity negative by no means says that the electricities in the one body should be of another nature than in the other; on the contrary! the same two different kinds of electricity are found in every body, save that in the body's so-called natural state they cancel one another: the negative electricity is in all bodies like itself as negative, the positive as positive. The prelude activity of form shows itself in this, that the difference between the universal and the individual is immediate just as much as the unity is. From the standpoint of the universal it must indeed look as though electricity were a certain common something that ran through the bodies, spread itself upon the surfaces according to their different form, passed from the one body over to the other, and so forth; but if we look more closely, all this points to activities and changes in the bodies concerned themselves. Electricity does not reduce bodies to mere members of transition, in order everywhere to set itself in peculiar relations to itself alone; it is just as much the bodies that, by means of electricity, set themselves in a peculiar relation to one another.

How deeply electricity is implicated in the peculiarity of the different bodies can already be seen from the manner in which it is awakened. It is awakened by friction; but friction is precisely what makes the bodily differences recognisable; it is a significant expression of the difference between the glass and the resin, that the two bodies, on being rubbed with the same stuff, wool or silk, become, the first positive, the second negative. It is awakened by distribution. If a body A be charged for example with positive electricity it will, across an insulating interval, act upon another body B in such a way that by distribution there arises in B negative electricity on those parts of the surface that are nearest to A (influence-electricity of the first kind) but in the surface furthest from A, on the contrary, positive electricity (influence-electricity of the second kind). To this there correspond again the stronger or weaker attraction, for since the distribution is greatest in the good conductors, the attraction to these must also be strongest. The peculiar difference, determined by electricity, between good, half-good and bad conductors, *insulators*, which is so significant for articulation, heat will not merely alter but at higher degrees even efface: one knows no body that at white heat remains an insulator. But when heat works not inhibitingly but conditioningly, the articulation is so pronounced that even the mere contact of two different bodies is enough to awaken the electricity. How qualitatively peculiar the opposition is can be seen from a series such as: zinc, lead, iron, copper, silver, gold, platinum, in which every preceding member becomes positively electric by contact with one that follows. The electric difference between zinc and copper and the electric difference between copper and platinum are together equal to the electric difference between zinc and platinum; or: when one lays a copper plate upon a zinc plate and upon this a platinum plate, the electric tension in the end plates is just as great as if one had laid the platinum plate and the zinc plate immediately upon one another. The peculiarity, tension and prickliness that bodies display in the polar opposition has led Hegel to inculcate with real emphasis what idealism, even in the very newest time, is still so glad to persuade itself of, namely that the elementary bodies are at bottom self-beings. "Reflexion," he says, "is accustomed to conceive the body-individual as something dead, which only comes into outward mechanical contact or enters into a chemical relation. — We, on the contrary, conceive the electrical tension as the body's own selfishness, which is physical totality and asserts itself in contact with another. It is its own wrath, the body's own effervescence, that we see; there is no one present but itself, least of all a foreign matter. Its youthful courage breaks out, it rears up on its hind legs; its physical nature gathers itself against entering into dealings with another, and that as an abstract ideality of light. Not only do we compare bodies, but they compare themselves and therein assert themselves as physical; it is a beginning of the organic," and so forth. — It is the eternal mania of the philosophy of immanence: if one cannot make Spirit into matter, then one must absolutely be able to make matter into Spirit; if one cannot explain the conscious will from mechanical attraction, then one must explain mechanical attraction from an

unconscious will; if one cannot derive the self from the selfless, then there is nothing else for it but to make the elementary bodies, the selfless individuals, into self-beings. The attempt is equally ingenious, whether it be “the physical totalities” or the naked atoms, the pure “force-points”, that are made into self-beings.

But if there is to be earnest with selfhood, then there must also be earnest with otherness; if Spirit is to be Spirit, then one had better let matter be matter and take the physical bodies for what in fact they are, lifeless, selfless realities. The peculiar union of the universally bodily with what is proper to bodies, which electricity displays, gives clearly to be known that both the bodies themselves and the fundamental parts of which they consist must have a selfless independence. For it is precisely the mark of the selfless that all utterances of a given body’s peculiarity can be exactly determined and measured by influences proceeding from other bodies, so that action and reaction, determined by an exact relation between inertia and force, correspond to one another down to the very smallest particulars. The mechanical laws are in the strictest sense laws of otherness; but the various exact measurements of electricity’s distributing force, its density, tension, current-strength and so forth put it beyond all doubt that the effects that are electric in the special sense are in every respect subject to the laws of otherness, just as much as the mechanical^{*)}.

By discharging a Leyden jar through the wire windings of a multiplier one gets an electric current, which however ceases instantly; by the voltaic pile (and the galvanic batteries) one gets a perennial current. Volta compared his pile with an extraordinarily large but weakly charged battery of Leyden jars that charged itself again instantly when it had been discharged: the electricity in the Leyden jar and the electricity in the pile is therefore essentially the same electricity. None the less the mode of operation is different^{**)}.

The different modes of operation do not lead straight off to an understanding of the inner how, but to the discovery of different laws and thereby to modes of representation which, with the laws’ help, must be brought into mutual agreement, reduced to fixed universal presuppositions and by degrees made clear in such a unity of connexion that they yield real moments for an ever richer development of electricity’s peculiarly articulated concept of nature: reduction and deduction condition one another mutually.

Of the current and of what flows one can doubtless form various representations; for although it is given that in the conducting wire there is + E at the pile’s positive

^{*)}In t seconds the quantity of electricity that has passed through the conductor at the current-strength s will be $e = st$; that is a law of otherness; Ohm’s law is a law of otherness, and so on.

^{**)“}When a Leyden jar is discharged, it is not only everywhere the same quantity of electricity that passes through the conductor; the quantity of electricity is also the same whether the conducting wire be short or long It stands otherwise with the electricity that a galvanic element produces; as a rule one needs only to close it with a tangent galvanometer in order to observe the effect upon the magnetic needle of the far more considerable quantity of electricity that such an element sends through the conducting wire each second; but if one inserts a long metal wire into the circuit, the galvanometer shows that the current-strength has thereby been diminished to a noticeable degree.” L. Lorenz. Lectures. pp. 213–14.

and $\div E$ at its negative pole; although it may be assumed that the current must be a double current, namely a motion of the opposite electricities in opposite directions: still the inner how. the unceasing conversion of positive into negative and of negative into positive, of repulsion into attraction and attraction into repulsion throughout the whole motion, is thereby in no way made clear^{*)}. The only decisive yield for the philosophy of nature is: that matter shows itself articulated down to the smallest parts; that the thoroughgoing articulation rests upon a punctually determined co-operation, fixed by the real concept, of the bodily individual with something universally bodily; that it is this co-operation which, in the form of articulation, secures to the elementary bodies, the selfless beings, an independence that in all relations, qualitative as well as quantitative, is independent.

c) *Electromagnetism.* The relation between the two kinds of articulating functions, the magnetic and the electric, is again determined by an articulation. It is the first impulse toward a further determination of this articulating reciprocal relation that gives H. C. Ørsted's discovery its world-historical significance.

"The likeness that in several respects obtains between electricity and magnetism had already long led physicists to surmise that these two forces must stand in a closer relation to one another. Yet all attempts to demonstrate such a relation were in vain, until Ørsted in 1820 showed that an electric current in the neighbourhood of a magnetic needle brings the latter out of its position. For if one leads an electric current from south to north beneath a magnetic needle, its north end will make a deflection toward the east,

^{*)}How Weber has sought to explain the electrodynamic forces by starting from the mutual repulsions and attractions of electricities at rest is an intricate investigation belonging to the special science. Only one point shall be particularly brought out. Against Helmholtz's objection that in Weber's formula there is a contradiction between the law of electric force and the law of the conservation of force, Weber has urged in his reply that the more closely determined principle of the conservation of energy finds application to two elements only when the potential of both is of like form. For two electric elements at the distance r , with a constant c , the potential is:

$$V = \frac{ee^1}{r} \left(\frac{1}{cc} \cdot \frac{dr^2}{dt^2} - 1 \right);$$

for two ponderable small parts it is, on the other hand:

$$V = \frac{mm^1}{r}.$$

If one now disregards the sign, one needs to ascribe to c not an infinitely great but only a very great value in order to obtain the same potential for the ponderable as for the imponderable elements. (For if the law of attraction be designated by $f(r) = \frac{\mu}{r^2}$, then, with respect to the potential, $\int f(r)dr = \varphi(r) = -\frac{\mu}{r}$; cf. C. Ramus. *Analytic Mechanics*. pp. 53–54; pp. 59–60). The point is that Newton's law of gravitation, so far as it is confirmed by facts, should be able to come under Weber's law. Cf. Zöllner, *Ueber die Natur der Cometen*. Vorrede. p. LXIV ff.; the piece: *Ueber die allgemeinen Eigenschaften der Materie*. pp. 313–41.

If a law for electricity has now really been found, or can be found, that takes Newton's law up into itself, then there will doubtless also be opened to the philosophy of nature a deeper insight into the inner connexion of the laws and thereby of the real concepts; but the fundamental relation between nature and Spirit, between object-power, real concept and comprehending self-power, will be and will remain unchanged the same.

and after some oscillations it will come to rest in a position that approaches the more nearly to being perpendicular to the magnetic meridian, and therefore to the current's direction, the stronger the current is. If the magnetic needle be brought beneath the current, the north end will make a deflection toward the west. By a counterweight one can bring a dipping needle to lie horizontal; if such a one be placed on the eastern side of the current, the north end will give a deflection downward; if it be placed on the western side, the north end will make a deflection upward. If the current's direction be reversed, the deflections go in the opposite direction, and the south end is always driven in the direction opposite to the north end.”^{*)}

It now became, as discovery followed upon discovery, gradually clear that “the electromagnetic force” was in its phenomenon a single member belonging to a widely embracing chain. Thus Ampère showed that in the relation between two current-conductors there was “an electrodynamic force” which uttered itself now in a repulsion, now in an attraction, from which it followed that two parallel current-conductors must attract one another when the currents go in the same direction, but repel one another when they go in opposite directions. He went further and demonstrated that two current-conducting spirals attract and repel one another like two magnets. He went further still, in seeking to prove that magnetism was really a peculiar phenomenon of electricity.

A screw-thread of copper through which an electric current is led acts in altogether the same way as a magnet of equal size. One can now regard a bar magnet as a bundle of linear bar magnets, which however, on account of their mutual influences, are not quite rectilinear but somewhat bowed toward the magnet's axis; these linear magnets are then again to be regarded as a series of elementary magnets of unequal strength. According to Ampère's explanation every elementary magnet will then behave as a circular electric current whose plane is perpendicular to the elementary magnets' axis, and the linear magnet will represent a series of such currents. Such a system of current-conductors Ampère called a solenoid, and the explanation is: that *every magnet can be regarded as a system of solenoids*.

If it could now merely be assumed that electric currents could maintain themselves, and that even in good conductors, without either giving off or receiving any work, then magnetism would at the same time be deduced from electricity; but the assumption is unreasonable. The investigation must then be carried on from another starting point.

By the help of strong electromagnets Faraday has instituted experiments upon the magnetic behaviour of different bodies and found that all bodies are in greater or lesser degree acted upon by magnetism, yet with this difference, that some are attracted, others — “the diamagnetic” — are on the contrary repelled by a strong magnetic pole. “The magnetic distribution in the diamagnetic bodies grows, at weaker magnetising forces, proportionally with these; but if these exceed a certain limit, the distribution approaches a certain maximum. This is therefore the same as takes place with the magnetic bodies;

^{*)}C. Holten. The Doctrine of Nature's Universal Laws. Copenhagen 1870. pp. 318–19.

but these latter reach this limit more quickly, and it may therefore happen that bodies which contain a mixture of magnetic and diamagnetic substances, for example charcoal or ferruginous bismuth, show themselves magnetic at weaker magnetising forces but diamagnetic at stronger ones, since the effect of the diamagnetic substance now gets the preponderance. Among the most strongly diamagnetic bodies belong bismuth, phosphorus and antimony. Water and quicksilver show themselves diamagnetic, whereas solutions of salts containing magnetic metals are as a rule magnetic. Oxygen shows itself to be magnetic; most other gases are on the contrary diamagnetic. All bodies are, according to Becquerel's assumption, more or less magnetic, and if a body shows itself diamagnetic, that is only because it is surrounded by a more magnetic medium. If one thus encloses a dilute solution of ferrous sulphate in a glass tube, then this tube is magnetic in air, but diamagnetic when it is held immersed in a concentrated solution of the same salt, because this solution is more strongly magnetic. — Since, however, a body in a vacuum can also behave diamagnetically, one is accordingly compelled also to assume that a magnetic distribution can take place in a vacuum.”^{*)} This last circumstance sets a question mark to the explanation instead of a full stop. If Ampère's theory be now set beside the great result of Faraday's investigations, then it is seen that magnetism has ceased to be anything particular: the magnetic articulation is in its kind just as thoroughgoing as the electric.

The ground both of the magnetic and of the electric polarity is essentially to be sought in the original relation between what is real in object-power and what is ideal in self-power. That the articulation cannot be explained from object-power, that is, from the material existence of things alone, is easy to see. If we begin with the elementary magnets, then every such magnetic fundamental part must be divided into three parts, of which the one is a north pole, the second a south pole and the third a neutral middle; but if we look more closely, each of these parts falls again into three parts. The part that forms the north pole is, when it is separated from the two others, itself the whole elementary magnet with north pole, south pole and neutral middle, and so on. The contradiction is that every part of the magnetic whole is to be a part existing in space, that every such part can again be divided, and that the divided part is itself a whole. The same contradiction repeats itself with electricity. In its natural state every part of the body is to have both electricities in indifferent unity. When the two electricities are one, then the body is not electric, which is to say: then electricity is not; when electricity is, then the neutral unity is not. But when actual electricity is only in the twoness, how can the possible be in the unity? Electricity's differenceless unity is precisely its impossibility! These contradictions can be solved only on the presupposition that the twoness's unity is in the real concept, and hence in power's ideality, and that the unity's twoness, precisely on that ground, comes to expression under given conditions in material existence, and hence in power's reality. This does not indeed make the mode of operation itself clear,

^{*)}Lorenz Work cited. pp. 2.7–79.

but it casts light on the essence of what is operative.

By referring what is in the phenomenon to what is operative in the essence, Ampère's theory is well fitted to illuminate the interior of the articulated totality and to lead to an explanation of terrestrial magnetism. "As is well known, a magnetic needle is acted upon by the earth as by a distant magnet, and quite in the same way the earth behaves also toward electric current-conductors. Thus a movable plane current-circuit sets itself perpendicular to the magnetic meridian, in that it behaves, with respect to the earth as well, as a magnetic plate whose north-magnetic side will turn toward the north; in the lowest part of the conductor the current will accordingly go from east to west. In the same way the current-conducting movable spiral will set itself with its axis in the magnetic meridian, and one can even get a movable part of a current-conductor to rotate by terrestrial magnetism alone. All these effects we can now ascribe with far greater probability to electric currents in the earth's interior than to really present magnetism. One has in fact already observed, in the part of the earth's surface accessible to us, now weak, now stronger electric currents, and it is therefore probable that further down in the earth's interior too there are electric currents, whose combined effects are what we observe on the surface." *)

Here, in these facts, is a peculiar interplay of forces, a remarkable alternation of the manifold and yet uniform. North pole and south pole, + E and – E in endless repetitions: it looks uniform. But what a manifoldness of qualifying activities and qualitatively determined differences comes to light upon this comparatively uniform ground! They are naturally necessary utterances of punctually determined presuppositions, regardless outbreaks of inner regard; the real ground is without regard, for the rational is nothing but regard.

That the functions of magnetic and electric polarity, with their mutual polar crossings, are not merely *telluric*, holding only for the terrestrial whole, but also *cosmic*, holding for the whole solar system; that one can speak of a magnetic storm just as well as of an electric one; that the northern lights^{*)}, the zodiacal light^{**)}, indeed even the relation of attraction between the sun and the comets^{***)} are phenomena of electric or electromagnetic origin

^{*)}Lorenz. Work cited. p. 270.

^{*)}That there must be a peculiar relation between magnetism and the northern lights is already indicated by this, that the luminous arc which it most often forms stands perpendicular to the magnetic meridian, and that the northern lights act upon magnetic needles even where they cannot be seen. Strong oscillations of the magnetic needle have been observed in Paris at the same time as the northern lights were seen in Christiania or in northern Scotland. C. Holten. *Nature's Universal Laws*. p. 285

^{**)On "Corona, Zodiacal Light, Northern Lights and the Phosphorescence of the dark ground of the heavens in its relation to the Moon" see Zöllner, Work cited. pp. 127–30.}

^{***)}On Olbers as founder of the electric theory of comets, cf. Zöllner p. 146 ff. On "the difference between gravitating and electric action at a distance upon ponderable masses" Zöllner lays down the following: "If a body stands at the same time under the influence of gravitation and of another's free electricity, then with increasing mass gravitation, but with decreasing mass electricity, will have the preponderance as moving force. For this reason the cometary nuclei, as liquid masses, stand under the influence of gravitation, the developed vapours, as aggregates of very small parts of mass, under the influence of the sun's free electricity." Wk. cit. pp. 117–120.

— all this may well in general be assumed, although physics is not yet in a position to demonstrate the causal connexions in detail. What is decisive for the real concept is that the thoroughgoing articulation yields an essential presupposition for all forming, body-shaping activity of nature.

C.

Body-Forming Functions: Chemism.

In all the utterances of articulation and polar opposition hitherto considered, the bodily members hold themselves immediately, and so far still mechanically, outside one another: tension is converted into living force, and living force into tension, heat into electricity, and electricity into work; but under all this the bodies concerned nevertheless assert their relative independence. Only in chemism does the articulating activity become so thoroughgoing that the members surrender their independence in their affinity: bodily existence is dissolved in order to be brought forth, and brought forth in order to be dissolved again. Phenomena of the polar functions' dissolving activity are seen in *electrolysis*; the reciprocal determination of dissolution and production, bound to peculiar relations of measure, upon which nature's body-forming capacity essentially rests, is made intuitable in *the electrochemical system*; the result is that nature, fruitful in real productions of form, is at bottom a great chemical laboratory, a self-working workshop for *elementary formations of bodies*.

a) *Electrolysis*. In and by articulation the natural cause has acquired a quite peculiar character, different from the mechanical and the merely dynamic. But articulation itself takes on in turn a differing stamp according as it is a magnetic, an electric or a chemical activity in which it gives itself to be known. The connexion between magnetism and electricity is so intimate that a physicist could be tempted to take magnetic polarity for a peculiarly conditioned expression of the electric, although the facts, hitherto at least, speak for a peculiar mutually corresponding co-operation in which the magnetic, so far as the comprehensive whole of nature (terrestrial magnetism) is concerned, is dependent upon the electric. The investigation now goes further, and the question becomes: *how does galvanism, particularly as streaming electricity, stand to chemism?*

From the ideal point of view — and every natural cause's reality is of course determined from within by its real concept, its ideality — different conceptions are possible. It is possible that electricity is one thing and chemical affinity another, as gravity is one thing and cohesion another. Affinity is indeed conditioned by a qualitative opposition after the manner of electricity; but the unlike electricities are found in every substance, whereas affinity's unlike members are separately existing substances. There is a free and a bound electricity, but there is not in the same sense a free and a bound affinity; one can

speak of an electric current, but not of a streaming affinity. Electric attraction can take place at a distance, chemical attraction only by immediate contact; under the equalisation of electric tension the bodies concerned hold themselves as existing members, under the equalisation of the chemical opposition the bodily members become moments for the formation of a new body. It is therefore possible that the chemical real difference has, in the system of power-concepts, another principle than the electric. But it is also possible that it is the same principle, only at a different stage of development, so that precisely what is preformed in electromagnetism becomes realised in chemism and receives its form of limitation in crystallisation. How the matter stands herewith cannot be settled by a priori speculation, but only by the categorical dictate of the facts.

What, in a voltaic pile, is the originally operative cause? Is it the electric current that is the cause of the chemical activity; or is it conversely the chemical activity that is the cause of the current; or is there here perhaps a peculiar and equal co-operation of both factors? According to Volta's assumption the mutual contact of the different metals is the sole source of the pile's electricity; Volta overlooked the chemical effects and regarded the contact. Since, however, the pile's chemical energy had shown itself in striking results, Faraday went to the opposite extreme and declared the chemical action that the fluids exercised upon the metals to be the only real cause of the electric current. Fechner has established the correctness of Volta's fundamental experiment, but Faraday has shown that galvanic currents can arise even without contact between different metals. Faraday's theory has brought with it a new terminology. The conductors at whose surfaces the chemical separation takes place are called *electrodes*; the one that is in connexion with the chain's positive pole is the *anode*, the way upward; the one that is in connexion with the negative pole is the *cathode*, the way downward; the body that lets itself be chemically separated by the current is an *electrolyte*, and the process itself an *electrolysis*.

As water is, so are all compound substances separated, when they — be it noted — conduct electricity and are at least in part in a molten or dissolved state. As regards what is immediately observable, there is no lack of exact determinations.^{*)} The electrolytic law says that the weights of the substances separated by the same quantity of electricity stand to one another as those substances' equivalent numbers.^{**)} How far can these investigations now open us an insight into the fundamental relation between galvanism and chemism?

It goes without saying that physics must from every side seek to give an account of bodies' molecular states and the possible changes these undergo. The separation of

^{*)}Lorenz. Wk. cit. pp. 223–234.

^{**)}For every part by weight of hydrogen set free between the pole plates there is dissolved in each "cell" 32.3 parts by weight of zinc. But now the weights of the chemical equivalents of hydrogen and zinc stand to one another as 12.48 to 403.32, or as 1 to 32.3. For every equivalent of hydrogen developed in the cell in which the electrolysis takes place, there must accordingly be dissolved in each of the chain's cells 1 equivalent of zinc. *The Chemical Part of the Science of Nature*, translated by C. L. Petersen. Copenhagen 1856. p. 81.

water, whose constituents can each be collected and measured in a voltameter, Grotthuss has explained in the following ingenious way. When hydrogen is combined with oxygen into water, the oxygen atoms become negatively and the hydrogen atoms positively electric through the intimate contact between the smallest parts; but on account of the uniform distribution of the small parts of oxygen and hydrogen the combination shows no free electricity. There now lie — one may represent it to oneself — in a given line a certain number of water-parts between the two poles of a galvanic chain. Each water-part consists of one oxygen and one (double) hydrogen atom. The positive pole acts upon the first water-part in such a way that the oxygen atom, which is negative, is attracted, the hydrogen atom, which is positive, on the contrary repelled. But the first water-part acts in turn upon the second, this upon the third, and so on. When now the attraction that the positive pole exercises upon the oxygen atom in the first water-part is sufficiently great, this is torn away from its hydrogen atom; but the separated hydrogen atom combines immediately thereafter with the oxygen in the second water-part; the hydrogen atom thereby set free draws in turn the oxygen from the third water-part, and so on. Throughout the whole stretch there takes place a continuing separation and composition of the water; only at the poles themselves can its constituents become free.

The separation of water by a galvanic current in a voltameter makes the voltameter itself a galvanic element. If the current be broken and the voltameter be connected immediately thereafter with a multiplier, one gets a short-lived deflection from a current in the opposite direction. That the current's direction has turned about rests on this, that the chemical activity has turned about; it is not a separation that now takes place, but a combination, a reunion of the water's constituents, oxygen and hydrogen, separated by the first current. The electric activity that the substances separated at the electrodes exercise is called polarisation, and the electrodes themselves are said to be polarised.

There can then be no doubt that galvanism and chemism stand to one another alternately as cause and effect. When therefore physics, although it finds itself in steady progress with the discovery of the laws, and although its investigations are far from completed, has already attained a manifold knowledge of the peculiar coming-forth of the phenomena, continues to assure us that it knows nothing, or as good as nothing, of the real essence either of magnetism or of electricity or of chemism: then a surmise is awakened that behind this alleged not-knowing there lies concealed a misinterpretation of the nature and essence of knowledge.

Every scientific knowledge of physical content is formed by a peculiar reciprocal action of sensation, representation and thought: to sensation there corresponds the fact, to thought the law, to representation the hypothesis. As representation, on the subjective side, stands mediating between sensation and thought, so the hypothesis, on the objective side, stands mediating between fact and law: the fact is what is sensed, the law what is thought, the hypothesis what is represented. Representation starts from what is sensed, prepares the resolution of the particulars into the universal and remodels

the whole for thought's benefit; it slips presuppositions underneath, that thought may draw consequences; in short, it unites itself with reflection and forms hypotheses. If one takes the word hypothesis in the most comprehensive sense, then one can say with truth that all physics' knowledge is in the last instance a knowledge of hypotheses. The mark of the hypothetical is then not the uncertain, but what is presupposed by representation: an unsensed fact set up behind what is sensed, and yet after the likeness of the sensible.

From this point of view we can understand that physicists hold the essence of light and heat to be more unknown than that of gravity, and the essence of magnetism and electricity in turn more unknown than that of light and heat. The cognition of the law of gravity rests upon only one universal presupposition (hypothesis), whose consequential consequences confirm themselves everywhere in the observation of the phenomena; the doctrine of light and heat, on the contrary, has a highly intricate hypothesis enclosing manifold presuppositions: the aether and its waves are for consciousness not immediately actual but merely possible, represented facts. Even if the hypothesis gradually gains in probability, even if the probability rise to a maximum that excludes all uncertainty, the hypothesis does not thereby change its nature, does not cease to be hypothesis. The highest that chemical physics can strive for is the formation of a hypothesis that explains the phenomena of magnetism, electricity and chemism in a way as many-sided and satisfying as the hypothesis of light at present explains the phenomena of light. But even if such a hypothesis could be set up, there is nevertheless a qualitative leap from the very likeliest hypothesis to a real insight into the essence of the matter: unsensed sensible facts, facts behind facts, can only be represented; but the representations must be resolved into concepts — and that is the law of all knowledge — the concepts must be thought. *b) The Electrochemical System.* The thinking in which the concepts of nature are clarified can as little dispense with representation as representation can dispense with the observation of facts. But in the concept, that is, in the logical essence of the matter, all representations determined by the facts must be resolved. This resolution is no thoughtless levelling that lets everything run out into one; it is on the contrary an ideal articulation, under which representation's elements are reflected and transformed into moments of content for the concept in question. As regards the essence in existence, the matter in fact, on the other hand, the concept's content must be separated out into representations, and what is represented exactly determined by what exists. To the subjective conversion of representation into concept and concept into representation there corresponds the objective conversion of ground into existence and existence into ground, of possibility into actuality and actuality into possibility, a conversion which nature accomplishes in a peculiar way in chemism and the chemical processes.

It is impossible to understand the objective conversion without being clear about the corresponding subjective one. Existence can be represented, but the ground cannot be represented, it must be thought; actuality can be represented, but possibility cannot be represented, it must be thought. What thought holds fast in this sense is something

more than formal determinations of connexion, combinations, relations; it is essence as essence, the essence of nature in its concept, an ideal essence of power, a substantial essence of thought. If one confuses what can merely be represented with what alone can be thought, if one supposes that by ever more ingenious combinations of bodies' invisible parts, by continued inventions of atoms, molecules and molecular states, one can work one's way forward to a penetrating cognition of the *essence* of things, then one deceives oneself; for it is as impossible to represent an essence as it is impossible to see a tone or hear a colour. The relation between the represented and the essential interior will show itself more clearly by degrees, when we mark how it has gone in the history of science with the interpretation of the chemical phenomena.

To represent to oneself facts behind facts, fundamental parts so small that not even the strongest microscope is able to make them visible, is by no means inadmissible. It is in such good agreement with the nature of things that the objections which speculative philosophy of nature has wished to raise against it betray a questionable leaning to the mystical. The mystical shows itself when thought, by a leap from possibility to actuality, skips the necessary intermediate determinations. Paracelsus assumed that the copper which came forth from "the blue vitriol" in which one had laid "metallic iron" came forth — not, as science now understands it, by a separating out from the vitriol (sulphate of copper oxide), of which it constitutes a component — but by a mode of formation that stood in essential consonance with the theological explanation of a *creatio* answering to an *annihilatio rerum*. There is in Schelling, and even in Hegel, much that is Paracelsian, save that in the latter negation and the negation of negation step into the place of *annihilatio* and *creatio*.

What is precarious in the hypothesis-forming representation is that its imaginary facts can so easily be false, and the false presuppositions settle fast as dogmas. A striking example of this is the hypothesis of phlogiston. The changes that all combustible bodies, the organic as well as the inorganic, undergo in fire, G. E. Stahl gathered under a common point of view, in that he derived combustion from a substance dwelling in all bodies, a universal combustibility-substance which he called phlogiston. Thus was formed the phlogistic theory, accepted from the latter half of the seventeenth to toward the close of the eighteenth century. The representation of phlogiston stands in close connexion with the mediaeval representation of the metals as compound substances. How the metallic calx, a designation that answers roughly to the metallic oxides of our own day, stands to phlogiston according to the theory can easily be made clear. The more phlogiston a body contains, the more combustible it is, and the less ash there is after the combustion. Carbon, which leaves behind only very little ash, is then especially rich in phlogiston. If now rusted, that is, burnt iron, the calx of iron, be heated with charcoal, the carbon will give up its phlogiston to the calx; there will thus come forth calx of iron with phlogiston, and dephlogisticated charcoal; the calx of iron will then, in its union with phlogiston, constitute the reduced metal, the pure iron, a compound substance.

Investigating science, which cannot dispense with a represented substratum, understands meanwhile with admirable shrewdness how to procure itself means for sifting the representations and testing their objective validity. This showed itself in a striking way when Lavoisier came forward with the balance and called the phlogistians to account.

If one weighs the iron before and after its combustion, one will convince oneself that the burnt mass weighs more than the unburnt. The combustion can therefore not take place by something going away; it must on the contrary rest on something being added; it consists not in a separating but in a combining. And what then is phlogiston? An arbitrary representation, irreconcilable with the nature of the matter! If phlogiston is to be the combustible, what then is the fire-sustaining? How can combustion rest upon a separating out of phlogiston, when the body increases in weight by combustion? When the adherents of the phlogistic theory had to answer these questions, they entangled themselves in contradictions and at last laid bare their weakness by resorting to such an assertion as this, that phlogiston must be a substance that made bodies lighter.*)

By combining quantitative with qualitative analysis chemistry can establish that matter maintains itself under all changes, that no creation and no annihilation takes place, that the weight of every chemical product is equal to the sum of the constituents' weights, and that every increase therefore signifies a combination, every decrease a separation. Within the circle of these fundamental determinations new representations may now form themselves for the grounding of the phenomena, new hypotheses be laid down; but they will all be subject to science's correctives, they must all pass through criticism's ordeal by fire.

The electrochemical theory, indicated by Ørsted, particularly developed by Davy and Berzelius, sets out in stoichiology to let both the qualitative and the quantitative come to full validity, the quantitative in the chemical proportions, the qualitative in the electric articulation. The mode of representation answering to this is in brief the following. As zinc and copper in contact with one another acquire opposite electricities, so, according to the electrochemical theory, the atoms of two elementary substances acquire opposite electricities. That the atoms are not in and for themselves electric, but first become so by mutual contact, is a presupposition of representation which explains the fact that

*)The phlogistic hypothesis has even been refuted by an exact mechanical proof. Tobias Meyer showed that pendulums of iron and of oxide of iron have the same time of oscillation, whereas the iron, according to the phlogistic theory a combination of calx of iron and phlogiston, ought to have less weight and greater mass than the calx of iron, and hence a greater time of oscillation; on the presupposition, of course, that the pendulums are of equal length. If the time of oscillation be called T , the moment of inertia I , the distance of the centre of gravity from the point of suspension a , and the weight P , there results:

$$T = \pi \sqrt{\frac{I}{aP}}.$$

For a pendulum with greater mass I , which contains the sum of all the parts of mass, becomes greater; for a pendulum with greater mass and less weight $\frac{I}{aP}$ consequently becomes greater, and consequently T also greater.

one and the same substance can, according as the combination's answering member is, be now positive, now negative. Thus sulphur in combination with oxygen forms the electropositive, but in combination with hydrogen the electronegative member. By the mutual contact of the fundamental parts the electricity becomes bound, so that the body is everywhere in its inner composition equally both positive and negative; but the articulation repeats itself in ever more concrete forms; for when the body produced by the combination is brought, under the requisite conditions, into chemical relation with another body, the neutrality ceases. As the elementary substances unite in pairs into binary combinations — combinations of the first order — so the binary combinations unite in turn in pairs into combinations of the second order and form *salts*, in which the one of the combinations, *the acid*, is the electronegative, the other, *the base*, the electropositive member; salts are combined in turn into double salts and so forth.. In the series of the elementary substances oxygen stands highest as the most negative, potassium lowest as the most positive substance; from this it can then be explained that an acid's character is in general the more strongly stamped the nearer its constituents lie to the negative, and a base's the more strongly the nearer its constituents lie to the positive end of the series: sulphuric acid is the strongest of all acids, potash the strongest of all bases. It is likewise in consequence of the theory, and a fine proof of the reciprocal determination of the quantitative and the qualitative, that a body which combines with oxygen in several proportions takes on now acid, now basic properties, according as the oxygen, the electronegative member, is more or less predominant. Thus $\ddot{\text{Mn}}$ is manganic acid and $\overset{\cdot\cdot\cdot}{\text{Mn}}_2$ permanganic acid, whereas $\dot{\text{Mn}}$, protoxide of manganese, and $\ddot{\text{Mn}}_2$, sesquioxide of manganese, are both bases, though of differing character. The quality of articulation is quantitative, its quantity qualitative: they are articulated relations of measure. It is finally intelligible from this that one and the same binary combination can in one composition play the part of the base, in another that of the acid. In silicate of alumina the alumina is base; in alumina-magnesia, $\overset{\cdot\cdot\cdot}{\text{Al}} \dot{\text{Mg}}$, the alumina is acid: the negative is negative only by its opposition to the positive, the acid only acid by its opposition to the base, and conversely.

Even if it be granted that the electrochemical theory at its present stage of development cannot sufficiently explain all chemical phenomena, that there is doubt about the place of individual elementary substances in the series, that chlorine for example shows itself now more positive, now again more negative than oxygen, and so forth, it must nevertheless also be granted that the system of presuppositions which the hypothesis-forming representation has here set up has a quite other scientific significance than the representation of phlogiston. But suppose even that the electrochemical theory were to develop into so comprehensive a many-sidedness that it at last took the other theories,

the theory of types^{*)}, the thermochemical theory^{**)}, spectral analysis and so forth up into itself as moments, and thus took on the stamp of a fully valid system of nature: there would yet show itself, for the hypothesis-forming representation, a limit, precisely that limit which qualitatively separates fact and essence.

In the exposition of the chemical forces' activity, the body-forming functions, the doctrine of atoms must necessarily play an essential part, from whatever point of view one otherwise regards it. An exact chemical analysis without any application whatever of atomistics may well be taken for an impossibility. But atomistics falls especially under representation. "There must," says Liebig^{*)}, "be a cause which makes the elements' coming together in other proportions impossible, which sets up an insuperable hindrance to their diminution or increase. The fixed proportions are utterances of this cause . . . the cause itself cannot be observed with the senses, and can become an object only for speculation, for the spiritual faculty of representation. In setting out," he adds, "to develop the view which has for the present become the prevailing one concerning the cause of the chemical proportions, I must beg that it not be forgotten that its falsity or truth has not in the least degree anything to do with the law itself; this last remains, as an expression of experience, always true and does not alter, however the representation of its ground may shift about". Herewith the hypothesis-forming speculation of representation is characterised in a telling way; but herewith too the limit of empirical knowledge is clearly and definitely indicated.

The philosophy of nature has now the task of making clear the relation between the represented atoms and the chemical real concepts; from this point of view the atoms are to be conceived as substantial points of function. That the atoms are in the chemical sense points of function means: 1) that their peculiar existing is determined by their peculiar acting; 2) that by their activity they both bring forth and themselves undergo a series of law-determined changes; 3) that under all changes they hold themselves unchanged, not in material immediacy, but in essential form as reflected constants.

From the first determination it follows that the real concepts let themselves be articulated by chemical analyses. Object-power's ideality, the power in the bodily essence, has its real expression in the atoms, oxygen's essence in atoms of oxygen, hydrogen's in atoms of hydrogen, water's in atoms of water. According to immediate object-power either the total concept (water's concept) must exist and the existence of the side concepts (oxygen's and hydrogen's) be cancelled, or conversely; this is power's doubly-sided expression, but in both cases the expression will rest upon the functioning atoms.

^{*)}By J. B. Dumas, who assumes that in the *organic* combinations there are certain types in which every element can be replaced by another element. The particular case in which the exchange of an equal number of equivalents takes place he calls *metalepsy*.

^{**)}J. Thomsen. Contribution to a Thermochemical System. Transactions of the Royal Danish Academy of Sciences. 5th Series. 3rd Volume. Copenhagen 1853.

^{*)}Chemical Letters. 5th Letter.

From the second determination it follows that the body formed by a chemical combination is a product of the atoms' arrangement only in so far as the arrangement is itself a conditioned effect of the atoms' nature. "With the number of the atoms that are united into a group, the direction of the attractions is multiplied; wherefore the strength of the attractions also decreases in the same degree as the manifoldness of the directions increases. A part of common salt, one of the smallest parts of cinnabar, presents a group of not more than two atoms; an atom of sugar, on the other hand, contains thirty-six, the smallest part of olive oil several hundred single atoms. In common salt the attraction utters itself in only one, in the atom of sugar in thirty-six different directions. Without anything being added or taken away, we can conceive the atom of sugar's thirty-six single atoms as ordered in a thousand different ways; with every change in the position of a single one of these atoms the compound atom ceases to be an atom of sugar, for the properties peculiar to it vary with the manner in which its atoms are arranged" (Liebig). But the manner in which the atoms are arranged must surely, under given conditions, have its ground in their mutual attraction, in their mutual disposition, in their nature and essence. In representation the cause can be separated from the conditions, but not in the concept. Sugar's concept and olive oil's concept are in the essential sense total concepts; it is in these total concepts that the ideal-real causality of the arrangement must originally be sought. The compound atoms are not aggregates — an aggregate of atoms is no atom — they are concrete points of function, newly formed atoms, for which the single atoms are substantial moments in power's expression.

From the third determination it follows that the body-forming functions become form-giving in that they effect a relative *transformation* in the ground, while they effect a corresponding *alteration* in existence.

If one denies the transformation, then there is no essential formation of body; no new body comes forth by the combination. Water, for example, is then, strictly speaking, only an illusory, not a substantial unity of body; the phenomenon of water as an immediate thing of sense is a semblance without essence: water as water is not there in actuality, but only in our imagination. It is, according to the atomistic mode of representation, not the water but the atom-groups of oxygen and hydrogen that is the thing. The elementary atoms are the things of things. Instead of a ground of things the atomist sees everywhere things of things, things within things, things behind things, the real things behind the unreal. Since now the real things are unsensed things of sense, the hypothesis-forming speculation of representation comes at last, out of sheer interest in sensation, into conflict with the senses.

If one overlooks the alteration conditioned by the arrangement, then one overlooks the articulation and loses oneself in a foggy speculative formalism. Oxygen and hydrogen then do not become, by mutual alteration, the fundamental parts of water, discrete moments in actual parts of water, but abstract side-forms of the concrete water-form; then one mediates form by form in formal infinity, but to an essential insight into any

actual, natural formation of body one does not attain.

c) *Elementary Formations of Bodies.* No formation of a body is in the physical sense to be conceived as a single effect of a single cause; every body comes forth, be it as educt or as product, from the co-operation of different causes. As it goes with the formations, so also with the transformations. In the reciprocal action of cohesion and heat by which the forms of state are determined, the mechanical, the dynamic, the magnetic, the electric and the electrochemical can take part in the most various ways. So the productions are highly unlike. What a swarm of simple and compound substances, of chemical formations, mechanical aggregates, amorphous lumps, crystalline masses and clear crystals!

Seen under the point of view of ideal causality, the elementary formations of bodies are, without regard to their apparently greater or lesser originality relatively to one another, all to be referred to a formative drive prevailing in nature, and this formative drive in turn to the fundamental will that utters itself in the system of real concepts. From the point of view of the physically real causes, on the other hand, the formations are to be conceived partly singly, partly in groups, and without exception to be referred to the forces operating in given substances under given circumstances. For a union of two such opposed points of view a third must necessarily be sought; this third is unambiguously designated by *the activity of form*. Form is the natural middle category between the determining real concepts and the corresponding material causes.

α) *Real Concepts and Forms.* Concepts *are*, but forms *become*. The concepts oxygen and hydrogen do not come to be as consequences of the formation of oxygen and hydrogen; but oxygen and hydrogen are formed in that the concepts are *realised*, and the concepts are realised in that the substances are *formed*. Water's concept does not result from the formation of water, but the formation of water results from the concept by means of the form. There are manifold forms of mixture, but there is, strictly speaking, not a single concept of mixture. Concepts can in new combinations unfold their consequences into new concepts, but they cannot be mixed. The quartz-form, the breccia-form, the gravel-form and the like are evident forms of mixture; but the concepts themselves, by which such formations are originally determined, are none the less to be conceived as fully valid categories of nature. The essence of mixture can indeed also reflect itself in a concept; but that proves just how far the concept of mixture is from being a concept of mixed kind. To mixture belongs accidentality, and accidentality too has its concept; but precisely that reflexion in which the concept of accidentality is demanded makes the accidental itself a necessary moment: the reflexion of essence cancels accidentality by positing it, and posits it as cancelled.

Only with existence does accidentality come to be, and with accidentality mixture. The form of nature, the middle category between concept and existence, has in itself this doubly-sidedness, that it is present both in the logical being of the concepts and in the real becoming of things; from the activity of form it will be understood that the accidentality cancelled in the essence can, in the elementary formations, step forth in

outwardly co-operating accidents, and that the real concepts, which in themselves permit of no mixture, are realised in mixed forms.

A naive running together of concepts and forms has only too often led the philosophy of nature astray. How great the blunders are that abstract speculation has committed in letting the activity of form fall metaphysically together with the form of body, and this in turn with the a priori determining concept of body, is well known. Plato's doctrine of ideas in particular has not seldom had to serve as a point of support for a conception, as poor in spirit as it is mystically obscure, of the divine archetypal forms' implantation in matter. (Cudworth). The plastic art of nature in the crystals has served as proof that crystallisation itself was subject to the magic of the types. From this point of view one would then have to represent to oneself that the whole manifoldness of stereometric figures which, according to crystallography's now prevailing systematics, occurs in the regular, tetragonal, hexagonal, rhombic, monoclinometric and triclinometric systems, has been from eternity in the region of the ideas; that it is these self-subsistent typical forms (τύποι, παραδείγματα) which — for the sake of the plastic art — everywhere press forward and, where it can be done, stamp their images in matter, much as a signet ring stamps its inscription in the wax or in the melted sealing-wax. Eagerly as Aristotle combats Plato's doctrine of the eternal archetypes of things, he is by no means himself free from harbouring an obscure accessory representation of the metaphysical pre-existence of the forms operative in matter. The schoolmen accordingly found no scruple in making use of the Aristotelian forms of nature as divine potentates that ordered matter according to their own demand. Thus matter was there only for form's sake and had necessarily to conform itself to the form.^{*)} The same one-sidedness prevails even in Hegel, only with this difference, that the dialectic of the concept is here so thoroughly carried through that the blunder must really strike the eye. "Form's activity," he says, "is none other than the concept's in general: to posit *the identical as different and the different as identical*. . . . To the extremes of subjectivity, which holds itself at one point, and of the fluid, which is only as continuum but in itself completely undetermined, magnetism yields *the middle*, form's abstract liberation, which in the crystal becomes a material product, as it already shows itself for example in *the ice-needle*. The activity that has passed over into its product is the shape, and determined as *crystal*. In this totality the different magnetic poles are reduced to neutrality, and the linearity of the place-determining activity is realised into the whole body's face and surface . . . Form, the one, works all this, in that it α) bounding the sphere, crystallises the body outwardly, and β) forming out the punctuality, crystallises its inner continuity in a thoroughgoing

^{*)}Non habent esse formæ propter materiam, sed magis materiæ propter formas . . . Neque igitur formæ ideo sunt diversæ, ut competant materiis diversis, sed materiæ ideo sunt diversæ, ut competant diversis formis.

Unde fit, ut qvædam formæ requirant materiam simplicem, qvædam vero materiam compositam; et secundum diversas formas diversam partium compositionem oportet esse congruentem ad speciem formæ et operationem ipsius. Thomas Aquinas.

manner in *the cleavage of the laminae*, that is, in *the nucleus-shape*.”

One sees how reflexes of material determinations of form — bounding surfaces, axes, fracture surfaces, inner and outer crystallisation — have hovered before the thinker in the development of the real concept of form; but one sees also how entirely the operative causes with their conditions are dissolved and vanished in the a priori formal analysis of the concept. The consequence of this is that elementary formations of bodies in the most outwardly concrete forms of mixture must all let themselves be derived with the same logical necessity from the forming concept: in granite, the earth’s real kernel, there is then revealed a “granite principle” with quartz, mica and felspar, and therein is recognised “the terrestrial trinity”.

β) Material Causes and Material Forms. That the physicist has from every side directed his attention to the actual forms of bodies goes without saying; but to conceive form as a substantial essentiality for itself, or to develop the universal real concept of the movements of form, does not lie in physics’ way at all. The crystal form is a physical result of a body’s passage from the fluid to the solid state. How little there can in this connexion be any talk of a peculiar principle of crystallisation is seen from this, that at the passage an amorphous or merely crystalline formation can come forth just as well as a formed-out crystal; in this respect the conditions rule. In order to bring bodies to crystallise one can employ different procedures — solution in water for example, melting and cooling, evaporation and condensation of vapours — according to the constitution of the bodies concerned; but the influence of the conditions is on the whole known by the rule that the crystals become the larger and the finer the more slowly and quietly the crystallisation takes place. This fact points in turn to a peculiar connexion of conditions and operative causes. It is the figure and magnitude of the mutually attracting atoms upon which the formation of the crystal essentially depends. Thus the material cause answers exactly to the condition, for the more slowly and quietly the crystallisation takes place, the better can the atoms get time to arrange themselves in full agreement with their inner mutual attraction.

By setting physical causes in the place of the metaphysical forms, investigating science has already attained great results. Haüy, crystallography’s real founder, started at the beginning of this century from the presupposition that to every peculiar crystal form there must answer a peculiar chemical composition. His theory of the primitive forms determined by *molécules intégrantes*, and of the lawfulness with which the layers of the molecules altered, a lawfulness from which the derived forms naturally let themselves be explained, actually led him to predict from the crystal form what analysis afterwards confirmed, a difference in the chemical composition of certain minerals; thus he distinguished heavy spar from strontian earth, diopase from emerald, epidote from actinolite, and so forth. His point of view was indeed one-sided, but the one-sidedness was in a scientific form. It is an honour to the founder that his theory let itself be exactly corrected; science went a step forward when Mitscherlich proved both that different

substances crystallise in the same form — isomorphism — and that one and the same substance crystallises in different forms — dimorphism, polymorphism.^{*)} If polymorphism can bring it about that from one substance (for example titanitic acid) several substances are formed (rutile, anatase and brookite), then isomorphism can conversely bring it about that different substances alternately form one and the same substance (garnet for example), the substantiality being essentially determined by a typical formula. These are fruitful glimpses into the atomic constitution of bodies that have opened up through this analysis of the activity of form. And yet, with all this, form as form has lost its essence's originality, its priority as that which determines from within.

Here there now shows itself this singular thing: metaphysics, which makes an essence of form, cannot explain the activity of form; physics, which gives so clear an account of the activity of form, cannot recognise form's a priori essence! The matter is, according to what has gone before, easy to clear up. As middle category between concept and existence, form — the *nature*-form, be it noted — must belong equally to both. In its unity with the concept form is ideal causality, its power an ideal power, in which material causes and conditions enter as reflected real determinations, cancelled in the reflexion of power. In its unity with existence form's real causality is dispersed and divided into a multitude of material causes and supervening conditions; form's ideal, original power disappears for the observer and loses itself in a parcelling out of determinations of power, under whose reciprocal action the actual, bodily form can first come forth as the result of outwardly conditioned activity of form. Both ways of seeing, the metaphysical and the physical, have their common ground in the fundamental medium, in the reciprocal determination of ideal and real power: nature's elementary activity of form is a presentation, articulated in matter, of Spirit's infinitely forming comprehension.

γ) The Reciprocal Action of the Elementary Forms. It is the formed substances, the existing bodies, that unceasingly transform one another. Every body has according to its essence an inward form, and so far the form-giving comes from within; but since bodily existence is outward and must necessarily have an outward form, bodies must find themselves in outward relations, and the form-giving must come from without. What is characteristic of elementary form-giving is a many-sided reciprocal determination of a from-within with a from-without, a reciprocal determination that answers in every respect to natural otherness. In mechanical form-giving — bounding forms by rubbing, cutting, breaking up, crushing and so forth — the inward is a moment for the outward; but that outwardness none the less has an inward for its substratum is evident. If a body is to take on a form by being cut by another, it must be of a peculiar constitution. The aptitude to receive the outer form rests upon cohesion, the inner form, and this in turn

^{*)}The fundamental forms for polymorphous substances belong as a rule to different systems; yet they can also belong to the same system. Titanitic acid crystallises dimorphously in the tetragonal system as rutile and anatase; but anatase's pyramids cannot be referred to rutile's, or conversely.

upon a peculiar relation to heat. In chemical form-giving the outward is a moment for the inward. The siliceous substance (silicon) and oxygen are mutually transformed into silicic acid; the mutual transformation is a consequence of their indwelling form, their interior. None the less an outward is altogether necessary for the development of this interior; the two elementary substances, each existing for itself, must by outward circumstances be brought together under such conditions that they can enter into combination. In the formation of the mass of which a granite is formed, the mechanical with its outward and the chemical with its inward have had an equal share. When it then seems as if the formation of quartz (crystallised silicic acid) were more original than the formation of granite, then a distinction must surely be made between existence and concept. Seen in the concept, there is exactly the same kind of reciprocal determination of an inward and an outward in the formation of quartz as in that of granite. From the side of existence, on the other hand, it is quite true that there can be quartz without granite, but not granite without quartz. That the more compound formations take on the stamp of the derived, while the uncompounded and less compounded look like the original ones, rests upon time and parcelling out, not upon concept and essence. The physically abstract can be derived from the concrete just as well as conversely. The coming forth of alumina, for example, may just as well be owing to a dissolution of felspar as the formation of felspar to a formation of alumina; minerals can come forth by the melting of mountain masses just as well as mountain masses by the heaping together of minerals.

There is in the essence the same relation between the real concepts and the material activity of form as between the laws and their fulfilment. In the formation of a sand-heap the laws are just as punctually fulfilled as in the formation of quartz or felspar. However wildly it may have whirled, and however much that is accidental there may be in a sand-heap's outer form and inner composition, no physicist will say that for every new accidental sand-heap a new law, accidental according to the circumstances, must have been formed; but he will say that it is the same eternal old laws that at every new formation take up the accidents they meet, master them as conditions and transform the accidentality into a moment in conditioned necessity. So too with the real concepts of form; they do not come to be, occasionally and according to circumstances, for they are in self-power's eternal comprehension; but they become realised in the course of the ages, now here, now there, according as the development of form takes place. The existential difference between the real concept sand-heap, the real concept quartz and the real concept granite reflects itself in existence as a difference of essence between accidental and necessary forms of formation; but whence does this come? Plainly from this, that in otherness's elementary world there must be a relative, actual difference between the more constant, the substratum, which expresses the necessary, and the inconstant, the single and particular, the incessantly varying, which presents the accidental. And yet necessity is accidental, and accidentality necessary: it is precisely a mark of elementary otherness that necessary and accidental are abruptly separated and abruptly fall together.

In the form of existence the necessary is antilogical just as well as the accidental, the relatively constant just as well as the variable. How the elementary functions look from this point of view must be seen more closely from the reciprocal determinations between the formed parts and the comprehending whole.

Chapter Three.

The Elemental Whole: the Body of the Earth.

A.

The Composition of the Body of the Earth.

The earthly whole, the earth-concept's totalised expression in power, encloses a compass, given in facts, of elements, physical causes, functions and elementary formations of bodies. Unfolding its rich content of masses and forms, object-power shows itself at once as power in the manifold and as unity-power; but from both sides as power in another. The unity-power that holds the body of the earth together has the existence of the parts for its actual other, comes by this otherness outside the unity, becomes to itself an other, a manifoldness-power that lets the whole subsist in outwardly independent parts. The air, the sea and the earth's crust with its fire-fluid interior are particular parts of mass of the great body. The whole is a composite; the composition, a unity in the form of otherness, is conditioned by the mutual opposition of the parts. But the parts' opposition is determined only by forms of state, elementary forms of otherness; the opposition is formal, not substantial: the whole body of the earth could, under certain conditions of pressure and temperature, be by turns a mass of air, a fire-fluid mass, a mass of water and a solid mass of earth; then the whole is as each part, and each part again as the whole. The masses opposed by the forms of state pass over into one another: the parts are totalised. Through the totalising of the parts the unity comes to light, and object-power shows itself from the opposite side as otherness-power. The other of the manifold is the unity itself, the mighty unity that lets the parts subsist in the wholeness and become moments for the articulation of the thoroughgoing whole. It is in the reciprocal action of *the comprehending parts and the articulated whole that earthly elemental nature* is to be known.

a) *Comprehending Parts.* With the body of the earth the concept of elementary corporeality becomes completely developed, not in the form of the concept but in the form of existence. The concept of corporeality has three principal sides, each of which presents corporeality whole, namely: *matter*, materialised possibility; the *body* determined by

fixed limits, materialised actuality; and the unity of possibility and actuality materialised as *mass*. Relative as this formal distinction is, it will nevertheless show itself decisive in the determination of the body of the earth's comprehending parts: *the atmosphere, the earth's crust and the sea*.

α) The Atmosphere. That the atmospheric air, regarded as mass, is an actual body just as well as the solid earth, no one will draw into doubt. None the less a clod of earth, a stone and the like make quite another palpable impression of corporeality than pure air does. If this difference rested on a subjective estimate alone, it would be without significance in science; but it has in itself an objective ground, for it depends on cohesion. It is cohesion that gives a body its mass-like unity and determines its independence: the weaker the cohesion, the less independent the body. The parts of the air — oxygen, nitrogen, carbonic acid, aqueous vapours — find themselves in mechanical mixture, and the mass of air is as good as free of cohesion. By the parts' being outside one another the mass is discrete, but by their interpenetration it is nevertheless a continuum. The form of air, the tensely fluid, discretely continuing form of otherness, makes air a materialised possibility, a unity of space and time equalising for existence, uniformly extended like space, fugitive as an element of time. A body disappears when it is dissolved into air; it disappears not merely for sight and feeling^{*)}, but it disappears in actuality itself by losing its own determining form. Air's form is a formless form, the hovering form of the being of matter, of materialised possibility, the form as well for the bodily existence that has been cancelled as for that which has not yet come to be.

As the being of matter existing in the universal, but none the less antilogical, form of possibilities, air finds itself in peculiar fundamental relations to the universally operative essentialities: gravity, light, heat and electricity. These fundamental relations are in turn specified by particular reciprocal action with the earth's other parts, mainland and sea. Under heat's influence air takes up vapours and thereby sucks in moisture; then the differences come to light: the formless matter clothes itself in form and makes its existence recognisable in phenomena and forces.

Pure, still air is in abstract generality a transparent and therefore invisible medium; but gravity, heat and moisture unite to impart to the layers such a difference that the rays of light passing through are refracted and bent — the observing astronomer must calculate the refraction. By the cooling of the vapours the being of matter announces its hidden corporeality: the invisible becomes visible in *mists and clouds*. The formative drive stirs as in dreams, the ideal causality of the activity of form, nature's indwelling fantasy, is set in motion, and the variable reflects itself antilogically in the constant. The clouds, these tossed-off pictures of matter's transformation, possibility's unripe foetuses, fugitive productions of improvised corporeality, change their shape with every moment;

^{*)}Chlorine, bromine and iodine are indeed visible as gases; but that does not concern the airy in general. One must on the whole remember that the marks of character adduced here acquire their relative significance only by particular opposition to the other forms of state.

and yet there is in their appearance, on account of the repetition of the conditions, a certain constancy: *feather-clouds, globe-clouds and trailing clouds* are typical forms for alterations and transitions.

Air big with vapour is not perfectly transparent. Under the states of transition the aqueous vapours let only the red and yellow rays pass through them; so there are produced such phenomena as the red of morning and of evening. From an optical point of view the region of clouds is on the whole to be regarded as the region of illusory formations, of shifting appearances, rich in refracting media, rich in colours, images of light and mirages; *the rainbow, mock suns, the Fata Morgana* and the like are phenomena, but not bodies.

That materialised possibility is itself a material actuality is known by its activity. The activity, mechanical in motion, is mechanically dynamic in polar tension. By indicating the atmosphere's pressure the barometer indicates the air's unequal warming, as the hygrometer indicates its unequal moisture. The unequal warming cancels the equilibrium and produces currents of air; this is what meteorology explains in detail, *the origin of the winds*. The air which on account of the strong warming in the earth's equatorial regions rises into the air lifts itself above the colder masses of air to both sides toward the poles, while the air below streams from the poles toward the equator. As the earth turns about its axis, masses of air are carried from the higher latitudes to the equator with a lesser velocity of rotation over lands that move with greater rapidity from west to east, and thereby acquire, in relation to the moving surface of the earth, a motion from east to west; thus there comes forth *the trade wind* with the regularly shifting *monsoons*. The electrically articulating activity that goes through all ponderable matter goes on peculiar conditions through the layers of air and sets them in peculiar activity. Currents in higher and lower layers cross one another; the masses of cloud set themselves against one another as positive and negative; the forces enter into tension, the discharge takes place in *the lightning* and is accompanied by the rolling of thunder.

In their general outlines the phenomena fall together with the fulfilment of the laws in such a way that the constant is predominant; but under the incessant variation of the conditions the constant passes over into the variable: the manifoldness of accidents in rules and exceptions makes the boundary of the law-bound possibilities a movable one.

For intuitive representation calm, transparent air is an incorporeal corporeality, a pure resistanceless passivity; but the forces slumber in the possibility; when the storm rises and the hurricane rages, when the lightnings crackle and the thunder rolls: then the passive has become active; then it is recognised that there are forces in possibility's womb, that the air, "the bearer of sound", has a voice to raise in earthly affairs.

β) *The Earth's Crust*. The solid form, the form of objectivity in materialised immediacy, the completely determining form for bodily actuality, is as opposed as possible to the form of air. Regarded generally, the earth's crust constitutes a solid continuum; but fixity itself conditions a breaking, whereby discreteness is actualised; the one continuum

becomes innumerable separate continua, a body of bodies, a whole in nothing but fragments. It is the hard otherness, outwardness posited in composition: unity-power and the power in the manifold are equally strong.

From this presupposition there follows a peculiar relation between the constant and the variable.

The variable is a constant; for the constituent parts are, each in its place, resting results of a completed activity of form: the great body that has come to be has its coming-to-be behind it. The widely spread combinations — combinations of silica, of clay, of lime, of talc, of iron and so forth — are found as existing; their becoming belongs to the past, to primeval time. In the infinite manifoldness of all this present, abiding, relatively constant, time is at once confirmed and cancelled. The time of the comings-to-be vanished in the unrest of the formations, in the manifold movements of the elementary transitions; now time stands still, contemplates its work and rests in its results. Such is the impression of the constant.

But the variable comes to its right just as emphatically; it comes first and immediately to light in space. Thus geognosy speaks of the occurrence of minerals, stones, strata of earth, mountain masses; and when it means the same, it is nevertheless always something else. The quartz that is *here* is in actuality another than the one that is *there*; even if the composition be the same, the existence cannot be: the same mass of quartz cannot possibly exist at the same time in two, let alone in a thousand different places. Nor is the same in the chemical combinations typical for minerals everywhere the same. Felspar is indeed always felspar; it has, like all minerals, its determinate chemical formula; but that the felspars vary, that there is a real difference between an orthoclase (potash felspar), an albite (soda felspar), a labradorite (lime felspar) and so forth, is well known.

The variable shows itself still more prominently in mechanically typical relations of mixture. Lava is indeed always lava; but there is nevertheless a considerable difference between the kinds of lava. "Etna's lava," says the geognost, "consists especially of a mixture of augite; in Vesuvius's, on the other hand, the felspar is predominant, and the augite plays a more subordinate part; in some of the older lava streams from this volcano (on Monte Somma) we find even a great quantity of leucite instead of the labradorite. The minerals named are, however, the predominant ones in the lavas of all volcanoes, though in different proportions of quantity, and according as one of them is preponderant the lava is termed *augite lava* (dolerite lava), *felspar lava* (trachyte lava) or *leucite lava*. It is true that now and then still other minerals^{*)} appear as subordinate constituents of the lavas, thus especially magnetic ironstone, which indeed is never quite wanting, mica, olivine, hornblende and so forth; but these may be regarded as unessential or as having arisen under peculiar circumstances."^{**)}

^{*)}Lava is on the whole, as may particularly be concluded from Bunsen's analyses, no determinate species of rock.

^{**)B.} Cotta. Geological Pictures, translated by C. Fogh. Copenhagen 1859. p. 127.

Still more outward is the variation in the manner in which the one species of stone is said to pass over into the other; *granite into syenite*, this in turn into *diorite*, “whose dense and schistose varieties are with difficulty distinguished from those of *diabase*, in which hornblende is replaced by augite and the felspar is either labradorite or oligoclase, wherefore these dense, schistose, porphyritic, porous or amygdaloidal varieties are often gathered under the common name of *greenstones*.” “These greenstones,” it is further said^{***}), “resemble again very much the melaphyres, *the dolerites and basalts*, in which pyroxene” (that is, augite) “forms the chief mass in combination with labradorite, olivine and magnetic ironstone, or with nepheline; but the dolerites and basalts appear already as lavas.” Such transitions, which all bear the stamp of antilogical accidentality, can be determined subjectively and yet scientifically by geognostic comparison; but so hard is the otherness, so dominant is manifoldness-power in the elementary outwardness, that the composition of the one species of stone cannot possibly be derived from the relation between the constituents of the other. That the varying formations have, each according to its conditions in its place, a fully valid cause, a sufficient ground, cannot be doubted; but the real ground has gone immediately together with its existence; the confusion of the parcelling out rules in the local causes, and the necessity is antilogical: a systematics of the formations in the logical sense there can be, but a systematics of the stones there cannot be.^{*)}

Has unity-power then given itself up for lost? By no means. It has everywhere, in the unbroken systems of composition, its material middle instances, peculiar central points for completed processes of formation, great groups of mass-like productions; in these productions the materialised universal has been co-operative, partly in passive, partly in active form. It is water, the passive-active solvent, and fire, the warm furtherer of elementary activity, that have each in its own way functioned in the service of the universal and thereby of the unity of power. There is, as regards constituents and composition, great difference between the stratified species of stone — transition limestone, Zechstein, shell limestone, Jura limestone, chalk, travertine and so forth — but they all have nevertheless a distinctive stamp of common origin, in so far as they are *neptunian*, formed, that is, by solution and deposition in the waters. In opposition to these the massive stones with their corresponding porphyries and lavas have the distinctive stamp of having passed through the process of fire, partly as older, *plutonic*, partly as younger, *volcanic* formations. As witness to a many-sided reciprocal action both between the process of water and that of fire and between the masses themselves we have the *metamorphic* species of stone.^{*)} “Exact observations comprehending great

^{***})Wk. cit’d. p. 137.

^{*)}Only from a general point of view can it be brought out that the species of rock richest in silicic acid (granite and syenite) are on the whole the older, and the more basic ones (dolerite and basalt) the younger.

^{*)}The greatest difficulty in the chemical investigations of metamorphism lies in the limitation of the means, and especially in the circumstance that a quantity of substances which, on account of the extraordinarily small quantities in which they occur, can hardly be demonstrated by the sharpest reagents,

mountain tracts have,” says A. Humboldt, “proved that eruptive masses of stone do not appear as any wild, unbound operating force. In the regions of the world most remote from one another one often sees granite, basalt or diorite exercise their transforming influence upon clay slate and dense strata of lime, as well as upon the grains of quartz in sandstone, and that in a manner alike down to the single utterances of force.”**)

A natural tension of unity-power and manifoldness-power, grounded in the strong elementary otherness, displays itself in a tense opposition between the earth’s active — fire-fluid? — interior and the solid, resting exterior. One who could survey at one glance the whole stiffened surface would surely discover that it is always, now at this place, now at that, in a trembling; these local tremblings then indicate what volcanic eruptions and earthquakes, with all the omens of subterranean storms, confirm in the strongest way, that even the firmest parts with their composition are in the power of the wholeness, subject to time and its vicissitudes, and that the earth’s crust is indeed a solid and comprehending part, but still only a part of the whole.

γ) *The Sea.* The drop-fluid element has in its form of state this peculiarity, that the matter-like possibility and the bodily actuality neutralise one another into existing mass. Air and earth form extreme members of opposition, immediate extremes; but the sea lays itself mediating between the two: it is especially the peculiarity of the form of mass, as elementary form of otherness, that allots to it the mediating role. In the cohesionless mass of air the boundary is so movable that what is becomes nothing but becoming, the actual nothing but possibility. The mass of earth, the mass of metal, of mineral, of stone, has so strong a cohesion that each piece for itself is a determinate body with fixed boundaries and existing properties. In water as water, on the other hand, corporeality has too much independence to be fugitive matter and too little independence to hold itself objectively fast as a body bounded in itself; so its characteristic form becomes a particular existing form of mass, undulating between air and earth, between matter and body.

Water evaporates and passes over into air; it freezes into ice and becomes as a part of the earth’s crust. It is water that in the masses of cloud gives air its phenomena, and it is water that in masses of snow and ice mingles with the strata of earth, loosens their connexion, tears them along with it, thaws and washes them away in dissolved masses. In mud, the thickly flowing mass, water can assimilate itself to earth, as in vapour it can assimilate itself to air; but according to its peculiar cohesion and specific gravity it holds itself between the two, resting upon the earth, akin to the air.

The sea, with its depths and currents, its relations of temperature, light and electricity, is, so to speak, a fluid stratum of earth, an embodied counter-image to the air, by whose winds it is moved, whose play of colour it borrows, whose shifting phenomena it mirrors.

none the less exercise a mighty effect in the course of the ages and sometimes appear in mighty unfoldings of mass. Carl Vogt. *Grundriss der Geologie*. Braunschweig. 1860. pp. 462–63.

**)Kosmos I. p. 224.

There are currents in the sea just as well as in the air, and there is on the great scale a certain correspondence between currents of air and currents of the sea. — In the ship's journal on his third voyage Christopher Columbus says^{*)}: "I take it for settled that the waters of the sea move from east to west, as the stars of heaven do (*las aguas van con los cielos*).” The equatorial current must in particular be regarded as a consequence of the trade winds and of the advancing tide.

The earth's crust can bulge and burst; but it has no ebb and flood: the cohesion is too strong in the body; the air can sink and rise, but not in ebb and flood: the cohesion is too weak in the fugitive matter. The tides, ebb and flood, are changes particularly characteristic of the mass as mass. The mass, which receives its form from its surroundings, is too little coherent to hold its bounding fast and withstand influences from without, and yet coherent enough to swell, gather and form itself under attractions from the moon and the sun.

Although the ocean covers about three quarters of the earth's surface, it might nevertheless, as regards the composition of the whole, seem as if the fluid element were more restricted than the solid and the aeriform; for while the air surrounds the whole exterior of the earth, and the earth's crust its whole interior, the sea must content itself with a bed assigned to it within narrower bounds. This restriction is, however, completely cancelled by the series of transformations — from water to vapour to clouds to rain and so forth — that is so characteristic of water's circulation.

Sea water is, as is well known, not a uniform unity of mass, merely a chemical combination of oxygen and hydrogen, but a concrete mass containing salt, with chlorine, sodium, magnesium and so forth.^{**)} The quantity of salt, which at all depths is about the same, exercises the most considerable influence upon the sea's physical properties. Salt water has its greatest density at -2.67°C , its freezing point at -2.55° , whereas fresh water has its greatest density at 4.5°C and its freezing point at 0° — an essential difference with regard to upper and lower layers during freezing. That the form of mass rules, that the constituents, however different they may be in other respects, flow away into the dissolving, neutralising unity of mass, is seen from this, that the differences which physics demonstrates in all the intermediate layers between the surface and the deep essentially concern heat and cold, density, pressure, motion and rest, and thus yield, not particular determinations of objectivity in particular bodies, but nothing but predicates of mass, constant-variable fundamental relations, material universal determinations in undulating transitions.

It goes without saying that all the parts of the body of the earth must themselves be bodies, the air and the sea just as well as the solid earth; it goes without saying that every body must unite in itself both possibility and actuality; but in that the fundamental difference utters itself in elementary forms of otherness, corporeality unfolds its three

^{*)}A. Humboldt. Kosmos I. p. 275.

^{**)}G. Forchhammer. On the Constituents of Sea Water. Copenhagen 1859.

principal sides in three different relations of possibility and actuality. Thus we have the three circles of corporeality: in the one circle actuality is dissolved into its moment of possibility, that is, in *the atmosphere*; in the second possibility is stiffened in its moment of actuality, that is, in *the earth's crust*; in the third an equilibrium of possibility and actuality is set in undulating masses for a mediating transition between air and earth; that is the circle of the waters, the tensionless yet fluid element, *the sea*.

b) The Articulated Whole. On account of the developed difference between the comprehending parts, the earthly whole has attained, not merely an articulation of the substances of matter in the abstract dynamic sense, as by magnetism and electricity, but, in consequence of the formation of bodies particularly conditioned by chemism, a concrete and actual articulation, so that the parts, uninterdependent as they are in and for themselves, appear *each in its own circle* with elementary independence. How the articulation is categorically carried through — partly by the composition's own relations, in *the opposition of land, sea and air*; partly by a union of telluric and cosmic relations, in *the zones of the earth and the climates*; partly, finally, by those individualisations which yield the object of the description of the earth (geography) and whose manifold particulars are all enclosed by the relation between *the ocean and the great continents* — shall be touched upon in what now follows.

α) Land, Sea and Air. If from the beginning there had been perfect equilibrium between unity-power and the power in the manifold, the somewhat flattened globe of the earth would, according to what has gone before, simply have had three concentric envelopes: the solid crust, the drop-fluid sea and the sea of air. But unity-power, which makes the parts uninterdependent, and the power in the manifold, which gives them independence, find themselves in mutual tension. So the equilibrium is unquiet. The power in the whole shows itself as a movable power of connexion; it goes with heat and cold, in great and small, through fire and water, and drives the elementary activity of form forward in all directions to a many-sided development of articulating consequences.

The unity of the body of the earth's interior and its exterior is, without regard to the differing constituents, an actual unity of opposition realised in actual forms of state. Since the emphasis falls on the opposition, the solid crust shrunk by cooling is too narrow to hold the warm interior, the hot vapours with the fire-fluid masses, and whatever else there may be, within fixed, immovable bounds. So the exterior must for all its hardness give way here and there; the earth's crust bulges and is broken; something raises itself while something else sinks: mountain peaks, islands, groups of islands come to light above the mirror of the sea. But it is precisely through the manifoldness that unity's way to wholeness goes: the separation of land and sea is an articulation of the whole.

In the formal respect the articulation is a bounding, and the bounding mutual: the mainland is the sea's, and the sea in turn the mainland's boundary. In reality bodily independence still holds good over against the mass, in so far as it is originally the mainland that sets a boundary to the sea, and not conversely. And yet, remarkably

enough, the qualitative difference between body and mass is lost in the mutual bounding: the body is mass and the mass body, for the articulating power is equally divided. The sea has power to alter the bounds of the lands, as conversely the lands have power to alter those of the sea. So one speaks of masses of land just as well as of masses of water; what is common is that the masses are formed in being bounded. For the form of bounding there is indeed an inward cause, but the inward is, as regards the articulation, everywhere a moment for outward division. Elevations, fractures, subsidences hold good just as well for that part of the earth's crust which constitutes the sea bottom, and hence for reciprocal actions of the solid and the fluid in the deep, as for configurations of the mainland's surface: there are valleys and cliffs *beneath* the sea just as well as *above* it.

Mountain tracts and valley regions appear not as separate bodies but as great forms of mass. As a whole the mountain is the materialised universal in which the difference of the constituents disappears, and hence mass. As part of the whole mass of land the mountain has particular existence only in the form, that is, in that bounding by which the mass is there: the mountain is a formed mass just as well as the wave and the cloud. The inner difference between independent and un-independent, which rests essentially upon the form of state, has under the articulation drawn itself into a mass-like independence and has thereby come to expression in the form of bounding: the mountain with its fixed and the cloud with its fugitive outlines stand to one another as the constant to the variable; but the constant form of bounding is a form of mass just as well as the variable one.

Mountains have, like clouds, typical forms: *massif mountains*, *chain mountains*, *plateau mountains*, *alpine mountains* and so forth. The causes of formation are for mountains as for clouds inward-outward (with the difference, of course, that lies in the opposition of the fixed and the fugitive); for repetition of something common, common causes with common conditions, is in the fixed as in the fugitive the only way in which the universal can be expressed in elementary existence. According to the figurative form of the bounding, the mountain masses are earth-fast, petrified clouds, the cloud masses airy, hovering mountains.

As regards the fugitive masses, the form of bounding is movable, and the change rests upon transformations in which they disappear in order to come to light again; it is otherwise with the fixed masses. The more fixed the masses are, the more resistance the form of bounding offers to influence coming from without, the more does the transformation take on the stamp of destruction and disturbance. Air in uproar threatens destruction; but it is something else that is threatened; air itself is not exposed to destruction. How should that element whose essence is sheer change be able to suffer by being changed? The sea in uproar threatens destruction, but destruction does not threaten the sea. The mighty sea can swell and roar, can seethe in breakers and lay itself to rest in cool transparency: the sea in all transitions is always the same. It is the mainland, what is immediately independent, that is threatened with destruction from

all sides, from the earth's interior, from air and sea. It is by these destructions that the earthly reciprocal action between wholeness and parts displays itself in striking relations of power. The parts can be destroyed; therein is seen the impotence of manifoldness-power; but the whole cannot be destroyed: unity-power comes to the manifold's relief on account of the wholeness. The destruction of the parts is the elementary activity of form by which new parts are produced; the more different these parts are, the more richly they group themselves into characteristically opposed whole masses, the more prominent is the articulation in the whole.

β) Zones of the Earth and Climates. As a whole for itself the body of the earth has a compass of telluric relations answering to its parts, relations of reciprocal action determined exclusively by its peculiar composition. As part of a greater whole, as planet, member in the solar system, the earth finds itself in cosmic relations, and is dependent upon the sun not merely mechanically with regard to gravity, but also dynamically with particular regard to light and heat. For an outward consideration the two systems of relation can doubtless be held apart and so far distinguished *in abstracto*; but in actuality they are neither parcelled out nor pieced together; they are in the most various ways fused into concrete, operative unities.

In so far as the telluric relations are taken up into the cosmic, the articulation is designated by *the zones of the earth* (zones); in so far as the cosmic are taken up into the telluric, it is carried through in the constant-variable differences of *the climates*.

In the relation of gravitation there is reciprocal action between the earth and the sun, they attract one another mutually; but as regards light and heat the reciprocal action ceases: the earth receives without giving.^{*)} The foundation of the receiving is its double motion, the daily about its axis and the yearly in its orbit. That the earth's orbit, the ecliptic, cuts the equator at an angle of 23° 28' is of rich consequence for the distribution of light and heat. The torrid zone, the belt between the tropics on both sides of the equator; the temperate zones, the belts between the tropics and the polar circles; and finally the frigid zones about both poles as far as the polar circles, are unequally placed.

At the equator day and night have each their twelve hours; with removal from the equator the day's length alters with the season; about the poles day and night have each their season. By day the earth's surface is warmed by the sun's rays, whose effect is as the sine of the angle at which they fall; after sunset the earth radiates heat toward the space of heaven without the loss being made good: the relation between warming and cooling depends upon the relation between the length of day and of night. Meteorology has derived from this the following propositions: 1) that heat must decrease from the equator toward the poles; 2) that heat in the neighbourhood of the equator must throughout the whole year be fairly evenly distributed; 3) that the seasons stand forth the more distinctly the further one removes from the equator, since the difference between summer and

^{*)}One can indeed say that the sun also gets its share of the heat that the earth radiates into the space of the world; but so far as the sun is concerned this may be regarded as a vanishing magnitude.

winter heat grows ever greater, so that the summer even in the neighbourhood of the polar circle can still be very warm.

The heat-absorbing and heat-radiating earth stands in an active relation to everything that it receives. The telluric conditions co-operate with the cosmic and, by taking these up into their own circle, form those particular relations of temperature which, in concretions of the constant with the variable, give the climates. "If the whole surface of the earth were covered with water, or if it were formed of solid land, everywhere flat, which was everywhere of the same constitution and had at all places the same capacity to receive and give off rays of heat, then a place's relations of heat would depend only upon its geographical latitude, and all places on the same degree of latitude would have the same climate. But as it is, the effect that the sun's rays can produce is altered by manifold causes, and a place's climate depends not only upon the direction of the sun's rays but also upon the circumstances under which they act; it depends upon the form of the land and of the sea, upon the direction and height of the mountain tracts, upon the direction of the prevailing winds and so forth."^{y)}

How the matter stands in detail with land and sea climate, with the mean heat of the year and of the seasons, with the curvatures of the isotherms; why isothermes and isochimenes wind like serpents across the circles of latitude, and so forth — all this belongs under that new circle of empirical observations which Alexander Humboldt has introduced and which the science of nature restlessly carries on. Philosophy must restrict itself to an indication of established fundamental relations.

The functionary of which the power of wholeness particularly avails itself in all elementary articulations is heat. Heat works for unity in that it works for the uniform, the vaporous. So it meets resistance in the different constitution of different bodies: the manifold and many-formed stands outwardly up against unity. But heat is prepared for the resistance; it has itself introduced it. Through degrees and gradations the intensive unity of heat enters into the manifoldness of the many-formed, takes up the bodily differences and constitutes itself into *temperature*. The elemental whole in all its parts, the oppositions of land, sea and air, the differences of the climates, the vertical ones in the heights of mountains and the horizontal ones in the tracts of the earth — all have their subsistence in and by temperature. Without a thoroughgoing unity-power conditioning all activities of form, the composite whole would be only a finite piecework; but heat, this inwardly intensive and yet so widely extended unity, cancels the piecework, melts the finite into the dynamic infinity of gradations, alterations, transitions: heat in its temperature is an articulating functionary.

y) *The Ocean and the Great Continents*. The body of the earth, this great whole divided into masses of water and of land, is a realised power-concept; if such a power-concept be straightforwardly conceived as a logical universal essence and its self-development

^{y)}The Chemical Part of the Science of Nature, after Joh. Müller's *Grundriss der Physik und Meteorologie*. Translated by C. L. Petersen. Copenhagen 1856. p. 205.

presented in the light of absolute necessity, then the blunder is evident.

“It is,” says Hegel, “the concept’s business to conceive as necessarily determined what appears to sensible consciousness as accidental.” On this presupposition he supposes he can deduce “the continents” from the concept of the body of the earth. — “In general the striving directed toward a more determinate shape opposed to the sphere goes out upon the pyramidal, in that within the sphere it forms a base, a breadth, which tapers toward the other side; and thence comes the land’s falling apart (*das Zerfallen des Landes*) toward the south. But the restless, revolving current hollows out this figure by degrees from the west inward toward the east, drives and presses the solid as it were toward the east, and lets the figure swell toward the eastern side like a bent bow, so that it is curved and rounded inward on the west. On the whole the land is torn asunder into two parts, the old and the new world . . . The new world presents on the whole the undeveloped twoness — a northern and a southern part after the manner of the magnet; the old, on the contrary, the complete division into three parts, of which the one, Africa, the solid metal, the lunar, stiffens with heat, where man is dully lost in himself — the mute Spirit that does not step into consciousness; the second, Asia, is the bacchantically cometary extravagance, the wildly big with young of itself alone, the formless begetting which yet could not become master of its own middle; but the third, Europe, forms consciousness, the rational part of the earth, the equilibrium of streams and valleys and mountains, whose middle is Germany. The continents are accordingly not accidental, divided merely for convenience’ sake; there is an essential difference between them.”

There is in this conception something of a grand kind; it passes like a breath of spirit over the ocean and the great continents; but it is nevertheless evident that the deduction of “what appears to sensible consciousness as accidental” does not here reach its goal; the immanent deduction does not attain to “the rational part of the earth”, but loses itself in a “bacchantically cometary extravagance” and cannot “become master of its own middle”.

To set out upon a deduction according to which the globe’s present shape — the five continents, Asia, Europe, Africa, America, Australia, together with the division of the world-ocean into the Atlantic, the Indian and the Pacific with their intermediate seas, gulfs, indentations and so forth — should proceed with logically absolute necessity from the concept of the globe, is to set oneself a superhuman and just therefore unscientific task. The mutual bounding of the ocean and the great continents is an outward object for intuition, observation, measurement, description and delineation; but it is in truth no object for immanent logical deduction. In the articulated whole the concept of the earth is — in accordance with its present stage of development — doubtless completely realised; but precisely by the realisation the concept has gone under as concept, its logical determinations have in all continents, on all boundaries, at all coasts, in all valleys and on all mountains become antilogical. It is not the concept of the earth with its indwelling sides of essence, but the body of the earth with its great limbs, that we

immediately have before us. Here, in this factually given whole, there is not a certain sum of accidents falling outside the concept alongside a certain number of determinations of necessity falling together with the concept; but *everything is necessary because everything is accidental*: antilogical necessity is the stamp, with those ineffaceable features that object-power has set upon the composite whole, upon the unity of connexion realised in the articulation.

It is not a logical a priori deduction, but an actual recognition and demonstration of the realised concept of corporeality, that matters with such a unity of connexion. The logical in the concepts of earth, air and sea glimmers everywhere as a reflex of what is antilogical in representation; only so can the development of the concept prepare the understanding of the actualised concept. Physics has, as regards the constitution of the body of the earth in the whole as in the particular, still many observations to make and many problems to solve; that is its interest in infinity in the midst of the finite: to know and know and know — the constant in the variable, the universal in the particular, the necessity of the laws in the many and different cases; but however great the advances made, one never gets the better of the accidental, the antilogical point of the fact. The philosophy of nature follows in physics' track and controls the development of the real concepts; but however many-sidedly thoroughgoing the connexion may be, it sees nevertheless everywhere the accident and accidentality peeping forth from every single member: it is in the unity of composition not an *absolute* but a finite, *conditioned* and therefore *accidental* necessity that, with object-power's immediacy, governs the earthly whole.

c) *Earthly Nature*. In so far as the body of the earth is an outward object for observing consciousness, we have in its composition the many things of nature but not the natural unity of things — the *geognostic* standpoint. If consciousness takes up impressions of objects and phenomena into reproductive fantasy, then the idea of nature comes to light in a double gleam of objective and subjective — the *aesthetic* standpoint. Only in the scientifically cognised reciprocal determination of ideal and real causality can the idea realised in the elementary formations step forth distinctly as *nature* — the *teleological* standpoint.

α) *The Geognostic Standpoint*. If we gather all those parts of the science of nature that belong to the investigation of the shape and composition of the body of the earth under the common name of *geognosy*, then geognosy, with astronomy, physical geography, chemical physics, chemistry, mineralogy and lithology for auxiliary sciences, is so widely ramified and comprehensive a science that it may well be said to span the whole of earthly elemental nature. Can this science, then, teach us in the proper sense what nature is? No! It investigates the things of nature, but not nature. It teaches, to be sure, that every peculiar body has its peculiar nature, and that the activity under which bodies are formed, grouped, altered, is a thoroughgoing activity of nature; but nature itself in the unlike natures, the many differing activities, stands before the beholder's thought

as an unopened, enigmatic essence; the only thing known with certainty about this unknown essence is that it works according to laws. “We do not know what force is, nor what light, what heat, what magnetism, what electricity is in itself”; those are the investigators’ own words; but those who acknowledge such a not-knowing must surely also, for consistency’s sake, grant that they do not know what nature in itself is.

It is the singular thing, the at once attractive and repellent thing, about the consideration of the body of the earth, that this whole of nature, so rich in manifold formations, stands so near to us and yet so far, is so knowable and yet so unknowable. It looks as if the whole outward nature lay open to the observer’s gaze and gave itself straightforwardly as it is in its facts, as if there were nothing at all behind; and yet there is something behind, something hidden, a secret that always draws itself further back the deeper the investigator penetrates. In so far as geognostic knowledge is clear about itself and its essential limit, there is in this no disappointment at all. The things of nature are all without distinction objects for scientific cognition. There may be in the uppermost of the region of air, in the innermost of the earth, in the bosom of the mountains and in the abysses of the sea manifold things of which no one, be he physicist or philosopher, has yet dreamed; but the investigator knows that what has hitherto been unknown, however surprisingly new it may be at first glance, must nevertheless always in one way or another let itself be ordered into series of the old, the already known. Only when the boundary of experience is overstepped, when one expects of the advance of geognostic science something which by its nature it neither can nor will yield, can disappointments set in and illusion begin its play. For even if all the strata of the earth in all five continents were examined, turned over and rummaged through; nay, if the most improbable thing of all were to happen, that a way opened into the earth’s middle and the forces from the secret workshops of darkness stepped forth into the light of day: we should nevertheless not come a hair’s breadth further with nature as nature. This doubly-sidedness — that nature in itself is an inner unity of essence which indivisibly penetrates everything, and yet in actuality a divisible essence, completely present in the connexion of the parts — is a riddle which geognosy, however much the masses of knowledge be increased and however strongly insight grow in empirical compass, is unable to solve. Investigating consciousness continues to have existing nature outside itself as a completed object, an independent other, something known and akin, and yet a stranger of which it dare not say: it is flesh of my flesh and bone of my bones — it is power in another, elementary otherness.

β) The Aesthetic Standpoint. The secret bond of kinship between the beholder of nature and the nature beheld is first of all discovered by fantasy; but the discovery is coloured by the gleam of light in which consciousness at different stages of development conceives its own essence. “When one,” says Schiller^{*)}, “recalls the beautiful nature that surrounded the ancient Greeks; when one considers how familiarly this people could

^{*)}Schillers sämtliche Werke. Zwölfter Band. Stuttgart und Tübingen 1838. pp. 187–88.

live with free nature under its happy sky, how very near its mode of representation, its mode of feeling and its manners lay to that simple nature of which its poetic works are so faithful a mirror, then one must be startled to remark that one finds so few traces of the sentimental interest with which we moderns can cling to scenes and characters of nature. "A. Humboldt does not seem rightly to have understood Schiller's thought when, with regard to the passage adduced, he instructs us that these individually excellent utterances can by no means be applied to all the peoples of antiquity, that "a deep feeling for nature utters itself in the oldest poems of the Hebrews and the Indians: hence among peoples of very different, Semitic and Indo-Germanic, descent."^{**}) Schiller's fundamental thought is briefly and tellingly uttered thus: "either the poet *is* nature or he will *seek* it. The former gives us the naive, the latter the sentimental poet So long as man is still pure — not, be it understood, raw — nature, he works as an undivided sensible unity and as a harmonising whole. Sense and reason, receptive and self-active faculty, have not yet parted in their office, still less do they stand in contradiction with one another If man has entered into the state of culture, and art has laid its hand upon him, then that *sensible* harmony in him is cancelled, and can now utter itself only as *moral* unity, that is, as striving after unity. The agreement between his feeling and his thinking which in the first state *actually* obtained now exists merely *ideally*; it is no longer in him but outside him, as a thought that has first to be realised, no longer as the fact of his life."^{*)}

If the thought uttered by Schiller be given a particular direction toward the doctrine of consciousness, then it will be understood how in principle the matter stands with the Hindu's, the Greek's and the Hebrew's consideration of nature.

For the Hindu there is no essential difference between the conscious and the unconscious; subjective nature in the soul and objective nature in the world of things are one and the same essence. The opposition between the self and the selfless has not yet come to be, or, in so far as it is immediately felt, the self as the finite is reduced to being the untrue, the evil, and the true, blessedness-bringing infinity is sought in the selfless. To sink oneself in oneself is to lose oneself in the unbounded; all the particular appearances of the infinite One are intuited in fantastic symbolism as changing forms of the universal life, dreamlike revelations of what stirs obscurely in the ground of the soul; no wonder that at this stage of consciousness there can utter itself a sympathy with the enigmatic, the unconscious, which has a deceptive likeness to the ambiguous pantheism of modern romanticism.

For the Greek the main thing is the essence of man in the harmoniously totalised form of contemplable individuality. If it is now man himself who is to be regarded as the nature that is in itself completed, filled with content within its bounding, individualised, then it is also intelligible from this that the delineations of nature occurring in the Greek poetic works, however full of fidelity, life and grace they may be, yield only the background

^{**})Kosmos. II. p. 9.

^{*)}Work cited. pp. 196–98.

or at most the middle ground in those pictures whose foreground is formed by human actions, sufferings and passions.

For the Hebrew the opposition of nature and Spirit is both posited and cancelled. Nature is created; only to the Creator's praise can the Hebrew be poetically inspired for nature. The Hebrew consideration of nature in antiquity and the physical one of the present day form extremes so hard that at first glance it looks as if they were irreconcilable. The Hebrew sees in the prominent events of nature nothing but miracles, visible signs of Spirit's unconditioned self-power. The almighty will is the cause of all causes: Spirit commands, nature obeys; Spirit does everything, nature nothing. For present-day physics the causes of nature are one and all, Spirit is gone; here are laws, but no lawgiver; here is unity of connexion, but no unity of self; here is necessity, but no freedom: nature does everything, Spirit nothing.

It will, however, according to what has gone before, be possible to see that the two extremes are in the natural-philosophical sense complements of one another. In conceiving self-power as the cause of causes it must surely be expressly understood that the almighty will goes through natural intermediate causes: the real causality is indispensable to the ideal. In bringing out the causes of nature it must be expressly understood that the endless series of causes all point back to the fundamental cause, and hence to the cause of causes, self-power: the ideal causality is indispensable to the real. Not thought alone, but fantasy too makes it evident from many sides that nature is nature only by means of Spirit. *γ) The Teleological Standpoint.* The aesthetic consideration of nature is by no means so irrelevant to science as, seen in passing, it might seem. Fantasy big with thought, however capriciously it may go to work in the particular, is essentially bound to the fundamental law of objectivation; and the law says: *there is nothing objective but what is objectified.*

Selfhood and otherness, subjectivity and objectivity, the conscious and the unconscious are to all eternity correlative. Take the self-being away, and there is no essence in another; for an essence that posits and represents itself by itself is, on account of this its self-positing and self-representing, not an other-being but a self-being. Take subjectivity away and then see what becomes of the objective. An object that is in a position to objectify itself for itself and for other objects is more than object, it is a self-being, a subjectivity. Take consciousness away and consider what becomes of the unconscious! It is one of the fundamental errors of the present day that the objective should be older than the subjective, and the natural world rest upon a substratum of eternally selfless, unconscious being. How so great an error has been able to enter upon science's way and remain standing as a dark, ingrown prejudice that shades the truth can be briefly shown.

That the investigator must use his consciousness in every investigation goes without saying; but two things especially contribute to dazzle him. In the first place he must see to it that the object can in each and all be presented in its own light, so that it is neither distorted nor coloured by subjectivity; thus the opposition of consciousness and object

is sharpened to the utmost. In the second place the investigator has to do only with the objects of experience in their finite relation to human consciousness. Since now it cannot occur to any rational person to make the actual existence of objects dependent upon human consciousness and to declare that there are no visible things when we do not see them, none audible when we do not hear them, none tangible when we do not feel them, there is thereby strengthened the in itself fundamentally false representation that the sensible in things can be without any sensation, the thinkable without any thinking. At this standpoint nature is and remains a riddle.

Elementary nature is the selfless essence that is grounded in eternal selfhood, a content rich in thought but unconscious of an infinitely self-thinking consciousness. Nature as nature cannot explain itself; it is immediate in its facts, and the fact of facts is in turn the blind, ungrounded facticity. Only when it is fully recognised that nature, bearing, forming, fruitful in formations, in all unconsciousness faithfully, that is, immediately, renders those plans, thoughts and determinations of will which an infinitely mighty and rich consciousness has laid in its constitution, is it recognised for what it in truth is: the teleologically real other-being whose end is set by a seeing self.

From an exclusively geognostic point of view the reflexion of the earthly whole in all its parts is outward, empty and formal: as the whole is, so must the parts be, and conversely; further one does not come. From a teleological point of view the development of the whole becomes the end that is set, and the parts means for the end. On the side of the facts, the things of nature, nothing is hereby altered; the essential alteration concerns only the conception, the understanding, the grounding: the teleological conception must go as one with the geological.

B.

The Historical Development of the Earth.

C.

The Body of the Earth as Elemental Organism.

Part Three.

Organic Nature.

Chapter One.

The Origin of Organic Life.

A.

Life and Organism.

B.

Basic Organic Forms.

C.

The Systematics of the Basic Forms.

Chapter Two.

Organic Natures.

A.

Plant Nature.

B.

Animal Nature.

C.

Human Nature.

Chapter Three.

Conclusion: Nature and Spirit.

A.

Nature is Created by Spirit.

B.

Nature is Sustained by Spirit.

C.

Nature is Explained by Spirit.